GREEN CLIMATE FUND

Meeting of the Board
16 – 19 March 2021
Virtual meeting
Provisional agenda item 15

GCF/B.28/02/Add.07/Rev.01

4 March 2021

Consideration of funding proposals - Addendum VII

Funding proposal package for FP160

Summary

This addendum contains the following seven parts:

a) A funding proposal titled "Monrovia Metropolitan Climate Resilience Project";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat's assessment;
e) Independent Technical Advisory Panel's assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

It is noted that the following parts have been slightly modified: a) funding proposal; and d) Secretariat's assessment.

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Funding Proposal

Project/Programme title:	*Monrovia Metropolitan Climate Resilience Project*
Country(ies):	*Liberia*
Accredited Entity:	*United Nations Development Programme*
Date of first submission:	*20 December 2019*
Date of current submission	*March 1, 2021*
Version number	*[V.001]*

GREEN CLIMATE FUND

Contents

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	ProjectProject	**A.2. Public or private sector**	PublicPublic
A.3. Request for Proposals (RFP)	Not applicable Not applicable		
A.4. Result area(s)	*Check the applicable GCF result area(s) that the overall proposed project/programme targets. For each checked result area(s), indicate the estimated percentage of GCF budget devoted to it. The total of the percentages when summed should be 100%.*		
	Mitigation: Reduced emissions from:	GCF contribution:	
	☐ Energy access and power generation:	Enter number%	
	☐ Low-emission transport:	Enter number%	
	☐ Buildings, cities, industries and appliances:	Enter number%	
	☐ Forestry and land use:	Enter number%	
	Adaptation: Increased resilience of:		
	☒ Most vulnerable people, communities and regions:	53.7%	
	☐ Health and well-being, and food and water security:	Enter number%	
	☒ Infrastructure and built environment:	46.3%	
	☐ Ecosystem and ecosystem services:	Enter number%	
A.5. Expected mitigation impact	***N/A***	**A.6. Expected adaptation impact**	*Direct: 250,000 people* *Indirect: 1 million people (population of Monrovia)* *25 % of country population*
A.7. Total financing (GCF + co-finance)	25,638,905___ USD	**A.9. Project size**	Small (Upto USD 50 million)
A.8. Total GCF funding requested	17,255,755____ USD		
A.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☒ Grant ☐ Equity Enter number ☐ Loan Enter number ☐ Results-based payment Enter number ☐ Guarantee Enter number		
A.11. Implementation period	6 years	**A.12. Total lifespan**	*30*
A.13. Expected date of AE internal approval	*12/20/2019*	**A.14. ESS category**	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes ☒ No ☐

A.19. Complementarity and coherence	*Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.* Yes ☒ No ☐
A.20. Executing Entity information	The Government of Liberia (GoL) represented by the Environmental Protection Agency (EPA) will serve as Executing Entity (EE) for the project.

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

1. Liberia's capital city, Monrovia[1], is extremely vulnerable to the climate change impacts of sea-level rise (SLR) and the increasing frequency of high-intensity storms, both of which contribute to coastal erosion and shoreline retreat. SLR is a significant contributor to accelerated coastal erosion, and along with the increasing intensity of offshore storms and waves, exacerbates coastal erosion, the impacts of which result in significant damage to buildings and infrastructure in Monrovia's coastal zone. Additionally, SLR is threatening the sustainability of ecosystem services provided by mangroves in the Mesurado Wetland[2] at the centre of the Monrovia Metropolitan Area (MMA), which is further exacerbated by urban encroachment into, and over-exploitation of the mangroves. These changes negatively impact the habitat for economically important fish species and the loss of these nursery areas will have a considerable impact on the fishery-based livelihoods of approximately 55,000 Monrovians, 46% of whom are women.

2. The most vulnerable part of the MMA coast is West Point, an impoverished and densely-populated informal settlement situated on a narrow spit between the coast and the Mesurado Wetland, with dwellings built up to the shoreline. In the last decade[3], coastal erosion has caused the shoreline to regress by 30 m, leading to the loss of 670 dwellings and threatening public spaces and boat launching sites that are critical to fishery-based livelihoods. Without intervention — and with the added impact of climate change — coastal erosion is expected to cause further shoreline regression of 190 m by 2100. This is equivalent to an additional 110% more than the coastal retreat expected under a non-climate change or baseline scenario[4].

3. To adapt to the severe impacts of climate change on Monrovia's coast, it is necessary to change the current approach to addressing the impacts of climate change from a focus on short-term solutions to long-term integrated and participatory planning that involves the public sector, private sector and communities at all levels of governance. The proposed project is requesting GCF support to address barriers to effective climate change adaptation in the coastal zone of Monrovia, and Liberia more generally, through interventions in three inter-related focus areas: i) coastal protection; ii) coastal management; and iii) diversified climate-resilient livelihoods. In this way, the proposed project will build the long-term climate resilience of coastal communities in Liberia by both addressing immediate adaptation priorities and creating an enabling environment for upscaling coastal adaptation initiatives to other parts of Monrovia and Liberia.

4. The project will address one of the most urgent adaptation needs in Monrovia by constructing a rock revetment to protect West Point against coastal erosion and storms. The revetment was selected as the preferred solution, because while a 'soft solution' in the form of beach nourishment with an associated groyne was considered technically feasible, the sustainability of this option would be limited, because the regular maintenance required was not feasible in the local context[5]. From an infrastructural perspective, the project will protect and build the climate resilience of approximately 10,800 people in West Point and avoid damages of up to USD 47 million to the individual and communal property of West Point residents as well as securing launch sites for fishing boats which

[1] In this proposal, 'Monrovia' and the 'Monrovian Metropolitan Area' (MMA) are used interchangeably to refer to the jurisdictional or administrative entity of the MMA.
[2] the estuary of the Mesurado River
[3] 2008 to 2018
[4] See Annex 2.B (Vulnerability Sub-assessment) for Economic and Financial Analysis of Monrovia Metropolitan Area, and specifically West Point.
[5] Stabilising or 'fixing' the shoreline by means of a rock revetment is the preferred solution to coastal erosion at West Point by both the Government of Libera and affected communities. This approach also represents the most socially sensitive design because it requires low-to-no maintenance while still accommodating boat launching and landing. A rubble mound revetment with rock armour, which is able to withstand extreme wave conditions and storm events, is proposed. The Engineering Sub-assessment Report (Annex 2.C) showed that the northern portion of the proposed revetment is a less dynamic wave environment, and the conceptual design for this portion of the intervention site consequently proposes lighter rock armour. The 'toe' of the structure will consist of a resistant geotextile and will be anchored in the existing beach sediment to a level of 5m below mean sea-level to account for future deepening of the area directly in front of the revetment. A six-metre wide promenade, for access to the shoreline and recreation activities, is proposed between the revetment and existing dwellings at West Point. Two boat launching and landing sites are proposed as part of the preferred option at the southern end and centre of the revetment, respectively. These launch and landing sites will be provided in addition to the open beach area to the north of the proposed revetment, where fishing boats are already launching and landing. Further details on the stakeholder engagement process that led to this decision is available in Annex 2.A Feasibility Study, Section *10.2 Analysis of coastal defence options*.

will have a positive impact on the fisheries sector. The construction of this coastal protection infrastructure will form part of a strategic, cohesive coastal adaptation strategy using an Integrated Coastal Zone Management (ICZM) approach.

5. The paradigm shift necessary for adopting an evidence-based and participatory ICZM approach across Liberia will be facilitated by the proposed project through initiatives to strengthen the technical and institutional capacity of the government and communities to adapt to the rapidly changing coastal landscape and to undertake long-term, climate-responsive planning on the coast. Based on quantitative, defensible scientific data in coastal management and planning, the proposed project will develop a national-scale high-resolution multi-criteria vulnerability map and design a national ICZM Plan (ICZMP) for Liberia in consultation with all relevant stakeholders, including the private sector. By fostering partnerships among government institutions and between the Government of Liberia (GoL), private sector actors, research institutions and communities, the project will improve coordination on coastal management and create an enabling environment for ongoing coastal adaptation beyond the project area and after the project implementation period.

6. The proposed project will increase local adaptive capacity by strengthening gender- and climate-sensitive livelihoods and protecting mangroves in the Mesurado Wetland within Monrovia. Specifically, adaptative capacity in Monrovia will be increased by: i) safeguarding ecosystem services provided by mangroves and increasing the resilience of these ecosystems to climate change, through community co-management agreements between government and communities; ii) improving community knowledge on climate change impacts and adaptation practices; and iii) strengthening climate-sensitive livelihoods and supporting the uptake of climate-resilient livelihoods. This is an important element of the integrated approach because while the development of ICZMP will improve coastal management at an institutional level, limited institutional capacity in Liberia means that capacitating communities to engage positive adaptation strategies is critical to ensure an increase in their long-term climate resilience. The latter two activities will be based at the innovation and education centre — to be established in West Point. In addition to being the focal point for climate-resilient livelihood development, the innovation and education centre will act as a hub for awareness-raising and other community-led actions being implemented under the project[6]. An exit strategy and O&M plan (Annex 21) will ensure that the proposed project activities will be sustained in the long-term[7].

7. These investments by the GCF and the Government of Liberia (GoL) will catalyse a paradigm shift in the management of Monrovia's coastal zone towards an integrated, transformative and proactive approach that addresses current and anticipated climate change risks and which mixes both infrastructure (where necessary) and coastal ecosystems in adaptation efforts. This will directly benefit a total of ~250,000 people in the communities of West Point through coastal defence and enhanced livelihoods; and through enhanced livelihoods and improved protection of mangrove ecosystems in the communities of Topoe Village; Plonkor and Fiamah; and Nipay Town and Jacob's Town. In addition, the project will indirectly benefit approximately one million[8] people through the adoption of a transformative, climate risk-informed ICZM approach for Liberia, with the first phase of implementation focused on the Monrovia Metropolitan Area (MMA). The combination of direct and indirect beneficiaries under this project will ultimately confer adaptation benefits on one quarter of the total population of Liberia.

[6] Recognising the risks of the COVID-19 pandemic, all project activities will operate strictly within government mandated regulations and best practices. All government directives, such as lockdowns and mandatory quarantine will be adhered to, as will any restrictions on travel, organisation of events or sizes of meetings and workshops.
[7] Further information on the exit strategy and sustainability of the proposed project can be found in Section B.6.
[8] Direct benefits will accrue at the site-specific scale, whereas indirect benefits will accrue at the municipal scale — i.e. the population of MMA, which is estimated at one million people.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate rationale and context (max. 1000 words, approximately 2 pages)

Context

8. Liberia's capital, Monrovia, has a population of over a million people, with many of the city's most vulnerable communities located along the highly exposed coastal zone. These communities are largely reliant on fishery-related livelihoods, which has resulted in a proliferation of impoverished people settling in densely populated neighbourhoods that extend right up to the shoreline[9]. Although the combination of accessible natural resources and proximity to central markets makes neighbourhoods in the coastal zone popular for local residents, their exposure to coastal hazards such as storm surges and coastal erosion makes these densely-populated settlements extremely vulnerable to the impacts of climate change. With the physical characteristics of the shoreline offering very limited natural coastal defence, extensive erosion already causes significant damage to coastal infrastructure[10]. In addition, the limited availability of land along the coast has also led to unplanned urban expansion into the mangrove forests within the Mesurado Wetland[11]. This urban expansion, along with the harvesting of natural resources — such as fuelwood — has increased the pressure on this estuary which provides ecosystem services to coastal communities, including the provision of critical breeding habitats for economically-important fish species. These problems of baseline erosion from natural, dynamic coastal processes that already threaten coastal communities in risk-prone areas and the degradation of mangrove ecosystems, are expected to be greatly exacerbated by climate change; see below and Section 4 of Annex 2.A: Feasibility Study (FS)[12].

9. Over the last decade, accelerated coastal erosion has caused shoreline retreat of ~30 m at the Monrovian community of West Point, leading to the loss of ~670 dwellings and threatening public spaces and boat launching sites that are critical to fishery-based livelihoods. The loss of dwellings and infrastructure due to accelerated coastal erosion between 2008 and 2018 is clearly visible in Figure 1. A detailed assessment of the different factors (both climate and non-climate related) contributing to coastal erosion at West Point can be found in Sections 2.10 and 3.1 of the FS[13].

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[9] Further information on Monrovia's coastal communities' socio-economic conditions is provided in Annex 2.A: Feasibility Study, Section 1.
[10] Section 3.1 of Annex 2.A: Feasibility Study provides details of the two major impacts of climate change in Monrovia, flooding and coastal erosion resulting in shoreline retreat.
[11] The Mesurado estuary has been declared a Ramsar wetland site.
[12] Section 4.3 of Annex 2.A: Feasibility Study provides details on the impacts of climate change on mangrove ecosystems in Liberia and the Monrovia area.
[13] Section 2.10 of Annex 2.A: Feasibility Study elaborates the on biophysical factors that have resulted in coastal retreat by summarising the bathymetric, tidal and wave conditions which characterise the nearshore and offshore environment of Monrovia and Section 3.1 provides details of the two major impacts of climate change in Monrovia, flooding and coastal erosion resulting in shoreline retreat.

Figure 1. Position of the coastline at West Point and the mouth of the Mesurado Wetland in 2008 (left image and in red) and 2018 (right image and in blue).

Climate change impacts

10. Accelerated coastal erosion, resulting from a combination of sea-level rise (SLR) and an increase in the frequency of high-intensity storms, poses a substantial threat to the homes, assets and livelihoods of coastal communities in Monrovia. In addition, SLR is likely to negatively impact the mangrove ecosystems in the Mesurado Wetland, on which many livelihoods depend. An analysis of these two climate change impacts, summarised below, has informed the design of the proposed project, which will support urgent coastal adaptation and reduce vulnerability to these impacts of climate change. Modelling of additional climate variables, related to precipitation and temperature, has also informed aspects of the project design.

Coastal erosion and sea-level rise

11. Climate projections under Representative Concentration Pathway (RCP) 8.5 predict SLR (Figure 2) of 75 cm by 2100 along Liberia's coast (a conservative estimate assuming no contribution from melting ice sheets), as well as an increase in the frequency of high-intensity storms resulting in an increased offshore significant wave height[14,15]. The combined effect of these climate impacts will rapidly increase the rate of coastal erosion in Monrovia, threatening local communities and coastal infrastructure (see FS Section 3.2). In addition to increased coastal erosion, climate change will impact vulnerable coastal communities in Monrovia through: i) degradation of the

[14] The significant wave height is the average height of the highest one-third of all waves measured which is equivalent to the estimate that would be made by a visual observer at sea. The significant wave height profile along Liberia's coast is shifting towards an increase in the occurrence of all large waves. The return period of extreme storm events that historically occurred once every 100 years are projected to decrease to 1 in 40 years under RCP4.5 and 1 in 25 years under RCP8.5 by the year 2100.

[15] Sections 3.4.1–3.4.3 of Annex 2.A: Feasibility Study provide details on the impacts of climate change on the forcing mechanisms of coastal erosion; sea-level rise, wave conditions and storm conditions, respectively.

mangrove ecosystems on which their livelihoods and food security depend[16]; and ii) inundation of vital infrastructure such as boat-launch sites, dwellings and socio-economic spaces and amenities such as fish markets[17].

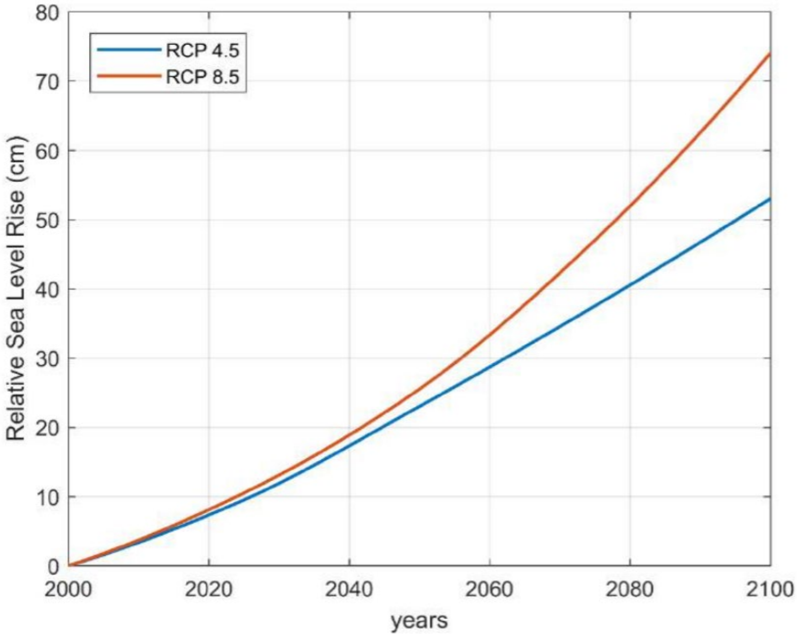


Figure 2. Projected relative sea-level rise for RCP4.5 and RCP8.5.

12. Section 3.3.2 of the FS — through modelling the impact of SLR, waves and sediment transport under future climate conditions — demonstrates that of the five most vulnerable sections of Monrovia's coastline (Figure 3), West Point (labelled Section 3)[18] is the most at risk in terms of both exposure and sensitivity[19]. In West Point, erosion induced through climate change factors (SLR, increased significant wave height, as well as changes in runoff and sediment transport) will result in coastal retreat of an additional 190 m by 2100 (Figure 4 and Figure 5). This is equivalent to an additional 110% more than the coastal retreat expected under a non-climate change, or baseline, scenario. Between 2008 and 2018, observed shoreline regression was 30 m at West Point, with further losses up to 10m occurring in early 2019. These losses are consistent with the losses modelled under the RCP8.5 climate scenario (note that expected changes under RCP4.5 are also provided in the FS) over the same time period (see FS, Section 3.6). Without intervention, an additional USD 40–48 million[20] worth of damage is anticipated to occur at West Point by 2100. Given its extreme exposure and vulnerability, West Point was selected as the main target site for coastal protection interventions under this project. Figure 4 below highlights how West Point (Section 3, orange line) is expected to experience the highest percentage of additional climate-induced coastal erosion between 2020 and 2100, while Figure 5 indicates the coastal profile at West Point under storm conditions with a return period of 100 years under RCP8.5 (see FS Section 3 for similar changes under RCP4.5) in 2100.

[16] Climate change degrades mangroves through inundation because of SLR and through changes in water temperature, salinity and sediment transport. For more information see GEF Project Document. 2016. Improve sustainability of mangrove forests and coastal mangrove areas in Liberia through protection, planning and livelihood creation – building blocks towards Liberia's marine and coastal protected areas. Available at: https://www.conservation.org/docs/default-source/gef-documents/liberia-mangroves/5712-liberia-mangroves-prodoc-20160311.pdf?sfvrsn=20715c6e_2.

[17] Further information on climate change impacts on Monrovia is provided in Annex 2.A: Feasibility Study, Sections 3 and 4.

[18] West Point is a densely populated neighbourhood situated between the ocean and the Mesurado estuary, with houses extending right onto the shore.

[19] Further information is provided in Section 4 of Annex 2.A: Feasibility Study and Chapters 4 and 5 of Annex 2.B: Vulnerability Sub-assessment.

[20] based on current value, optimistic and pessimistic socio-economic scenarios, Annex 2.B: Vulnerability Sub-assessment, Chapter 5

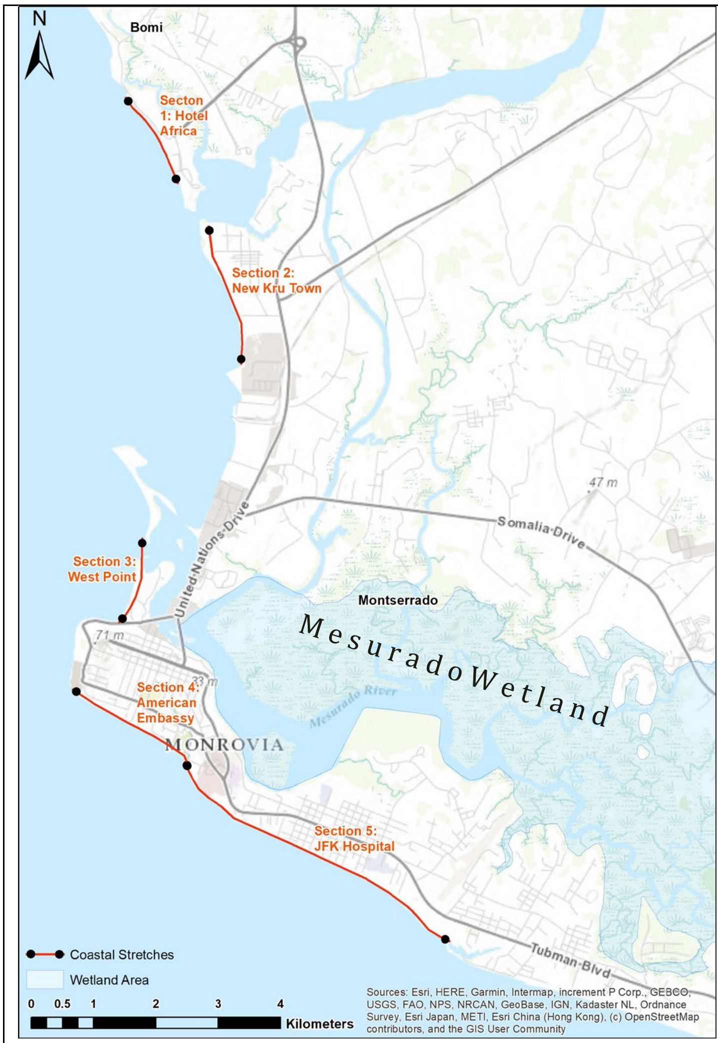


Figure 3. Map of Monrovia's coastline depicting five vulnerable areas identified under the Vulnerability Sub-assessment (Annex 2.B). The Mesurado Wetland is shown in the centre of the map.

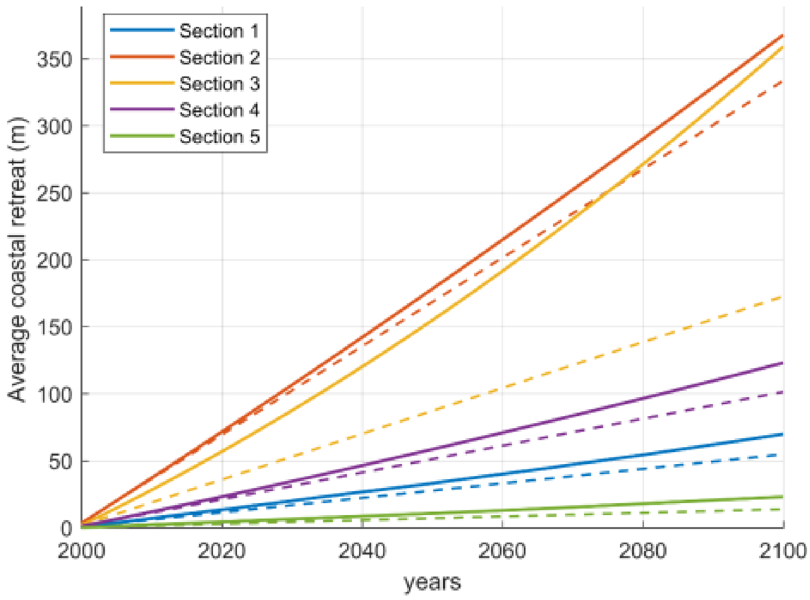


Figure 4. Average coastal retreat between 2000 and 2100 for five coastal sections in Liberia with (continuous lines) and without (dashed lines) climate change under RCP 8.5 (Annex 2.B: Vulnerability Sub-assessment). West Point is shown in orange, i.e. Section 3.

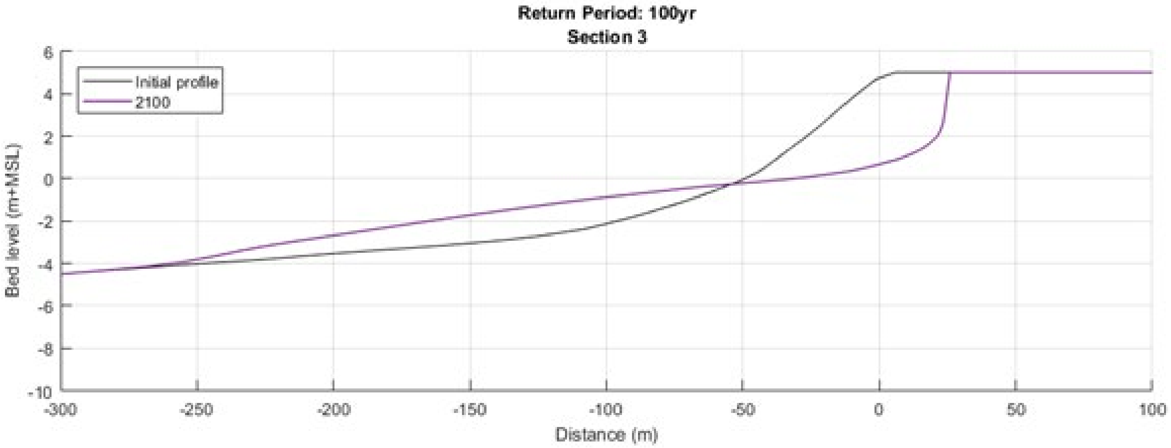


Figure 5. Modelled coastal profile for storm with return period of 100 years for RCP 8.5 in 2100 at West Point (Section 3).

13. In addition to the damage to infrastructure, climate change will also affect the livelihoods of communities in West Point and beyond, of which fishing is the most important. This makes coastal protection measures only part of the solution needed to adapt to climate change in Monrovia, with enhancing climate-resilient livelihoods being important for medium- and long-term climate resilience. Mangroves in the Mesurado Wetland play a critical role as the spawning ground for the fish stocks sustaining the local fishing industry[21], as well as in supporting other livelihood activities, including charcoal production and fish smoking[22]. Development pressure and inadequate land-use planning have, however, resulted in people settling in and around the mangroves. This, along with other anthropogenic impacts such as unsustainable harvesting of fuelwood, has led to the degradation of mangrove ecosystems in parts of the Mesurado Wetland (Figure 6)[23].

14. Annex 2.D provides more comprehensive baseline information and analysis for the mangrove components of the project. This analysis is based on the best available data, employing a combination of quantitative analysis and qualitative validation methods[24], including: i) identification of degradation trends and hotspots using time-series spatial data; ii) problems and root causes of mangrove degradation; iii) relevant past and ongoing projects; and iv) lessons learned and best practices.

[21] Mangrove systems around Monrovia are important breeding grounds for various commercially viable aquatic species, including fish, crabs, shrimps and water snail.
[22] For further details on the importance of the mangrove ecosystems in the Mesurado Wetland to Monrovians, see Section 3.1 of Annex 2.D: Mangrove Sub-assessment
[23] The results of a comprehensive spatio-temporal al analysis of mangrove degradation in the Mesurado Wetland are provided in Section 2 of Annex 2.D: Mangrove Sub-assessment while a summary is provided in Section 5 of Annex 2.A. Feasibility Study.
[24] A first draft of this mangrove-focused annex has been included as part of this revised submission as Annex 2.D. Following the analysis of secondary data, proposed project activities will be revised. The revised activities will then be presented to relevant Liberian stakeholders for validation and review. Relevant stakeholders to this end include various technical experts from inter alia Conservation International (having implemented a number of mangrove-related projects in Liberia), UNDP, and the EPA. This analysis will be used to redesign Output 3 based on the feedback received from GCF, as well as the validation described above.

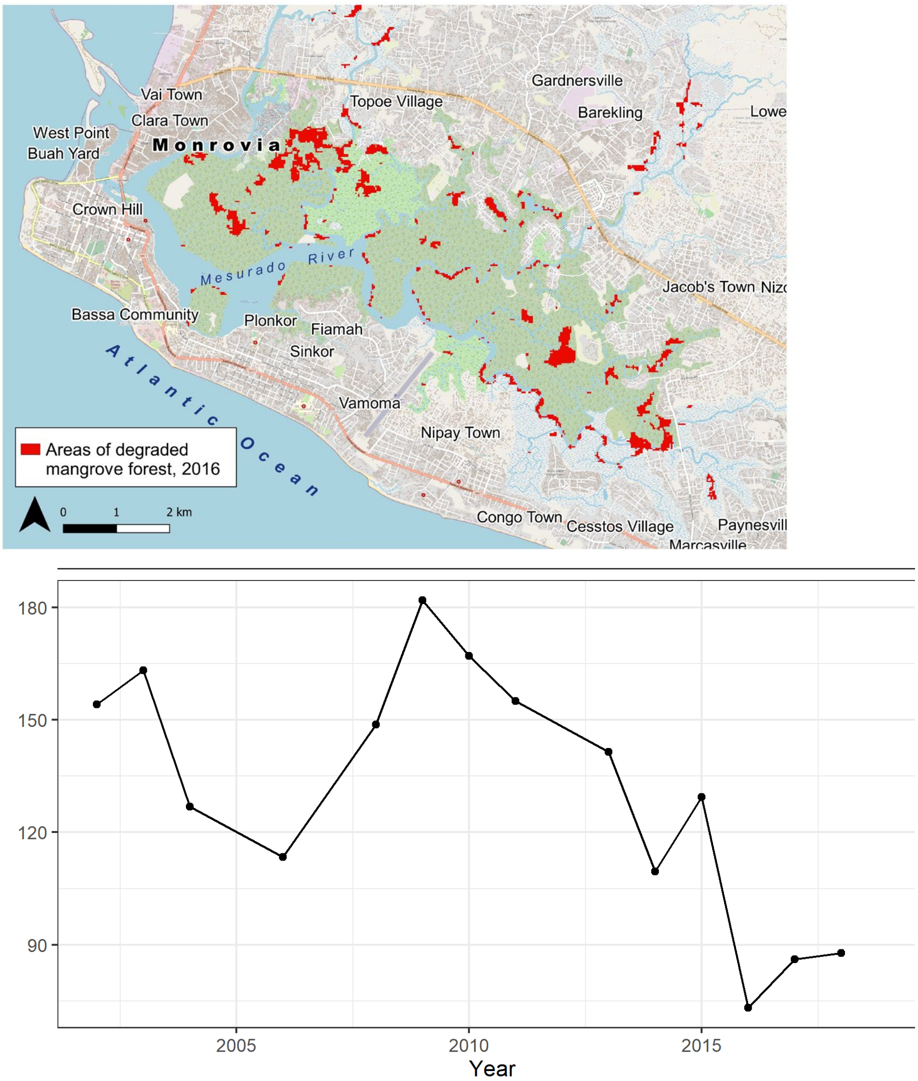


Figure 6. Map showing spatial location and graph showing loss of mangrove cover (ha) over time in areas of the Mesurado Wetlands between 2002 and 2019. Data supplied by Global Mangrove Watch.

15. Climate change is likely to exacerbate the threats of baseline degradation and compromise the ecosystem services provided by the wetlands. SLR (Fig. 2) presents a substantial threat to mangroves in the Mesurado Wetlands, as it is likely to increase the depth of the water beyond the threshold at which they can survive, thereby reducing the area in which mangrove ecosystems are viable. In addition, the increasing number of hot days projected under future climate conditions is likely to place physiological strain on the mangrove and reduce their ability to photosynthesise. Further details on the impacts of climate change on mangroves in the Mesurado Wetlands are provided in Section 4.3 of the FS (Annex 2A). As a result of these impacts on mangroves, climate change will impact the sustainability of the fishery industry, on which approximately 3,000 fisherfolk and 9,000 fishmongers in Monrovia depend for their livelihoods. Given that most fishmongers in Monrovia are women, the loss of the mangroves will disproportionately affect women and exacerbate existing gender inequality.

Figure 7. Proposed revetment intervention at Section 3 - West Point (Annex 2.C: Engineering Sub-assessment).

Temperature

16. Projections indicate that maximum temperatures are likely to increase towards the end of the century. An increase in the number of hot days (where the maximum temperature exceeds 30°C) is also expected under future climate conditions (Figure 8). Like SLR, the temperature increase is projected to have an impact on the mangrove ecosystems of the Mesurado Wetland. The optimum leaf temperature for mangrove photosynthesis is 29–32°C, while the threshold above which photosynthesis ceases is 38–40°C. Therefore, warmer temperatures may benefit mangrove growth below this threshold. Similarly, the impacts of temperature changes on fisheries and other livelihoods are likely to be mixed and dependent on the physiological thresholds of economically important species (further details are provided in Section 4 of the FS). The impacts of changing temperatures on mangrove ecosystems have been considered in the design of mangrove conservation initiatives under Output 3 of the project.

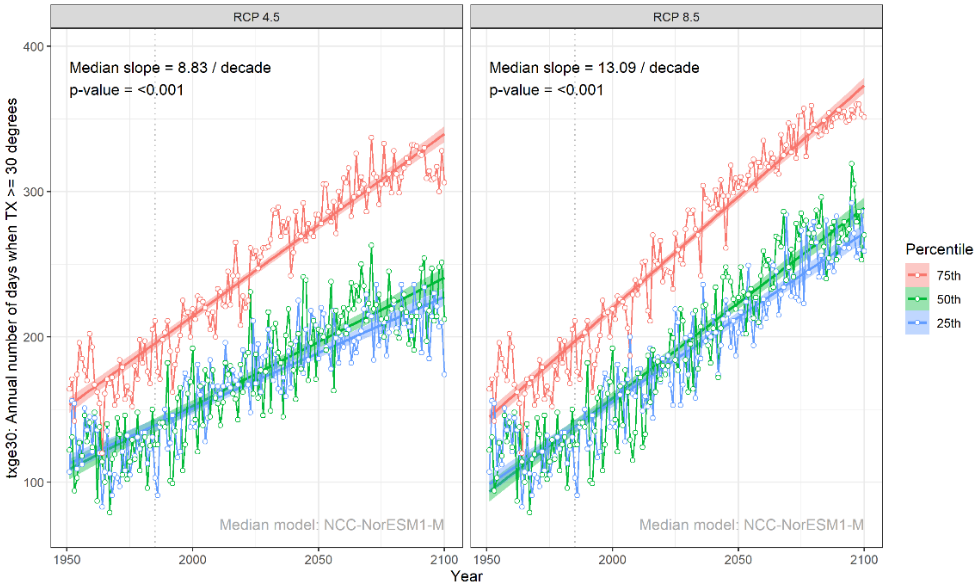


Figure 8. Annual number of hot days where maximum temperature exceeds 30°C.

Precipitation

17. An increase in the frequency of high-intensity storms is projected under future climate conditions (Figure 9). This is likely to increase coastal erosion, as described above. In addition, increased frequency of high-intensity storms will increase flood risk in parts of Monrovia and reduce the ability of fishers to safely operate by making landing sites unsafe and decreasing the number of days where fishers are able to safely operate. The design of the coastal protection measures under Output 1 (Figure 7) include drainage systems to reduce flood risk and protection of boat launching sites for fishers. Unlike the increases in temperature, increased precipitation will not necessarily have a substantial, direct impact on the mangrove ecosystems of the Mesurado wetland — as these mangroves are not wave facing and are partially protected from storm surges. Projections for overall trends in precipitation vary between models and scenarios. Further details of these projections and impacts are provided in Sections 3 and 4 of the FS.

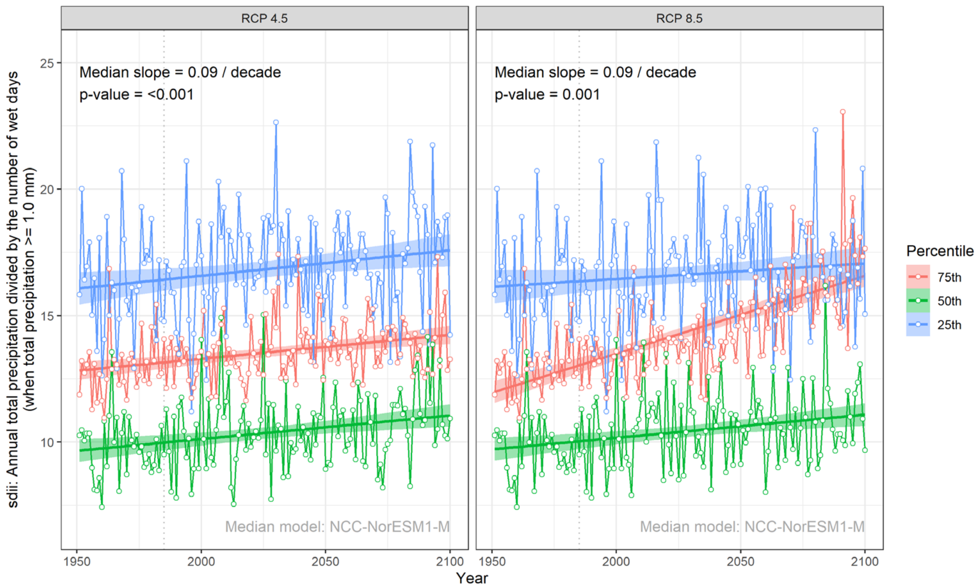


Figure 9. Annual total precipitation divided by the number of wet days where total precipitation exceeds 1.0 mm.

Related Interventions

18. The proposed project is complementary to three recent projects currently under implementation by UNDP and Conservation International (CI) in the MMA under the Global Environment Facility (GEF). Similarly, a recently approved World Bank project with comparable and complementary objectives is also included as an entry point for collaboration with organisations working in Liberia, which will promote economies of scale and improve the efficiency of both projects. The table below summarises the above-mentioned related interventions by showing key results and outcomes, lessons learned, and complementarity of the proposed project with each of the projects.

Table 1. Related projects and interventions.

Proponent/Funder/Date	**Project title**	**Key results**	**Key lessons learned**	**Complementarity/upscaling**
UNDP/Global Environment Facility (GEF)/2017	Enhancing Resilience of Liberia Montserrado County Vulnerable Coastal Areas to Climate Change Risks II	Reduced the vulnerability of physical assets and ecosystems as well as protecting coastal areas from SLR and coastal erosion in Monrovia	Robust, sustainable coastal protection measures are needed to adequately protect exposed infrastructure and vulnerable communities	The proposed project will complement the installation at New Kru Town and address the gap in spatial coverage by establishing a similar, improved intervention at West Point, drawing on the lessons learned and thereby extending the proportion of Monrovia's coastline that is robustly defended against accelerated coastal erosion and SLR.
Conservation International (CI)/GEF/2016	Improve sustainability of mangrove forests and coastal mangrove areas in Liberia through protection, planning and livelihood creation – building blocks towards Liberia's marine and coastal protected areas	Strengthen the conservation and sustainable use of mangrove forests through effective participatory land-use planning and the establishment of marine and coastal protected areas.	Community-led conservation initiatives are fundamental to measurable gains in the protection of biodiversity and ecosystem services.	While this CI-GEF Trust Fund project aims to establish protected areas in at least 35% of Liberia's mangroves, neither land-use plans, nor community conservation agreements have been established for the Mesurado mangroves. Building on the lessons learned and successes achieved by the above-mentioned GEF project, the proposed GCF project will: i) develop community-led co-management agreements for the mangroves ecosystems in Monrovia; ii) reduce the pressure on the Mesurado Wetland for natural resources by enhancing the resource-efficiency of livelihood activities that depend on the ecosystem; and iii) incorporate the management of mangrove ecosystems into the Integrated Coastal Zone Management Plan (ICZMP).
UNDP/GEF	Cross-cutting Capacity Development	Developing the capacity of the GoL to meet its global environmental obligations. One component of this project is the development of an integrated environmental knowledge	The provision of reliable, defensible information to inform decision-making is fundamental to achieving project impact	The proposed GCF project will build on the activities relating to the EKMS by: i) developing the capacity of the system to serve as an information repository for Integrated Coastal Zone Management (ICZM); ii) capacitating ten government institutions to access and use the system, including the provision of technical equipment; and iii)

		management system (EKMS) to collect, collate and disseminate information relating to environmental priorities, with an emphasis on climate change		developing knowledge products to raise awareness about the EKMS.
World Bank/IDA/2020	Liberia First Inclusive Growth Development Policy Operation	This project is designed to address weather- and climate-related risks by supporting national adaptation policies, as well as supporting social protection.	Recent national commitments emphasize the socio-economic importance of improved planning, adaptation, resilience, and enhanced disaster-risk management.	The proposed GCF project will align with the World Bank project by including key project personnel from the World Bank team in the cross-sectoral working groups (CSWGs) developed under Output 2. This arrangement will enable the sharing of relevant project information, lessons learned and promote efficient use of project resources. For example, where community and stakeholder consultation is proposed, a unified approach to consultation that includes both projects will be adopted to save costs and time.
African Development Bank (AfDB)/GCF/2021	Enhancing Climate Information Systems for resilient development in Liberia (Liberia CIS)	Strengthen Liberia's multi-hazard impact-based forecasting and early warning systems (MH-IBF-EWS) through improved meteorological and hydrological systems	Reliable, timeous and extensive data is needed for effective monitoring, forecasting and early warning for sector- and location-specific response planning and disaster management	The proposed project will utilise information generated by the monitoring systems improved under the AfDB/GCF project to inform the development of a national ICZM Plan for Liberia. Spatial and biophysical coastal data generated under the proposed project will also be incorporated into the MH-IBF-EWS developed under the AfDB/GCF project..
UNDP/GEF/ project under development	Enhancing the resilience of vulnerable coastal communities in Sinoe County of Liberia	Protect assets and enhance livelihood diversification of Liberian coastal communities through the implementation of sea and river defense management	Incorporating sea and river defense management into ICZM is necessary to ensure adequate climate change adaptation planning and addressing of climate change impacts and risks.	The proposed GCF project contributes to and aligns with the proposed GEF project interventions to increase climate change adaptation and resilience through innovative technologies and initiatives to promote the sharing of climate-related information. In addition, the proposed GCF and GEF projects both contribute to increasing the resilience of coastal communities and ecosystems from climate change-induced sea-level rise (SLR) and subsequent coastal erosion. The GEF project will focus on Sinoe County for coastal protection measures and all other coastal counties for policy and capacity building thereby upscaling initiatives under the GCF project in Monrovia.

19. In addition to the projects listed above, Liberia's approved Readiness Proposal supports the Government of Liberia to advance its National Adaptation Planning (NAP) process in climate-sensitive sectors and is underpinned by four key outputs. These outputs are summarised below, along with text demonstrating complementarity with the activities proposed under this project:

20. (A) Strengthening institutional frameworks and coordination for implementation of the NAP process: the proposed project will make a strong contribution to strengthening Liberia's adaptation-related institutions, particularly the national ministries responsible for coastal zone management, i.e. the Ministry of Mines and Energy (MME), Ministry of Finance and Development Planning (MFDP) and the Ministry of Public Works (MPW). Specifically, the project will strengthen Liberia's existing ICZM framework by working with relevant government institutions to develop a national Integrated Coastal Zone Management Plan (ICZMP) that addresses climate change risks through a participatory and consultative process. Development of the plan will be led by a service provider with experience in developing and implementing climate risk-informed ICZM. To ensure the plan is effective and implementable it will be developed through a phased approach, whereby the first phase will entail producing a high-resolution, multi-criteria vulnerability map of the entire Liberian coastline. The second phase will entail the development of a national climate-responsive ICZMP through a collaborative process led by international and national experts. The plan will be developed based on the high-resolution vulnerability map and extensive stakeholder consultations. To ensure effective implementation of the ICZMP across the relevant government sectors, an ICZM Committee and Cross-Sectoral Working Group (CSWG) will be established under this phase, with the explicit mandate of ensuring the integration of climate risks and the alignment and conformity of the ICZMP across the relevant government sectors. The CSWG will develop a detailed action plan for the implementation of the ICZMP, identifying sector-specific regulations and by-laws for updating and integrating ICZM principals. Additionally, during the second phase, the ICZMP will be implemented in Monrovia with support from the GCF grant financing of the project. The final phase of the ICZMP development will entail updating the ICZMP three years after the initial plan has been developed. This will ensure that lessons learned during the implementation process in Monrovia are incorporated into the second iteration of the ICZMP, which will facilitate the upscaling of this activity through ICZMP implementation across Liberia.

21. (B) Expansion of the knowledge base for scaling up adaptation: cross-sectoral capacity development and knowledge empowerment are fundamental informants of the activities under Output 2 of the proposed project. Specific aspects that address the need to expand the national knowledge base and create an enabling environment to scale up adaptation include, but are not limited to: strengthen the technical capacity of the CSWG established under Sub-activity 2.1.3 to implement the ICZMP through long-term training via a Training-of-Trainers (ToT) approach. To support the effective implementation of the ICZMP developed under Activity 2.1, the project will host workshops on ICZM planning and implementation to build the capacity of the CSWG and additional representatives from all relevant government institutions. As part of this process, a detailed action plan for sector-specific implementation of the ICZMP will be developed by the CSWG to facilitate the integration of ICZM into relevant by-laws and regulations. Capacitating technical personnel responsible for overseeing the uptake of implementation of ICZM across all relevant sectors will support the effective execution, management and uptake of ICZM within the MMA as well as facilitating upscaling across the country.

22. (C) Building capacity for mainstreaming climate change adaptation into planning, and budgeting processes and systems: similarly, activities within the project that promote the mainstreaming of climate change adaptation into Liberia's national processes and systems include a training and action programme for members of the CSWG to mainstream ICZM into relevant government entities and support mainstreaming of ICZM and coastal adaptation into national policies, programmes and plans. Sub-activity 2.2.5 has been designed with this need in mind: the international expert contracted under Sub-activity 2.2.3 will conduct a 16-week, phased training programme for the members of the CSWG and two additional technical personnel from each of the 10 government institutions. This long-term training programme will include instruction on the implementation of the ICZMP, important considerations for cross-sectoral collaboration and will also include training on how to conduct the course for additional members within each institution. This training programme will emphasise knowledge and tools for integrating climate risks. mainstream action plans and will support the different institutions to perform their roles and meet targets as specified in the ICZMP. This will build the capacity of institutional representatives to effectively implement the ICZMP, further capacitate government technical staff and ensure alignment between all relevant institutions,

thereby supporting cross-sectoral collaboration and the uptake of ICZM across Liberia. The international expert will also provide support to the different institutions for mainstreaming and implementing the ICZMP.

23. (D) Formulation of financing mechanisms for scaling: The project's main contribution to innovative finance mechanisms that create an enabling environment for upscaling is under Output 3, where the adaptive capacity of vulnerable communities will be strengthened and innovation enabled by supporting local fishery-based livelihoods through sustainable co-management of mangrove areas and providing training to catalyse the uptake of diversified climate-resilient livelihoods, thereby implementing ICZM at the municipal scale. Capacity development of this nature will be achieved through informing community members on how to use solar-powered cold storage facilities (Annex 2.D: Mangrove Assessment Section 6.1.1) and training those involved in small-scale informal enterprises to manufacture and sell eco-friendly products such as energy-efficient cookstoves. To this end, Activity 3.4. is particularly relevant. This activity will establish small-scale manufacturing facilities and develop training material to capacitate community members to manufacture and sell cookstoves to support alternative climate-resilient livelihoods and improve management of mangrove ecosystems[25]. The facilities will be incorporated into the education and innovation centre and will focus on producing energy-efficient products with a specific emphasis on activities suitable for women and other vulnerable groups, in collaboration with the CSC established under Activity 3.1. The focus on energy-efficient cookstoves was selected because there is a market for these stoves in Monrovia and the construction of the stoves requires minimal material input or training. Training materials will be made publicly accessible to support the upscaling of the activity to other parts of Monrovia.

24. While past and ongoing initiatives have made some progress in limiting the impact of climate change and coastal erosion on vulnerable communities, several gaps in the successful implementation of climate-responsive coastal adaptation are still prevalent in Liberia[26]. These gaps include limited: i) financial capacity of the public sector to invest in climate-resilient development; ii) technical capacity of both local- and national-level governments to interpret climate information and integrate ecosystem-based approaches into decision-making; iii) institutional and financial capacity for climate-resilient land-use planning and enforcement; iv) policy support, as well technical and financial capacity for adopting and implementing ICZM at all levels of government; v) awareness of predicted climate change impacts among coastal communities and government-level decision-makers; vi) awareness about the importance of ecosystem services for livelihoods; and vii) opportunities for alternative, climate-resilient livelihoods. Additional investment from the GCF is sought to enable interventions to address these gaps and ensure that planning and development along the Liberian coastline is integrated, climate-responsive and supports the country's adaptation priorities.

B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

Problem statement

25. The proposed project will address the extremely urgent climate change-induced problem of accelerated coastal erosion and vulnerability of the climate-sensitive fishery industry in Monrovia, as well as mitigating the ongoing degradation of mangrove ecosystems in the Mesurado Wetlands. The accelerated rate of coastal erosion is directly linked to SLR and the increasing frequency of high-intensity storms, which enhance the erosion potential of waves (Figure 10)[27]. The combined effect of SLR and changes in storm intensity is exacerbating current erosion — and expected to increase shoreline retreat along the densely-populated coastline by 110% over the next 80 years, with significant impacts in terms of damage to buildings and infrastructure. The absolute urgency of this scenario is illustrated by the loss of ~ 10m of shoreline at West Point between January and July 2020, according to anecdotal sources. In addition to accelerating coastal erosion, climate-induced SLR will lead to the loss of mangroves in the Mesurado Wetland (Annex 2.A: Feasibility Study, Section 4.3), exacerbating ongoing baseline degradation and

[25] While energy-efficient cookstoves are expected to facilitate a reduction in greenhouse gas emissions, the emission reduction co-benefits of this activity during the project lifespan are expected to be negligible. Upscaling of cookstove production after project implementation will increase the potential for emission reductions. Further detail on expected emission reductions is provided in Section B.3, paragraphs 101 and 102.

[26] Further details of the gaps summarised here are given in Annex 2.A: Feasibility Study, Section 7.

[27] Climate forcings will contribute to an accelerated sediment loss along the Monrovian coastline. Anticipated changes in the future climate conditions in the near shore environment include enhanced wave energy, continuing local sea level rise, increases in river run-off, and more destructive storm surges from enhanced storm intensity (Figure 10). These climate forcings will enhance longshore sediment transport and increase the sediment flux of rivers and basins. These activities are the most critical processes leading to the significant in coastal erosion currently present and anticipated along the Monrovian coast. A further detailed analysis of climate change and coastal erosion projections and expected impacts can be found in Annex 2.A: Feasibility Study, Section 2 as well as in Annex 2.B: Vulnerability Sub-assessment, Section 4.

impacting on local climate-sensitive livelihood activities[28]. These mangroves provide important ecosystem services, including providing breeding grounds for economically important fish species. SLR and the impacts associated with this process also threaten the entire coastline of Liberia to varying degrees. To date, interventions to address these impacts have been *ad hoc* and uncoordinated, resulting in limited effectiveness in building resilience. The preferred solution to the above-mentioned problems, described below, focuses on addressing the ad hoc and inadequately-designed nature of previous interventions to address coastal erosion in Liberia. Lessons learned concerning the benefits of adequately resourced and robustly designed infrastructure, paired with an integrated and long-term approach to coastal planning is at the heart of the preferred solution.

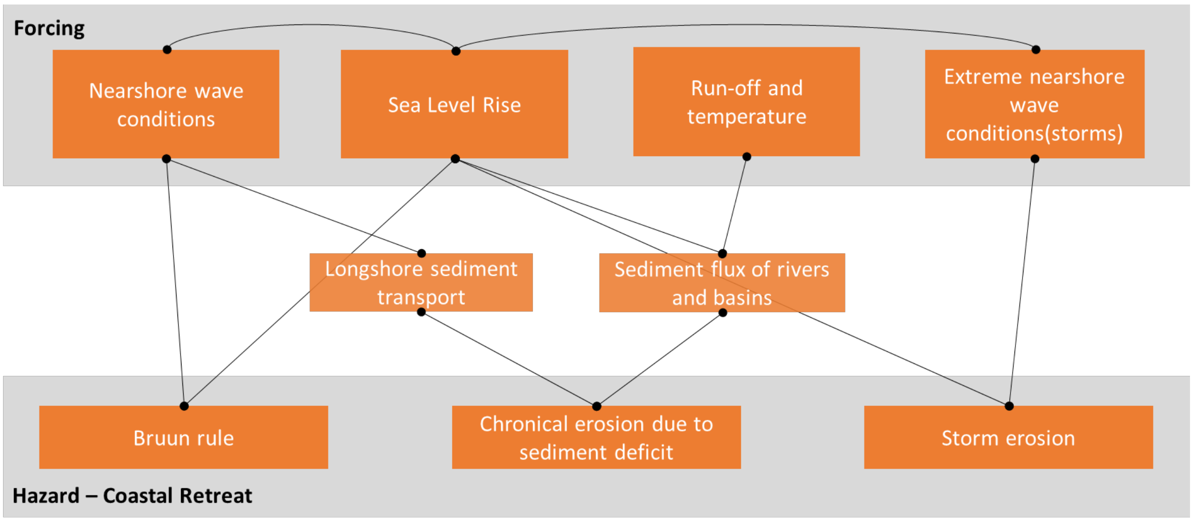


Figure 10. The interrelation between forcing mechanisms and coastal retreat.

Preferred solution

26. The preferred solution to the above-mentioned climate hazards is to: i) address the immediate needs of vulnerable communities in Monrovia by protecting highly exposed areas of the Monrovian coastline from accelerated coastal erosion and SLR and preventing the forced relocation of vulnerable peoples through the establishment of coastal defence infrastructure[29]; ii) shift the coastal management paradigm towards an integrated, evidence-based and climate-responsive approach by adopting ICZM in Monrovia and Liberia to improve coordination among government institutions as well as with communities and the private sector, and incorporate long-term considerations into adaptation efforts; iii) secure the livelihoods of vulnerable communities who rely on climate-sensitive fishing practices through sustainable co-management of mangrove ecosystems; and iv) enable community-level innovation and learning for the promotion of climate-resilient livelihoods and development of long-term adaptive capacity.

27. Securing the livelihoods of the 12,000 people[30] within the MMA who rely directly on the climate-sensitive fishing sector is a particularly important aspect of the preferred solution because of the relationship between the fisheries and food security in Monrovia. These livelihoods are underpinned by the ecosystem services provided by mangroves within the Mesurado Wetland, which are under threat from climate change-induced SLR, unsustainable extraction and pollution. The most cost-effective approach to adapting to the impacts of climate change on these mangroves will be to engage communities in facilitating the sustainable use of mangroves to reduce anthropogenic pressure and increase the resilience of mangroves to the impacts of climate change-induced SLR. Similarly, empowering Monrovian coastal communities to develop alternative, climate-resilient livelihood activities will strengthen their adaptive capacity.

[28] Further information about the impacts of climate change on mangroves in the Mesurado Wetland and an assessment of livelihood activities dependent on these ecosystems is presented in Annex 2.A: Feasibility Study, Sections 2 and 5 as well as in Annex 2.D: Mangrove Sub-assessment.
[29] A rock revetment represents the most comprehensive option to address coastal erosion at the target community of West Point, addressing the problem for the entire West Point area through installation of the rock revetment at a highly active and exposed site, which will protect tens of thousands of vulnerable people.
[30]Comprising approximately 3,000 fisherfolk and 9,000 fishmongers, who in turn form part of the ~250,000 residents of the target communities within and adjacent to the Mesurado Wetlands.

28. The project will catalyse a paradigm shift in Liberia's approach to coastal spatial planning and the sustainable management of coastal ecosystems that addresses current and future climate change risks, incorporating long-term, climate-responsive planning. This shift will be realised through the use of quantitative, scientifically-defensible data in the development of a national climate risk-informed ICZMP, using infrastructure- ecosystem- and community-based adaptation approaches, which will be implemented in the MMA and be used as a platform to develop public-private partnerships for coastal management. In addition, the paradigm shift in coastal management will be realised by supporting the sustainable use of resources to address the impacts of climate change on livelihoods in the MMA specifically.

Barriers

29. There are three main barriers to addressing the above-mentioned problem statement. These are summarised below and expanded on in Section 8 of the FS.

Limited technical and financial capacity within the public and private sector for the establishment of coastal protection infrastructure in the short- to medium-term.

30. The effectiveness of long-term adaptation planning for coastal communities in Monrovia under future climate change conditions will be increased by improving land use plans and effectively enforcing these plans. The GoL — and private sector actors in Liberia — however, lack the technical and financial capacity for designing and constructing the infrastructure necessary to protect the most vulnerable communities in Monrovia in the short-term because of insufficient government revenue and numerous development priorities. These communities are already settled in areas that will be severely affected by projected climate change impacts. The private sector operating in the most vulnerable areas is largely made up of small and informal enterprises with limited capacity for long-term planning. Lack of access to finance and support for development also limit the financial capacity of these enterprises for investing in climate-adaptive activities. Addressing this barrier effectively requires a combination of short-term — hard infrastructure — and long-term — ICZM, land use planning and strengthening informal value chains — interventions. While there has been no lack of small-scale, ad hoc interventions to protect coastal infrastructure and vulnerable communities at West Point in the past decades, the sustainability of interventions of this nature is inherently limited.

Limited knowledge and technical capacity within government institutions for the adoption and implementation of Integrated Coastal Zone Management (ICZM).

31. There is currently limited capacity within relevant government institutions in Liberia to promote sustainable and climate resilient coastal development through ICZM. Responsibility for the management of different aspects of the coastal zone in Liberia are currently distributed across 10 government institutions[31]. As a result, the management of coastal resources currently occurs largely on an *ad hoc* and sectoral basis in Monrovia, which in turn detracts from the sustainability of management structures and interventions and limits the potential for systematically developing public-private partnerships to support coastal management. In addition, there is limited knowledge of the climate change vulnerability of different coastal areas and no existing plan to support an integrated approach to coastal management and climate change adaptation, despite this being a legal requirement in Liberia. This barrier is further exacerbated by insufficient climate information systems (CIS). Limited knowledge and technical capacity as well as the lack of an approved ICZM plan is a barrier that will need to be overcome in order to promote sustainability of coastal settlements, food security and climate-resilient planning.

Limited awareness about the impacts of climate change, including climate-sensitive nature of fishery-based livelihoods, mangrove ecosystems and food security, and limited development of opportunities for adapting to these impacts.

32. A significant barrier to building the climate resilience of coastal communities in Monrovia is limited recognition and awareness of: i) climate change; ii) the projected impacts of climate change; and iii) the climate-sensitive nature of coastal livelihoods (i.e. the unsustainability of current extractive practices), particularly in the fisheries sector, which are supported by climate-sensitive mangrove ecosystems. This barrier results in the unsustainable use of mangrove ecosystem services in Monrovia, which contributes to degradation and reduced ecosystem services

[31] These institutions are: Ministry of Mines and Energy (MME); ii) Environmental Protection Agency (EPA); iii) National Fisheries and Aquaculture Authority (NFAA); iv) Ministry of Public Works (MPW); v) National Disaster Management Agency (NDMA); vi) Liberian Hydrological Service (LHS); vii) Liberian Meteorological Service (LMS); viii) Liberian Maritime Authority (LMA); ix) Ministry of Finance and Development Planning (MFDP); and x) National Port Authority (NPA). For further details on the roles of these institutions see Annex 2.A: Feasibility Study, Section 5.

provision. In addition, there has been limited investment in the development of alternative climate-resilient livelihood opportunities as a result of the limited understanding of their importance. Overcoming this barrier will require increasing the awareness of decision-makers, planners and communities regarding the projected impacts of climate change on coastal livelihoods and developing opportunities for the uptake of alternative climate-resilient livelihoods. There is also a need for decision-makers and communities to collectively agree to sustainably manage mangrove areas, taking into consideration the linkages between functional mangroves and the viability of the fishery sector that supports the livelihoods of coastal communities in Monrovia.

Proposed interventions

33. This project will address the climate change impacts of SLR and increasingly frequent and intense storms through three outputs, namely: i) the protection of coastal communities and infrastructure at West Point against erosion caused by sea-level rise and increasingly frequent high-intensity storms; ii) institutional capacity building and policy support for the implementation of Integrated Coastal Zone Management (ICZM) across Liberia; and iii) protecting mangroves and strengthening gender- and climate-sensitive livelihoods to build local climate resilience in Monrovia.

34. Under Output 1, coastal protection infrastructure will be constructed to protect critical sections of the coast on a local scale (see Sections 8 and 9 of the FS for a description of site selection and the options analysis for different interventions[32]). This will address the technical and financial barriers to establish coastal protection infrastructure that is urgently needed in Monrovia. Protection for the most vulnerable areas on the Monrovian coastline — West Point in particular — will require the construction of the 'hard intervention' strategy of rock revetments to stabilise the shoreline, as opposed to 'softer', ecosystem-based adaptation (EbA) solutions. Such a rock revetment will be constructed at West Point to protect homes and infrastructure against coastal erosion and storm surges. Stabilising or 'fixing' the shoreline by means of a rock revetment is the preferred and only feasible solution to coastal erosion at West Point because of a range of factors as described in the technical studies undertaken for this project (Annex 2.C: Engineering Sub-assessment). These include: i) the exposed nature of the coastline; ii) the steep gradient of the beach and nearshore profile; and iii) the high energy wave environment[33]. The hard infrastructure approach also represents a socially sensitive design that will be sustainable beyond the project lifespan because it requires low-to-no maintenance[34] and accommodates launching and landing of fishing boats. A rubble mound revetment with rock armour — which is able to withstand extreme wave conditions and storm events — is proposed. The Engineering Sub-assessment showed that the northern portion of the proposed revetment is a less dynamic wave environment, and the conceptual design for this portion of the intervention site consequently proposes lighter rock armour. The revetment will be anchored in the existing beach sediment to a level of five metres below mean sea level to account for future deepening of the area directly in front of the revetment. A six-metre wide promenade, for access to the shoreline and recreation activities, is proposed between the revetment and existing dwellings at West Point. Two boat launching and landing sites are proposed as part of the design at the southern end and centre of the revetment, respectively. These launch and landing sites — which will create a locally safe environment with mild wave action — will be provided in addition to the open beach area to the north of the proposed revetment, where fishing boats are already launching and landing. This intervention has been developed through an integrated approach and will serve as basis for upscaling coastal protection to the rest of the MMA, as well as nationally.

35. Output 2 will facilitate the paradigm shift necessary for adopting an evidence-based ICZM approach across Liberia. The technical and institutional capacity of the government and communities to adapt to the rapidly changing coastal landscape and to undertake long-term, climate-responsive planning will be strengthened. Two core components of ICZM are the promotion of reliable knowledge — based on well-established practices and procedures — and good governance. The adoption of an ICZM approach entails the advancement and facilitation of: i) the integration of a variety of stakeholders and stakeholder interests, including private sector stakeholders; ii) long-term climate change adaptation and planning — with ecosystem-based adaptation (EbA) as a fundamental concept of the ICZM

[32] Details of the options analysis and selection process, including the reasoning for selecting the revetment over nature-based solutions, are given in Chapter 6 of Annex 2.B: Vulnerability Sub-assessment.

[33] Further information on the wave dynamics and limited feasibility of alternative interventions is provided in Annex 2.A: Feasibility Study, Section 4 and 9 as well as in Chapter 4 and Appendices 8.A-8.F of Annex 2.B: Vulnerability Assessment. Details of the options analysis and selection process are given in Chapter 6 of Annex 2.B: Vulnerability Sub-assessment

[34] Maintenance requirements were identified through stakeholder engagements as a critical consideration in the selection of coastal protection measures and the revetment was therefore designed to minimise operations and maintenance costs and requirements. Details of the selection and design processes can be found in Annex 2.B: Vulnerability Sub-assessment, Chapter 7, and on pg. 63 of Annex 6: ESAR.

approach; iii) proactive planning based on reliable knowledge and consideration of long-term resilience; and iv) co-governance of coastal ecosystems. A shift to using quantitative, defensible scientific data in coastal management and planning will be promoted through the development of a national-scale high-resolution multi-criteria vulnerability map and the design a national ICZMP for Liberia in consultation with all relevant stakeholders, including the private sector[35]. This plan will include spatial planning guidelines in the context of climate change and will describe adaptation actions along the coast, such as livelihood diversification as well as hard and soft coastal protection measures, enabling the upscaling and replication of specific coastal defence interventions undertaken during the project. Given the severity of the expected impacts of climate change in the medium- and long-term, the ICZMP will also assess the feasibility of 'managed retreat' where appropriate. To ensure that the ICZMP can be implemented effectively, the project will focus on building the capacity of the 10 government institutions responsible for coastal management and will facilitate the implementation of the ICZMP in Monrovia. This focus on capacity development and coordination will ensure that ICZM continues beyond the lifespan of the project.

36. Under Output 3, the adaptive capacity of vulnerable communities will be strengthened and innovation enabled by supporting local fishery-based livelihoods through sustainable co-management of mangrove areas and providing training to catalyse the uptake of diversified climate-resilient livelihoods. These activities, including the co-management of mangroves, will be underpinned by the strengthened institutional framework for ICZM developed under Output 2 and will serve to implement ICZM at the municipal scale. Lessons learned from these activities will be used to inform the revision of the ICZMP, ensure that policy and planning are informed by the experiences of target communities. Capacity development for community-level ICZM will be achieved through informing community members on how to use solar-powered cold storage facilities (Annex 2.D: Mangrove Sub-assessment Section 6.1.1) and training those involved in small-scale informal enterprises to manufacture and sell eco-friendly products such as energy-efficient cookstoves[36]. Please see the Mangrove Sub-assessment for more information on energy-efficient cookstoves and cold storage facilities.

37. Taken together, the three project interventions of coastal protection, comprehensive, long-term planning for coastal zone management and the strengthening of local livelihoods — in conjunction with strong awareness-raising and knowledge management considerations — will address both the immediate and long-term impacts of climate change on the coast of Monrovia and facilitate the potential for upscaling these initiatives across Liberia. These three outputs have been designed to incorporate sustainability considerations from the outset, with minimal maintenance required beyond the project period. An O&M Plan (Annex 21) has been developed to cover the minimal maintenance needed and will be part of an effective exit strategy that will ensure the long-term sustainability of the project interventions and their associated benefits[37.] The Theory of Change (ToC) diagram (Figure 811) illustrates the impact pathways of project interventions in overcoming the abovementioned barriers to achieve the paradigm shift in coastal zone management. The ToC diagram outlines how the progression from Project Results (Outputs) to Outcomes will achieve the Project Goal — to address priority adaptation needs through the adoption of an ICZM approach, supporting the strengthening of climate-resilient livelihoods and the development of coastal protection infrastructure. Specifically: i) Outcome 1 will see a reduced exposure to coastal erosion as a result of Output 1 — the protection of coastal communities and infrastructure — and Output 2 — the building of institutional capacity and policy support for the implementation of ICZM; ii) Outcome 2 will see institutional and regulatory systems strengthened for climate responsive ICZM as a result of Output 2; and iii) Outcome 3 will see an increase in income from climate-resilient livelihoods as a result of Output 3 — protecting mangroves and strengthening gender- and climate-sensitive livelihoods to increase local adaptive capacity. Detailed descriptions of these Outputs, and the activities they consist of, are provided in B.3 below.

[35] Several important industries in Liberia are located in the coastal zone and are vulnerable to the impacts of climate change in these areas. These industries include hotels, tourism, mines and manufacturing. During the development of the ICZMP, engagement with these stakeholders will be undertaken and the project will be used as a platform to leverage private sector support for coastal management.

[36] The manufacturing and sale of energy-efficient cookstoves has been selected for this activity to promote the adoption of alternative climate-resilient livelihoods and improve mangrove ecosystem management. While energy-efficient cookstoves are expected to facilitate a reduction in greenhouse gas emissions, the emission reduction co-benefits of this activity during the project lifespan are expected to be negligible. Upscaling of cookstove production after project implementation will increase the potential for emission reductions. Further detail on expected emission reductions is provided in Section B.3, paragraphs 101 and 102.

[37] Detailed information on the exit strategy can be found in Section B.6 of this proposal.

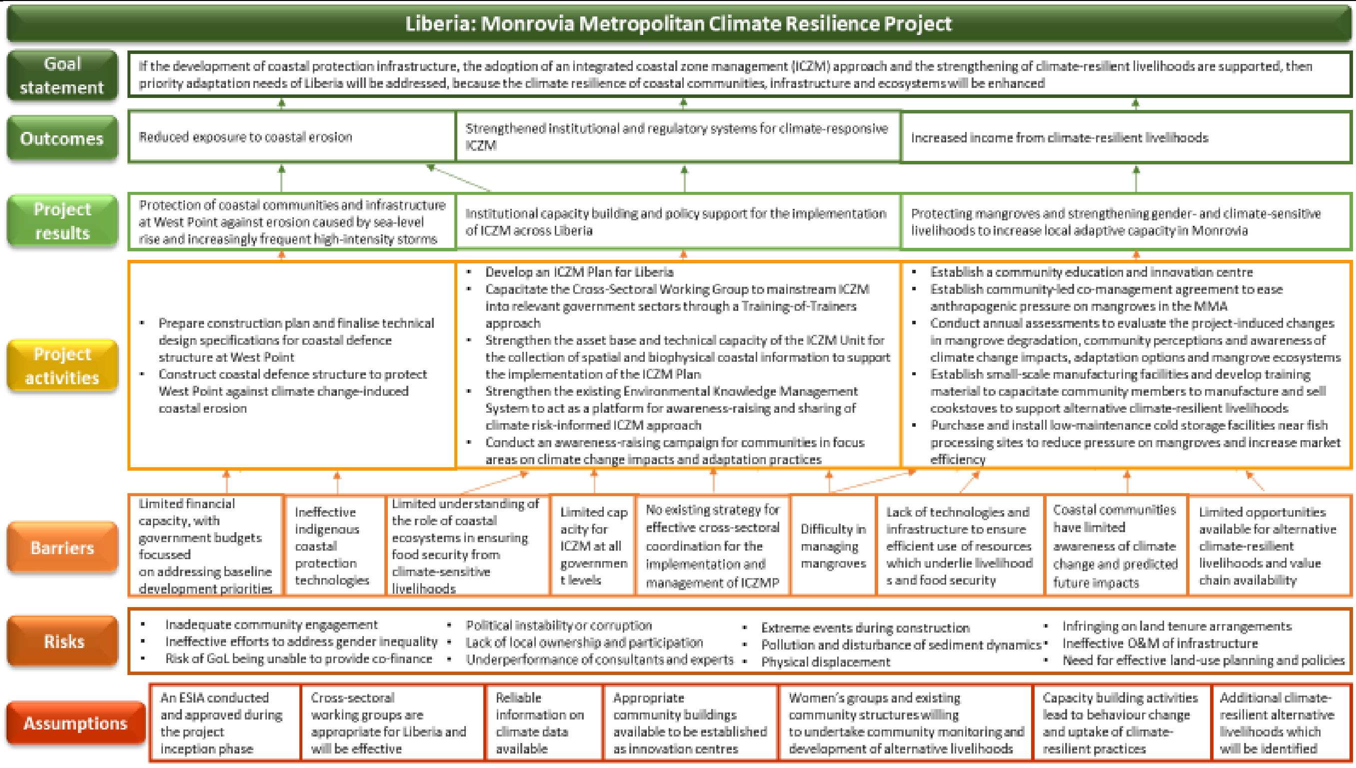


Figure 11. Theory of Change.

B.3. Project/programme description (max. 2000 words, approximately 4 pages)

38. The proposed project will enhance the resilience of vulnerable coastal communities within the Monrovia Metropolitan Area (MMA) to climate change-induced sea-level rise (SLR) and an increase in the frequency of high-intensity storms. This overarching objective of the project will be achieved through three integrated outputs, which will be implemented at three distinct spatial scales — national, municipal and site-specific. These outputs will focus on: i) establishing effective and sustainable coastal protection measures for Monrovia's most vulnerable communities at West Point (Output 1); ii) strengthening institutional capacity to manage coastal areas of Liberia in a climate-resilient manner and raising community awareness on climate change impacts and adaptation practices (Output 2); and iii) protecting mangroves and strengthening livelihood practices in vulnerable communities across the MMA to enhance people's adaptive capacity and contribute to long-term transformational change (Output 3). Implementing a robust, sustainable coastal defence solution at West Point under Output 1 as well as the institutional and community-level changes in Outputs 2 and 3 will trigger transformational change in terms of how Liberia manages its coastline in the future. The investment at West Point will serve as a demonstration site of the value of robust and sustainable coastal protection measures, which will be used to leverage future investments in coastal protection infrastructure for the most vulnerable Liberian communities. In addition, the comprehensive technical assessments used to inform the design of the West Point revetment provide a foundation for the design of further coastal protection interventions. To complement these measures, the integrated approach to mangrove management, fisheries management, and coastal planning and development taken in Outputs 2 and 3 will promote gender-responsive policy reforms that will apply across the whole Liberian coastline. In this way, the project will effect major transformational change.

Output 1. Protection of coastal communities and infrastructure at West Point against erosion caused by sea-level rise and increasingly frequent high-intensity storms.

39. Coastal communities within the MMA are at risk from accelerated coastal erosion and increased incidence of extreme weather events, which has already resulted in the displacement of at least 670 households[38]. This risk is disproportionately borne by the poorest and most vulnerable communities in Monrovia. Under this output, GCF grant funding will be used to construct coastal defence infrastructure — in the form of a rock revetment — to protect the West Point settlement against accelerated coastal erosion and extreme storm events induced by climate

[38] Further details on the displacement of households are available in Annex 2.A: Feasibility Study, Section 3.

change. West Point is one of the most densely populated and poorest areas on the Monrovian coast and is the most vulnerable section of the coast to climate change-induced acceleration of coastal erosion (see Figure 2). Co-finance will facilitate the finalisation and validation of the detailed design and construction plans, and rock material for the construction of the revetment at West Point will be provided as in-kind co-finance.

Activity 1.1. Prepare construction plans and finalise technical design specifications for coastal defence structure at West Point.

40. Detailed technical design specifications, based on the designs contained in the Engineering Sub-assessment (Annex 2.C[39]), will be finalised and an in-depth construction plan developed to ensure that the construction of the West Point revetment and green promenade under the project (Activity 1.2) are implemented appropriately and with the inputs of communities (50% women) in the final design and plan. An Environmental and Social Impact Assessment (ESIA) will be conducted to inform the final design, in accordance with the requirements of Liberian law, following the Environmental and Social Assessment Report (ESAR). In addition, a project site-specific hydro-engineering study of West Point will be conducted by qualified engineers to ensure that the risks of potential localized and broader flooding at incorporated in the detailed design of the defence structure. The final technical design will be based on the existing conceptual design provided in the Engineering Sub-assessment funded by GCF PPF resources and will consider best practices and lessons learned from other coastal protection interventions in Monrovia. This conceptual design has been informed by detailed geo-physical and social assessments, including an options analysis of nature-based solutions and other hard infrastructure options (Annexes 2.A: Feasibility Study, 2.B: Vulnerability Sub-assessment, 2.C: Engineering Sub-assessment and 2.D: Mangrove Sub-assessment). This approach will be integrated into the GEF-LDCF project which is currently being developed to strengthen ICZM in Sinoe County, Liberia, highlighting that nature-based solution should be used along with infrastructural interventions to reduce the vulnerability of coastal communities. Preventing both heavier water loads in the Mesurado Estuary and saline intrusion into freshwater aquifers were critical considerations in this initial design of this intervention[40]. Concurrent to the development of the technical design plans, the construction plan will be developed. The construction plan will identify all necessary inputs, procurement procedures, contracting guidelines, relevant legislation, ESS processes and gender-responsive stakeholder consultations necessary for the construction of the coastal defence infrastructure[41]. The final detailed technical design and construction plan will be developed with input from community and government stakeholders and will be validated prior to implementation. The Ministry of Mines and Energy (MME) will be the Responsible Party for this activity and co-finance will be provided by UNDP.

41. Sub-activities under this activity will include:

Sub-activity 1.1.1. Refine and detail the design and construction plan for the coastal defence structure and green promenade at West Point.

Under this sub-activity, a service provider will be contracted to develop an ESIA, considering the impacts associated with the planned revetment at West Point as well as livelihood activities under Output 3. This will be done in accordance with the requirements of Liberian law for the construction of the revetment, and will include an assessment of the risk of economic displacement as a result of: i) the revetment restricting access to boat launching sites for fisherfolk; and ii) the development of the mangrove co-management agreement under Activity 3.2. If required based on this assessment, a Livelihood Restoration Plan (LRP) will be developed as part of the ESIA process and included in the Environmental

[39] Chapter 3 of the Engineering Sub-assessment (Annex 2.C) provides the detailed engineering assessment of the protection measures (infrastructure). It includes the design of the proposed measures. Section 3.5.2 (page 38) of Annex 2.C provides the description and detailing of the design for the revetment at West Point.

[40] The development of a hard revetment at West Point is very unlikely to affect water loads in the estuary. This is because the revetment will be developed at the ocean front and will not affect the flow field inside the wetland, which is dependent on tidal water level fluctuation. The revetment, like the existing beach at West Point, will not affect this tidal water level fluctuation which is responsible for water levels in the estuary. Inundation of West Point would be a potential concern if there was insufficient drainage through the revetment during heavy rains. However, the revetment structure will be permeable as it will be constructed out of rocks. Additional drainage facilities (such as drainage from promenade and/or culverts) have been accounted for in the costing of the revetment (both capital and operational expenditure) and the drainage capacity and requirements will be assessed during the finalisation of the revetment designs. The revetment will not be proximate to freshwater aquifers and is therefore highly unlikely to cause saltwater intrusion. Further details on how these risks are taken into account in the design of the revetment can be found in Annex 2.C.

[41] Guidelines for the construction plan have been compiled during the feasibility assessment of the project and are available in Annex 2.A: Feasibility Study, Section 10.

and Social Management Plan (ESMP) for the project. A consultant will be contracted to undertake a hydro-engineering study and develop the final technical design and construction plan for the coastal defence infrastructure to be developed at West Point, based on the ESIA. This final design will draw on the hydro engineering study, existing, prior work and the completed technical feasibility designs (see Annex IIC: Engineering Sub-assessment) and will be developed in consultation with local communities and government stakeholders, with 50% women representation, to ensure that sufficient access to beach landing sites is retained and that adequate drainage facilities and amenities are incorporated into the promenade.

Sub-activity 1.1.2. Host a validation workshop to present and validate final design and construction plan to government and community stakeholders.

42. Under this activity, an official validation workshop will be held to present the finalised design and construction plan to relevant government and community stakeholders and to explain how stakeholder concerns and input have been incorporated into the design. In addition to government and community stakeholders, other interested parties and civil society groups such as development organisations and NGOs will be invited to attend the validation. All consultative and workshop processes will be gender-responsive[42] and include 50% women.

Activity 1.2. Construct coastal defence structure to protect West Point against climate change-induced coastal erosion.

43. To protect the community of West Point from the impacts of accelerated coastal erosion and extreme weather events resulting from climate change, an engineering firm will be contracted to construct a rock revetment at West Point. The revetment will be 1,050 m long and will include a green promenade[43] as well as associated amenities[44]. The promenade will be utilised as an open community space and will compensate for the loss of beach areas that are used for numerous social, domestic and economic purposes[45] but are currently dissipating from erosion. The design of the revetment[46] is such that three areas of beach will be retained and protected to ensure that the ocean is accessible for social and domestic purposes as well as for the launching of fishing boats and the processing of fish. Constraints on capacity for maintaining physical infrastructure were considered in the selection and design of the revetment (see the Engineering Sub-assessment). As a result, the revetment is designed to be a low maintenance intervention[47] with a minimum lifespan of 30–50 years. To further increase the sustainability of the intervention, labourers from the West Point area engaged in the construction of the revetment will be trained on the best maintenance techniques for the revetment. Women will constitute at least 40% of waged labour on project-related activities, including construction. MME will be the Responsible Party for this activity, which will be financed by GCF, UNDP and GoL.

44. Sub-activities under this activity will include:

Sub-activity 1.2.1. Construct a rock revetment with a green promenade and community amenities at West Point.

45. A firm will be procured, and construction will be undertaken according to the detailed final design and construction plan developed under Sub-activity 1.1.1. The firm implementing the protective infrastructure at West Point will engage local wage labourers from the West Point area to work on the construction of the infrastructure. While most of the beach is inaccessible by cars because of the dense housing layout at West Point, bollards will be provided for the limited vehicular entry points to eliminate any risk of theft of site materials. The cost of bollards will be minimal and will be absorbed under construction costs. The type and location of access control infrastructure will

[42] Taking into account appropriate meeting times and places, gender- and cultural-sensitivities, so that women are able to attend and fully participate.
[43] Further information on the green promenade, including an artist's impression, are provided in Annex 2.C: Engineering Sub-assessment.
[44] Associated amenities will include, but not be limited to, waste bins, benches or other seating, boat storage facilities and designated open areas for the drying and smoking of fish
[45] Domestic purposes include utilisation of the beach for washing, cooking and drying or smoking of fish,
[46] Validated concept designs for the revetment are included in Annex 2.C: Engineering Sub-assessment
[47] The GoL has signed an MoU stating their intention to finance any required maintenance. The government entity responsible for undertaking this maintenance will be identified during the development of the ICZMP under Activity 2.1.2. Please see Annex 21: O&M Plan for further information.

form part of the detailed design phase during[48]. Equal opportunities and equal pay will be provided to men and women, and wages will comply with Liberian labour laws. Women will constitute at least 40% of waged labourers and the implementing firm will make provision for gender-specific safety protocols and personal protective equipment. The labourers engaged for the construction of the revetment will be trained by the firm on best maintenance techniques for the revetment. To foster sustainability and local ownership, regular monitoring of the structure and its performance will be done in collaboration with the local communities through local leadership. During the construction phase, training will be provided to representatives of local communities and community leaders on basic monitoring systems, data collection and reporting to the lead government institution as per the ICZMP. This will include capacity to check for wear and tear on a regular basis and after minor weather events, maintenance of the promenade, planting local vegetation species under supervision of relevant authority. This training will also be provided to representatives of fisher folk in terms of the maintenance of the fish landing sites. Additionally, during the construction phase, awareness-raising on health and safety with regards to the construction of coastal infrastructure will be facilitated during the daily safety debriefs, which is an important consideration for the construction phase given the close proximity of the densely populated West Point community. In addition to ensuring the representation of women as wage labourers, at least 50% of those providing training will be women.

Output 2. Institutional capacity building and policy support for the implementation of Integrated Coastal Zone Management (ICZM) across Liberia.

[48] The risk of theft of sand and rock materials at West Point is low. Extraction of materials during and following construction is therefore unlikely. However, security personnel can be hired to reduce the risk of theft of equipment and materials during the construction phase if the contractors deem it necessary. Security has been employed for safeguarding site equipment and materials on other projects and can be made available during the construction of the West Point revetment. After project completion, the risk of materials being stolen is considered minimal. In addition, the project has been designed with the West Point community's ongoing engagement and will continue to play a large role in ensuring the project is effectively implemented.

46. The project will strengthen the capacity of the national government in Liberia to implement climate risk-informed Integrated Coastal Zone Management (ICZM) and to support cross-sectoral coordination among the 10 government institutions responsible for climate-resilient coastal management (Figure 912)[49]. The ICZM approach considers land-use planning in relation to climate risks, the interconnectivity of coastal systems, stakeholder sectors, as well as the intrinsic dependence of communities on coastal resources for their livelihoods. It is aligned with the GEF project for ICZM in Sinoe County planning. It also promotes evidence-based decision-making using scientifically defensible data. Implementing such an approach in the MMA — and across Liberia — will transform the existing coastal management paradigm and reduce the vulnerability of coastal communities to the impacts of climate change-induced SLR, accelerated coastal erosion and increased incidence of extreme weather events. The adoption of a climate-resilient ICZM approach will be facilitated by the development of a high-resolution, multi-criteria vulnerability map of the Liberian coastline and a national Integrated Coastal Zone Management Plan (ICZMP), as well as by equipping and capacitating relevant stakeholders to effectively implement the plan. Additionally, activities under this output will increase the knowledge of institutions and communities on climate-change related coastal hazards, potential adaptation strategies and the importance of an integrated approach to coastal zone management (Figure 12). Through knowledge sharing and engagement, activities under this Output will support the development of public-private partnerships for coastal management. This will be achieved by developing an awareness-raising campaign targeting four communities in the MMA and upscaling the existing Environmental Knowledge Management System (EKMS) which was established under a previous GEF-funded initiative. The participatory approach to the implementation of ICZM will improve coordination: i) between governmental institutions; and ii) between GoL, communities and the private sector for coastal management. The knowledge management, awareness-raising and vulnerability mapping components of this output will be facilitated through co-financing, while GCF grant funding will be used to develop the climate-responsive ICZMP as well as the procurement of biophysical data collection systems and spatial data.

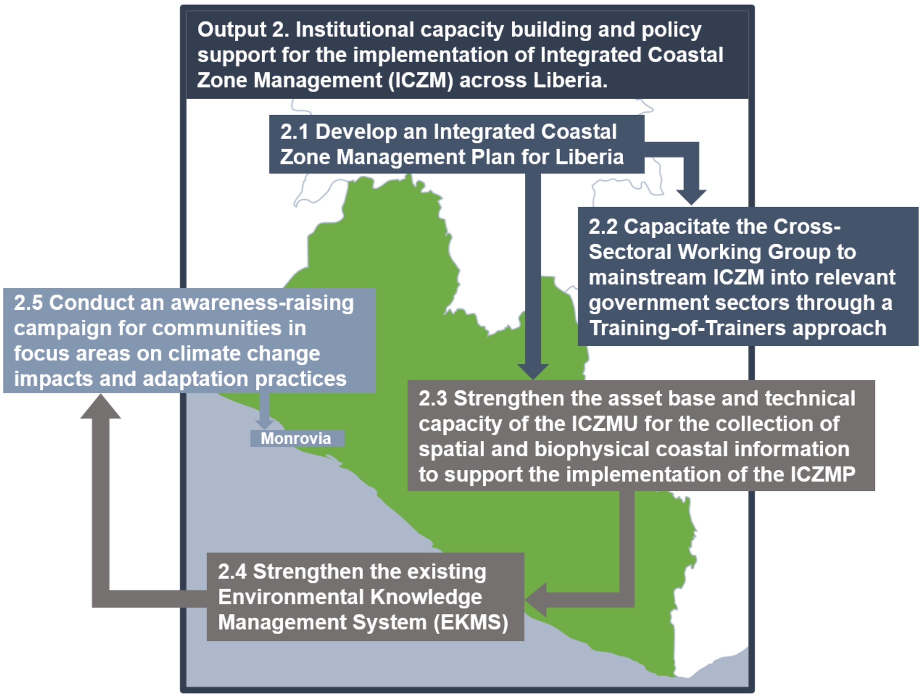


Figure 12. Linkages between activities under Output 2 of the proposed project.

Activity 2.1. Develop an Integrated Coastal Zone Management Plan for Liberia.

[49] These institutions are: Ministry of Mines and Energy (MME); ii) Environmental Protection Agency (EPA); iii) National Fisheries and Aquaculture Authority (NFAA); iv) Ministry of Public Works (MPW); v) National Disaster Management Agency (NDMA); vi) Liberian Hydrological Service (LHS); vii) Liberian Meteorological Service (LMS); viii) Liberian Maritime Authority (LMA); ix) Ministry of Finance and Development Planning (FNDP) and x) National Port Authority (NPA).

47. Under this activity, the project will work with relevant government institutions[50] to develop a national Integrated Coastal Zone Management Plan (ICZMP) that addresses climate change risks, through a participatory and consultative process. This ICZMP will use an integrated approach to infrastructure-, community- and ecosystem-based adaptation in the coastal zone of Liberia. This approach will inform and be informed by lessons learned from collaboration between communities, government, private sector and NGOs for: i) strengthening climate-resilient livelihoods and ecosystem-based adaptation under Output 3; and ii) implementing protective coastal infrastructure under Output 1. Development of the plan will be led by a service provider with experience in developing and implementing climate risk-informed ICZM. To ensure the plan is effective and implementable it will be developed through a phased approach, whereby the first phase will entail producing a high-resolution, multi-criteria vulnerability map of the entire Liberian coastline. The second phase will entail the development of a national climate-responsive ICZMP through a collaborative process led by international and national experts. The plan will be developed based on the high-resolution vulnerability map and extensive stakeholder consultations with government institutions and private sector entities in the coastal zone[51]. To ensure effective implementation of the ICZMP across the relevant government sectors, an ICZM Committee and Cross-Sectoral Working Group (CSWG) will be established under this phase, with the explicit mandate of ensuring the integration of climate risks and the alignment and conformity of the ICZMP across the relevant government sectors. The CSWG will develop a detailed action plan for the implementation of the ICZMP, identifying: i) sector-specific regulations and by-laws for updating and integrating ICZM principals; ii) opportunities for strengthening public-private partnerships to support ICZM; and iii) strategies for financing coastal management interventions, particularly relating to climate change adaptation. Additionally, during the second phase, the ICZMP will be implemented in Monrovia with support from the GCF grant financing of the project. The final phase of the ICZMP development will entail updating the ICZMP three years after the initial plan has been developed. This will ensure that lessons learned during the implementation process in Monrovia are incorporated into the second iteration of the ICZMP, which will facilitate the upscaling of this activity through ICZMP implementation across Liberia. Furthermore, the ICZMP will draw on data generated from Liberia’s multi-hazard impact-based forecasting and early warning systems (MH-IBF-EWS), which will be strengthened under the proposed AfDB/GCF project “Enhancing Climate Information Systems for resilient development in Liberia (Liberia CIS)”. The ICZMP developed under this activity will be used as a base from which the GEF project will build on for developing county level ICZMPs.

48. This process will result in a dynamic ICZMP that is continuously adaptive, iterative and a resource for the sustainable and climate-resilient management of the Liberian coastal zone. The ICZMP will: i) provide guidelines for integrating climate change risks into planning and coastal resource use; ii) support integrated development planning; iii) develop linkages between the 10 government institutions, local research bodies, the private sector and civil society groups through the ICZM Committee and CSWG; and iv) facilitate the incorporation of climate change considerations into coastal zone management. This will support a shift in the approach to managing the coastal zone in Liberia. To address the local gap in expertise required to develop and implement this ICZMP, the project will facilitate long-term training for relevant institutional representatives on best practice for ICZM through a Training-of-Trainers (ToT) approach under Activity 2.2. MME will be the Responsible Party for this activity, which will be financed by GCF, UNDP and GoL.

49. Sub-activities under this activity will include:

Sub-activity 2.1.1. Develop a high-resolution, multi-criteria vulnerability map of the Liberian coastline.

50. To build the knowledge base for climate risk-profiling in Liberia and to support the generation and implementation of an ICZMP, an international firm will be contracted to work alongside local experts to develop a high-resolution, multi-criteria vulnerability map that assesses exposure, sensitivity and hazard parameters within coastal areas. Data collected from vulnerability studies conducted under the NAP Readiness process will be included in the development of the map, as will the high-resolution spatial datasets procured under Sub-activity 2.3.1. The vulnerability map will incorporate socio-economic and gender considerations in addition to biophysical and climate change factors to ensure long-term decision-making beyond the lifespan of the project. The resulting map will

[50] These institutions are: Ministry of Mines and Energy (MME); ii) Environmental Protection Agency (EPA); iii) National Fisheries and Aquaculture Authority (NFAA); iv) Ministry of Public Works (MPW); v) National Disaster Management Agency (NDMA); vi) Liberian Hydrological Service (LHS); vii) Liberian Meteorological Service (LMS); viii) Liberian Maritime Authority (LMA); ix) Ministry of Finance and Development Planning (FNDP) and x) National Port Authority (NPA).

[51] including hotels, mines and other coastal industries

provide municipalities within the MMA — and across Liberia — with the knowledge base required to incorporate climate change considerations into future urban planning and ICZM. This is an integral step to inform the development of the ICZMP by providing a baseline of scientifically defensible information on coastal vulnerability in Liberia and which will be enhanced by the improved national multi-hazard impact-based forecasting and early warning systems (MH-IBF-EWS).

Sub-activity 2.1.2. Develop the climate-responsive ICZMP in collaboration with stakeholders, incorporating information from the map developed under Sub-activity 2.1.1.

51. Under this activity, the project will develop the ICZMP through a participatory and consultative process. An appropriately qualified service provider will be contracted to engage with relevant stakeholders through a gender-sensitive approach[52] — including the leaders of vulnerable communities, NGOs, private sector actors located in the coastal zone[53], the 10 relevant government institutions involved with coastal management in Liberia as well as the Ministry of Gender, Children and Social Protection (MoGCSP). The inputs generated by these engagements will be considered in relation to the vulnerability map produced under Sub-activity 2.1.1 and will inform the design of the ICZMP. Additional engagements with these stakeholders will be undertaken to inform the revision of the ICZMP three years after the initial development. The ICZMP will be developed at a national scale but will only be implemented in Monrovia under the project. Approaching implementation in a phased manner — whereby the project supports implementation on a limited scale — will provide an opportunity for newly capacitated institutional staff to exercise the skills gained through the long-term training under Activity 2.2, while providing upscaling potential for the ICZMP across Liberia. Additionally, under this activity, the service provider developing the ICZMP will: i) define the mandate of the ICZM Committee and the CSWG based on consultations with all government stakeholders[54]; ii) define the roles and targets for each of the 10 institutions; and iii) work collaboratively with the GoL to mandate responsibility for the maintenance of the revetment — constructed under Activity 1.2 — to a relevant government entity. These mandates will be presented along with the ICZMP under Sub-activity 2.1.3.

Sub-activity 2.1.3. Host validation workshops to present and secure ownership of the ICZMP for representatives from the 10 government institutions responsible for coastal management in Liberia.

52. Under this activity, the two workshops will be held to present the newly developed ICZMP in Year 2 and the updated ICZMP in Year 6 to high-level representatives from the 10 government institutions responsible for coastal management in Liberia as well as the Ministry of Gender and Social Protection. The objective of the workshops will be to secure the endorsement and joint ownership of the plan, including commitment to the roles and targets for each institution. During the workshops, any outstanding points regarding the plan will be addressed and the members of the 10-person ICZM Committee will be elected during the first workshop. The ICZM Committee[55] will be comprised of high-level government representatives, comprising 50% women, from different government departments. The Committee will be mandated to strengthen cross-sectoral collaboration for ICZM and support the mainstreaming of ICZM into sectoral policies and plans.

53. A further action undertaken during this initial validation workshop will be the nomination of government staff from each department to serve on a Cross-Sectoral Working Group (CSWG). The CSWG will include technical specialists from each of the 10 government institutions responsible for coastal management and will be capacitated under Activity 2.2 to mainstream ICZM into their respective institutions. The members of the CSWG will also be mandated to provide technical information to their counterpart representatives on the ICZM Committee. This structure will contribute to information flow between decision-makers and technical personnel within as well as between different government institutions.

Activity 2.2. Capacitate the Cross-Sectoral Working Group to mainstream ICZM into relevant government sectors through a Training-of-Trainers approach.

[52] Which will include collecting sex and age disaggregated data, supporting the trainer-of-trainers, and coordinating with relevant ministries on gender focus-points during the ICZMP development.
[53] including hotels, mines and other coastal industries
[54] This will include considerations on drivers and implications of illegal sandmining along beaches and rivers causing erosion and exacerbating the effects of climate change.
[55] The terms of reference/mandate for this committee, as well as the Cross-Sectoral Working Groups, will be defined as part of the ICZMP under Activity 2.2.

54. Under this activity, the technical capacity of the CSWG established under Sub-activity 2.1.3 will be strengthened to implement the ICZMP through long-term training via a Training-of-Trainers (ToT) approach. To support the effective implementation of the ICZMP developed under Activity 2.1, the project will host workshops on ICZM planning and implementation to build the capacity of the CSWG and additional representatives from all relevant government institutions. As part of this process, a detailed action plan for sector-specific implementation of the ICZMP will be developed by the CSWG to: i) facilitate the integration of ICZM into relevant by-laws and regulations; ii) identify opportunities and develop plans for strengthening public-private partnerships for ICZM; and iii) identify financing strategies for climate change adaptation priorities in the coastal zone. Capacitating technical personnel responsible for overseeing the uptake of implementation of ICZM across all relevant sectors will support the effective execution, management and uptake of ICZM within the MMA as well as facilitating upscaling across the country. The GEF ICZM project will build on the lessons learned and best practices noted under this activity in its CSWG undertakings for mainstreaming ICZM.

55. The use of a ToT approach will support the various government institutions to capacitate their staff beyond the lifespan of the project. In addition, ensuring that representatives from all 10 institutions receive the same training will allow for the requisite technical capacity to be introduced into each institution and will enable the alignment of ICZM across all government sectors. MME will be the Responsible Party for this activity, which will be financed by GCF, UNDP and GoL.

56. Sub-activities under this activity will include:

Sub-activity 2.2.1. Host an inception workshop to present and provide orientation to the CSWG on the finalised ICZMP.

57. An inception workshop will be held under this activity to introduce the ICZMP to the members of the CSWG nominated under Sub-activity 2.1.3. The CSWG will include one representative from each of the 10 government institutions involved in coastal management as well as representatives from the Ministry of Gender, Children & Social Protection (MoGCSP) and the Integrated Coastal Zone Management Unit (ICZMU). This inception workshop will provide an opportunity for the members of the CSWG to engage with each other, become familiarised with the ICZMP and provide initial input for the training to be developed under Sub-activity 2.2.2.

Sub-activity 2.2.2. Host a planning workshop for the CSWG to develop an action plan for implementing the ICZMP, including plans for changes and updates required for sector-specific regulations and by-laws

58. Once the CSWG has undergone initial training, a two-day planning workshop will be hosted to develop an action plan for the implementation of the ICZMP. This action plan will include the identification of regulations and by-laws in each of the relevant sectors[56] that need to be updated and changed to align with ICZM as well as a stocktake of job descriptions for staff involved in ICZM to assess where formal roles and responsibilities need to be assigned to support ICZM implementation. The institutions responsible for each of these regulations and by-laws will also be identified, along with the processes for updating each of them. A plan for strengthening partnerships between government institutions and private sector stakeholders — including hotels, mines and other industries with an interest in coastal management — will also be developed. In addition, a session in the workshop will be dedicated to the identification of possible financing strategies for coastal climate change adaptation interventions and the development of a financing plan to leverage investment in coastal protection. The effectiveness of the financing plan will be monitored as part of the Impact Assessment conducted under Activity 3.3 and the collation of lessons learned under Activity 2.4. This information will be used to inform the revision of the ICZMP in Year 6 of project implementation. The possible role of private sector investment in coastal adaptation will be considered as part of this plan. Roles, responsibilities and timelines for the required changes, updates and actions will be set out in the action plan. Through engagement between the representatives of the 10 government institutions on the CSWG and those on the ICZM Committee, support for the implementation of the action plan will be developed within each of the relevant institutions.

Sub-activity 2.2.3. Develop a curriculum and training program to capacitate relevant technical personnel on cross-sectoral coordination and the effective implementation of the ICZMP.

[56] Including, for example, land-use management, forestry and fishing

59. Under this activity, an international expert with experience in coastal management and the ToT approach will be contracted to develop a training curriculum and program to capacitate the CSWG and additional technical staff from each of the 10 government institutions in ICZM. The expert will develop the framework of the training curricula with input from the consultants engaged under Sub-activity 2.1.2 and will finalise the curricula through a participatory process with the ICZM Committee under Sub-activity 2.2.4. The consultant engaged under this activity will use the finalised curricula and program to train the CSWG in best practice for cross-sectoral coordination and the implementation of ICZM as well as capacitate members of the CSWG to conduct the training within each of their respective departments.

Sub-activity 2.2.4. Host a validation workshop with the CSWG to finalise training curricula.

60. To facilitate a participatory approach for defining the training curricula, a three-day co-creation workshop will be held with all members of the CSWG and ICZM Committee to present the framework of the training program and work collaboratively to: i) define the various roles and responsibilities for each of the 10 institutions engaged in coastal management; and ii) finalise the curricula to be included in the training program. The national consultants engaged under Sub-activity 2.1.2 will support this activity by attending the workshops and guiding the development of the curricula to ensure it incorporates the most important considerations of effective ICZM.

Sub-activity 2.2.5. Conduct a training and action programme for members of the CSWG to mainstream ICZM into relevant government entities and support mainstreaming of ICZM into national policies, programmes and plans.

61. Under this activity, the international expert contracted under Sub-activity 2.2.3 will conduct a 16-week, phased training programme for the members of the CSWG and two additional technical personnel from each of the 10 government institutions. This long-term training programme will include instruction on: i) the implementation of the ICZMP; ii) financing strategies for ICZM priorities; iii) important considerations for cross-sectoral collaboration; and iv) training on how to conduct the course for additional members within each institution. This training programme will emphasise knowledge and tools for integrating climate risks, mainstream action plans and will support the different institutions to perform their roles and meet targets as specified in the ICZMP. This will build the capacity of institutional representatives to effectively implement the ICZMP, further capacitate government technical staff and ensure alignment between all relevant institutions, thereby supporting cross-sectoral collaboration and the uptake of ICZM across Liberia. The international expert will also provide support to the different institutions for mainstreaming and implementing the ICZMP.

Activity 2.3. Strengthen the asset base and technical capacity of the ICZMU for the collection of spatial and biophysical coastal information to support the implementation of the ICZMP.

62. A significant knowledge gap in the form of limited spatial, meteorological and oceanographic data was identified during the project feasibility process. This stems from the current reliance by the Liberian Meteorological Services on remotely-sensed data as opposed to *in situ* data collection by means of wave buoys. In addition, access to remote sensing data is limited to low-resolution publicly available satellite data. Under this activity, the project will procure high-resolution satellite imagery for the Mesurado Wetland and coastline of Monrovia. These data will enable detailed mapping and monitoring of coastal ecosystems in the Mesurado Wetland as well as coastal erosion and will be used in the vulnerability mapping under Sub-activity 2.1.1 and project impact monitoring under Sub-activity 3.3.1. An international firm will also be contracted under this activity to develop an integrated solution for the collection, processing and synthesis of meteorological and oceanographic data offshore of Monrovia. This integrated system will be hosted by the ICZMU and will support the implementation of the ICZMP by providing a local knowledge base on ocean dynamics for the 10 relevant government institutions involved with the implementation of the ICZMP. The system will be linked to existing climate information and early warning systems. Furthermore, the system will complement the GCF/AfDB project "Enhancing Climate Information Systems for resilient development in Liberia (Liberia CIS)" currently under development to improve Liberia's meteorological and hydrological systems. MME will be the Responsible Party for this activity, which will be financed by GCF.

63. Sub-activities under this activity will include:

Sub-activity 2.3.1. Procure high-resolution remote sensing data to monitor coastal erosion and mangrove ecosystem health and degradation in the MMA, to be made publicly accessible and used for the vulnerability map under Sub-activity 2.1.1.

64. Under this activity, high-resolution satellite imagery[57] will be procured for the MMA. First, five historical datasets — one for every second year for the past ten years — will be procured along with a new dataset in Year 2 of the project. Additional datasets will be acquired in Years 4 and 6 of the project. The historical datasets will be used to inform the development of the vulnerability map under Sub-activity 2.1.1. In addition, all data procured under this activity will be processed and analysed under Sub-activity 3.3.1 to assess the health and degradation of mangrove ecosystems and evaluate the impact of the project in terms of reducing mangrove degradation. Processed satellite imagery and datasets will be made publicly accessible through the Environmental Knowledge Management System (EKMS) under Activity 2.4.

Sub-activity 2.3.2. Contract a specialist firm to provide an integrated, near-shore wave buoy-based data collection and processing system to support the collection of meteorological and oceanographic data for improved generation of information on parameters relevant to ICZM and meteorological decision-making in Liberia.

65. The above-mentioned integrated solution will include: i) four wave buoys that are capable of capturing, recording and transmitting data relating to *inter alia* wind direction, windspeed, wave height, sea-level pressure, sea surface temperature and tides; ii) associated information and technology equipment; iii) the development and installation of a database for collecting, processing and cataloguing data; iv) training courses on the installation, use and maintenance of the system; and iv) on-site and off-site technical support in integrating the system with the existing early warning system and weather forecasting system established under projects funded by the GEF as well as the AfDB/GCF project currently under development. The training courses on the installation, use and maintenance of the system will be delivered to technical staff at relevant government institutions, including the EPA, NDMA, LMS and ICZMU and training materials will be made available to the trained staff to facilitate training of additional staff.

The specialist firm will also be contracted on an annual basis to provide off-site technical support to ensure that the system is fully operational and that technical staff trained in the installation, use and maintenance of the system (including staff of the EPA, NDMA, LMS and ICZMU) benefit from technical assistance for the duration of the project. This will assist technical staff within relevant institutions to continue to maintain the system beyond the project lifespan.

Activity 2.4. Strengthen the existing Environmental Knowledge Management System (EKMS) to act as a platform for awareness-raising and sharing of climate risk-informed ICZM approach.

66. The project will support the newly implemented Environmental Knowledge Management System (EKMS) in Liberia by extending accessibility to the system, incorporating resources on climate-risk informed ICZM and contributing to awareness on the application of the system. The EKMS will be hosted by the EPA to act as a platform linking the 10 institutions responsible for addressing climate change impacts within the coastal zone and to provide information relating to climate risks and ICZM to the private sector. Additionally, the project will help establish a partnership between the EPA and research institutions, including the University of Liberia, to develop case studies, and further the national understanding of coastal protection. Information and communication (ICT) equipment will be procured to link these institutions and provide a platform to access and share climate information to support climate-responsive ICZM. This information will include: i) climate dynamics; ii) best practice for ICZM; iii) the training material for ICZM developed under Sub-activity 2.2.3; iv) the ICZMP developed under Sub-activity 2.1.2; v) the spatial data procured under Sub-activity 2.3.1; vi) the high-resolution multicriteria vulnerability map developed under Sub-activity 2.1.1; and vii) lessons learned for upscaling and replicating project activities, which will be reported on under Activity 3.3. A national service provider will also be contracted under this activity to regularly update the system and to collect reports on the implementation of the ICZMP on an annual basis, which will be collated into a synthesis report and included as lessons learned in the database. Systems established under this activity for knowledge sharing will be used for upscaling through collaborative efforts with the GEF ICZM systems to provide climate information, products and services relating to coastal flood and erosion early-warning and risk management. The cross-project collaboration will contribute to effective coastal protection and adequate climate

[57] The datasets will have a resolution of 2 m or higher and will include red, green, blue and near-infrared bands of satellite images.

change adaptation planning that will limit the impact of climate change hazards on vulnerable communities and reduce their vulnerability.

67. To further support uptake of the system, the project will support the distribution of knowledge products to government entities and private sector stakeholders to increase awareness on the application of the EKMS. Improved awareness of the system in conjunction with increased ease of access to the EKMS will support improved access to environmental and climate change information for all government institutions responsible for coastal management. The EPA will be the Responsible Party for this activity, which will be financed by GCF and UNDP.

68. Sub-activities under this activity will include:

Sub-activity 2.4.1. Procure ICT equipment for each of the relevant government institutions to access and use climate and environmental information through the platform.

69. Under this activity, the project will procure computer equipment, remote servers and on-site data storage to facilitate access to the EKMS for the 10 government institutions responsible for management of the coastal zone and associated climate change impacts in Liberia. The ICT equipment procured under this activity will also be linked to the near-shore data-collection and monitoring system established under Activity 2.3. This will enable technical staff in the relevant government departments to access a shared resource on ICZM and facilitate access to data on current ocean conditions. As a result, cross-sectoral coordination as well as planning will be improved and early warnings for extreme storm surges or other storm-related disasters will be enhanced.

Sub-activity 2.4.2. Collect lessons learned on ICZM on an annual basis and incorporate this information into the EKMS.

70. A national service provider will be contracted under this activity to collect lessons learned, prepare a report and update the EKMS on an annual basis. The report will provide suggestions on best practice for effectively addressing climate change risks by implementing the newly developed ICZMP in Liberia and by scaling up project activities. This will be informed by project monitoring initiatives under Activity 3.3. Collecting lessons learned on an annual basis will not only support future updating of ICZM frameworks in Liberia in an effective and iterative way but will also ensure that all relevant government institutions benefit from lessons learned within each separate institution, as well as enabling research institutions to develop case studies and increase understanding about coastal protection and management in Liberia.

Sub-activity 2.4.3. Design and disseminate knowledge products on climate-responsive ICZM in government institutions and to the private sector.

71. Under this activity, a national service provider will be contracted to develop an awareness-raising strategy to increase knowledge on the EKMS and facilitate greater uptake of the EKMS within government institutions and among private sector partners[58]. Knowledge products developed under this strategy will include posters and brochures which will provide information on the EKMS as well as on how to access ICZM and climate change information on the system. These knowledge products will be tailored to identified user-groups, which will include staff members of the relevant government institutions and private sector stakeholders in the coastal zone. The knowledge products will be designed in a gender-sensitive manner and will ensure that they are equally accessible for all government employees and private sector stakeholders. Furthermore, the products will depict the benefits of ICZM and the EKMS for different economic activities and social groups practicing different livelihoods strategies.

Activity 2.5. Conduct an awareness-raising campaign for communities in focus areas on climate change impacts and adaptation practices.

72. A specialist media or communication firm will be contracted to design a contextually appropriate campaign using a participatory process. The awareness-raising campaign will include developing visual communication materials and

[58] Relevant private sector partners will be identified by the service provider in consultation with the 10 government institutions. These private sector partners will be targeted because of their location in the coastal zone and the relevance of coastal climate change adaptation to their businesses.

knowledge products as well as facilitating local radio programmes and community meetings, which will serve as the primary platforms to engage communities. The communication materials — for example, billboards, posters, brochures and stickers — will be distributed within the four communities prioritised by the project[59] — West Point, Jacob's Town, Fiamah and Topoe Village — and will be erected in public spaces. These four communities have been selected based on their vulnerability to the impacts of climate change, linked to their proximity to the coastline (West Point) and degraded areas of the Mesurado Wetland (Jacob's Town, Fiamah and Topoe Village). The campaign will be designed in a gender-sensitive manner with a specific emphasis on differentiated impacts of climate change on social groups practicing a variety of livelihoods, as well as including information on a range of potential adaptation practices. The development and dissemination of knowledge products under this campaign will be further supported by upscaling the EKMS under Activity 2.4, which will serve as a repository for the knowledge products developed by the contracted firm.

73. To support further awareness-raising on coastal climate change impacts under this activity, knowledge-sharing groups will be established in each of the four communities prioritised for awareness-raising, making use of existing community structures where they exist. These groups will be overseen by the Community Stewardship Committee (CSC) established under Sub-activity 3.1.2 and linked to the knowledge and innovation centre established under Sub-activity 3.1.1. These groups will facilitate on-the-ground awareness-raising events, conduct visits to schools and various community organisations to distribute the knowledge products and raise awareness on climate change impacts and the actions that can be taken by communities to help reduce the pressures placed on local environments, such as the use of energy-efficient cookstoves introduced under Activity 3.4 over traditional wood fuel cookstoves. This activity will be implemented by the EPA and financed by GCF, UNDP and GoL.

74. Sub-activities under this activity will include:

Sub-activity 2.5.1. Design an awareness-raising campaign on climate change impacts and adaptation practices.

75. Under this activity, a media firm with experience in West Africa and expertise in communications, climate change and gender will be contracted to design an awareness-raising campaign on climate change impacts and adaptation practices. The firm will develop the initial campaign material through a participatory approach with relevant institutions and will update the campaign on an annual basis based on lessons learned — collected by the knowledge-sharing groups engaged under Sub-activity 2.5.3. The awareness-raising campaign and knowledge products will prioritise the impacts of climate change on mangroves and on the coastline, gender-specific climate change vulnerabilities and a range of adaptation practices, including activities that reduce pressure on mangrove ecosystems.

Sub-activity 2.5.2. Host radio programmes on climate change impacts, vulnerability and adaptation practices in coastal and wetland areas.

76. Under this activity, radio shows will be hosted every three months on climate change impacts, vulnerability and adaptation practices relating to coastal and wetland areas. The radio shows will each be one hour long and will be facilitated by government or project employees with experience and knowledge on climate change and adaptation practices. Additionally, the shows will include provision for men and women in communities to call-in and present questions and/or input, capitalising on local ecological knowledge as well as ongoing innovative and sustainable adaptation strategies.

Sub-activity 2.5.3. Establish, train and engage community-based knowledge-sharing groups to support on-the-ground awareness-raising and knowledge-sharing.

77. Under this activity, knowledge-sharing groups will be established in each of the four communities selected for awareness raising under the project. The knowledge-sharing groups will consist of four members in each of the four areas — at least half of whom will be women. The members of the knowledge-sharing groups will be nominated by their communities and the selection process will be managed by the Community Stewardship Committee (CSC) established under Sub-activity 3.1.2. The roles and responsibilities of the knowledge sharing groups will be set out in the constitution of the CSC and all members of the knowledge-sharing groups will sign agreements with the CSC

[59] West Point, Jacob's Town, Fiamah and Topoe Village

when they are selected. As the Executing Entity, the EPA will oversee this process and ensure and verify that the selection of the members of the knowledge-sharing groups is consistent with the needs of the project. The knowledge-sharing groups will be trained through annual five-day workshops and will be paid a monthly stipend for the duration of the project. The stipend will cover the time and transport costs of the group members and will be transferred to each group member using mobile money services. The EPA will also oversee the payments of stipends to the knowledge-sharing groups, which is included in the project budget and co-financed by the GoL. The groups will be mandated to assist the project in raising awareness by facilitating workshops organised under Sub-activity 2.5.5 as well as to collect lessons learned on the awareness-raising campaign and success of the knowledge products under Activity 3.3. The training of the knowledge-sharing groups will include links with the AfDB/GCF project focusing on climate information and early warning systems (EWS). In collaboration with the project management team from that project, synergies will be developed, including supporting last mile coverage of EWS through the knowledge-sharing groups. The community members engaged under this activity will be overseen by the Community Steering Committee (CSC) established under Sub-activity 3.1.2 and will be based at the education and innovation centre established under Sub-activity 3.1.1. In addition to assisting the project with awareness-raising workshops, the groups will conduct community outreach on a monthly basis, including undertaking visits to schools and other community gatherings for the dissemination of information on climate change, adaptation practices and co-management of natural resources.

Sub-activity 2.5.4. Install and distribute awareness-raising knowledge products within the four communities prioritised for awareness-raising interventions.

78. The visual media and other knowledge products designed under Sub-activity 2.5.1 will be produced, installed and disseminated under this activity. These products will include billboards, posters, brochures, stickers, caps and t-shirts. Large visual products, such as billboards and posters will be installed in open public spaces and around communal areas such as schools and community buildings. The smaller products — including stickers, brochures, caps and t-shirts — will be distributed through the education and innovation centre established under Sub-activity 3.1.1 as well as by the knowledge-sharing groups established under Sub-activity 2.5.3. These products will be refined annually by the firm contracted under Sub-activity 2.5.1 taking into account lessons learned by the knowledge-sharing groups and incorporating information from relating to early warning systems from the AfDB/GCF project.

Sub-activity 2.5.5. Organise community meetings to raise awareness around climate change adaptation and the benefits of proposed interventions.

79. Under this activity, gender and socially sensitive community meetings will be held to raise awareness around climate change impacts, adaptation practices and the benefits of the interventions being implemented under the project. A total of three community meetings will be held annually in each of the four areas prioritised for awareness-raising under the project. Two of the three meetings will be accessible to any members of the community and one will be exclusively accessible to women to provide a forum for raising awareness on gender-specific climate change impacts and adaptation practices, as well as to disseminate information on women's rights, women's empowerment and strategies for reducing gender-based violence.

80. These community meetings will be held at venues in the communities to ensure that they are accessible to community members and will be co-facilitated by the knowledge-sharing groups established under Sub-activity 2.5.3. The meetings will also serve as events to distribute the smaller knowledge products designed under Sub-activity 2.5.1 and to implement other awareness-raising actions — including setting up billboards in public spaces and disseminating information posters to schools.

Output 3. Protecting mangroves and strengthening gender- and climate-sensitive livelihoods to build local climate resilience in Monrovia.

81. Under Output 3, the proposed project will put measures in place to reduce the anthropogenic pressure on mangrove ecosystems in the Mesurado Wetland. By reducing this pressure, the health of the ecosystems will be improved and their resilience to sea-level rise (SLR) and the impacts of climate change will increase, so increasing the resilience of the communities in Monrovia that rely on these ecosystems for their livelihoods. In addition, reducing the degradation of mangroves will increase the likelihood of their long-term survival and adaptation to climate

change through migration. These mangroves are likely to be negatively impact by climate change-induced SLR which will decrease the area of the Mesurado Wetland in which they can grow. In addition, urban encroachment into the mangrove forests and the harvesting of mangroves for fuelwood, charcoal production and fish smoking have resulted in degradation of the mangrove ecosystems in Monrovia[60]. In order to develop long-term climate resilience within vulnerable, mangrove-dependent communities there is a need to: i) support community conservation of mangroves in the MMA to secure the provision of critical ecosystem services in the face of observed and expected climate change impacts; and ii) develop local climate-resilient livelihood activities and strengthen market access for small and informal enterprises engaged in these activities. These community level approaches to adaptation will benefit from improved coastal protection measures being implemented under Output 1 and will integrate with the long-term adaptation planning being developed under Output 2, to support transformative change at both institutional and community levels.

82. Activities under this output will target four communities adjacent to degraded areas of the Mesurado Wetland. West Point, Jacob's Town, Fiamah and Topoe Village are particularly vulnerable to the impacts of climate change as a result of their proximity to and reliance on the coast and mangroves for their livelihoods (Figure 13)[61]. These communities will also play an important role in ensuring the long-term protection of these mangrove ecosystems as they surround degradation hotspots in the Mesurado Wetlands. These activities will focus on four integrated thematic areas for building climate resilience, namely: i) safeguarding ecosystem services provided by mangroves; ii) improving community knowledge on climate change impacts and adaptation practices; iii) reducing pressure on mangrove forests from fuelwood harvesting by reducing demand for fuelwood; and iv) strengthening climate-sensitive livelihoods and supporting the uptake of climate-resilient livelihoods. To capitalise on the experience of NGOs based in Monrovia, the project will seek to engage NGOs — with experience in the four aforementioned thematic areas — as service providers for relevant activities under this output.

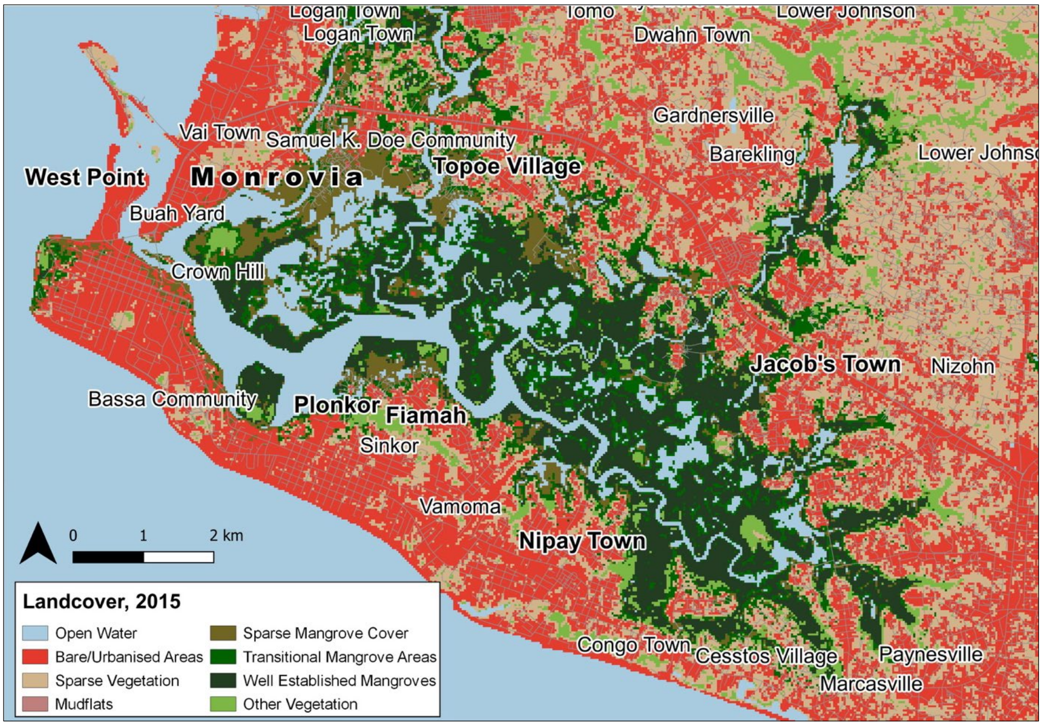


Figure 13. Landcover map of Monrovia, showing the Mesurado Wetland and focus areas (in bold text) for the proposed project — West Point, Topoe Village, Jacob's Town, Plonkor and Fiamah.

83. To ensure the safeguarding of mangrove ecosystem services, a community-centred co-management agreement will be developed based on the principles of Community-Based Natural Resource Management (CBNRM), developed through extensive stakeholder engagement. An awareness-raising campaign will be conducted to inform the focus area communities about the co-management agreement, alternative livelihood opportunities, climate change impacts and the importance of mangrove ecosystems. The impact of these awareness-raising activities will

[60] Annex 2.D: Mangrove Sub-assessment provides details of how the Mesurado Wetland supports livelihoods of communities in Monrovia as well as an analysis of the threats to these ecosystem both from climate change and anthropogenic pressure.
[61] See Section 3 in Annex 2.D: Mangrove Sub-assessment.

be assessed through annual state-of-knowledge assessments. Finally, to reduce pressure on the mangrove ecosystems from fuelwood harvesting, facilities for the manufacturing of energy-efficient cookstoves will be established at the education and innovation centre and solar-powered cold storage facilities will be established near major fish processing sites to reduce the need for fish smoking. The cold storage facilities will be managed by women in the community and will significantly reduce food wastage, which will contribute to making fisheries more resilient to climate change impacts. In this way, the facilities will support the vulnerable coastal communities to increase their income and improve their adaptive capacity. The incorporation of solar self-sustaining cold storage facilities into fishing communities is a priority within the MMA because of the limited infrastructure and capacity for generating electricity required to prolong the shelf-life of fish prior to transport to the market.

Activity 3.1. Establish a community education and innovation centre to function as a community knowledge generation and learning hub, repository on climate change adaptation practices and to host community activities under Output 3.

84. Under this activity, an education and innovation centre will be established in West Point, through co-financing, to host and coordinate activities under Output 3. Because of its central location, the centre will be accessible to communities across Monrovia and will act as a repository for material on awareness-raising, knowledge-sharing, livelihood development and innovation related to climate change[62]. An existing building on community- or government-owned land will be identified and renovated to make it fit for purpose and accessible to people with disabilities. Multi-purpose facilities, including cooking areas, open training areas and appropriate sanitation facilities will be installed as part of the renovations, thereby ensuring that the centre serves as a multi-purpose space and receives buy-in from the actively engaged local communities. Additionally, the centre will host training on the community-level maintenance of the revetment build under Output 1 (see Annex 21: O&M Plan). All activities and initiatives implemented through the centre will incorporate gender-responsiveness as a primary consideration.

85. Management of the centre will be undertaken by the community knowledge-sharing groups established under Sub-activity 2.5.3 with oversight provided by the project management team and the Community Stewardship Committee (CSC) established under Sub-activity 3.1.2. The centre will be governed by a set of by-laws agreed upon by the community or government entity that owns the centre and the CSC, in consultation with the MPW and MoGCSP. The by-laws will be subject to review and acceptance by the Executing Entity. The CSC will be legally registered as a community-based organisation (CBO). Two bank accounts will be opened by the CSC, in association with the centre to facilitate: i) the management of income from the cookstoves sold under Activity 3.4 and purchase of further materials for manufacturing the cookstoves; and ii) the management of the nominal fees from the usage of the cold storage facilities established under Activity 3.5 and maintenance of these facilities. The income derived from these activities will be managed by the CSC and reported on quarterly to the PMU for the duration of the project. All income will be used to support the sustainability and upscaling of project activities and will not be considered profit. Indicative estimates of the income managed by the CSC are provided in the descriptions of Activity 3.4 and 3.5 below. After project completion, the funds will also be used to continue to pay a monthly stipend to the CSC members[63]. In addition to facilitating the various community training workshops, the centre will act as the home base for the CSC and for consultants conducting assessments under Activities 3.2; 3.3 and 3.5. The establishment of the centre as a physical location will significantly contribute to the impact of Output 3 in developing alternative livelihoods, increasing community awareness on climate change and shifting the community's perception of the value of mangroves. This activity will be implemented by the EPA and financed by GCF, UNDP and GoL.

86. Sub-activities under this activity will include:

Sub-activity 3.1.1. Locate, renovate and equip existing structure situated on community or municipal-owned land in West Point.

[62] The project proposes to upgrade existing buildings/structures to establish the innovation centres, which will minimise the cost of this intervention.

[63] The expenditure to pay the CSC members is USD500 per month. If the income is insufficient to cover this cost, the EPA will support the CSC to collectively decide whether to reduce the monthly stipend for each member or other options that will enable the CSC to continue functioning effectively. It is expected that the time commitment expected from the CSC will be reduced after the project has been completed.

87. Under this activity, a site will be identified by the PMU in consultation with community and government stakeholders, to serve as the education and innovation centre and an agreement will be reached with the respective community, government department or municipality overseeing it[64]. The site will be selected according to: i) availability and willingness of owners for it to be used by the community; ii) size and suitability for use as an education and innovation centre; and iii) location and accessibility to the community. The centre will be renovated by the Ministry of Public Works, overseen by the EE, to ensure that is a safe environment to host community engagements and accessible to people with disabilities. It will be outfitted with basic cooking, training and sanitation facilities to ensure that it acts as a community gathering area. The newly renovated centre will also be equipped with basic furniture or with other equipment — for example, tables and seating — to facilitate the hosting of workshops and other training activities. Maintenance costs for the centre are expected to be minimal as only basic and durable facilities will be installed. A Memorandum of Understanding (MoU) will be reached with the community or government entity that owns of the renovated centre to undertake any required maintenance for the duration of the project.

Sub-activity 3.1.2. Establish and commission the ten-person Community Stewardship Committee with representatives from each of the four areas.

88. Under this activity, the Community Stewardship Committee (CSC) will be established and registered as a community-based organisation to provide high-level oversight for all community-based activities under Output 3 and 1. The members of the CSC will be selected every 3 years based on community nominations and each member will enter into a contractual agreement with the EPA which will outline their roles and responsibilities and the conditions of their stipends[65]. They will include at least 50% women and be representative of the four focus areas (West Point, Topoe Village, Jacob's Town, and Fiamah and Plonkor) and include the women managing the cold storage facilities under Activity 3.5. The CSC will provide community oversight and support for each of the activities under Output 3 and community-based activities under Output 1 (revetment maintenance). The education and innovation centre will act as a headquarters for the CSC who will host regular meetings with the knowledge-sharing groups to provide input and guidance. The PMU will facilitate the development of a formal constitution for the CSC, which will be finalised and validated during the workshop under Sub-activity 3.1.4. The constitution will set out the roles and responsibilities of the CSC and the knowledge-sharing groups established under Activity 2.5. In addition, two bank accounts will be opened and governed by the CSC to manage the income and expenditure relating to: i) the energy efficient cookstoves developed under Activity 3.4.; and ii) the use and maintenance of the cold storage facilities developed under Sub-activity 3.5.2. These funds will also be used to pay monthly stipends to CSC members after the end of the project lifecycle. The CSC will report to the PMU every quarter on the income and expenditure relating to these two activities for the duration of the project.

Sub-activity 3.1.3. Host an inauguration event to open the centre to the public and present the members of the Community Stewardship Committee.

89. To promote the education and innovation centre and secure community buy-in, an inauguration event will be held at the centre. This event will incorporate awareness-raising activities and during the event community members nominated for the CSC — as well as the by-laws governing the education and innovation centre — will be presented by local officials and community authorities.

Sub-activity 3.1.4. Host a collaboration workshop for government representatives, CSC members and NGOs to meet and define the guiding principles for the education and innovation centre.

90. Under this activity and subsequent to the launch of the education and innovation centre, a two-day collaboration workshop will be organised for all members of the CSC, as well as representatives from relevant government ministries, NGOs, development, conservation and community organisations. This workshop will provide an opportunity for all relevant stakeholders to: i) finalise and validate the constitution of the CSC; ii) define a set of guiding principles for the education and innovation centre; iii) develop linkages between similar ongoing initiatives; and iv) share innovative ideas and lessons learned from past experiences. Additionally, during the workshop, guest speakers on each of the three thematic areas prioritised under Output 3 — CBNRM, local livelihood development

[64] The buildings will not be purchased under the project – instead existing community-owned structures (structures owned by community groups or local government). There will be no issues with property rights since the project will be utilising community or public land and/or buildings.
[65] These contractual agreements between the CSC members and the EPA will apply both during and after the project period.

and awareness-raising — will be invited to give presentations as well as experts on climate change, ICZM and gender mainstreaming.

Activity 3.2. Establish community-led co-management agreement to ease anthropogenic pressure on mangroves in the MMA.

91. To safeguard ecosystem services and livelihoods that depend on mangroves, the project will facilitate the development of a community-led co-management agreement for mangroves in the Mesurado Wetland[66]. The agreements will be underpinned by the existing land-use management framework and the improved institutional and regulatory framework for ICZM developed under Output 2. They will also be supported by awareness-raising activities and incentives for alternative livelihood practices developed under Activities 3.4 and 3.5. The development of the co-management agreements will be facilitated by expert negotiators and will be undertaken through a participatory approach — engaging the CSC, relevant government institutions, NGOs and civil society. Participation of these stakeholders in the design process will ensure that community needs and concerns are incorporated into the agreement and that community buy-in and ownership of the agreement are developed. The agreement will prioritise equitable access and will seek to shift the paradigm of community members from considering themselves as users of mangrove ecosystems to participating in the co-management of the ecosystems. These objectives will be supported by the engagement and relationship-building between stakeholders. The co-management agreements will be developed in concert with the development of the ICZMP under Activity 2.1 and, like the ICZMP, will be reviewed and revised during the project period. This will ensure that both processes have opportunity for learning and that the CBNRM co-management model developed under Activity 3.2 can be incorporated into the ICZMP. The development of the co-management agreement will draw on lessons learned from similar initiatives elsewhere in Liberia[67] and will provide a foundation for future initiatives to conserve mangroves in Monrovia. This activity will be implemented by the EPA and financed by GCF.

Sub-activities under this activity will include:

Sub-activity 3.2.1. Develop the community-led co-management agreement through a participatory, gender-sensitive process.

92. Under this activity, two national consultants with experience in facilitation and natural resource management will be contracted to draft the co-management agreement. The consultants will undertake extensive consultations with conservation organisations, government institutions and community members to develop an understanding of the various pressures currently exerted on mangroves, as well as the various uses of mangroves and mangrove products by communities. These engagements will provide the consultants with the information necessary to draft the co-management agreement in such a manner as to ensure that the concerns of all relevant stakeholders are taken into account and incorporated into the agreement and provide an opportunity for stakeholders to build relationships and share information related to the complexities of managing urban mangrove ecosystems. Building on the institutional and regulatory framework for ICZM (Output 2), and the initiatives targeting alternative livelihood practices in Output 3, the co-management agreement will include the synthesis of positive incentives and for the sustainable use and conservation of the mangrove ecosystems in the Mesurado Wetland. The primary parties to the co-management agreement will be the communities of West Point, Topoe Village, Jacob's Town, Plonkor and Fiamah and the EPA. Through the consultation under this sub-activity, the national consultants will identify and confirm additional parties to the agreements, including, for example, NGOs like Conservation International and government agencies like the Ministry of Gender, Children and Social Protection (MoGCSP). Details of the monitoring of conservation actions (linked to Activity 3.3) and delivery of the agreed benefits will also form part of the co-management agreement.

Sub-activity 3.2.2. Host a three-day co-design and validation workshop for the community-led co-management agreement.

[66] Local buy-in to proposed co-management agreements is critical to this activity and will be facilitated by an analysis of lessons learned from similar co-management projects undertaken in Liberia. The analysis on mangrove ecosystems in the Mesurado Wetland (Annex 2.D) and engagement with Conservation International in particular will contribute to further inform this activity.
[67] These include the GEF-funded Conservation International project entitled "Improve sustainability of mangrove forests and coastal lamangrove areas in Liberia through protection, planning and livelihood creation – building blocks towards Liberia's marine and coastal protected areas", and the GEF-funded World Bank project entitled "Consolidation of Liberia's Protected Area Network"

93. Under this activity, a three-day co-design and validation workshop will be hosted for relevant government ministries, NGOs, conservation organisations and CSC members to engage on the draft co-management agreement developed under Sub-activity 3.2.1. Women and vulnerable groups will be represented at the workshop to ensure the inclusion of their specific concerns in this process. The workshop will be facilitated by the national consultants contracted under Sub-activity 3.2.1 and will provide an opportunity for any outstanding concerns or points of contention to be raised by concerned parties. The final day of the workshop will be used as a validation session to ensure that the final version of the co-management agreement incorporates the needs of all relevant stakeholders and secures buy-in from communities, government and NGOs alike.

Sub-activity 3.2.3. Design and implement an awareness-raising campaign on sustainable Community-Based Natural Resource Management (CBNRM) and the co-management agreement.

94. Two national consultants or service providers with experience in community awareness-raising and visual media design will be contracted under this activity to develop an awareness-raising campaign to disseminate information on the benefits of Community-Based Natural Resource Management (CBNRM) and the sustainable management of mangroves. The campaign will also include information on detrimental environmental practices, such as sand mining along beaches and rivers, and information on practices that help reduce the pressure on the environment, such as the use of the energy-efficient cookstoves introduced under Activity 3.4 over traditional wood fuel cookstoves. The service provider will work in collaboration with the national consultants contracted under Sub-activity 3.2.1 to ensure that the knowledge products developed: i) align with the co-management agreement and are gender-sensitive in their design; ii) are informed by local ecological knowledge; iii) depict the uses of mangroves by different social groups and sectors; and iv) provide an appropriate level of information for dissemination within communities. The knowledge products produced under this activity will include posters for each of the four areas, brochures to be distributed through the education and innovation centre and large signboards that will be installed within communities and at commonly-used entry points to access the mangroves, particularly in degradation hotspots. The awareness-raising campaign strategy and materials will be updated annually and informed by the state-of-knowledge assessments conducted under Sub-activity 3.3.2. This intervention will build on links with the AfDB/GCF project to strengthen Last Mile coverage of the early warning systems (EWS) developed under that project. This will be done by including information on EWS in the awareness-raising materials, in collaboration with the project management team of the AfDB/GCF project, and raising awareness about how to access EWS.

Sub-activity 3.2.4. Host a workshop to assess the effectiveness of the co-management agreement (Activity 3.2.1) and re-negotiate it in Year 5 based on findings from Activity 3.3.

95. Under this activity, a two-day workshop will be hosted for the CSC, relevant government institutions, NGOs and civil society in Year 5 of the project implementation period. At this workshop, the impact of the co-management agreement will be assessed based on information collected under Activity 3.3 and the experience of the relevant stakeholders. Drawing on these lessons learned, the co-management agreement will be revised and validated by the stakeholders at the workshop. Outcomes of this process will also inform the revision of the ICZMP in Year 6 under Activity 2.1.

Activity 3.3. Conduct annual assessments to evaluate the project-induced changes in mangrove degradation, community perceptions and awareness of climate change impacts, adaptation options and mangrove ecosystems throughout the project implementation period.

96. To assess the adaptive capacity of the four communities prioritised for awareness-raising and livelihood development, state-of-knowledge assessments will be undertaken periodically under this activity. These assessments — building on preliminary work undertaken during the PPF phase of the project — will consider community awareness on climate change impacts and adaptation options, CBNRM and the role of mangrove ecosystems in the MMA as well as the different types of climate-resilient livelihoods currently being practiced within the MMA. These surveys will be conducted periodically to determine the annual change-in-state, which will help to refine the awareness-raising campaigns and support the uptake of additional climate resilient livelihoods. The state-of-knowledge assessments will be complemented by periodic assessments of mangrove health and degradation in the Mesurado Wetland, conducted using the high-resolution spatial data procured under Sub-Activity 2.3.1. The reports produced from this analysis will assess the impact of the project on mangrove protection in Monrovia. The

primary purpose of this activity is not to inform the development of any interventions, but to act as a method to collect data on what interventions are effective as well as how successful community awareness raising, the co-management agreement and livelihood development activities are on an annual basis. Furthermore, the collection of data will enable research institutions, including, for example the University of Liberia, to strengthen their knowledge and understanding of coastal livelihoods and ICZM in Liberia. This will contribute to the development of national capacity for driving ICZM. In addition to the long-term benefits, ongoing monitoring and assessment will support the adaptive management of the project — including through the revision of the ICZMP and co-management agreement — and contribute to the development of further effective awareness raising activities and livelihood development initiatives beyond the lifespan of the project. To this end, lessons learned collated under this activity will be made available through the EKMS under Activity 2.4. This activity will be implemented by the EPA and financed by GCF, UNDP and GoL.

Sub-activities under this activity will include:

Sub-activity 3.3.1. Using high-resolution remote sensing data procured under Sub-activity 2.3.1, conduct analyses of mangrove extent and degradation in the Mesurado Wetland to assess the impact of project interventions

Under this activity, a report on trends in mangrove health and degradation in the Mesurado Wetland will be produced based on the high-resolution remote sensing data procured under Sub-activity 2.3.1. First, the report will provide an analysis of historical trends in mangrove degradation over the ten years prior to the project implementation period. This report will provide a baseline for measuring project impact on mangrove protection. The baseline report will be updated based on new datasets acquired in Years 4 and 6 of the project period to assess the impact of the project on mangrove degradation trends. These updates will be used in concert with those produced under Sub-activity 3.3.2 to evaluate the effectiveness of the co-management agreement and awareness-raising activities (Activity 3.2) and to collate lessons learned from the project interventions.

Sub-activity 3.3.2. Conduct a baseline state-of-knowledge survey on climate change impacts, adaptation options, CBNRM and mangrove ecosystems and undertake annual surveys to determine the change-in-state over the duration of the project.

97. Under this activity, a service provider will be contracted to conduct community-level assessments in each of the four areas prioritised under the project to develop indicators that determine baseline community perceptions and awareness on climate change, adaptative strategies, as well as the role and importance of mangrove ecosystems. Surveys carried out in the first year will be collated into a report that will act as a baseline on the state of community knowledge on climate change, adaptation and mangroves. Further annual surveys will be conducted throughout the project period to determine the effectiveness of the awareness-raising programme. Reports produced from the annual surveys will act as lessons learned and will be used to by consultants engaged under Sub-activity 2.5.1 to update the awareness-raising campaign on an annual basis, as well as to inform revisions to the co-management agreement under Sub-activity 3.2.4, and by tertiary institutions to further research on coastal protection and management in Liberia. The reports produced under this activity will be made publicly accessible through the EKMS (Activity 2.4) to support the replication and upscaling of successful awareness-raising strategies.

Sub-activity 3.3.3. Undertake periodic assessments of climate-resilient livelihood practices and new opportunities to inform the design of new options and determine their uptake as a result of the project.

98. An important function of the education and innovation centre established under Sub-activity 3.1.1 will be to provide community members — particularly women — with alternative climate-resilient livelihood activities to increase their resilience to climate change. To support livelihood development and the uptake of climate-resilient livelihoods, a national service provider with expertise in the development of alternative, climate-resilient livelihoods will be contracted to assess the baseline of existing livelihood strategies (including income levels) in the four target communities. Building on the preliminary information generated as part of the project PPF phase[68], the service provider will detail a menu of potential alternative livelihood strategies and/or strategies to increase the climate-resilience of existing livelihood practices. During the project PPF phase, the manufacturing of eco-friendly

[68] See Annex 2.A: Feasibility Study and Annex 2.D: Mangrove Sub-assessment; Annex 6: Environmental and Social Assessment Report; and Annex 8: Gender Assessment and Action Plan.

cookstoves (Activity 3.4) was identified as a climate-resilient livelihood activity. Under Sub-activity 3.3.3, further alternative climate-resilient livelihoods will be identified by the service provider and incorporated into the training of community members under Activity 3.4, to support the ongoing adoption of climate-resilient livelihood practices beyond the implementation period and geographical scope of the project. An alternative livelihoods option targeted under the LDCF UNDP Liberia project in Sinoe County aims to stimulate the Compressed Stabilised Earth Blocks (CSEB) industry as alternative to beach and river sand mining. This will be assessed for viability in Monrovia, along with the viability of agricultural residue for use as fuel-pellets. The assessment of agricultural residue will identify whether the fuel-pellet industry can foster and promote a circular economy while also reducing pressure on mangroves and forests. The livelihoods being assessed by the service provider will specifically focus on practices that can be taken up by women and other vulnerable people. In addition to the identification of alternative livelihoods, the assessment will identify the viability of improved fish smoking kilns[69] as complimentary interventions to the demonstration site at the EIC and improved cookstoves under Activity 3.4 and as support to the cold storage units introduced under Activity 3.5.

99. The service provider will additionally undertake annual surveys to determine the uptake of these livelihoods within the four communities — as well as outside of the project focus areas —, monitor the effectiveness of the financing plan developed under Activity 2.2.2 and prepare annual reports. In addition, the annual surveys and reports should be used to provide information on whether the livelihoods promoted by the project are appropriate and are able to secure community buy-in. Further upscaling of climate-resilient livelihoods will be supported by presenting the reports, which will act as a resource on what types of livelihoods are seen as appropriate as well as which of these gain the greatest buy-in by local communities.

100. The dissemination of information from these assessments — which will be collated into a report and associated knowledge products — will make use of the EKMS (Activity 2.4) and will contribute to the uptake of further alternative and diversified livelihood opportunities facilitated by the education and innovation centre. Furthermore, by providing access to these knowledge products the project will enable tertiary institutions to explore and support the replication and upscaling of successful strategies.

Activity 3.4. Establish small-scale manufacturing facilities and develop training material to capacitate community members to manufacture and sell cookstoves to support alternative climate-resilient livelihoods.

101. Under this activity, the project will support the uptake of diversified climate-resilient livelihoods and improved mangrove management by developing training courses, establishing small-scale manufacturing facilities, and providing material inputs for the fabrication of value-added products — specifically energy-efficient cookstoves[70]. The manufacturing facilities will be incorporated into the education and innovation centre in West Point (established under Activity 3.1) and will be accessible to communities from around Monrovia, including the other project focus areas of Topoe Village, Jacob's Town, and Fiamah and Plonkor. The activity will focus on developing alternative climate-resilient livelihood opportunities for women and other vulnerable groups, in collaboration with the CSC established under Activity 3.1. The activity will build on lessons learned from the *African Improved Cooking Stoves Programme of Activities*, and training materials will be made publicly accessible to support the upscaling of the activity to other parts of Monrovia. This activity will be implemented by the EPA and financed by UNDP.

102. Energy-efficient cookstove production was selected as a focus for this activity because there is a market for these stoves in Monrovia and the construction of the stoves requires minimal material input or training. In addition to providing alternative livelihood opportunities, supporting the uptake of energy-efficient stoves across the MMA will support the implementation of the mangrove co-management agreement (Activity 3.2) by reducing existing anthropogenic pressure on the MMA's mangroves, specifically relating to harvesting of mangroves for fuelwood[71]. In the awareness-raising campaign under Activity 3.2, alternative livelihood practices such as the use of improved, energy-efficient cookstoves will be promoted and links to Activity 3.4 will be made as a way to support behaviour change within the target communities. Although the largescale adoption of energy-efficient cookstoves has potential

[69] An example of an improved kiln is the FAO Thiaroye Processing Technique kiln, details of which are available online at: http://www.fao.org/3/a-i8301e.pdf
[70] See Section 7.2 of Annex 2.D: Mangrove Sub-assessment for further information on the proposed design of these cookstoves.
[71] See Section 2 of Annex 2.D: Mangrove Sub-assessment for details of fuelwood harvesting in the Mesurado Wetland.

benefits in terms of greenhouse gas (GHG) emission reduction[72], the mitigation co-benefits of this project activity are very small. It is expected that the use of project-supported stoves[73] will reduce GHG emissions by ~4,750 tCO_2e[74,75] during the project period and by ~1,580 tCO_2e/year (less than 0.0002% of Liberia's annual GHG emissions[76]), for the lifespan of the stoves. Because of the small scale, emission reduction benefits of the activity will not be measured under the project. Training under Sub-activity 3.4.1 and workshops under Sub-activity 3.4.3 will, however, develop skills and capacity for community members to upscale energy-efficient cookstove production beyond the lifetime and geographical scope of the project. Additionally a larger scale fish smoking kiln system (which can be used for larger volumes of fish, for example by collectives of fish mongers)will be installed at the EIC as a demonstration site, allowing communities to test this technology, and training will be provided on the benefits of the technology for potential up-scaling[77].

Sub-activities under this activity will include:

Sub-activity 3.4.1. Develop a training programme and publicly available training materials and conduct trainings to capacitate community members to manufacture energy-efficient cookstoves.

103. In order to provide communities with immediate benefits — as well as to contribute to increasing energy efficiency and reducing demand for fuelwood — a national service provider will be contracted under this activity to develop a training course and provide training on the construction of energy-efficient stoves through a training-of-trainers approach. The service provider will prepare a training manual for constructing these stoves, which will be accessible at the education and innovation centre (Activity 3.1) and through the EKMS (Activity 2.4). The service provider will also organise trainings at the centre to ensure that interested community members have the requisite information to construct the energy-efficient cookstoves. The service provider will undertake further annual trainings at the education and innovation centre to capacitate community members to undertake a variety of diversified livelihoods or develop a range of value-added products, based on the results of the assessments conducted under Sub-activity 3.3.2. This training, and specifically the training-of-trainers approach, will contribute to the sustainability of the livelihood activities beyond the project lifespan and facilitate the replication of the activities in other communities in Monrovia.

Sub-activity 3.4.2. Establish a small-scale manufacturing workshop and procure equipment and material for the education and innovation centre.

104. Under this activity, basic facilities — including open-air roofed structures — will be constructed and installed in close proximity to the education and innovation centre. These structures will be used during the training conducted under Sub-activity 3.4.1. Additionally, under this activity material inputs will be provided by the project to facilitate the initial construction of the energy-efficient cookstoves[78]. These materials will provide the initial financial input for manufacturing the first batch of stoves, with additional material inputs for the construction of further cookstoves to be entirely provided by income gained from selling the stoves[79]. The sale of the cookstoves will be managed by the CSC. Through the bank account established under Sub-activity 3.1.2, management of this income and

[72] The African Improved Cooking Stoves Programme of Activitieswill have a measurable impact on GHG emissions in Liberia and has been registered through the Clean Development Mechanism (CDM). Further information about this project is available at: https://cdm.unfccc.int/filestorage/7/T/S/7TSH21WOQ6CKNDXAZE48RJPG3FL5I9/PoA%205342%20CPA%2000014%20Liberia%20v4.0%2029032019?t=SE18cWwwZHZvfDALXuqkklfrkBFTXxfpsKD-

[73] US$16,000 is included in the project budget to purchase materials for the cookstoves and will support the manufacture of ~640 stoves (half in Year 2 and half in Year 4).

[74] Assuming the emissions reductions from the project-supported cookstoves will be equivalent to those of the *African Improved Cooking Stoves Programme of Activities* registered under CDM, each cookstove will reduce GHG emissions by ~2.47 tCO_2e per year.

[75] It is assumed that cookstoves manufactured in Year 2 will be operational from Year 3 and those manufactured in Year 4 will be operational from Year 5.

[76] calculated using national GHG emission information from the Intended Nationally Determined Contribution, available at: https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Liberia%20First/INDC%20Final%20Submission%20Sept%2030%202015%20Liberia.pdf

[77] Mindjimba, K., Rosenthal, I., Diei-Ouadi, Y., Bomfeh, K. and Randrianantoandro, A. 2019. FAO-Thiaroye processing technique: towards adopting improved fish smoking systems in the context of benefits, trade-offs and policy implications from selected developing countries. FAO Fisheries and Aquaculture Paper no. 634. Rome. FAO. 160 pp. Licence: CC BY-NC-SA 3.0 IGO.

[78] See Section 7.2 of Annex 2.D: Mangrove Sub-assessment for further information on the proposed design of these cookstoves.

[79] The income from manufacturing cookstoves is estimated at US$8,000/year (US$25/cookstoves x 320 cookstoves manufactured per year), where US$8,000 is budgeted for the initial purchase of material inputs and cookstoves are sold at cost price to incentivize uptake.

purchasing of additional material will be overseen by the CSC, in collaboration with the Project Management Unit and UNDP. Additional limited funds will be provided two years after the inception of this activity to purchase further material inputs based on the assessments undertaken under Sub-activity 3.3.2. Furthermore, the project will install an improved communal fish smoking kiln system, modelled after the FAO-Thiaroye fish processing technique (FTT), for use as a demonstration and education tool[80]. Introducing communal smoking facilities could change the way this informal economy operates. By demonstrating this technology, its potential up-take in the community can be assessed.

Sub-activity 3.4.3. Host an annual workshop at the education and innovation centre to capacitate women's groups on business strategies and connect them with value chain actors to upscale the development of energy efficient value-added products.

105. To support the uptake and upscaling of the livelihoods being supported under the project, annual workshops and consultations will be hosted at the education and innovation centre to act as information sessions and networking events for women's groups, fishmongers, value-chain actors and the domestic private sector, informal finance institutions and agencies. By facilitating relationship-building between community members and value chain actors, these events will provide practical support for the development of small cookstove manufacturing businesses. In addition to creating networking opportunities, the events will: i) enable business skills development; ii) provide training on marketing strategies and links to the awareness-raising campaign under Activity 3.2; and iii) facilitate access to finance for investments into, and upscaling of, the value-added products being developed at the education and innovation centre, including the energy-efficient cookstoves. These workshops will also support access to higher value markets for fishmongers — who will be benefiting from improved access to higher-value fresh fish as a result of the cold storage facilities established under Activity 3.5 — by connecting the fishmongers to hotels, restaurants and other local private sector actors. Informed by the livelihood assessment (Sub-activity 3.3.3), the events under this sub-activity in the later years of project implementation will also create space to explore other potential innovations and alternative livelihood opportunities, including for example, the upscaling of communal fish smoking facilities like those demonstrated at the education and innovation centre (Sub-activity 3.4.2).

Activity 3.5. Purchase and install low-maintenance eco-friendly cold storage facilities near fish processing sites to reduce pressure on mangroves and increase market efficiency.

106. The fishing industry in Monrovia accounts for a significant percentage of the food consumed by the population of the MMA, however limited access to cold storage means that most fish needs to be smoked to increase shelf life. This contributes to the demand for fuelwood from mangrove forests in the Mesurado Wetland[81]. Despite smoking, a significant proportion of the fish caught by communities does not make it to market (see Annex 2.D). To reduce waste and the reliance on fuelwood sourced from the mangroves as well as to improve food security for the MMA, cold storage units will be designed and installed in proximity to the major fish processing site in the MMA at West Point[82]. The proposed cold storage facilities under this activity are intended to be accessible to all fisherfolk in the area and to be provided at a relatively small scale, based on a preliminary assessment of the costs of these interventions. The units will be designed to require limited maintenance and any required maintenance will be facilitated by designated women on the CSC — who will be mandated with overseeing the operation and condition of the cold storage units as well as facilitating access for local community members. Funding for required maintenance will be supported by charging those who use the units a small fee, to be paid into a community fund established under Sub-activity 3.1.2. This will ensure the sustainability of the units and the ability to meet their minimum maintenance requirements. This activity will be implemented by the EPA and financed by UNDP.

Sub-activities under this activity will include:

[80] Mindjimba, K., Rosenthal, I., Diei-Ouadi, Y., Bomfeh, K. and Randrianantoandro, A. 2019. FAO-Thiaroye processing technique: towards adopting improved fish smoking systems in the context of benefits, trade-offs and policy implications from selected developing countries. FAO Fisheries and Aquaculture Paper no. 634. Rome. FAO. 160 pp. Licence: CC BY-NC-SA 3.0 IGO.
[81] See Section 2 of Annex 2.D: Mangrove Sub-assessment for further information on anthropogenic pressures on the Mesurado Wetland.
[82] See Sections 6.5 and 7.1 of Annex 2.D: Mangrove Sub-assessment for further information on the proposed design of these cold storage facilities.

Sub-activity 3.5.1. Develop site-specific eco-friendly, solar-powered cold storage units for the West Point fish processing site based on the needs of fishers and fishmongers

107. Under this activity, a service provider will be contracted to develop site-specific designs for the cold storage units based on the needs of the community, building on prior community engagements that have established the basic requirements for facilities of this type[83]. The service provider will also be responsible for developing a management system, in consultation with the CSC, that aligns with the preferred method for community management of the units and focuses on empowering women. Two representatives of the fishing community will be selected for the CSC and will be responsible for overseeing the management of the cold storage units[84]. This information will be generated through stakeholder consultations undertaken with fisherfolk in the area. All stakeholder engagements will be conducted in a gender-sensitive manner and will take into consideration the needs of vulnerable people and women for utilising and accessing the cold storage units[85]. The same service provider will be contracted to undertake annual assessments to evaluate the differentiated use and impact of the cold storage units by fishmongers and to determine the financial feasibility of upscaling this initiative across Monrovia and other regions of Liberia.

Sub-activity 3.5.2. Host a workshop in the targeted community to validate the designs (Sub-activity 3.5.1) and capacitate fisherfolk in the use and management of the cold storage units.

108. The designs developed under Sub-activity 3.5.1 will be validated by community members at a training and validation workshop. The service provider contracted under Sub-activity 3.5.2 will also provide training on the use, management and maintenance of the cold storage facilities as well as making community members aware of the benefits of using these facilities. The workshop will be conducted in a gender-sensitive manner, to ensure that the needs and inputs of both women and men are considered in the finalisation of the design for the cold storage units. This will also ensure that all user groups receive the appropriate training and information.

Sub-activity 3.5.3. Install and operate the low-maintenance eco-friendly solar-powered cold storage units.

109. A firm with experience in developing similar solar-powered cold storage units will be contracted under this activity to construct and install two eco-friendly cold storage units at the West Point fish processing site based on the designs validated in Sub-activity 3.5.2. The cold storage units will be constructed from upcycled shipping containers and will not require regular maintenance. In addition to developing the units, the contracted firm will be responsible for providing technical specifications, designs and training to relevant government representatives and the CSC to support future maintenance of the units and the potential for upscaling this system across other coastal areas of Liberia[86]. These designs will be publicly accessible through the EKMS (Activity 2.4).

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

110. The project will be implemented following UNDP's National Implementation Modality (NIM), according to the Standard Basic Assistance Agreement (SBAA) between UNDP and the GoL, the Country Programme Action Plan (CPAP), and as per the policies and procedures outlined in the UNDP POPP.

111. UNDP provides oversight and quality assurance involving UNDP staff in Country Offices and at regional and headquarters levels. The quality assurance role involves objective and independent project oversight and monitoring functions. This will include overseeing the achievement of project management milestones that are established as per the board-approved project proposal. Project assurance is independent of the Project Management function. As such the National Project Steering Committee cannot delegate any quality assurance responsibilities to the National Project Coordinator and/or anyone paid for by the project resources. The project assurance role is covered by the Accredited Entity fee provided by the GCF. As an Accredited Entity to the GCF, UNDP is required to deliver GCF-specific oversight and quality assurance services including: i) day-to-day oversight and supervision; ii) oversight of project completion; and iii) oversight of project reporting.

[83] Refer to Annex 2.D Section 6.1.1 for details on possible service providers
[84] As the EE, the EPA will approve the management structure, which will be incorporated into the constitution of the CSC and the contractual agreementss between the two relevant CSC members and the EPA.
[85] See Section 5.6.1 of Annex 2.D and Annex 8 for further information on gender sensitivity in the context of this activity.
[86] The income from the cold storage facilities will come from charging those who use the units a small fee. The income is estimated at 1,500$/month (10$/month x 150 users).

Executing Entity

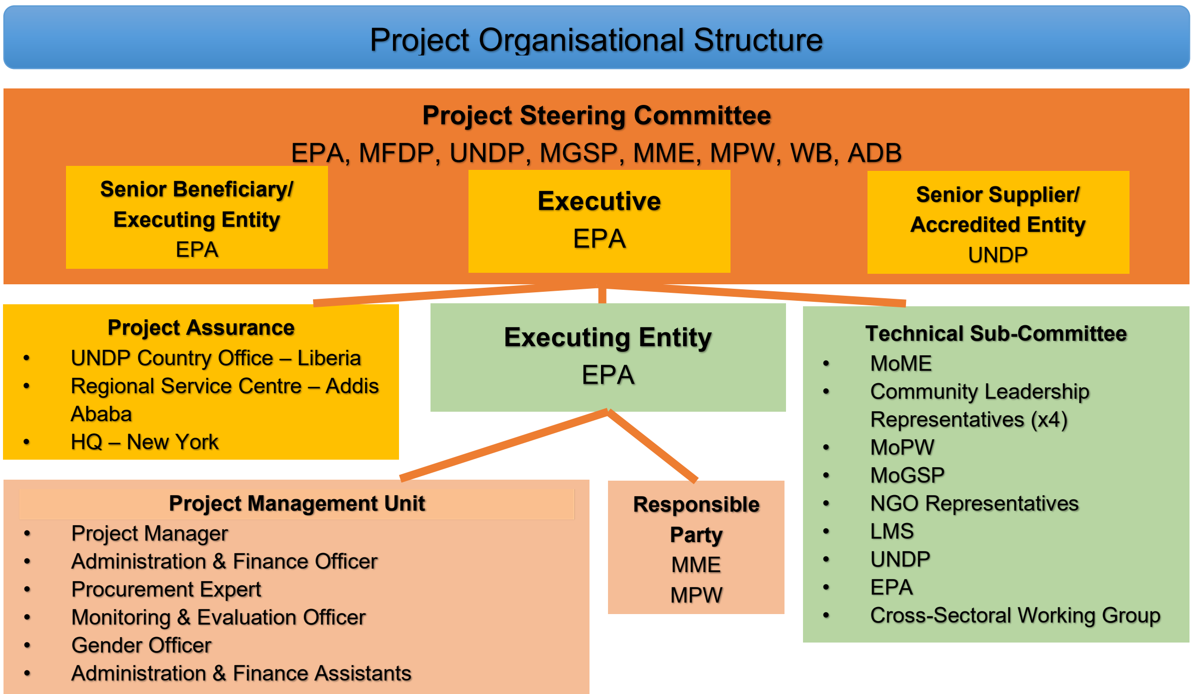
112. The national Executing Entity (EE) for this project is the Government of Liberia represented by the Environmental Protection Agency (EPA)[87]. The EE is accountable to UNDP for managing the project, including the monitoring and evaluation of project interventions, achieving project outcomes, and for the effective use of UNDP resources. Accredited Entity shall sign a project document with the EE. The Project Document will be legally binding and outline the detailed financial, procurement and implementation plan of the Project, as well as the relevant provisions for the compliance by the Executing Entity and Responsible Parties with the requirements from the AMA and FAA. The Ministry of Mines and Energy (MME), as a Responsible Party will enter into a legally binding Memorandum of Understanding (MoU) with the EE to assist in successfully implementing project activities and are directly accountable to the EPA as outlined in the terms of their agreement. Specifically, the Ministry of Mines and Energy will be responsible for Activities: 1.1 Prepare construction plans and finalise technical design specifications for coastal defence structure at West Point; 1.2 Construct coastal defence structure to protect West Point against climate change-induced coastal erosion; 2.1 Develop an Integrated Coastal Zone Management Plan for Liberia; 2.2 Capacitate the Cross-Sectoral Working Group to mainstream ICZM into relevant government sectors through a Training-of-Trainers approach; and 2.3 Strengthen the asset base and technical capacity of the ICZMU for the collection of spatial and biophysical coastal information to support the implementation of the ICZMP (Table 2). The EE will delegate to the MME as a Responsible Party — government entities with the technical competence in their respective areas — to ensure that the activity is implemented according to national requirements. A Memorandum of Understanding (MoU) will be established between the EE and the MME outlining their respective responsibilities under Activity 1.2. The funds for selected activities, as requested by the GoL, including Activity 1.2 (due to their higher value and high risk nature) will be channeled directly to service providers by UNDP as part of UNDP support services to GoL. Any other funding requests will be channeled through the EPA. The Ministry of Public Works (MPW) will assist the EE and the MME to implement Activity 1.2 and will be represented on the Project Steering Committee and Technical Sub-Committee. No funds will be channeled through the MME. Similarly, the Ministry of Gender, Children and Social Protection (MoGCSP) will be included on the Project Board to support gender sensitivity and oversight during the project implementation period.

113. The Project Board/Steering Committee will be chaired by the Director of the EPA and will be comprised of representatives from the EPA, MFDP, UNDP and MGSP, including at least 30% women. The World Bank and the African Development Bank will also be invited as members of the Steering Committee in order to realize synergies with their similar investments in the country, particularly the AFDB GCF-funded project. The Project Board is responsible for making, by consensus, management decisions when guidance is required by the Project Manager. Project Board decisions will be made in accordance with standards that shall ensure management for development results, best value for money, fairness, integrity, transparency and effective international competition. In case consensus cannot be reached within the Board, the UNDP Resident Representative (or their designate) will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed. The Project Board will meet according to established practices.

114. The Project Management Unit (PMU) will implement the project with support from the EPA and UNDP and will report to the Project Board. The Project Manager will be appointed to lead the PMU which will run the project on a day-to-day basis on behalf of the EPA within the constraints laid down by the Project Board. The Project Manager function will end when the final project terminal evaluation report and other project closure activities and documentation required by the GCF and UNDP have been completed and submitted to UNDP. The Project Manager is responsible for day-to-day management and decision-making for the project. The Project Manager's prime responsibility is to ensure that the project produces the results specified in the project document, to the required standard of quality and within the specified constraints of time and cost. The Executing Entity (Implementing Partner in UNDP terminology) will designate one of its employees as a National Project Director responsible for the overall direction, strategic guidance and timely delivery of the Project. Implementation of the project will be further supported by a technical sub-committee constituted of institutional, community and civil society representatives. The members of this committee will be called on to provide input for relevant project activities. The project management arrangements are summarised in the diagram below (Figure 14).

[87] The Environmental Protection Agency of Liberia (EPA) is a semi-autonomous body under the legal personality of the Government of Liberia (GoL). The EE for all activities under this project is the GoL represented by EPA.

Funds Flow: Donor funding, including the GCF funding, will be received by the administrative agent (UNDP) on behalf of the Executing Entity (GoL represented by the EPA) based on a standard letter of agreement (LOA) signed between the donors and UNDP and an MOU between UNDP and the Executing Entity separately. As per UNDP's National Implementation policy (NIM), the Project Management Unit's (PMU) capacity assessment, determined that funding may be advanced to entities for lower risk activities. UNDP will support the handling and channelling of funds for higher value activities that exceed a budget threshold of US$250,000 (for example, Activity 1.2) as a support service provisioned for in the LOA. For these higher value activities, a 'Direct Payments' option will be utilised where UNDP needs to ensure value for money and minimise any risks associated with exceeding the capacity of government agencies to manage procurement and deliver activities timeously. Through this option, transfers will be made directly from UNDP to the goods and service providers upon the request of the EE, and to the PMU for its day-to-day running functions. The Direct Payments modality ensures: i) that the request has come from an authorised official; ii) verification that the requested payment is in accordance with the project workplan; and iii) verification that payment is made to the designated party. For activities where a Direct Payments modality is being used, GCF Proceeds and the Accredited Entity's Co-financing will not flow through the Executing Entity. Instead UNDP (as administrative agent of the Executing Entity) will channel such funding via a Direct Payments option whereby transfer will be made directly to goods and services providers hired by the Accredited Entity and upon request of the Executing Entity. On the other hand, the Co-financing from the Government of the Host Country will flow to the EPA as budget allocations and used directly by such ministry to pay procured parties and/or transfer such funds to the relevant Responsible Party. For lower risk activities, funds will be advanced to the Executing Entity by UNDP and the Executing Entity will procure and transfer funders to the goods and service providers hired to implement those activities.



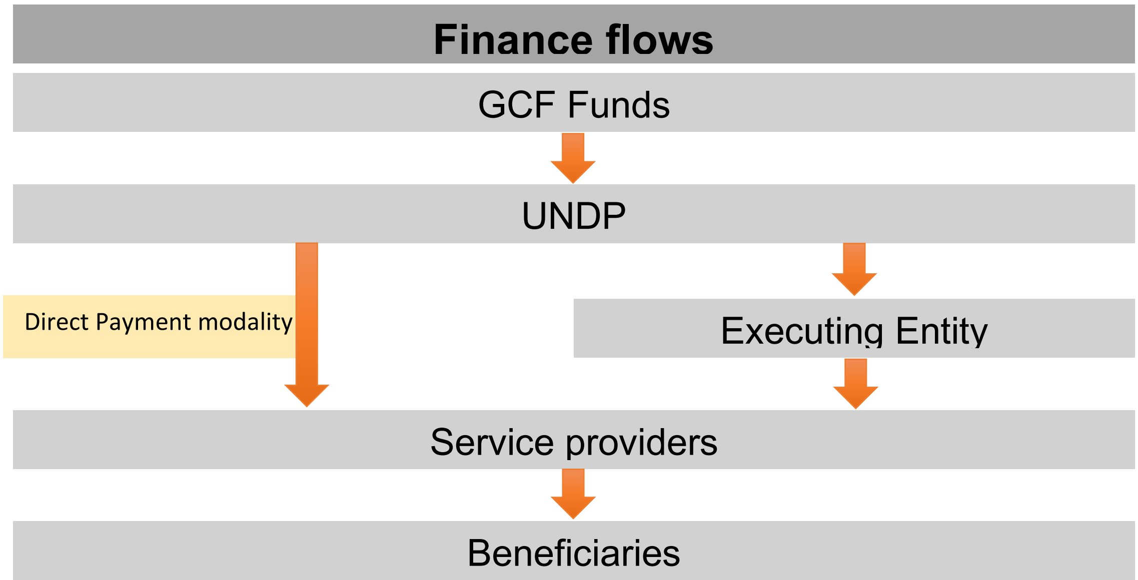


Figure 14. Institutional arrangements and financial flows.

Experience and track record of the Accredited Entity: UNDP

115. In Liberia, UNDP has led environmental protection and climate change responses through the facilitation of policy formulation, strategic planning, law making, coordination and information sharing. Through this role, UNDP has built strong relationships with decision-makers and proven its strengths as an impartial provider of technical advice and support. With its central role in the UN system, UNDP facilitates a multi-sectoral approach to assist the GoL to respond to complex issues such as climate change and green growth and is recognized as an experienced agency in institutional development and capacity building, bringing a long-term institutional and people-centred focus to capacity development.

116. As a multilateral organisation, UNDP can promote the dissemination of international norms and standards, bring technical assistance, experience and good practices to bear in Liberia. UNDP has demonstrated its long-term commitment to the provision of technical assistance to affect and sustain the institutional changes required in realising tangible improvements in institutional capacity. Over the last five years UNDP has implemented a coastal adaptation project, a climate change adaptation project in the agricultural sector, an early warning system project and an extractive industry sustainable development project.

Experience and track record of the Executing Entity: Government of Liberia represented by the Environmental Protection Agency (EPA)

117. Under the National Environmental Policy of Liberia (2003), the GoL set up the Environmental Protection Agency (EPA) as an independent authority for the management of the environment. The EPA is an autonomous body tasked with enforcing the National Environment Protection and Management Law under the Executive Branch of Government and is overseen by a nine-member board of directors from specific government agencies and the private sector. This board of directors is required to establish Technical Committees to advise the EPA on priority issues including climate change and marine and coastal ecosystems.

118. As the National Designated Authority (NDA) for Liberia, the EPA has been involved in the submission of several Concept Notes and Readiness Proposals for the GCF. These projects include: i) the Simplified Approval Process (SAP) Concept Note: *Resilient and Low Carbon Tree Crop Extension Project II*; and ii) *Liberia Adaptation Readiness Proposal*. Similarly, the EPA has a strong track-record of implementing climate adaptation projects for the Global Environment Facility (GEF). Between 2008 and 2018, the EPA implemented 12 GEF-funded projects with a focus on climate change adaptation in Liberia. The most recent of these projects are:

- *Building and Strengthening Liberia's National Capacity to Implement the Transparency Elements of the Paris Climate Agreement*, 2018 (GEF-ID 9923);

- *Conservation and Sustainable use of Liberia's Coastal Natural Capital*, 2017 (GEF-ID 9573); and
- *Improve Sustainability of Mangrove Forests and Coastal Mangrove Areas in Liberia through Protection*, *Planning and Livelihood Creation- as a Building Block Towards Liberia's Marine and Costal Protected Areas*, 2016 (GEF-ID 5712).

Table 2. Responsible Party, Legal Agreement, Beneficiaries and Co-finance for each Activity.

Activity	Responsible party	Legal agreement	Beneficiaries	Co-finance
Activity 1.1. Prepare construction plans and finalise technical design specifications for coastal defence structure at West Point.	MME	MoU between EPA and MME Contract with service providers	West Point	UNDP
Activity 1.2. Construct coastal defence structure to protect West Point against climate change-induced coastal erosion.	MME	MoU between EPA and MME MoU between EPA and MPW Contract with service providers	West Point	UNDP and GoL
Activity 2.1. Develop an Integrated Coastal Zone Management Plan for Liberia.	MME	MoU between EPA, and MME Contract with service providers	MMA	UNDP and GoL
Activity 2.2. Capacitate the Cross-Sectoral Working Group to mainstream ICZM into relevant government sectors through a Training-of-Trainers approach.	MME	MoUbetween EPA and MME Contract with service providers	MMA	UNDP and GoL
Activity 2.3. Strengthen the asset base and technical capacity of the ICZMU for the collection of spatial and biophysical coastal information to support the implementation of the ICZMP.	MME	MoU between EPA and MME Contract with service providers	MMA	No
Activity 2.4. Strengthen the existing Environmental Knowledge Management System (EKMS) to act as a platform for awareness-raising and sharing of climate risk-informed ICZM approach.	EPA	Contract with service providers	MMA	UNDP
Activity 2.5. Conduct an awareness-raising campaign for communities in focus areas on climate change impacts and adaptation practices.	EPA	Agreement between the Knowledge Sharing Groups and the CSC CSC Constitution and contractual agreements between the EPA and CSC members Contract with service providers	West Point, Jacob's Town, Fiamah and Topoe Villages	UNDP and GoL
Activity 3.1. Establish a community education and innovation centre to function as a community knowledge	EPA	CSC Constitution and contractual agreements	West Point, Jacob's Town, Fiamah and Topoe Villages	UNDP and GoL

generation and learning hub, repository on climate change adaptation practices and to host community activities under Output 3.		between the EPA and CSC members MoU between the owners of the center and EPA Contract with service providers		
Activity 3.2. Establish community-led co-management agreement to ease anthropogenic pressure on mangroves in the MMA.	EPA	Co-management agreement between target communities and EPA Contract with service providers	West Point, Jacob's Town, Fiamah and Topoe Villages	No
Activity 3.3. Conduct annual assessments to evaluate the project-induced changes in mangrove degradation, community perceptions and awareness of climate change impacts, adaptation options and mangrove ecosystems throughout the project implementation period.	EPA	Contract with service providers	West Point, Jacob's Town, Fiamah and Topoe Villages	UNDP and GoL
Activity 3.4. Establish small-scale manufacturing facilities and develop training material to capacitate community members to manufacture and sell cookstoves to support alternative climate-resilient livelihoods.	EPA	Contract with service providers CSC Constitution and contractual agreements between the EPA and CSC members	West Point, Jacob's Town, Fiamah and Topoe Villages	UNDP
Activity 3.5. Purchase and install low-maintenance eco-friendly cold storage facilities near fish processing sites to reduce pressure on mangroves and increase market efficiency.	EPA	Contract with service providers CSC Constitution and contractual agreements between the EPA and CSC members	West Point, Jacob's Town, Fiamah and Topoe Villages	UNDP

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

119. GCF grants are critical for this proposed project given the limited economic resources of Liberia and its numerous development priorities. While some government funds have been made available as co-financing, the current economic situation and borrowing profile of the Government of Liberia (GoL) is limited. Liberia has been classified as a Least Developed Country (LDC) with a 2018 GDP per capita of USD 674 — significantly lower than the LDC average GDP per capita of USD 1,042[88]. In addition, Liberia's Human Development Index (HDI) of 0.465 ranks the country 176th out of 189 countries assessed by the UNDP Human Development Report 2018. Economic and social gains made by Liberia in the post-civil war period from 2000–2010 were severely curtailed by the outbreak of the Ebola virus and subsequent epidemic between 2014 and 2015. The current COVID-19 pandemic poses a new threat to the Liberian economy. External demand for Liberian exports is expected to decrease and

[88] World Bank Data Portal. 2019. Available at: https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?cid=GPD_31&locations=LR-XL. Further information on the socio-economic context of Liberia is available in Annex 2.A: Feasibility Study, Section 1.

may disrupt investment in the export-oriented mining, agriculture, and forestry sectors[89]. A recent study by IMF[90] observed that macroeconomic stability in Liberia was already elusive with a faltering GDP growth and a spike in inflation when the pandemic emerged. The International Monetary Fund had cut down Liberia's 2019 growth outlook with the exchange rate depreciating by 26% over the year, and inflation accelerating to 28% at end-December. Growth in GDP for 2018 was 1.2%, while the projection for 2019 is a 1.4% contraction in the economy. Liberia also does not have an established borrowing programme of its own and its primary financing[91] includes concessional capital from IMF, WB, AfDB and other bilateral agencies. Due to these economic risks that persist and the IMF's cautionary stance on economic growth and inflation, Liberia is not currently able to raise any external financing in the form of concessional loans to replace GCF's grant for this proposed project. In addition, due to the public good nature of the benefits derived from the project, the serviceability of any such loans is uncertain. Without intervention, the estimated cost of accelerated coastal erosion at West Point alone is USD 47 million.

120. This project's proposed structure includes three outputs and twelve activities[92]. However, none of the activities or outputs clearly result in direct and quantifiable earnings or direct and quantifiable savings to the project owners or the project beneficiaries in a predictable time period. All the activities and outputs of this proposed project result only in non-attributable savings that are public goods and will benefit vulnerable communities in Liberia. Hence, a financial modelling-based analysis has not been conducted and a Financial Internal Rate of Return (FIRR) has not been calculated for any of these activities.

121. However, considering the GCF's minimum concession policy, this project's proposed activities have been analysed from a macro-economic and government perspective to assess the need for GCF grant as the only feasible financial instrument to fund the project activities. Hence, taking into consideration the factors such as the: i) climate-resilient sustainable development benefits arising out of the project; ii) identified climate change impacts; iii) fragile economic nature of the target population and Liberia's challenging economic situation; iv) persistent inflationary risks; v) lack of borrowing programmes; and vi) catalytic nature of the GCF grants, we recommend the following –

- The nature of the benefits does not accommodate repayment of capital in whatever form or serviceability of a loan instrument.
- The vulnerability of beneficiaries and the level of essentiality of the service do not accommodate repayment of capital in whatever form or serviceability of a loan instrument.
- There is co-financing from GoL and there is no incremental ability to stretch their contributions owing to budgetary and fiscal consolidation reasons.
- Hence, it is recommended that in order to reduce / close the existing financing and knowledge gaps and barriers to improve resilience of Liberia's population to climate change-induced hazards, GCF grants are essential. GCF's approach to financing only the incremental costs of adaptation have been given full consideration during the development of this proposal. Under Output 1, GCF will be financing only the construction phase of the protective infrastructure at West Point, with additional on-the-ground preparatory work[93] receiving co-finance directly from UNDP. Under Output 2, the bulk of GCF funding will be directed towards awareness raising on climate change adaptation as well as the development of an ICZMP, technology transfer and technical support

[89] World Bank. 2020. The COVID-19 Crisis in Liberia: Projected Impact and Policy Options for a Robust Recovery. Available at: https://openknowledge.worldbank.org/bitstream/handle/10986/34271/Liberia-Economic-Update-The-COVID-19-Crisis-in-Liberia-Projected-Impact-and-Policy-Options-for-a-Robust-Recovery.pdf?sequence=4&isAllowed=y
[90] International Monetary Fund, 2019 - https://www.imf.org/en/News/Articles/2019/03/08/pr1971-imf-staff-completes-2019-article-iv-mission-to-liberia
[91] International Monetary Fund Staff Report, 2018 - https://www.imf.org/external/pubs/ft/dsa/pdf/2018/dsacr18172.pdf
[92] Hard infrastructure as well as EbA measures were considered in the suite of possible interventions for coastal protection along the coastline of the MMA: i) Chapter 6 of the Vulnerability Sub-assessment (Annex 2.B) provides an overview of all the potential options considered; ii) Section 6.1 provides an overview of the objectives and requirements for protection and 6.2 provides overview of ICZM strategies; iii) Section 6.3.3 (page 101) provides an overview of the feasibility of all the protection options that have been considered including below will be incorporated into the revised FS and appropriately referred to in the FP: i) Chapter 6 of the Vulnerability Sub-assessment (Annex 2.B) provides an overview of all the potential options considered; ii) Section 6.1 provides an overview of the objectives and requirements for protection and 6.2 provides overview of ICZM strategies; iii) Section 6.3.3 (page 101) provides an overview of the feasibility of all the protection options that have been considered including NBS (Sections 8.J, 8.K and 8.L provide a complete list of options); iv) Chapter 7 provides a description of which options are proposed and why; v) Section 7.4 (page 118 to 121) gives a full description of the alternative options that are considered for West Point and which options are favoured as well as the respective advantages and disadvantages of each approach.
[93] Preparatory work to be financed by UNDP includes the development of a specific and focused ESIA and associated ESMP in accordance with Liberia law.

to capacitate the GoL to implement a proactive, climate-sensitive approach to coastal zone management. Additional activities under Output 2 — which represent a composite of adaptation and development interventions — will be co-financed by UNDP and GoL. Similarly, under Output 3, GCF funds will only be used for awareness raising on adaptation strategies and the provision of technical oversight, with the remainder of Output 3 — which focuses on conserving resources, supporting the development of climate-resilient livelihoods and strengthening food security — exclusively funded by a combination of UNDP and GoL co-finance.

122. The GCF's approach to financing only the incremental costs of adaptation have been given full consideration during the development of this proposal. Under Output 1, GCF will be financing only the construction phase of the protective infrastructure at West Point, with additional on-the-ground preparatory work[94] receiving co-finance directly from UNDP. Under Output 2, the bulk of GCF funding will be directed towards awareness raising on climate change adaptation as well as the development of an ICZMP, technology transfer and technical support to capacitate the GoL to implement a proactive, climate-sensitive approach to coastal zone management. Additional activities under Output 2 — which represent a composite of adaptation and development interventions — will be co-financed by UNDP and GoL. Similarly, under Output 3, GCF funds will only be used for awareness raising on adaptation strategies and the provision of technical oversight, with the remainder of Output 3 — which focuses on conserving resources, supporting the development of climate-resilient livelihoods and strengthening food security — exclusively funded by a combination of UNDP and GoL co-finance.

123. GCF is being requested to finance the cost of the construction of protective infrastructure at West Point under Output 1, as climate change is clearly contributing to increased rates of coastal erosion[95]. As evidence of its commitment to coastal adaptation and protection, the Government of Liberia (GoL) will contribute rock material[96] for the construction of the revetment as in-kind co-finance. The request for GCF support is being made to address the limited technical capacity of the GoL to design and implement an intervention of this type while considering the latest climate information and models. Additionally, as indicated in this section, the GoL has extremely limited financial resources and numerous urgent development priorities. GCF financing of this intervention will, however, contribute significantly to the knowledge base in Liberia on developing climate-resilient infrastructure and facilitate the strengthening of partnerships which will support the expansion of robust coastal protection and ICZM elsewhere along Liberia's vulnerable coastline, thereby creating an enabling environment for future country-led initiatives of a similar nature.

124. Financial support from GCF with maximum concessionally is consequently required to capacitate the GoL to protect vulnerable coastal communities against accelerated climate change-induced coastal erosion and safeguard livelihoods that are reliant on climate-sensitive mangrove ecosystems. In the absence of GCF involvement, the coastal community in West Point and across the MMA will become increasingly exposed to the impacts of accelerated coastal erosion, SLR and reduced provision of mangrove ecosystem services that support local livelihoods.

B.6. Exit strategy and sustainability (max. 500 words, approximately 1 page)

125. The proposed project has been designed with specific focus on an effective exit strategy that will ensure the long-term sustainability of the project interventions and their associated benefits. The specific elements of the exit strategy are presented below.

Post-project operations and maintenance

126. Project interventions have been designed to incorporate sustainability considerations from the outset, with minimal to no maintenance required beyond the project period. Through stakeholder consultations, the burden of operations and maintenance was identified as a risk to the sustainability of potential coastal protection interventions[97]. The rock revetment (Activity 1.2), for example, was therefore selected and designed to have low maintenance requirements once installed (detailed in Annex 2.C: Engineering Sub-assessment). For what

[94] Preparatory work to be financed by UNDP includes the development of a specific and focused ESIA and associated ESMP in accordance with Liberia law.
[95] Refer to Figure 4 in section B.1 which shows average coastal retreat between 2000 and 2100.
[96] A joint mission was undertaken by GoL (Ministry of Mines and Energy, EPA) and UNDP, to assess availability of rock at a quarry site originally used by GoL to source rocks for past GoL projects.
[97] Details of these stakeholder consultations are given in Chapter 6 of Annex 2.B: Vulnerability Sub-assessment and pg. 63 of Annex 6: ESAR.

maintenance is required, the Ministry of Mines and Energy (MME) has committed to financing and undertaking O&M after project completion[98]. The required financial commitments for maintenance of protective infrastructure beyond the project period are outlined in Annex 21: Operations and Maintenance (O&M) Plan. To further increase the sustainability of the intervention, community leaders, construction workers and fisherfolk from West Point will be trained on techniques for monitoring the revetment by the service provider responsible for its construction. A dedicated communication channel between the community and MME to report on revetment maintenance requirements will be established during project implementation. MME will oversee the monitoring of the revetment and perform regular technical inspections (ass detailed in the O&M Plan), using the information provided by the community and from these inspections to determine and meet all maintenance requirements. Workers from the West Point area engaged in the construction of the revetment will be trained by the construction firm on the technical skills needed to perform revetment maintenance, to ensure that local labour with the required training is available to MME when they need to perform maintenance activities. Overseen by the MME, the CSC will be responsible for the routine upkeep of the education and innovation centre, and will have access to small amounts of funds for this purpose through the income generated for the centre under Activity 3.4. During the collaboration workshop for government representatives, CSC members and NGOs under Sub-activity 3.1.4, the specific roles and responsibilities of the CSC members in terms of maintaining the education and innovation centre will be allocated among the CSC members. Maintenance for the education and innovation centre is expected to be low-cost as facilities are planned to be basic and durable. The MME will provide support for additional maintenance needs in cases where these costs go beyond the capability of the community, including the replacement of the roofing and window fittings. Similarly, any maintenance required for the cold storage facilities (Activity 3.5) will be funded through the nominal fee paid by community members who use the facilities and managed by the CSC, overseen by the MME. This will ensure the sustainability of the facilities and enable communities to hold the CSC accountable for their maintenance. Public sector investments into O&M will build ownership by relevant stakeholders, maximising the likelihood of effective, sustained maintenance beyond the project's implementation period. Co-management arrangements between relevant stakeholders, will also ensure cooperation and sustainable management of project interventions in the long term.

Integrated Coastal Zone Management (ICZM) approach

127. The sustainability of project interventions will be underpinned by the introduction of an integrated approach to coastal zone management in Liberia that emphasises capacity-building, cross-sectoral coordination and ongoing learning. The development of a vulnerability map and an ICZMP will form the basis of the ICZM approach in Monrovia and across Liberia. This process will be informed by active engagement with local stakeholders — including private sector partners and local communities in the coastal zone, with a focus on women — to ensure relevance to the local context. Training on the implementation of the ICZMP will also be operationalised through a Trainer-of-Trainers approach that will ensure that the skills are retained within relevant institutions and can be shared both nationally and regionally, facilitating the upscaling of ICZM and the sharing of knowledge. The use of a Training-of-Trainers approach in capacity-building activities will also foster buy-in from stakeholders and facilitate continued use of project tools, plans and policies beyond the project's lifespan. By bringing together government institutions across sectors, the project will facilitate the harmonisation of policies (Sub-activity 2.2.2) and planning relating to infrastructure, ecosystems and communities, creating an enabling environment for integrated adaptation and further investment in coastal adaptation. In particular, the harmonisation of policies and improved cooperation across government institutions will enable these institutions to: i) collaborate more effectively to leverage government finance for coastal adaptation; and ii) engage more systematically with the private sector and development partners to leverage additional investments. In addition, updating the ICZMP during the project period to reflect lessons learned from its initial implementation and from the implementation of the mangrove co-management agreements under Activity 3.2 will ensure that ICZM is implemented adaptively and grounded in the experiences of local communities.

128. The ICZM Committee and CSWG will be well-established before the end of the project period and will continue to function in a decentralised manner after the period of project implementation. Similarly, recommendations will be made during development of the ICZMP for: i) the assignment of departmental budget allocations for continued funding of ICZM activities outlined in the plan; and ii) the inclusion of ICZM into the job descriptions of relevant

[98] The revetment has been designed to have a lifespan of 30–50 years and is only expected to require maintenance if it is damaged by extreme events. The O&M costs are estimated between 0% and 0.5% per year for the rock works. Further information on the O&M requirements and plan are provided in Annex 21.

officials across government institutions. Moreover, strategies for financing coastal climate change adaptation initiatives will be identified by the CSWG, their success monitored over the course of the project and updated to inform the ICZMP. This government buy-in to the ICZM approach will ensure its sustainability beyond the project's lifespan. Additionally, the establishment of a wave buoy-based coastal data processing system and upscaling the existing EKMS will strengthen the information base underpinning the effective implementation of ICZM and will ensure that lessons learned on ICZM are actively collected, collated and disseminated to relevant stakeholders in accessible formats. The continued use of these tools will facilitate the ongoing application of the innovative approach to coastal zone management in Liberia beyond project completion.

Participatory approach to managing ecosystems

129. To ensure the sustainability of ecosystem management activities, the development of community-led co-management agreements for mangrove ecosystems will be undertaken with the close involvement of communities bordering the Mesurado Wetland. Their needs and concerns will be carefully considered, facilitating community buy-in and ownership of the agreements. The agreements and incentives for mangrove co-management will be underpinned by: i) the existing land-use management framework; ii) the strengthened institutional and regulatory systems for ICZM developed under Output 2; and iii) activities within the project that will increase awareness of the importance of mangrove ecosystems and how to sustainability manage them (Activities 2.5 and 3.2) and provide opportunities for the adoption of alternative livelihood practices (Activities 3.4 and 3.5). Lessons learned from the implementation of the mangrove co-management agreements will be incorporated into the revision of the ICZMP. Embedding participatory ecosystem management within the broader institutional framework will ensure ongoing support for mangrove conservation and management beyond the project period through the established institutional and regulatory systems. Conversely, incorporating lessons from community-based implementation of ICZM into the planning framework will increase the effectiveness and applicability of the ICZMP going forward. Similarly, the establishment of a Community Stewardship Committees (CSC) will promote community involvement and engagement in ecosystem management. Attention will be paid to the local context within which this committee will operate, in the interests of ensuring that the committee is self-sustaining. beyond the project lifespan Increased knowledge resulting from this involvement in the project will catalyse a shift in the way communities perceive and value mangroves, highlighting the link between mangrove conservation and health, livelihoods and food security. This will encourage continued participation and ongoing commitment to the safeguarding of these ecosystems. In addition, the small income streams developed through the cold storage facilities (Activity 3.5) and eco-friendly livelihood products (Activity 3.4) will enable the community facilities and the CSC to operate beyond the project lifespan.

Climate-resilient, gender-responsive and sustainable community livelihoods

130. A fully participatory approach will be used in the design of activities to build Monrovian communities' resilience to climate change and to generate gender-responsive, sustainable livelihoods. In particular, the long-term benefits of sustainable livelihoods will be promoted as an alternative to timber extraction from mangroves. This will have a sustained impact, as communities are likely to continue to use practices that are beneficial to them after project implementation.

131. Project activities are predominantly targeted at Monrovia's coastal fishing communities, approximately 60% of which are comprised of women. The long-term benefits of project activities will therefore inherently contribute to sustained gender equality. In addition, all committees established during project implementation will prioritise the inclusion of women, ensuring that future decision-making incorporates the interests of all genders. This will contribute to a shift in gender relations, which will additionally reduce inherent gender-based vulnerability in future. It also anticipated that, because of women's roles as primary caregivers, there will be externalities on future generations that will contribute to the sustainability of project interventions.

Scaling up and continuation of best practice

132. All project activities have been designed to incorporate sustainability considerations from the outset, including the potential for scaling up and continuation of best practice. Information generated through the development of the high-resolution vulnerability map — as well as best practice and lessons learned through the implementation of the ICZMP in Monrovia — will be shared widely and used to scale up project interventions across all coastal counties in Liberia. During the implementation period, collaboration between development partners — which have been initiated during project development — will be consolidated, including between UNDP, CI, the World Bank and JICA. These partnerships and experience will provide a robust platform for the implementation of other projects

in the coastal zone. Extensive engagement with the private sector during the development and implementation of the ICZMP will strengthen public-private partnerships for coastal management, increasing support for future coastal protection and adaptation initiatives. The design of these activities will, in conjunction with the training-of-trainers approach used, ensure that their implementation will continue successfully post project implementation. In addition, activities promoting uptake of climate-resilient livelihood activities — including the manufacturing of energy-efficient cookstoves and demonstrations on the benefits of improved community fish smoking systems — will focus on facilitating the development of small businesses, to enable the upscaling and replication of these alternative livelihood activities beyond the project implementation period and geographical scope. This will be done by: i) facilitating relationship building between potential community entrepreneurs and value chain actors; ii) providing training on business management skills; iii) supporting access to finance for initial investments in business start-ups; iv) supporting market analysis through the livelihood assessment under Activity 3.3 and marketing through the awareness-raising campaigns under Activities 2.5 and 3.2; and v) using the education and innovation as a platform for testing new technologies and innovations in the local community.

C. INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount	Currency
	17,255,755	USD ($)

GCF financial instrument		Amount	Tenor	Grace period	Pricing
(i)	Senior loans	Enter amount	Enter years	Enter years	Enter %
(ii)	Subordinated loans	Enter amount	Enter years	Enter years	Enter %
(iii)	Equity	Enter amount			Enter % equity return
(iv)	Guarantees	Enter amount	Enter years		
(v)	Reimbursable grants	Enter amount			
(vi)	Grants	17,255,755			
(vii)	Result-based payments	Enter amount			

(b) Co-financing information	Total amount	Currency
	8,383,150	USD ($)

Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
UNDP	Grant	1,577,750	USD ($)	Enter years Enter years	Enter%	Options
Government of Liberia	Grant	2,540,000	USD ($)	Enter years Enter years	Enter%	Options
Government of Liberia	In kind	4,265,400	Options	Enter years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options

(c) Total financing (c) = (a)+(b)	Amount	Currency
	25,638,905	million USD ($)

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	*Please explain if any of the financing parties including the AE would benefit from any type of guarantee (e.g. sovereign guarantee, MIGA guarantee).* *Please also explain other contributions such as in-kind contributions including tax exemptions and contributions of assets.* *Please also include parallel financing associated with this project or programme.*

C.2. Financing by component

Output	Activity	Indicative cost USD ($)	GCF financing		Co-financing		
			Amount USD ($)	Financial Instrument	Amount USD ($)	Financial Instrument	Name of Institutions
Output 1. Protection of coastal communities and infrastructure at West Point against erosion caused by sea-level rise	Activity 1.1: Prepare construction plan and finalise technical design specifications for coastal defence structure at West Point.	**705,712**	**333,012**	Grants	**372,700**	Grants	UNDP
	Activity 1.2: Construct coastal defence structure to protect West Point against climate	**17,442,309**	**13,169,909**	Grants	**7,000** **4,265,400**	Grants In-kind	UNDP GoL

and increasingly frequent high-intensity storms.	change-induced coastal erosion.						
Output 2: Institutional capacity building and policy support for the implementation of Integrated Coastal Zone Management (ICZM) across Liberia	Activity 2.1: Develop an Integrated Coastal Zone Management Plan for Liberia.	**2,426,014**	**912,014**	Grants	**1,508,500** **5,500**	Grants Grants	GoL UNDP
	Activity 2.2: Capacitate the Cross-Sectoral Working Group to mainstream ICZM into relevant government sectors through a Training-of-Trainers approach.	**322,460**	**128,360**	Grants	**181,100** **13,000**	Grants Grants	GoL UNDP
	Activity 2.3: Strengthen the asset base and technical capacity of the ICZMU for the collection of spatial and biophysical coastal information to support the implementation of the ICZMP.	**383,250**	**383,250**	Grants	-	-	
	Activity 2.4: Strengthen the existing Environmental Knowledge Management System (EKMS) to act as a platform for awareness-raising and sharing of climate risk-informed ICZM approach.	**186,500**	**40,250**	Grants	**146,250**	Grants	UNDP
	Activity 2.5: Conduct an awareness-raising campaign for communities in focus areas on climate change impacts and adaptation practices.	**787,400**	**185,000**	Grants	**320,400** **282,000**	Grants Grants	GoL UNDP
Output 3. Protecting mangroves and strengthening gender and climate-sensitive livelihoods to build local climate resilience in Monrovia	Activity 3.1: Establish a community education and innovation centre to function as a community knowledge generation and learning hub, a repository on climate change adaptation practices and host community activities under Output 3.	**689,982**	**323,482**	Grants	**330,000** **36,500**	Grants Grants	GoL UNDP
	Activity 3.2: Establish community-led	**185,750**	**185,750**	Grants	-	-	

	co-management agreement to ease anthropogenic pressure on mangroves in the MMA.						
	Activity 3.3: Conduct annual assessments to evaluate the project-induced changes in mangrove degradation, community perceptions and awareness of climate change impacts, adaptation options and mangrove ecosystems throughout the project implementation period.	**1,010,000**	**780,000**	Grants	**200,000** **30,000**	Grants Grants	GoL UNDP
	Activity 3.4: Establish small-scale manufacturing facilities and develop training material to capacitate community members to manufacture and sell cookstoves to support alternative climate-resilient livelihoods.	**134,550**			**134,550**	Grants	UNDP
	Activity 3.5: Purchase and install low-maintenance eco-friendly cold storage facilities near fish processing sites to reduce pressure on mangroves and increase market efficiency	**148,250**	-	Grants	**148,250**	Grants	UNDP
Project Management Cost		**1,216,728**	**814,728**	Grants	**402,000**	Grants	UNDP
Indicative total cost USD ($)		**25,638,905**	**17,255,755**		**8,383,150**		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

133. One of the major barriers to increasing the uptake of climate-resilient planning for coastal zones in Liberia is limited technical capacity to implement ICZM as well as a lack of access to real-time oceanographic monitoring systems. GCF financing will be used to address this barrier by developing an ICZMP (Activity 2.1) and capacitating relevant technical staff across 10 government institutions to implement ICZM (Activity 2.2). To further support the GoL to implement ICZM across the country and better understand climate change impacts in real-time, an integrated real-time oceanographic monitoring system will be procured under the project (Activity 2.3), and relevant institutional representatives will be capacitated to use the system. The benefits of transferring technology of this type into the Liberian context are numerous as the system will: i) increase the capability of the GoL to implement the ICZMP; ii) provide real-time data on ocean conditions, which will contribute to improved meteorology and EWS; and iii) provide

a means to measure and quantify climate-change impacts in the coastal zone, thereby supporting the country to develop a long-term data-set on the coastal impacts of climate change in Liberia.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

134. The proposed project will contribute to the achievement of the GCF's Paradigm Shift objective of increased climate-resilient sustainable development. This will be achieved by actively protecting infrastructure against climate change impacts, improving climate-responsive coastal planning, strengthening local livelihood practices and promoting the sustainable use of mangrove ecosystems to reduce the climate vulnerability of coastal communities in Monrovia. In so doing, the project will contribute to several indicative assessment factors in the GCF Performance Measurement Framework, including Fund-level Impacts:

- A1.0 Increased resilience of most vulnerable people, communities and regions; and
- A3.0 Increased resilience of infrastructure and the built environment to climate change.

Project Outcomes

135. Activities under the project will contribute to the achievement of two GCF Fund-level Outcomes (as per the Performance Measurement Framework) that will be used to assess the contribution of this project to climate-resilient sustainable development, as detailed below.

- *A5.0 Strengthened institutional and regulatory systems for climate-responsive planning and development.* Training on mainstreaming climate adaptation and improved management of coastal systems delivered through the project (Activities 2.1 - 2.2) will enhance the technical capacity of the GoL to plan for and respond to the impacts of climate change. Specifically, a Cross-Sectoral Working Group CSWG) will be established and trained to facilitate this mainstreaming process. An ICZM approach will also be integrated into national institutional processes to facilitate climate-responsive planning.
- *A7.0 Strengthened adaptive capacity and reduced exposure to climate risks.* The coastal protection measures implemented through this project will reduce Monrovian communities' vulnerability to climate risks. Specifically, the risk of accelerated coastal erosion and associated damage to infrastructure and houses in the West Point community will be substantially reduced.

Adaptation impact

136. The project is expected to directly benefit 250,000 people[99] (49% women) in the target communities of: i) West Point (through coastal defence and enhanced livelihoods); ii) Topoe Village; iii) Plonkor and Fiamah; and iv) Nipay Town and Jacob's Town (through enhanced livelihoods and improved protection of mangrove ecosystems). It is also expected to indirectly benefit approximately 1 million[100] people in Monrovia (49% women). This will be achieved by: i) reducing the impacts of climate change-induced coastal erosion; ii) improving the resilience of climate-sensitive livelihoods; and iii) developing a national-scale ICZMP and improving the capacity for ICZM in Liberia.

137. Through the construction of coastal protection measures, approximately 1,050 m of coastline at West Point will be defended against coastal erosion (Output 1), preventing 150 m of shoreline retreat by 2050 and 360 m by 2100[101] (RCP8.5). These interventions will directly benefit the coastal population of West Point (approximately 10,800 people, 49% women) whose homes, lives and livelihoods would otherwise be lost to coastal erosion. This will reduce damages to exposed and vulnerable infrastructure by USD 47 million[102]. The project will also improve the adaptive capacity of Monrovia's citizens who reside within and alongside the Mesurado Wetlands by strengthening the climate resilience of fishery-based livelihoods through the protection of mangrove ecosystems and supporting the uptake of diversified climate-resilient livelihoods (Output 3). When counted with the West Point beneficiaries described above, these interventions will directly benefit approximately 250,000 people living in the areas prioritised for interventions, many of whom are involved in fishing and fish processing[103]. The entire population of Monrovia (approx. 1 million people, 49% women) will benefit indirectly from the project through the establishment of climate-responsive ICZM practices and related capacity development (Output 2) in the MMA.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

[99] See Annex 2.B (Vulnerability Sub-assessment), calculation based on number of houses counted using Open Street Map data
[100] Calculated based on 2008 census data, with a population growth rate based on national population growth from 2008–2018
[101] See Annex 2.B (Vulnerability Sub-assessment)
[102] Present value of predicted damages up to 2100, RCP8.5; see Annex 2.A: Feasibility Study, Section 3
[103] 46% of whom are women: see Annex 2.B (Vulnerability Sub-assessment)

138. The project will catalyse a paradigm shift in the approach to coastal management in Liberia, shifting from a focus on immediate needs and short-term planning to incorporating medium- and long-term climate change considerations into coastal and development planning. This shift will be realised through the construction of robust coastal defence measures protect one of Liberia's most vulnerable communities (Output 1), the development of a climate-responsive ICZMP and the establishment of cross-sectoral coordination mechanisms (Output 2), as well as novel approaches to address the impacts of climate change on livelihoods in the Monrovia Metropolitan Area (Output 3). The use of quantitative data and scientific information in the development of the technical sub-assessments for the proposed project will strengthen the case for adopting the ICZM approach in Liberia and the strengthening of data collection and management under the project (Activities 2.1, 2.3, 2.4, and 3.3) will improve the evidence-base to support future ICZM decision-making. The project itself will serve as a model for integrated investment in coastal adaptation, as it incorporates infrastructure-, planning-, ecosystem- and community-based approaches to addressing complex adaptation challenges in the coastal zone, informed by robust analyses. The proposed project has the further potential to leverage alternative sources of finance (such as the local private sector and other development partners) to support similar coastal protection infrastructure and broader adaptation initiatives, as it will create a robust enabling environment for investment in the coastal zone. This potential will be explored as part of the implementation of the ICZMP and development of a financing plan to leverage additional investment in coastal adaptation (Activities 2.1 and 2.2).

Potential for scaling up and replication

139. The three project outputs will contribute to replicable and scalable climate-resilient development pathways in Monrovia, and elsewhere in Liberia, including building on institutional capacity development under the Liberia NAP readiness process — where assessments are being conducted on the economic impacts of climate change on vulnerable people and sectors — and lessons learned from previously implemented GEF-funded coastal defence projects[104]. Stabilisation of the shoreline at West Point under Output 1 lends itself to replication because other parts of Monrovia's coast (and indeed other coastal counties of Liberia) face similar risks in similar contexts. In particular, lessons learned (collected under Activity 2.4) and technical and project management capacity developed through the successful stabilisation of the shoreline at West Point will form the basis for replicating this process in other priority areas in Monrovia (assessed by the Feasibility Study) and elsewhere in Liberia. Similarly, under Output 2, the framework for implementing effective ICZM and coastal adaptation will be strengthened through: i) the development and revision after 3 years of a national ICZMP; ii) improved coordination and relationship-building between relevant government institutions (Activity 2.2); iii) harmonisation of policies and regulations across sectors to support coordinated ICZM; and iv) extensive (16-week) training of technical officials within the 10 government institutions involved in coastal zone management through a ToT approach. This strengthened framework and experience from the implementation of the ICZM in Monrovia will facilitate the expansion of this planning framework to the eight remaining coastal municipal areas in Liberia as well as the ongoing improvement of the national ICZM strategy, thereby realising the shift to including long-term climate change considerations into in coastal planning at municipal level across the country. Output 3, which focuses on strengthening livelihoods for vulnerable coastal communities in Monrovia will yield valuable lessons that can inform and be replicated through projects/interventions in similar contexts — i.e. vulnerable urban settlements and fishing communities — in Monrovia and across Liberia, as well as contributing a to stronger, more implementable ICZMP. This will include supporting community members to develop small businesses based on the construction of improved cookstoves, enabling the upscaling of more energy efficient cookstoves across Monrovia. The demonstration of an improved communal fish smoking system, at the education and innovation centre, will provide a further opportunity, along with the livelihood assessment (Activity 3.3.), forcommunity members to test a new technology with potential to upscale that technology and realise substantial economic and environmental benefits in fishing communities. In the short-term, lessons learned from all three outputs will be used to inform the implementation of a GEF-funded coastal resilience project in Sinoe County[105]. Like the proposed project, this project in Sinoe County includes improving coastal protection, strengthening cross-sectoral coordination and developing alternative, climate-resilient livelihoods.

Potential for knowledge management and learning

[104] Specifically, the GEF-funded Enhancing Resilience of Vulnerable Coastal Areas to Climate Change Risks in Liberia, which was originally implemented in the city of Buchannan and recently replicated in the vulnerable New Kru Town community in Monrovia.
[105] Further information on this project is provided in Section 6.1 of Annex 2A: Feasibility Study.

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140. A number of activities under the project will contribute to knowledge management and learning. First, Activity 2.2 will contribute strongly to learning and knowledge management through the development of ICZM capacity and expertise. To this end, technical officials from 10 government institutions will be capacitated through a ToT approach to adopt and implement climate-responsive ICZM for sustainability of coastal settlements and growth under climate change conditions. Technical skills will be imparted to government officials, including but not limited to: i) the synthesis and analysis of beach profile data; ii) coastal protection feasibility assessments; and iii) maintenance of coastal protection infrastructure. Second, Activity 2.3 will establish a system for data collection, analysis and management relating to coastal dynamics and ecosystems. This will help to increase the evidence base to support ICZM decisions-making, Third, Activity 2.4 will contribute to the improved knowledge management by upscaling the existing Environmental Knowledge Management System (EKMS) established under a GEF-funded initiative. This will include facilitating greater access to the system across 10 government institutions and private sector partners, as well as incorporating information on climate change adaptation and ICZM, which will be particularly important in promoting the sharing of project lessons and knowledge. Based on the EKMS, targeted knowledge products will be developed and disseminated to government institutions and the private sector to maximise the benefit of the information gathered under the project. Effective knowledge management will ensure that long-term adaptation planning incorporates context-specific lessons learned and best practices and that knowledge is shared across sectors and between partners to strengthen coastal management. The involvement of research institutions in the knowledge management and learning process will also ensure that the project supports the development of a growing evidence base for ICZM. Fourth, the establishment of an education and innovation centre under Activity 3.1 will contribute to increased community knowledge on alternative climate-resilient livelihoods and adaptation practices, particularly for the manufacturing of eco-friendly, energy-efficient products. Similarly, members of the communities of West Point, New Kru Town, Hotel Africa and the Atlantic Seaboard will be engaged to inform the development of a co-management agreement for the sustainable use of mangroves under Activity 3.2. Fifth, the annual alternative livelihood assessment undertaken under Activity 3.3 will increase the knowledge base to support the development of climate-resilient livelihood opportunities. Using the education and innovation centre as a platform to share the information collected and test new technologies, the project will foster community innovation. Finally, the project will support sharing of lessons learned and best practice through continuous monitoring and evaluation under Activities 2.4, 2.5 and 3.3. The generation of lessons learned and best practice under these activities on an annual basis will not only support adaptive project management, but also inform learning and best practice across community, sub-national and national levels as well as regional initiatives such as WABICC.

Contribution to the creation of an enabling environment

141. The project will create an enabling environment for climate-resilient coastal planning, the sustainable management of mangrove ecosystems and the uptake of climate-resilient livelihoods beyond the project lifespan. Output 2 has been specifically designed to create enabling conditions for the integrated approach to coastal management and governance that are necessary to undertake long-term, climate-resilient planning in vulnerable coastal areas such as Monrovia. This enabling environment will be created by improving coordination and policy harmonisation among government institutions and building capacity for the implementation of ICZM. The introduction of ICZM as an iterative and cyclical management approach will ensure that institutional arrangements and strategies established by the project will endure beyond its duration. A focus on strengthening public-private partnerships for ICZM under the project will also contribute to enabling further private sector involvement and support for coastal adaptation in Liberia in future. Similarly, embedding the mangrove co-management activities under Activity 3.2 in the institutional and regulatory framework strengthened under Output 2 will create an enabling environment to systematically improve mangrove conservation efforts throughout and beyond the project implementation period. The involvement of community, government, private sector and NGO stakeholder in this process will ensure that feedbacks between top-down and bottom-up approaches strengthen both and that initiatives incorporate the expertise and experience of a broad range of stakeholders. This and other activities under Output 3 have been designed as iterative activities that result in the ongoing development of diversified and climate-resilient livelihoods, as well as contributing to the sustainable use of mangrove ecosystems.

Contribution to regulatory frameworks and policies

142. The project will make a strong contribution to the development of ICZM policy in Liberia and assist the country to meet is legislative requirement of developing an ICZMP and updating this plan every three years. This will be achieved through the development of two iterations of an ICZMP during the project lifespan. ICZM is by definition an integrative framework and will consequently: i) take existing legislation and policy into account both strategically

and operationally; ii) contribute new knowledge and best practice to the policy landscape in Monrovia, and Liberia; and iii) influence changes in plans and policies at a county and district level. The regulatory landscape in Liberia will be further strengthened by ICZM capacity development within all relevant government institutions and through the harmonisation of policies and regulations related to ICZM across government institutions.

Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans

143. The proposed activities have been designed to remove specific barriers that impede the achievement of fund-level impacts and project outcomes as per the GCF Performance Measurement Framework (refer to Section D1). Project outputs lead to longer-term outcomes that include reduced vulnerability to future impacts of climate change, reduced loss from potential natural disasters, and enhanced livelihoods. The project will ultimately result in increased capacity of the GoL to undertake ICZM both locally, and at the national level. Through this capacity development process, climate change adaptation planning will be mainstreamed to increase resilience of coastal ecosystems and communities to climate change. Output 2, which will establish the ICZMP for Liberia, will be well-aligned with, and contribute to, the operationalisation of the GoL's key legislation, strategies and policies. Additionally, through Output 3, the project is in line with, and contributes to, the: i) Pro-poor Agenda for Prosperity and Development; ii) Republic of Liberia Agenda for Transformation: Steps Towards Liberia Rising 2030; and iii) Liberia Climate Change Policy, which all guide the country's efforts in both adaptation and mitigation. Furthermore, through instilling climate-resilient planning and protecting climate-sensitive livelihoods, the project will contribute to the achievement of the objectives of the Liberia National Adaptation Program of Action (NAPA) and the National Strategic Action Plan for Climate Change and Disaster Risk Management (NSAP).

D.3. Sustainable development (max. 500 words, approximately 1 page)

144. The proposed project will contribute to the achievement of seven Sustainable Development Goals (SDGs), namely: i) SDG 1 – No poverty; ii) SDG 2 – Zero Hunger; iii) SDG 3 – Good health and well-being; iv) SDG 11 – Sustainable cities and communities; v) SDG 13 – Climate action; vi) SDG 14 – Life below water. Apart from its contribution to the SDGs, the project will yield several environmental, economic, social, and gender co-benefits. These are described below.

Environmental co-benefits

145. The project will contribute to the protection of ecosystems and biodiversity in Monrovia through strengthening and promoting climate-resilient livelihoods (Outcome 3). This will yield the environmental co-benefits listed below:

- Reducing pressure on natural resources, including mangrove ecosystems, by promoting the manufacture and sale of energy-efficient livelihood products.
- Conserving Monrovia's mangrove ecosystems by engaging communities in an effective co-management system, thereby supporting the provision of ecosystem services including: i) stabilising sediments; ii) providing breeding habitats for a range of marine and terrestrial species; iii) improving and maintaining water quality; and iv) acting as carbon sinks.
- Reducing greenhouse gas emissions from burning fuels for cooking, by facilitating the production of energy-efficient cookstoves. The emissions reductions during the project are expected to be negligible[106]. However, project activities will promote the upscaling of cookstove production beyond the project implementation period and geographical scope, which will lead to more substantial emission reductions.

Economic co-benefits

146. Through the coastal protection defence interventions implemented at West Point (Outcome 1), the project will reduce replacement costs of economic and social assets damaged by erosion and storm surges. In addition to this direct benefit, several economic co-benefits are expected through the implementation of ICZM (Outcome 2) and the promotion of climate-resilient livelihoods (Outcome 3). These co-benefits are described below:

- Reducing economic losses in key sectors on the coast of Monrovia and West Point in particular — including industry, transport and housing — will benefit the city through *inter alia* circumventing loss and replacement costs.

[106] Assuming equivalent emission reductions to the *African Improved Cooking Stoves Programme of Activities,* registered under the Clean Development Mechanism, the use of project-supported cookstoves will reduce GHG emissions by ~4,750 tCO_2e during the project implementation period.

- Improving management of mangroves in the Mesurado estuary will benefit the fishing sector as these ecosystems act as critical breeding habitats for economically important fish species. The economic value of mangroves to fisheries has been estimated at USD 12,800/ha/year on the Atlantic coast of Central Africa[107].
- Improving community co-management of mangroves will contribute to the sustainable use of these ecosystems for diversified livelihood activities.
- Generating short- and long-term work opportunities for the construction, operations and maintenance of coastal defence infrastructure established by the project, as well as through the Community Stewardship Committee (CSC), will benefit the population of Monrovia. The use of local communities for casual labour, with specific provisions for the employment of women, has been incorporated into the design of the project.
- Installing cold storage facilities in fishing communities will reduce spoilage of fish, thereby and increasing the income/earning-potential of fishery-based livelihoods without increasing demand on fish stocks.
- Establishing facilities for the manufacture and sale of energy-efficient livelihood products will provide income-generation opportunities for community members.

Social co-benefits

147. The project will contribute to social development and cohesion by building the resilience of vulnerable coastal communities and their livelihoods to the impacts of climate change. This will yield multiple social co-benefits, including those described below:

- Enhancing coastal defence infrastructure (Outcome 1) will protect coastal communities, reduce mortality and reduce damage to the dense concentrations of economic assets and critical social infrastructure associated with coastal zones — including homes, schools, clinics and roads, as well as livelihood assets such as boats and fish processing equipment — from SLR and storm surges in the low-income area of West Point. The defence infrastructure will also protect hundreds of households from being displaced, thereby avoiding significant social disruption that would occur without the project.
- Improving the climate resilience of fishery-based livelihoods by working with communities to improve the management of mangrove ecosystems will contribute to the nutritional health, well-being, resilience and social cohesion of these communities.

Gender-sensitive development

148. The project will contribute to the implementation of both Liberia's Gender Policy (2009), which aims to reduce gender inequality, and the Climate Change Gender Action Plan for Liberia (2012), which promotes gender-responsive interventions to address climate change-related coastal erosion. This contribution will be achieved through the implementation of gender-sensitive activities, including: i) gender-responsive training for the management of mangroves; ii) assessing gender-specific climate vulnerabilities; iii) developing gender-specific adaptation options and alternative livelihood opportunities; and iv) increasing the resilience of Monrovia's fishing community which includes 9,000 fishmongers, most of whom are women.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Vulnerability of the country and beneficiary groups

149. Liberia's population is extremely vulnerable to the impacts of climate change, ranking 4th in the 2017 Climate Change Vulnerability Index[108]. This vulnerability is largely linked to the country's exposure to SLR and extreme weather events on its 560 km coastline. Increases in the frequency and intensity of extreme weather events are expected under future climate conditions, and the socio-economic impacts of such events, coupled with SLR and the resulting coastal retreat, are likely to be severe. The extent of these impacts is exacerbated by the high population density and associated concentrations of economic and social assets in the country's coastal zone as well as the dependence of this population on climate-sensitive livelihoods. Indeed, Liberia ranks 8th in the world for the percentage of the population living in low elevation coastal zones — mostly concentrated in the capital city of Monrovia. The target site for Output 1 — West Point — is located on a large sand spit on the western fringe of the Mesurado Wetland, making it one of the most vulnerable communities to SLR and extreme weather events in Liberia, with approximately 10,800 people — 49% women — projected to be directly at risk from coastal retreat by 2050[109].

[107] Ajonina GJ, Kairo G, Grimsditch G, Sembres T, Chuyong G, Mibog DE, Nyambane A and FitzGerald C. 2014. Carbon pools and multiple benefits of mangroves in Central Africa: Assessment for REDD+.
[108] Verisk Maplecroft. 2017. Climate Change Vulnerability Index.
[109] Annex 2.B: Vulnerability Sub-assessment (page 9)

150. The growth of Monrovia's population — at 3% per year — coupled with the ongoing shoreline loss, is leading to an increase in the density of the city's population, putting considerable pressure on the environment and ecosystems on which many livelihoods depend. The fisheries sector contributes approximately 10% of the national GDP and employs at least 37,000 people in Monrovia alone. The vulnerability of these livelihoods is being exacerbated by the impacts of SLR and the increasing frequency of intense storms. The low-income areas along the coast of Monrovia, including West Point, are home to the traditional artisanal fisherfolk of the Kru and Fanti communities who derive livelihoods from coastal waters and fish stocks. The mangroves in the Mesurado Wetland serve as critical habitat and nursery grounds for these fish stocks, strongly linking the livelihoods of fisherfolk to the health of mangrove ecosystems. In addition, these mangroves are an important source of fuelwood and other resources for the surrounding communities[110]. However, the loss of land and mangroves associated with SLR and extreme weather events is threatening the supply of these resources, including nutrient-rich fish which serve as the major protein source for much of the MMA's population. In turn, the decreasing supply of fish negatively affects: i) the food security of these communities; and ii) revenues from fisheries — especially for women, as 60% of persons active in the fishing sector are women[111].

151. The proposed project addresses these vulnerabilities by: i) building a coastal revetment on 1,050 m of coastline at West Point — ensuring landings for fishing boats (Output 1); ii) improving the management of Liberia's coastal zone by developing capacity for ICZM (Output 2); and iii) supporting the development of gender- climate-resilient livelihoods and improving the resilience of climate-sensitive livelihoods in Monrovia (Output 3).

Domestic finance, social and economic status quo

152. Since 1990, Liberia has been classified as a Least Developed Country (LDC) with a 2018 GDP per capita of USD 674 — significantly lower than the LDC average GDP per capita of USD 1,042[112]. In addition, Liberia's Human Development Index (HDI) of 0.465 ranks the country 176th out of 189 countries assessed by the UNDP Human Development Report 2018. Economic and social gains made by Liberia in the post-civil war period from 2000–2010 were severely curtailed by the outbreak of the Ebola virus and subsequent epidemic between 2014 and 2015. The Department of Economic and Social Affairs of the United Nations Secretariat (UN/DESA) compiles a snapshot report of all LDCs every three years based on three composite indices[113], namely the Economic Vulnerability Index (EVI, where lower is better), the Human Assets Index (HAI, where higher is better) and the Gross National Income (GNI, where higher is better). Figure 15 shows that in 2018, Liberia: i) was more economically vulnerable than the average of all LDCs in terms of the EVI; ii) had a GNI index equivalent to only 30% of the LDC average; and iii) had a lower HAI than the average figure for LDCs.

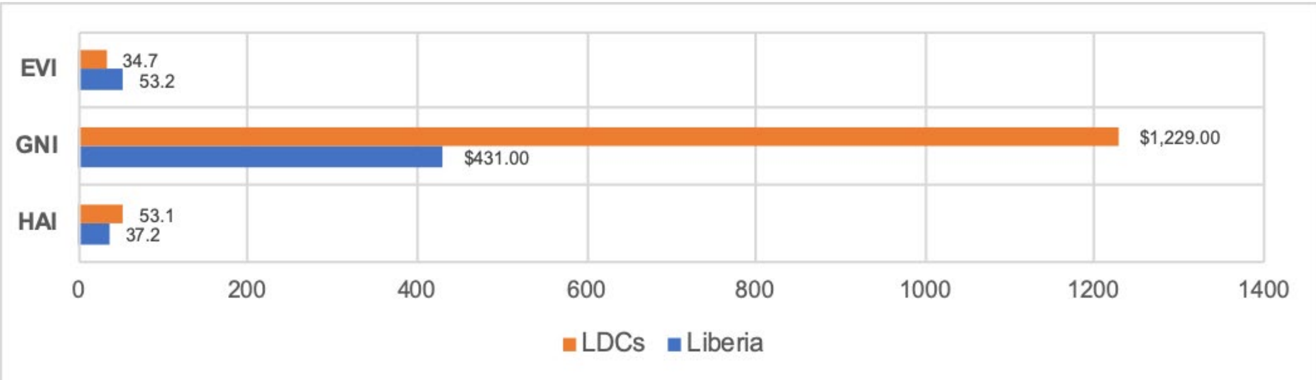


Figure 15. Comparison between Liberia and averages of all Least Developed Countries (LDCs) in 2018 in terms of the Economic Vulnerability Index (EVI), Human Assets Index (HAI) and Gross National Income (GNI) index[114].

[110] A detailed assessment of the role of these mangrove ecosystems in supporting livelihoods and the threats to the ecosystems is given in Annex 2D: Mangrove Sub-assessment, Section 3.
[111] Annex 2.B: Vulnerability Sub-assessment
[112] World Bank Data Portal. 2019. Available at: https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?cid=GPD_31&locations=LR-XL.
[113] UN/DESA. 2018. The Least Developed Country Category: 2018 Country Snapshots. Available at: https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/Snapshots2018.pdf.
[114] UN/DESA. 2018. The Least Developed Country Category: 2018 Country Snapshots. Available at: https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/Snapshots2018.pdf.

153. Liberia is disproportionately reliant on its primary production sectors for economic growth and development, with the agriculture, forestry and fishing sectors accounting for approximately 70% of GDP in 2018[115]. Reliance on the export of primary commodities — with little to no value-addition — results in the Liberian economy being extremely vulnerable to global commodity price fluctuations and market shocks. This economic vulnerability in turn compromises the ability of the GoL to deliver basic services such as water, housing, transport and energy to its citizens in a sustainable manner. The necessary emphasis by the GoL on basic development needs — including poverty reduction and health — means that funding climate change adaptation interventions to protect vulnerable communities is not considered as a priority, despite the urgent need for such interventions. Consequently, the GoL does not have the financial capacity to adequately fund adaptation initiatives at the scale required to significantly reduce climate change impacts on vulnerable communities.

Institutional capacity needs

154. To date, the GoL has established several laws, policies, strategies and institutions to address the increasing impact of climate change on the lives and livelihoods of people in Liberia (Section D.5). However, limited technical and financial capacity within most government departments is a significant barrier to the implementation of climate change adaptation, and ICZM in particular. The structural economic problems faced by the country — described above — as well as financial resources being directed towards the provision of social services, such as education and health, result in limited national sources of financing being available for adaptation interventions. In addition, limited understanding of the economic benefits of coastal adaptation interventions restricts the opportunities and entry points for private sector investment. Financial limitations constrain the implementation of adaptation interventions resulting in insufficient: i) work facilities; ii) access to technical equipment; iii) training to use and maintain technical equipment; and iv) data collection, analysis and management. Furthermore, there is limited technical capacity within government institutions and NGOs at both the national and local level to effectively use and enhance knowledge management systems as well as to disseminate information products to coastal communities for reducing their vulnerability to climate change.

D.5. Country ownership (max. 500 words, approximately 1 page)

155. The GoL considers the management of coastal erosion and the impacts of extreme climate events to be a high priority, as evident from the development of several national policies and plans and the commitment of the GoL to providing rock material as in-kind co-finance for Output 1 of the proposed project, as well as the commitment of the Ministry of Mines and Energy to ensuring the effective operatin and maintenance of the project assets beyond the project implementation period. The proposed project's alignment with these plans and policies is described below.

156. The **Pro-poor Agenda for Prosperity and Development** (PAPD, 2018-2023) is Liberia's five-year National Development Plan which underpins strategies and plans across multiple sectors of government. The project contributes to three of the four pillars of the PAPD, namely: i) 'Power to the People' by contributing to lifelong environmental education and reducing gender inequalities; ii) 'Economy and Jobs' by enhancing the resilience of climate-sensitive livelihoods; and iii) 'Governance and Transparency' by building the capacity of the GoL for effective ICZM.

157. The **National Policy and Response Strategy on Climate Change** (2018) is the central policy underpinning Liberia's response to climate change and serves as the basis for all future adaptation and mitigation planning. It identifies the sectors most vulnerable to climate change impacts and names five enabling pillars to support the safeguarding of these sectors. The project will contribute to four of these pillars, specifically cross-sectoral coordination, capacity-building, integrated planning and the development of infrastructure.

158. The **National Adaptation Plan of Action** (NAPA; 2006) emphasises addressing coastal erosion as a critical national adaption priority. Along with addressing coastal erosion in vulnerable areas of the MMA, the project further aligns with Liberia's NAPA by supporting: i) improved urban planning capacity in government institutions for incorporating climate change considerations; and ii) the dissemination of information to improve climate change knowledge at local and national levels.

159. Liberia's **Climate Change Gender Action Plan** (ccGAP; 2012) prioritises the mainstreaming of gender equality into national climate change policies, programmes and interventions. The project contributes to one of the ccGAP's six priority areas for gender mainstreaming, namely the improvement of gender mainstreaming within coastal areas. This will be achieved by: i) protecting vulnerable homes through establishing hard infrastructure; ii) incorporating

[115] *Ibid.*

gender considerations into the ICZMP; and iii) increasing participation of women in decision-making at community and institutional levels, including involvement in CBNRM.

160. The **Initial National Communication** (INC) to the UNFCCC (2013), provides a broad overview on the projected climate impacts on Liberia and an indication of the country's current GHG inventory. The adaptation priorities in the INC focus on the need to adapt to the impacts of projected SLR and increasingly frequent extreme climate events. The project is aligned with the approach recommend by the INC for reducing coastal impacts of climate change, including through the construction of protective infrastructure within the MMA.

161. Liberia's statement of its **Intended Nationally Determined Contribution** (INDC; 2015) identifies adaptation actions focusing on reducing the impacts of climate change on seven critical sectors, including agriculture, energy, health, forestry, coastal zones, fisheries and infrastructure — conditional on the provision of international financial assistance and aid. The project will align with one of the three adaptation priorities identified under the INDC, namely the construction of coastal defences to reduce the vulnerability of urban coastal areas.

162. In addition, the project will contribute to the implementation of and creating an enabling environment for land-use planning and management in Liberia, through the strengthening of policies and coordination among government institutions for ICZM. This will support the implementation of the **Land Use and Management Framework** developed by the Liberian Land Authority, as well as existing zoning ordinances in Monrovia and elsewhere.

Capacity of Accredited and Executing Entities to deliver:

163. UNDP, as the Accredited Entity of the project, provides a qualified team of national and international experts that are equipped to assist the GoL to respond to complex issues such as climate change and green growth. Through their support, UNDP will ensure that the project adopts a comprehensive, multi-sectoral approach that is centred around results-based activities. This support will include equipping Executing Entities with the knowledge and skills required to achieve expected project outputs. The strong relationships that exist between UNDP and local decision-makers, as well as UNDP's proven track record as an impartial provider of technical advice and support, will facilitate the strengthening and expansion of analytical work in key sectors and the advancement of disaster risk management within the broader context of sustainable development.

164. The EPA [116] – an autonomous body tasked with enforcing the National Environment Protection and Management Law under the Executive Branch of Government – is the National Designated Authority (NDA) for Liberia and is responsible for the overall management of GCF engagement in Liberia. Under the supervision of UNDP and a multi-sectoral Board of Directors – represented by public and private sector stakeholders – the GoL represented by the EPA will serve as Executing Entity (EE) for the project. To date, the EPA has implemented projects with a focus on climate change in Liberia, demonstrating their capacity to successfully implement the project. The Board of Directors has also established sector-specific Technical Committees to advise the EPA on priority issues, including climate change as well as marine and coastal ecosystems. These committees will provide strategic oversight during the implementation of the project.

Stakeholder Engagement[117]

165. The project was developed following an extensive stakeholder engagement process, beginning with initial institutional and community engagements during the first phase of project design in 2016, prior to the initiation of the PPF-financed Feasibility Study. The majority of the first phase engagements focussed on identifying broad needs and vulnerabilities within the MMA, which were incorporated into early project design. Numerous engagements were also organised with relevant government departments to validate the design of the project and ensure alignment with national priorities.

166. Further stakeholder engagements focused on beneficiary communities were conducted in 2019 as part of the second phase of development. In total, over 40 community stakeholder engagements were conducted from February to May 2019 to ensure local ownership of the project and to capture the needs of communities more clearly. In this context, stakeholder engagement was primarily intended to:

[116] The EPA was established under the National Environmental Policy of Liberia (2003) as an independent authority for the management of the environment

[117] Further details on the stakeholder engagement process are presented in Annex 7.

- ensure that expectations and concerns of local stakeholders were incorporated into project design and implementation;
- establish rapport with beneficiary communities to encourage local buy-in for the project;
- identify a range of desired gender-related development impacts of climate change projects in the target sites and to make sure that such impacts are incorporated into the design framework;
- support executing entity in preparing gender assessments and action plans; and
- ensure that all stakeholders, especially women, are effectively involved and equally represented throughout the project design and implementation process.

167. Final institutional stakeholder engagements and validation were held during the concluding phases of project development, between July and September 2019. These engagements included a three-day workshop in Monrovia that was attended by institutional representatives from relevant government institutions[118], UNDP and the prioritised communities, as well as several consultants and technical experts involved with the development of the project. These engagements were organised primarily for government institutions and communities to voice concerns and provide input. The final project design was validated at a workshop held in Monrovia on the 5th of September 2019 and attended by government and representatives from the West Point, New Kru Town, Hotel Africa and Atlantic Seaboard communities.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

168. The proposed project is requesting grant finance from the GCF to address the climate change impacts of SLR, accelerated coastal erosion, and increasingly frequent and intense storms on vulnerable coastal communities. The grant funds requested from GCF will be reinforced by additional co-finance from UNDP and the GoL.

Incorporation of best practice

169. The project incorporates best practice and lessons learned from relevant ongoing interventions and initiatives[119] that focus on the protection of vulnerable communities and ecosystems in the coastal zone — particularly two GEF-funded projects in Monrovia (refer to Section B.1. for further detail). Best practice and lessons learned from these initiatives have guided the development of the project's methodologies, tools and techniques for: i) the construction of coastal protection measures — including labour practices appropriate for the Monrovian context —; ii) the management of mangrove ecosystems, particularly through liaising with communities that rely directly or indirectly on mangroves for their livelihoods and/or for raw materials; and iii) the development of a contextually-relevant ICZMP for Monrovia. Specific aspects of best practice that have been incorporated into the design of the project are summarised below.

- Designing the coastal protection measures to have minimal maintenance requirements to ensure their sustainability;
- Incorporating updated technical information such as beach profile data, bathymetry and sediment dynamics into local planning and management of coastal activities;
- Adopting an integrated approach to coastal zone management by applying a systemic and coordinated approach to local planning in risk-prone coastal areas;
- Improving governance in ICZM planning through establishment and management of the cross-sectoral working group (CSWG) alliances (institutional connectivity);
- Capitalising on the wealth of local ecological knowledge held by civil society by establishing and sustaining community-based participation in improved co-management and monitoring of natural resources;
- Collaborating with development partners[120] working in Liberia, throughout the design and implementation of the project.

170. The project also incorporates the principles of adaptive management by making provision for iterative design and revision under several activities. For example, the ICZMP (Activity 2.1) will be revised three years after it is initially developed to incorporate lessons learned from its initial implementation. In addition, the awareness-raising campaign (Activity 2.5) and state-of-knowledge assessments (Activity 3.3) will be updated annually for the duration of the project. These iterative processes will maximise the effectiveness of the project interventions by ensuring that they are specific and responsive within the Liberian context.

[118] Government representatives included assistant ministers and technical personnel from the Environmental Protection Agency (EPA), Ministry of Mines and Energy (MME), Ministry of Public Works (MPW), Ministry of Finance and Development Planning (MFDP).
[119] These investments include both public expenditure and donor-funded initiatives
[120] During project development, UNDP engaged with Conservation International, the World Bank and JICA to validate project design and explore opportunities for synergy. Engagement with these and other development partners will continue throughout project implementation.

E

Cost-effectiveness

171. A conservative estimate of the expected net present value (ENPV) of the project is USD17.73 million using a 10% discount rate, with an economic internal rate of return of 21%. In addition to the above damages estimated, benefits also include an increase in income growth and the social benefit of the project through improved infrastructure.

172. Of the USD25 million of project costing, 34% of the necessary budget resources or USD8.4 million is contributed in the form of Co-financing with majority funded by GoL in spite of its economic and budget constraints mentioned above, bringing in co-finance resources — in the form of grants and in-kind contributions — to the project demonstrates its commitment to the project and more strongly its moral responsibility to invest in building climate resilience of its population.

173. This project proposes the construction of coastal protection measures along approximately 1,050 m of exposed and vulnerable coastline as well as measures to protect and enhance the climate-resilience of the livelihoods of ~25% of Monrovia's population. The cost of these measures is USD104 / beneficiary. This is a reasonable cost, especially given, the fact that the indirect beneficiaries are excluded.

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the GCF's Performance Measurement Frameworks under the Results Management Framework to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Paradigm shift objectives

Please select the appropriated expected result. For cross-cutting proposals, tick both.

☐ Shift to low-emission sustainable development pathways

☒ Increased climate resilient sustainable development

E.2. Core indicator targets

Provide specific numerical values for the GCF core indicators to be achieved by the project/programme. Methodologies for the calculations should be provided. This should be consistent with the information provided in section A.

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO_2 eq) to be reduced or avoided (mitigation and cross-cutting only)	Annual	Click here to enter text. t CO_2 eq
	Lifetime	Click here to enter text. t CO_2 eq
E.2.2. Estimated cost per t CO_2 eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total project financing	____ Choose an item.
	(b) Requested GCF amount	____ Choose an item.
	(c) Expected lifetime emission reductions	____ t CO_2eq
	(d) Estimated cost per t CO_2eq (d = a / c)	____ Choose an item. **/ t CO_2eq**
	(e) Estimated GCF cost per t CO_2eq removed (e = b / c)	____ Choose an item. **/ t CO_2eq**
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	(f) Total finance leveraged	____ Choose an item.
	(g) Public source co-financed	____ Choose an item.
	(h) Private source finance leveraged	____ Choose an item.
	(i) Total Leverage ratio (i = f / b)	____
	(j) Public source co-financing ratio (j = g / b)	____
	(k) Private source leverage ratio (k = h / b)	____
E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	250,000[121] 122,500 female 127,500 male
	Indirect	1 million[122] (total population of Monrovia) 490,000 female 510,000 male
	For a multi-country proposal, indicate the aggregate amount here and provide the data per country in annex 17.	
E.2.5. Number of beneficiaries relative to total population (disaggregated by sex)	Direct	6% (~3% of the total population of women and ~3% of the total population of men)
	Indirect	21% (~10% of the total population of women and ~11% of the total population of men)
	For a multi-country proposal, leave blank and provide the data per country in annex 17.	

[121] Direct benefits will accrue at the site-specific scale. Project interventions will directly benefit West Point's communities through coastal defence and enhanced livelihoods; and through enhanced livelihoods and improved protection of mangrove ecosystems in the communities of Topoe Village; Plonkor and Fiama; and Nipay Town and Jacobs Town.

[122] Indirect benefits will be realised through the adoption of a climate risk informed ICZM at the municipal scale within the project area. Therefore, indirect beneficiaries have been calculated as the total population of Monrovia Metropolitan Area (MMA) in Liberia.

E.3. Fund-level impacts

Select the appropriate impact(s) to be reported for the project/programme. Select key result areas and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected impact result. The result areas indicated in this section should match those selected in section A.4 above. Add rows as needed.

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions	*A1.1 Change in expected losses of lives and economic assets (US$) due to the impact of extreme climate-related disasters*	Impact assessment report completed by ICZM Unit, including: i) on the ground longitudinal survey augmented with transect walks assessing damages and wave/storm ingress; and ii) GIS assessment of change over time of coastline	Damage to infrastructure at West Point in the 2008-2018 period totals USD5.23 million 0 lives reported as lost as a result of coastal erosion in West Point in the 2008-2018 period[123].	Damage to infrastructure and assets vulnerable to climate change impacts at West Point expected to be reduced by USD0 by 2024[124], compared to a scenario with no intervention [125].	Damage to infrastructure and assets vulnerable to climate change impacts at West Point expected to be reduced by USD1.23 million by 2027[126], compared to a scenario no intervention [127].	Rock revetment will protect infrastructure, assets and communities against ongoing coastal erosion as well as extreme climate related disasters. No new settlements in exposed areas that are not protected by revetment during project period. Direct tangible damage without the project was estimated for 2030 in the Economic Analysis (Annex 3A) [128]. Damage from coastal retreat and storms will only be reduced by the

[123] The AE will not report on losses of lives due to the fact that no loss of life has been reported in West Point as a result of coastal erosion to date and no loss of life is expected. Coastal erosion in this area is primarily a result of wave action and is therefore incremental rather than sudden. It is, therefore, mostly possible to avoid loss of life.
[124] At mid-term, the construction of the revetment will be underway, and it will not yet be an effective coastal protection measure. The change in expected loss from coastal retreat and storms is therefore USD0.
[125] In a scenario with no intervention, loss and damage of USD7.84 million is expected by 2024. This is calculated as 40% of the expected loss and damage of USD19.6 million by 2030 presented in Annex 3A: Economic Analysis.
[126] The construction of the West Point revetment will be completed in Year 5 of the project period. It is therefore expected that the revetment will reduce the expected loss and damage from coastal retreat and storms over one year during the project period. The target given is the sum of the reduced loss and damage from coastal retreat (~USD920,000) and storms (~USD310,000) in the first year after the construction of the revetment is complete. Further detail is provided in Annex 3B: Economic Analysis Spreadsheet.
[127] In a scenario with no intervention, loss and damage of USD13.72 million is expected by 2027. This is calculated as 70% of the expected loss and damage of USD19.6 million by 2030 presented in Annex 3A: Economic Analysis.
[128] Details of these estimates are provided in Annex 3A: Economic Analysis. Avoided losses as a result of the completed West Point revetment have been projected to 2030 (USD19.6 million), 2050 (USD85.1 million) and 2100 (USD154.9 million).

						revetment after construction is completed. Average damage for houses at West Point due to coastal erosion between 2008 and 2018 is assumed to be 50% of the value of the houses, which is valued at USD15,600 per dwelling[129].
	A1.2 Number of males and females benefitting from the adoption of diversified, climate-resilient livelihood options (including fisheries, agriculture, tourism, etc.)	Household and affected stakeholder surveys compiled by ICZM Unit Remote sensing surveys compiled by ICZM Unit to monitor the extent of the mangrove forest degradation from human activity	0 beneficiaries with improved climate-resilient livelihoods	Total beneficiaries with improved, climate-resilient livelihoods: Males = 1,625 Females = 2,425 Mangrove conservation Males= 1,600 Females= 2,400 Adoption of solar-powered cold storage facilities Males= 0 Females= 0 Adoption of energy efficient cookstove	Total beneficiaries with improved, climate-resilient livelihoods: Males = 4,867 Females = 7,267 Mangrove conservation Males= 4,800 Females= 7,200 Adoption of solar-powered cold storage facilities Males= 720 Females= 1,080 Adoption of energy efficient cookstove	Mangrove protection and conservation efforts will have a positive impact on fish stocks, improving the climate resilience of fishery-dependent livelihoods in Monrovia. All fisherfolk and fishmongers in Monrovia will benefit from this. At least 25% of West Point fisherfolk and fishmongers will use the cold storage facilities. These beneficiaries will also benefit from mangrove conservation. 100% of West Point fisherfolk will benefit from more climate-resilient boat launching sites. These beneficiaries will also benefit from

[129] Annex 3: Financial and Economic Assessment, Appendix J – Damage and CBA model Monrovia base case

				manufacturing Males= 50 Females= 50 Decreased exposure of boat-launching sites to dangerous conditions induced by climate change Males= 0 Females= 0	manufacturing Males= 125 Females= 125[130] Decreased exposure of boat launching sites to dangerous conditions induced by climate change Males= 1,800 Females= 0	mangrove conservation. Annual training workshops will be held for 150 participants from Year 2 to Year 5. It is assumed that there will be 33% uptake of training on cookstove manufacturing, and at least 50% of the uptake will be among people who are not fisherfolk or fishmongers and therefore are not included in the beneficiaries of mangrove conservation initiatives.
A3.0 Increased resilience of infrastructure and the built environment to climate change	*A3.1 Number and value of physical assets made more resilient to climate variability and change, considering human benefits*	Impact assessment report compiled by ICZM Unit[131]	No protected boat launching sites, and zero metres of resilient public amenity space at West Point[132].	No boat launching sites and 400 m of public amenity space resilient to coastal erosion and wave action, with a value of USD6.03 million	2 boat launching sites and 1050 m of public amenity space resilient to coastal erosion and wave action, with a value of USD17.22 million	Boat launching sites will be considered resilient if fisherfolk are able to continue to launch boats at the same sites despite increased wave action and storm surges. Public amenity space will be considered resilient if the space is still accessible and suitable for community use despite increased wave action and storm surges.

[130] There are a total of ~12,000 fisherfolk and fishmongers in Monrovia, ~60% of whom are female. See Section B.1 for further details.

[131] Sources of data for the impact assessment will include: i) household and affected stakeholder surveys augmented with remote sensing surveys; ii) an on the ground longitudinal survey augmented with transect walks assessing damages and wave/storm ingress; and iii) GIS assessment of change over time of coastline

[132] Fishing boats are currently launched from informal beach sites exposed to coastal erosion and extreme wave action. The beach is also currently used as public amenity space and is exposed to coastal erosion and dangerous waves.

						Resilience of infrastructure, boat launching sites and public amenity space will be increased from reduced exposure to damage-causing waves, SLR and coastal retreat. Value of resilient boat launching sites and public amenity space calculated from the cost of construction of the revetment and green promenade, including boat launching sites[133]. Mid-term target is ~35% of the total target for the length of the revetment.

E.4. Fund-level outcomes

Select the appropriate outcome(s) to be reported for the project/programme. Select key expected outcomes and corresponding indicators from GCF RMF and PMFs as appropriate. Note that more than one indicator may be selected per expected outcome. Add rows as needed.

Expected Outcomes	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term)	Final	
A5.0 Strengthened institutional and regulatory systems for climate-responsive	*A5.2 Number and level of effective coordination mechanisms*	Coordination report produced by the project M&E Specialist[134, 135]	One coordination mechanism (ICZM Unit) at Level 1	3 coordinatio n mechanism s for ICZM (ICZM Unit, ICZM	3 coordination mechanisms for ICZM (ICZM Unit, ICZM Committee	Political and institutional stability enable establishment and functioning of

[133] As represented in Annex 4: Detailed budget plan.

[134] It is necessary for this Coordination Report to be project generated to avoid bias in performance measurement which may result from one of the entities being assessed also being responsible for its own assessment.,

[135] The report will be informed by a scorecard incorporating effectiveness level indicators: Level 0 = coordination mechanism for ICZM has yet to be established. Level 1 = coordination mechanism for ICZM is established, but not operational. Level 2 = Coordination mechanism for ICZM is partially operational, engaging effectively across government institutions, but implementation of the ICZMP is not fully effective. Level 3 = coordination mechanism for ICZM working effectively to implement the ICZMP in Monrovia.

planning and development			Two coordination mechanisms (ICZM Committee and CSWG) at Level 0	Committee and CSWG) at Level 2	and CSWG) at Level 3	ICZM coordination mechanisms
A7.0 Strengthened adaptive capacity and reduced exposure to climate risks	*A7.1 Use by vulnerable households, communities, businesses and public-sector services of Fund-supported tools instruments, strategies and activities to respond to climate change and variability*	Coastal management assessment conducted by ICZM Unit[136]	No government departments or agencies using ICZMP (Level 0)	10 target government institutions at Level 2	10 target government institutions at Level 4	Capacity building activities will lead to behaviour change and uptake of climate-resilient practices Political will and staff retention CSWG and the ICZM Committee will have the authority to plan updates to regulations and by-laws ICZMP will be used to change management practices

[136] This assessment will use indicator levels for the integration of ICZM: Level 0 = No government institution or agencies implementing ICZM; Level 1= Institutions engaged in developing an ICZMP; Level 2 = ICZMP developed using a participatory approach; Level 3 = Institution-specific action plan for the implementation of ICZMP developed; Level 4 = A portion of each target government institution's budget allocated to implementing the ICZMP.

		Annual state-of knowledge surveys compiled by contracted service provider (Activity 3.3) Household surveys conducted by the ICZM Unit	Targeted communities have limited understanding of climate change and its likely impacts on livelihoods in Monrovia[137]. 0% of targeted communities have made changes to livelihood practices and have adopted alternative livelihood opportunities to increase their climate resilience[138]	10% of targeted communities have an improved understanding of climate change and its likely impacts on livelihoods in Monrovia 0% of targeted communities have made changes to livelihood practices and have adopted alternative livelihood opportunities to increase their climate resilience	25% of targeted communities have an improved understanding of climate change and its likely impacts on livelihoods in Monrovia 15% of targeted communities have made changes to livelihood practices and have adopted alternative livelihood opportunities to increase their climate resilience	Knowledge and resources accessed via the education and innovation centre will improve climate-resilient practices amongst targeted communities and diversify livelihood options

[137] The baseline understanding of climate change and its impacts will be defined from the baseline state-of-knowledge survey
[138] These opportunities include use of cold storage facilities, and the manufacture and use of efficient cookstoves developed under the project

E.5. Project/programme performance indicators

The performance indicators for progress reporting during implementation should seek to measure pre-existing conditions, progress and results at the most relevant level for ease of GCF monitoring and AE reporting. Add rows as needed.

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Output 1. Protection of coastal communities and infrastructure at West Point against erosion caused by sea-level rise and increasingly frequent high-intensity storms.	Length of formal coastal protection revetment at West Point constructed during project lifespan	Implementation report from contractor on construction of coastal defence infrastructure Site survey of coastal protection infrastructure conducted by the ICZM Unit	No formal coastal defence structure at West Point	400 m of revetment constructed at West Point	1050 m of revetment constructed, and green promenade established at West Point	No extreme natural or social economic events delaying construction.
	Number of dwellings and non-residential buildings and assets in West Point behind coastal protection revetment infrastructure constructed during project lifespan	Spatial survey of damages to buildings/signs of ocean ingress. GIS assessment of coastal erosion rate compiled by ICZM Unit Household surveys assessing community perceptions of coastal erosion and risk compiled by ICZM Unit	0 dwellings in West Point protected from coastal erosion 0 non-residential buildings and assets in West Point protected from coastal erosion	500 West Point dwellings protected from coastal erosion by revetment 65 non-residential buildings and assets[139] in West Point protected from coastal erosion by	1,812 West Point dwellings protected from coastal erosion by revetment 217 non-residential assets and buildings in West Point protected by revetment	Number of dwellings and non-residential buildings in area to be protected remains constant or increases. Dwellings will be 'protected' as rock revetment will prevent damage or destruction from coastal erosion and storm surges. No new settlements in exposed areas that are not protected by revetment during project period.

[139] Non-residential buildings and assets in West Point include community buildings, small shops, markets, religious and cultural centres, and schools and government buildings. A list of all assets to be protected under Output 1 is provided in Annex 3: Economic Analysis.

				revetment		
Output 2: Institutional capacity building and policy support for the implementation of Integrated Coastal Zone Management (ICZM) across Liberia.	Change in how ICZM is integrated into roles and responsibilities of government staff involved in coastal zone management	Annual performance reviews for staff of government institutions Stocktake of job descriptions conducted by the CSWG	Responsibility for ICZM implementation not formally assigned	ICZM implementation not formally integrated into the roles and responsibilities of all relevant staff However, at least 10 staff members contributing to ICZM implementation	ICZM implementation is formally integrated into the job descriptions of relevant staff At least 40 staff members contributing to ICZM implementation	No government restructuring, or major political changes occur that impede adoption of ICZMP. The availability of staff time and assignment of roles and responsibilities will result in improved implementation of ICZM. The details of how ICZM will be implemented will differ substantially between government entities.
Output 3: Protecting mangroves and strengthening gender- and climate-sensitive livelihoods to build local	Change in income levels for community members using the education and innovation centre from the adoption of	Baseline survey Participant surveys compiled by CSC	TBD[140]	No change in income levels	30% increase from the baseline in income	Interventions for improving climate-resilient livelihoods from the target communities will be taken up and

[140] The baseline will be determined under Activity 3.3 as part of the baseline data collection on climate-resilient livelihoods.

climate resilience in Monrovia.	climate-resilient livelihoods	Annual state-of-knowledge assessments compiled by CSC			compared to the baseline for at least 50% of community members engaged in livelihood enhancement activities	will result in a measurable increase in income.
	Consideration of mangrove conservation in decisions about livelihood activities for members of the four targeted communities	Impact assessment reports compiled by ICZM Unit Annual state-of-knowledge assessments compiled by CSC	Mangrove conservation not explicitly considered in decisions about livelihood activities for members of four targeted communities	25% of surveyed community members understand mangrove conservation and its benefits and are aware of the links between mangrove conservation and livelihood security	25% of surveyed community members are able to give at least one example of where consideration of mangrove conservation has changed their decisions about livelihood activities	No social or political developments that will lead co-management parties to not honour their commitments in the agreements. Consideration of mangrove conservation in livelihood-related decisions will reduce anthropogenic pressure on mangroves. Behavioural change will occur after an understanding of mangrove conservation is evident.
	Rate of degradation of mangrove ecosystems	Impact assessment reports compiled by ICZM Unit	TBD[142]	Degradation of mangrove ecosystems	Degradation of mangrove ecosystems	Implementation of co-management agreements will result in a measurable decrease in

[142] The baseline will be determined under Activity 3.3 as part of the baseline mangrove spatial analysis.

		Remote sensing and GIS assessment of mangrove extent compiled by ICZM Unit. Baseline established at start of project[141]		reduced by 10% compared to the baseline in area covered by mangrove co-management agreement	reduced by 30% compared to the baseline in areas covered by mangrove co-management agreement	mangrove degradation.
	Percentage of fisherfolk and fishmongers in West Point using cold storage facilities	CSC report on use of the cold storage facilities Impact Assessment Report for the project produced by the ICZM Unit	0% of fisherfolk and fishmongers at West Point using project-developed cold storage facilities	0% of fisherfolk and fishmongers at West Point using project-developed cold storage facilities	25% of fisherfolk and fishmongers at West Point making use of cold storage facilities	Cold storage facilities will open access to refrigerated fish markets for fisherfolk and reduce spoilage by preserving fish for longer

E.6. Activities

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in the implementation timetable. Add rows as needed.

Activity	Description	Sub-activities	Deliverables
1.1. Prepare construction plan and finalise technical design specifications for coastal defence structure at West Point.	A construction plan will be developed and detailed design finalised for a revetment and green promenade at West Point, based on the existing conceptual design in the	1.1.1. Refine and detail the design and construction plan for the coastal defence infrastructure and green promenade at West Point.	• Detailed engineering designs and construction plan produced by engineering firm • ESIA conducted for the coastal defence revetment to inform

[141] The extent of mangrove coverage in the Mesurado Wetland of Monrovia is ~2320 ha.

	Engineering Sub-assessment (Annex 2.C). This detailed design and construction plan will identify all required inputs and guidelines for procurement procedures, contracting, ESS processes and gender-sensitive consultations. The design and construction plan will be validated by the community and government stakeholders prior to implementation.	1.1.2. Host a validation workshop to present and validate final design and construction plan to government and community stakeholders.	design and construction plans at West Point. • 1 workshop report based on validation workshop occurring in year 2 for 100 participants
1.2. Construct coastal defence structure to protect West Point against climate change-induced coastal erosion.	An engineering firm will be contracted to build the revetment and green promenade according to the design and construction plan developed under Activity 1.1. The revetment will protect the West Point community from accelerated coastal erosion and provide an open community space as well as preserving launching sites for fishing boats.	1.2.1. Construct a rock revetment with green promenade and community amenities at West Point.	• Revetment (coastal defence measures) of approximately 1,050 m at West Point with green promenade constructed • Report on training for monitoring and maintenance of the revetment
2.1. Develop an Integrated Coastal Zone Management Plan for Liberia.	A national Integrated Coastal Zone Management Plan (ICZMP) will be developed in consultation with relevant government institutions and stakeholders. This will be done in phases by: i) producing a multi-criteria vulnerability map of the Liberian coastline; ii) collaboratively developing an ICZMP; iii) establishing an ICZM Committee and Cross-Sectoral Working Group (CSWG) to implement the plan; and iv) revising and updating the ICZMP after three years to incorporate lessons learned.	2.1.1. Develop a high-resolution, multi-criteria vulnerability map of the Liberian coastline. 2.1.2. Develop the climate-responsive ICZMP in collaboration with stakeholders, incorporating information from the map developed under Sub-activity 2.1.1. 2.1.3. Host validation workshops to present and secure ownership of the ICZMP for representatives from the 10 government institutions responsible for coastal management in Liberia.	• High-resolution multi-criteria vulnerability map of Liberian coastline produced • ICZMP developed and adopted by year 3 and updated in year 6 • Workshop reports for 2 validation workshops with 50 participants including the ICZM Committee and the CSWG, one conducted in year 2 and one in year 6 for the development and updating of the ICZMP respectively • 10-member CSWG established • 10-member ICZM Committee established
2.2. Capacitate the Cross-Sectoral Working Group to mainstream ICZM into relevant government sectors through a Training-of-Trainers approach.	The Cross-Sectoral Working Group (CSWG) will be trained through a Training-of-Trainers (ToT) approach to increase their capacity to implement the ICZMP. This will include hosting an extensive series of training workshops for members of the CSWG and representatives of relevant	2.2.1. Host an inception workshop to present and provide orientation to the CSWG on the finalised ICZMP. 2.2.2. Host a planning workshop for the CSWG to	• Action plan developed during two-day meeting for integrating ICZM into by-laws and regulations for the 10 relevant institutions • Workshop reports based on two inception workshops organised,

	government institutions. The ToT approach will enable the 10 government institutions responsible for ICZM to further capacitate their staff. To support the implementation of the ICZMP, the CSWG will also develop an action plan for integrating ICZM into sector-specific regulations and by-laws.	develop an action plan for implementing the ICZMP, including plans for changes/updates required for sector-specific regulations and by-laws. 2.2.3. Develop a curriculum and training program to capacitate relevant technical personnel on cross-sectoral coordination and the effective implementation of the ICZMP. 2.2.4. Host a validation workshop with the CSWG to finalise training curricula. 2.2.5. Conduct a training and action programme for members of the CSWG to mainstream ICZM into relevant government entities and support mainstreaming of ICZM into national policies, programmes and plans.	one in year 2 and one in year 6, for 25 participants • 2 co-design workshops for 39 members of the CSWG and the ICZM Unit (one in year 2 and one in year 5) to introduce the ICZMP and collaboratively define curricula • Training curricula for capacitating relevant technical personnel on ICZM developed • Long-term (16-week) training programme conducted to capacitate 36 technical staff from relevant government institutions (3 members from each of the 10 relevant institutions and 6 members of the ICZMU) • Stocktake of job descriptions assessing integration of ICZM into roles and responsibilities of government staff involved in coastal zone management
2.3 Strengthen the asset base and technical capacity of the ICZMU for the collection of spatial and biophysical coastal information to support the implementation of the ICZMP.	The capacity of the ICZM Unit (ICZMU) for the collection, processing and synthesis of spatial, meteorological and oceanographic data will be developed. This will include: i) the procurement of high-resolution spatial data for Monrovia; ii) the acquisition of wave buoys and other relevant equipment; iii) the installation of a database for collecting and processing the data; and iv) training and support for technical staff to use and maintain the system.	2.3.1. Procure high-resolution remote sensing data to monitor coastal erosion and mangrove ecosystem health and degradation in the MMA to be made publicly accessible and used for the vulnerability map under Sub-activity 2.1.1. 2.3.2. Contract a specialist firm to provide an integrated, near-shore wave buoy-based data collection and processing system to support the collection of meteorological and oceanographic data for improved generation of information on parameters relevant to ICZM and meteorological decision-making in Liberia.	• High-resolution spatial datasets for eight years (five historical and three during the project) • Integrated, near-shore wave buoy-based data collection and processing system operationalised through • the installation of 4 wave buoys and associated ICT equipment; • development of database, data collection and processing systems; and • training courses on use and O&M
2.4. Strengthen the existing Environmental Knowledge Management System (EKMS) to act as a platform for awareness-raising and	The newly implemented EKMS will be upscaled by improving its accessibility and incorporating information on ICZM into the system. ICT equipment will be procured for the 10 government institutions	2.4.1. Procure ICT equipment for each of the relevant government institutions to access and use climate and environmental information through the platform.	• ICT equipment for the EKMS procured, including: ○ 20 laptops for 10 ministries ○ Storage capacity installed and

sharing of climate risk-informed ICZM approach.	responsible for coastal management to enable them to access and operate the knowledge management platform. In addition, knowledge products on ICZM and the EKMS will be developed and disseminated to government institutions.	2.4.2. Collect lessons learned on ICZM on an annual basis and incorporate this information into the EKMS. 2.4.3. Design and disseminate knowledge products on climate-responsive ICZM in government institutions and to the private sector.	bandwidth capacity improved • Annual reports on lessons learned for ICZM produced • Knowledge products on ICZM and the EKMS produced and disseminated over 5 years, including: • 500 posters per year • 1,000 brochures per year
2.5. Conduct an awareness-raising campaign for communities in focus areas on climate change impacts and adaptation practices.	An awareness-raising campaign on climate change impacts and adaptation practices will be designed and conducted among four priority communities in Monrovia — West Point, Jacob Town, Fiamah and Topoe Village. The campaign will include community meetings, radio programmes and visual communication materials. The campaign will be supported by knowledge-sharing groups in each of the four target communities and will be developed to take gender-differentiated vulnerabilities to climate change into consideration. The campaign will be reviewed and updated annually for the project duration.	2.5.1. Design an awareness-raising campaign on climate change impacts and adaptation practices. 2.5.2. Host radio programmes on climate change impacts, vulnerability and adaptation practices in coastal and wetland areas. 2.5.3. Establish, train and engage community-based knowledge-sharing groups to support on-the-ground awareness raising and knowledge-sharing. 2.5.4. Install and distribute awareness-raising knowledge products within the four communities for awareness-raising interventions. 2.5.5. Organise community meetings to raise awareness around climate change adaptation and the benefits of proposed interventions.	• Awareness-raising strategy developed and updated annually • 4 radio programmes per year for 5 years • 20 5-day workshops organised (one for each of the 4 areas annually for 5 years) • Knowledge-sharing groups consisting consist of 4 members for each of the 4 areas established and trained • 12 Billboards (3 for each of the 4 targeted areas) in public spaces for 5 years • Awareness raising products distributed per year for 5 years (400 posters, 400 brochures, 800 stickers, 400 caps and 400 t-shirts) • Minutes from 3 community meetings for 100 participants organized per year in each of the 4 areas for awareness raising
3.1. Establish a community education and innovation centre to function as a community knowledge generation and learning hub, a repository on climate change adaptation practices and to host community activities under Output 3.	An education and innovation centre will be established in an existing community or government-owned building to host and coordinate community-based activities (Activities 3.2 and 3.4) and all community-level awareness-raising being implemented under Output 3. The centre will be managed by a Community Stewardship	3.1.1. Locate, renovate and equip existing structure situated on community or municipal-owned land in West Point. 3.1.2. Establish and commission the ten-person Community Stewardship Committee (CSC) with	• Existing building identified and renovated to be established as an education and innovation centre • Detailed operation and maintenance plan for education and innovation centre • 10-person Community Stewardship Council (CSC) with at least 50%

	Committee (CSC) and will serve as the focal point for all community events and capacity building activities.	representatives from each of the four areas. 3.1.3. Host an inauguration event to open the centre to the public and present the members of the CSC. 3.1.4. Host a collaboration workshop for government representatives, CSC members and NGOs to meet and define the guiding principles for the education and innovation centre.	women representatives commissioned • Inaugural event (1,500 participants) to launch the education and innovation centre organised • Workshop report based on one 2-day collaboration workshop organised for CSC members as well as for representatives from relevant government ministries, NGOs, and development, conservation and community organisations • CSC Constitution
3.2. Establish community-led co-management agreement to ease anthropogenic pressure on mangroves in the MMA.	A co-management agreement will be developed through a participatory process to facilitate the sustainable use of — and reduce anthropogenic pressures on — mangroves in the Mesurado Wetlands. This agreement will be developed taking into consideration: i) uses of mangroves by different social groups; ii) gender considerations; and iii) guidance provided by government stakeholders and NGOs involved in the conservation of mangrove areas. Roles and responsibilities, including for the CSC and community members will be clearly outlined. The agreement will be revised after three years to incorporate lessons learned during the project.	3.2.1. Develop the community-led co-management agreement through a participatory gender-sensitive process. 3.2.2. Host a three-day co-design and validation workshop for the community-led co-management agreement. 3.2.3. Design and implement an awareness-raising campaign on sustainable Community-Based Natural Resource Management (CBNRM) and the co-management agreement. 3.2.4. Host a workshop to assess the effectiveness of the co-management agreement (Activity 3.2.1) and re-negotiate it in Year 5 based on findings from Activity 3.3.	• Co-management agreement for reducing anthropogenic pressure on mangroves finalised • Workshop report from one 3-day co-design and validation workshop organised for relevant government ministries, NGOs, conservation organisations and CSC members • Awareness-raising campaign on CBNRM designed and implemented (100 posters and 400 brochures distributed per year for 5 years and 20 signboards erected) • Workshop report from one 2-day workshop organised for the assessment and re-negotiation of the co-management agreement, for relevant government ministries, NGOs, conservation organisations and CSC members
3.3. Conduct annual assessments to evaluate the project-induced changes in mangrove degradation, community perceptions and awareness of climate change impacts, adaptation options and mangrove ecosystems throughout the project implementation period.	Spatial analyses of the health and degradation of mangrove ecosystems in the Mesurado Wetland will be undertaken using the high-resolution data acquired under Activity 2.3. An initial assessment will be conducted and updated twice during the project period to assess the impact of the	3.3.1. Using high-resolution remote sensing data procured under Sub-activity 2.3.1, conduct analyses of mangrove extent and degradation in the Mesurado Wetland to assess the impact of project interventions	• 3 reports on mangrove extent and degradation in the Mesurado Wetland incorporating spatial analysis to assess the impact of project interventions. Reports to be delivered in years 2, 4 and 6. • Baseline report with annual updates on state-

	project. In addition, a baseline state-of-knowledge assessment will be undertaken on community awareness on: i) climate change impacts and adaptation options; ii) CBNRM and the role of mangrove ecosystems in the MMA; and iii) different types of climate-resilient livelihoods currently being practiced within the MMA. The baseline surveys will be collated into a report and annual follow-up surveys will be conducted to determine the annual change-in-state, which will help to refine the awareness-raising campaigns and support the uptake of additional climate-resilient livelihoods.	3.3.2. Conduct a baseline state-of-knowledge survey on climate change impacts, adaptation options, CBNRM and mangrove ecosystems and undertake annual surveys to determine the change-in-state over the duration of the project. 3.3.3. Undertake periodic assessments of climate-resilient livelihood practices and new opportunities to inform the design of new options and determine their uptake as a result of the project.	of-knowledge developed based on four community-level assessments conducted in each of the four focus areas targeted under the project to develop indicators that determine baseline • Baseline reports and annual updates on current and potential climate-resilient livelihoods practices produced based on alternative, climate-resilient livelihoods strategies developed for the four target communities • Baseline and final project impact assessment conducted
3.4. Establish small-scale manufacturing facilities and develop training material to capacitate community members to manufacture and sell cookstoves to support alternative climate-resilient livelihoods.	The uptake of diversified climate-resilient livelihoods will be supported by developing training courses, facilities and providing material inputs for the fabrication of value-added products, specifically for energy-efficient cookstoves. The facilities will be incorporated into the education and innovation centre and will focus on producing energy-efficient products with a specific emphasis on activities suitable for women and vulnerable social groups. Different livelihood activities will be supported throughout the project lifespan depending on the results of the assessment conducted under Activity 3.3 and all training materials will be publicly available to support the upscaling of activities.	3.4.1. Develop a training programme and publicly available training materials, and conduct trainings to capacitate community members to manufacture energy-efficient cookstoves. 3.4.2. Establish a small-scale manufacturing workshop and procure equipment and material for the education and innovation centre. 3.4.3. Host an annual workshop at the education and innovation centre to capacitate women's groups on business strategies and connect them with value chain actors to upscale the development of energy efficient value-added products.	• Specification for energy-efficient cookstoves publicly available • 5 annual workshops at the innovation centre designed and organised for implementing the community-based training program • 1 vocational training centre (simple workshop area) established with equipment and material inputs procured for production of energy efficient cookstoves • Workshop reports from 5 workshops (1 per year for 5 years) organised for 100 participants to capacitate women on business practices
3.5. Purchase and install low-maintenance eco-friendly cold storage facilities near fish processing sites to reduce pressure on mangroves and increase market efficiency.	Cold storage facilities will be designed and installed in proximity to the West Point fish processing site. These cold storage facilities will be designed according the community requirements as determined by a needs assessment. They will be powered by solar energy and	3.5.1. Develop for site-specific eco-friendly, solar-powered cold storage units for the West Point fish processing site, based on the needs of fishers and fishmongers.	• Detailed report on community needs for cold storage produced based on needs assessment conducted. • Validation workshop conducted to endorse the final cold-storage designs

	will be managed by women in the community in collaboration with the CSC established under Activity 3.1.2. The facilities will be designed be accessible to all fisherfolk in the area and to require limited maintenance and any required maintenance will be provided by the service provider contracted to develop the storages, with support from UNDP.	3.5.2. Host a workshop in the targeted community to validate the designs (Sub-activity 3.5.1) and capacitate fisherfolk in the use and management of the cold storage units 3.5.3. Install and operate the low-maintenance eco-friendly solar-powered cold storage facilities.	• 2 eco-friendly, solar-powered cold storage units installed • Report on trainings provided to the members of the facility managers, CSC and representatives of fishmongers to operate, maintain and manage the solar-powered cold facilities.

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)
174. Project-level monitoring and evaluation will be undertaken in compliance with the UNDP POPP and the UNDP Evaluation Policy. 175. The primary responsibility for day-to-day project monitoring and implementation rests with the Project Manager. The Project Manager will develop annual work plans to ensure the efficient implementation of the project. The Project Manager will inform the Project Board and the UNDP Country Office of any delays or difficulties during implementation, including the implementation of the M&E plan, so that the appropriate support and corrective measures can be adopted. The Project Manager will also ensure that all project staff maintain a high level of transparency, responsibility and accountability in monitoring and reporting project results. 176. The UNDP Country Office will support the Project Manager as needed, including through annual supervision missions. The UNDP Country Office is responsible for complying with UNDP project-level M&E requirements as outlined in the UNDP POPP. Additional M&E and implementation quality assurance and troubleshooting support will be provided by the UNDP Regional Technical Advisor as needed. The project target groups and stakeholders including the NDA Focal Point will be involved as much as possible in project-level M&E. 177. A project inception workshop will be held after the UNDP project document has been signed by all relevant parties to: i) re-orient project stakeholders to the project strategy and discuss any changes in the overall context that influence project implementation; ii) discuss the roles and responsibilities of the project team, including reporting and communication lines and conflict resolution mechanisms; iii) review the results framework and discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E plan; iv) review financial reporting procedures and mandatory requirements, and agree on the arrangements for the annual audit; and v) plan and schedule Project Board meetings and finalize the first year annual work plan. The final inception report will be cleared by the UNDP Country Office and the UNDP Regional Technical Adviser and approved by the Project Board and submitted to the GCF within 6 months after the Funded Activity Agreements effectiveness date. 178. A project implementation report will be prepared for each year of project implementation. The Project Manager, the UNDP Country Office, and the UNDP Regional Technical Advisor will provide objective input to the annual GCF APR. The Project Manager will ensure that the indicators included in the project results framework are monitored annually well in advance of the GCF APR submission deadline. Progress against the gender action plan, stakeholder engagement, social and environmental safeguards updates, challenges and delays must also be monitored by the Project Manager and reported in the GCF APR. The GCF APR will be shared with the Project Board and other stakeholders. 179. An independent interim revaluation will be undertaken, and the findings and responses outlined in the management response will be incorporated as recommendations for enhanced implementation during the final half of the project's duration. The terms of reference, the review process and the final evaluation report will follow the standard templates and guidance available on the UNDP Evaluation Resource Centre. The final evaluation report will be cleared by the UNDP Country Office and the UNDP Regional Technical Adviser and will be approved by the Project Board. The final evaluation report will be available in English. 180. An independent final evaluation will take place when project activities have concluded. The terms of reference, the evaluation process and the final evaluation report will follow the standard templates and guidance available on the UNDP Evaluation Resource Centre. The final evaluation report will be cleared by the UNDP Country Office and the UNDP Regional Technical Adviser and will be approved by the Project Board. The TE report will be available in English.

181. A key tool for MRV for the project is an independent impact assessment which will be commissioned to a third party. While the detailed scope of the assessment will be further developed, it is expected that the assessment will involve data collection, at least twice in the course of the project at the baseline and end-line of a representative sample of households in the four prioritized communities in Monrovia. These surveyed households will include both the project target and non-target beneficiaries with controlled comparison of project impact at various levels. Adoption rate will also be measured with test of options to improve the rate of adoption of some of the interventions. Importantly, the outcome of interest in this assessment includes the changes in productivity; access to alternative livelihoods, fishing activities; importance of cold storages and the extent to which climate information influences households such as use of mangroves and settlement planning. The execution of activities such as livelihood options, co-management of mangroves, climate advisory services that are disseminated to a larger population could possibly permit investigation of impacts of these three interventions independently as well as in combination (the feasibility to be confirmed). With a higher number of vulnerable households than the project can serve, there will be potential to objectively select who will receive the program and who will not. This will be based on vulnerability and need at the beginning of the project with a phase-in design because the program may not be rolled out within one year. Moreover, this assessment will make use of satellite data to assess the changes in settlement patterns, the efficacy of the revetment and reduced rate of mangrove degradation etc. before and after interventions.

182. The UNDP Country Office will include the planned project evaluations in the UNDP Country Office evaluation plan, and will upload the evaluation report in English and the management response to the public UNDP Evaluation Resource Centre (ERC) (www.erc.undp.org).

183. The UNDP Country Office will retain all M&E records for this project for up to seven years after project financial closure in order to support ex-post evaluations.

184. A detailed M&E budget, monitoring plan and evaluation plan will be included in the UNDP project document.

185. Monitoring, reporting and evaluation arrangements will comply with the relevant GCF policies and Accreditation Master Agreement signed between GCF and UNDP.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

186. A summary of the identified risks to the proposed project's implementation and sustainability, associated impacts and mitigation measures are presented in the risk tables below. Overall, the risk assessment concludes that the proposed project's overall risk rating is low. However, this is predicated on successful mitigation, particularly with regards to amending governmental practices to ensure project effectiveness.

Table 3. Risk matrix[143].

			Impact		
			Low	Medium	High
			1	2	3
Likelihood	Low	1	1	2	3
	Medium	2	2	4	6
	High	3	3	6	9

Risk
Very Low Risk
Low Risk
Medium Risk
High Risk
Extreme Risk

187. Selected Risk Factors were assessed according to the risk matrix (Table 3), initially without and then with mitigation measures being undertaken to express the baseline risk and mitigated risk. These results are shown in Table 4 below.

Table 4. Project risk factors and mitigated risks.

Factor #	Likelihood	Impact	Risk	Mitigation Potential	Mitigated Risk
1	Medium	Low	Low Risk	High	Very Low Risk
2	Low	Medium	Low Risk	High	Very Low Risk
3	Medium	Medium	Medium Risk	Medium	Low Risk
4	High	High	Extreme Risk	Medium	High Risk
5	Low	Medium	Low Risk	Medium	Very Low Risk
6	Medium	High	High Risk	High	Low Risk
7	Medium	High	High Risk	High	Low Risk
8	Low	Medium	Low Risk	High	Very Low Risk
9	High	Low	Medium Risk	High	Very Low Risk
10	Medium	Low	Low Risk	High	Very Low Risk
11	Medium	High	High Risk	High	Low Risk
12	Medium	Medium	Medium Risk	High	Low Risk
13	Low	Low	Low Risk	High	Very Low Risk
14	High	High	Extreme Risk	Medium	Medium Risk

188. Factor 4 (governance) is essential to the overall success of the project but presents the highest risk (Table 4). Commitment to effective land use planning by the GoL is therefore critical to ensure that this risk is mitigated. Factor 14 — Land Tenure infringement — is also noted as an Extreme Risk. Government structures will need to support the project in addressing these concerns and mitigate the risk to medium.

Selected Risk Factor 1

Category	Probability	Impact
Technical and operational	Medium	Low

Description

Inadequate engagement with local-level stakeholders and project partners. This may occur as a result of insufficient coordination between implementing agencies during project implementation and/or limited integration of local knowledge into project activities.

Mitigation Measure(s)

- The project design has involved extensive engagement with local stakeholders to ensure support for project activities.
- Extensive engagements with project partners, other agencies operating in the area and stakeholders will continue during project implementation.
- Project implementation activities will build on and/or draw from existing local committees, groups, systems and procedures to ensure local stakeholder participation.

Selected Risk Factor 2

[143] United Nations Office for Disaster Risk Reduction. 2017. National Disaster Risk Assessment, Governance System, Methodologies, and Use of Results.

Category	Probability	Impact
Technical and operational	Low	Medium
Description		
Inadequate incorporation of gender sensitivity as well as social safeguards considerations and livelihoods into the implementation of the project activities. The management structures and systems may have adverse impacts on gender equality and may reproduce discrimination based on gender.		
Mitigation Measure(s)		
• A Gender Assessment and Action Plan (Annex 8) has been developed and will be implemented to ensure that gender equality is central to all interventions. • Activity design will include the full participation of women in consultations and capacity development and be sensitive to gender vulnerability disparities in all decision-making. The project design will consider the livelihood capacity development of women in the fishing industry and in the local communities. • A Social Management Plan as part of the ESMP has been designed based on a detailed screening process. The ESMP provides a framework for ensuring that project activities do not have a negative impact on the environment or local communities (Annex 6).		
Selected Risk Factor 3		
Category	Probability	Impact
Governance	Medium	Medium
Description		
The failure to commit and provide the co-financing commitment from "new money" or sustainable “payment-in-kind” for project development resulting in funds from other fundamental services/development being compromised.		
Mitigation Measure(s)		
UNDP has committed to providing co-financing for the project and has worked with the GoL to identify further sources of co-financing and in-kind activities. The availability of in-kind co-finance for Activity 1.2 in the form of rock materials has been verified by the GoL and UNDP.		
Selected Risk Factor 4		
Category	Probability	Impact
Governance	High	High
Description		
The long-term project success requires a commitment to coastal and land-use planning policies and actions to safeguard communities/infrastructure. It is also necessary to ensure that there is no overlap and/or misalignment between project activities and other actions in the MMA.		
Mitigation Measure(s)		
• The project has been designed with extensive stakeholder engagement at all levels of government and affected parties to ensure that there is synergy between project activities and any past and ongoing initiatives in the MMA. • Regular meetings will be held between the GoL and relevant individuals/organisations responsible for the management of the project. This will ensure that future projects or initiatives are aligned with the project’s interventions. • Government commitment to address land tenure arrangements during project implementation in coastal and tidal areas agreed as a condition of the project. This will include a commitment to new legal instruments and enforcement where necessary.		
Selected Risk Factor 5		
Category	Probability	Impact
Governance	Low	Medium
Description		

Changes in political leadership or focus could result in delays in project implementation or even abandonment of some activities. Additionally, turnover of government staff may impede capacity building, the retention of skills and knowledge, and knowledge management across the relevant institutions to ensure coastal and land use policy and actions are aligned with sustainable practices.		
Mitigation Measure(s)		
• Regular engagement with non-political actors – such as government agency officials, staff and local community members – will be prioritised during project implementation. In so doing, knowledge regarding the project planning and implementation processes will be shared amongst a number of individuals who can hold the implementing agencies accountable for delivery in the case of key individuals leaving their respective organisations. • The formulation of local ordinances, mainstreaming of project interventions into local planning and budgeting systems and establishment of accountable centres/offices will ensure that political changes will have negligible impact on the sustainability of project interventions.		
Selected Risk Factor 6		
Category	Probability	Impact
Governance	Medium	High
Description		
Corruption (at any level) hampers activities and the delivery of outputs and risks project cost escalation or delivery overrun.		
Mitigation Measure(s)		
• The UNDP Country Office will support implementation and provide oversight to minimise the risk of corruption. The overall risk level assessed for this project using UNDP's Partner Capacity Assessment Tool (PCAT) is "low." The PCAT is used in conjunction with the Harmonized Approach to Cash Transfers (HACT), undertaken by an independent third party for every UNDP Programme Cycle (every 4 years), accompanied by regular spot checks and specific project audits (annual in the case of this GCF project). UNDP has undertaken a HACT micro-assessment of the Environmental Protection Agency (EPA) to assess the risks associated with the entity as an Executing Entity (EE) in Liberia. The micro-assessment provides an overall assessment of the EE's programme, financial and operations management policies, procedures, systems and internal controls. In addition to the HACT micro assessment, and as part of this project, UNDP Country Office will undertake the following actions to detect and address any issue related to the risks identified in the project: i) Spot checks will be carried out once every year by an external independent party and by UNDP; ii) NIM Audit – A project audit will be undertaken annually as budgeted for under this project; and iii) Review of annual work-plans to ensure that the planned expenditures are in line with the budgeted project activities.		
Selected Risk Factor 7		
Category	Probability	Impact
Governance	Medium	High
Description		
Government officials and local communities do not take ownership of the project's community-based interventions. This could lead to limited commitment by communities to achieve the project outcomes and objectives, as well as limited sustainability of project interventions after the project.		
Mitigation Measure(s)		
• Stakeholder engagement is essential to community ownership and will enable communities to be involved in the planning of the project and the ongoing positive benefits. • Target communities will be active participants in the design and implementation of interventions focusing on improving the resilience of climate-sensitive livelihoods and supporting CBNRM. Social marketing and awareness-raising campaigns will be developed to promote community support of the project activities. These strategies will be developed to ensure all community members understand and benefit from the establishment of the community education and innovation centre and the activities associated with them.		
Selected Risk Factor 8		
Category	Probability	Impact

Technical and operational	Low	Medium
Description		
International experts/consultants/developers do not perform according to expectations.		
Mitigation Measure(s)		
• The services of consultants and contractors will be procured as per the guidelines in the UNDP manual. This will ensure that only suitably qualified applicants are contracted for project activities. • The UNDP Country Office will have an active role in project oversight ensuring that standards are not compromised in any aspect of the project.		
Selected Risk Factor 9		
Category	Probability	Impact
Technical and operational	High	Low
Description		
Limited long-term operations and maintenance of hard interventions to effectively combat coastal erosion.		
Mitigation Measure(s)		
• Initial local consultations concluded that maintenance of hard structure interventions was unlikely to be undertaken regularly or appropriately given budget and capacity constraints of local authorities. The designs of the revetments and groyne were done with this limitation in mind and are intended to function as designed for a minimum of 50 years. • Operations and Maintenance costs are estimated to average ~USD50,000 per year (Annex 21: Operations and Maintenance Plan). The ICZMP process is intended to address the institutional responsibilities and budgets for its implementation, including the maintenance of the coastal defence structures. The ICZMP itself and the institutional responsibilities associated will be endorsed at the highest level of government as part of the government policy processes.		
Selected Risk Factor 10		
Category	Probability	Impact
Technical and operational	Medium	Low
Description		
Extreme ocean weather events/hazards could negatively affect or delay project on-the-ground project interventions.		
Mitigation Measure(s)		
• The revetment and groyne material will be resilient to most weather and ocean conditions. • The final construction plan will incorporate careful planning of activities informed by ocean conditions, weather and climate briefs. Interventions will be scheduled in conjunction with this forecasting information to reduce the potential for extreme weather events to negatively affect the implementation project interventions.		
Selected Risk Factor 11		
Category	Probability	Impact
Technical and operational	Medium	High
Description		
The potential unintended environmental and dynamic changes to shore and nearshore sediment erosion and deposition as a result of physical coastal interventions.		
Mitigation Measure(s)		
• An Environmental and Social Assessment Report (Annex 6) has been developed to specifically mitigate the potential changes that may result from the implementation of coastal physical interventions.		
Selected Risk Factor 12		

Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
Inadequate incorporation of environmental safeguard considerations into the implementation of the project activities as well as its management structures and systems. The scale of pollution in the creeks, estuaries and mangrove swamps around Monrovia proves too extensive for control by existing agencies.		
Mitigation Measure(s)		
• An Environmental Management Plan has been designed based on a detailed screening process. This will provide a framework for ensuring that project activities do not have a negative impact on the environment. • Project design will include surveys of drainage and pollution sources, focussed on identifying curable problems to be addressed by the project, with others raised for longer-term action by the authorities.		
Selected Risk Factor 13		
Category	Probability	Impact
Reputational	Low	Low
Description		
Project might involve temporary or permanent and full or partial physical displacement.		
Mitigation Measure(s)		
• The project has been designed to avoid physical displacement. • Should it prove to be necessary for a small number of people, then a full UNDP- (and IFC-) compliant resettlement approach will be used.		
Selected Risk Factor 14		
Category	Probability	Impact
Governance	High	High
Description		
Project development may infringe on land tenure arrangements in marginal coastal areas.		
Mitigation Measure(s)		
• The GoL has committed to address land tenure arrangements in coastal and tidal areas. This will include a commitment to new legal instruments and enforcement where necessary. Adopting the ICZM approach and developing the ICZMP with a specific section dedicated to coastal land tenure regulation in high-risk areas will serve to mitigate this risk. The above-mentioned component of the ICZMP will be fully aligned with the World Bank-funded land use masterplan for Monrovia currently under development[144]. • Project design will support government actions by including messages on land tenure in its community awareness raising programmes. • Project design will also include support to government agencies through its capacity strengthening programme.		
Selected Risk Factor 15		
Category	Probability	Impact
Technical and Operational	Low	High
Description		

[144] The enforcement of the ICZMP will not be specifically dependent on the wider adaptation policy development processes supported by the World Bank project. Alignment and collaboration will rather ensure that there is no duplication of efforts, and that stakeholders, especially communities, are clear on the two processes.

Construction of coastal protection measures has the unintended consequence of exacerbating existing flood risk in West Point
Mitigation Measure(s)
• The revetment at West Point has been designed to armour the vulnerable shoreline and fix it in place. The revetment will use porous materials, a permeable design and include a drainage system to enable water to drain through the structure into the sea, rather than creating a levee which would contain flood water. • During the development of the detailed design and construction plan under activity 1.1, a site-specific hydrological study of West Point, to be conducted under the activity, will also be used to assess the impact of the revetment on existing localised and broader flooding and mitigate this risk by enabling the design of adequate drainage capacity.
Other Potential Risks in the Horizon
• Currency fluctuations (both nationally and internationally) could negatively influence the project's budget, including the implementation costs. This may result in: i) suboptimal and incomplete implementation; and ii) a reduction in sustainability and upscaling potential. • Project development might be delayed by the COVID-19 pandemic as national and regional responses to the pandemic may involve restrictions including lockdowns. The project and its personnel will abide by all rules and regulation mandated by the Government of Liberia in response to COVID-19. Project staff will take additional precautions to ensure that stakeholders and beneficiaries are not exposed to and that project activities do not in any way, allow spreading of the virus. Awareness among project staff and stakeholders, including communities will be embedded in all interactions. Project staff will distribute awareness materials and personal protective gear that is made available by parallel efforts in the region. If and when possible, project staff and consultants who require movement between regions and internationally will immunise themselves.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

189. The environmental and social risk associated with the project was evaluated in accordance with GCF's Environmental and Social Safeguards (ESS) as well as UNDP's Social and Environmental Standards (SES). The project is considered to be Category B or Medium Risk (GCF ESS) and Moderate Risk (UNDP risk category). This initial categorisation was determined during the PPF-financed Feasibility Study conducted from 2018–2019. UNDP's Social and Environmental Screening Procedure (SESP) and the Environmental and Social Assessment Report (ESAR) developed for the project substantiate this categorisation. The ESAR complies with the broad ESS requirements of the GCF and includes an Environmental and Social Management Plan (ESMP). It is, however, important to note that the ESAR is an interim report and in order to comply with national legislation, an Environmental and Social Impact Assessment (ESIA)[145] will need to be conducted and approved by the EPA — as the environmental authority in Liberia — prior to the implementation of Activity 1.2. The primary environmental and social risks that may be associated with the project — as identified by the ESAR — are presented below.

Environmental impacts

190. Overall the project is likely to result in some negative environmental impacts. These impacts are, however, expected to be temporary and mitigated effectively through appropriate management measures. During the construction phase of Output 1, substantial earthworks will need to take place along the West Point beachfront. Potential negative impacts that are associated with the establishment of coastal infrastructure include local changes in water quality in the short-term as well as a shift in sediment transport patterns in the long-term. Additionally, there are inherent pollution risks[146] associated with the use of earth-moving equipment in coastal environments. These risks have, however, been accounted for in the ESAR and ESMP, and appropriate risk mitigation procedures will be included in the Terms of Reference for contractors. In addition, any long-term shifts in sediment transport patterns are expected to exhibit minor impacts on a local scale as a result of an existing perpendicular breakwater associated with the port of Monrovia.

Social impacts

191. Two main social risks were identified during the Feasibility Study as a result of the construction of a revetment at WestPoint — and an associated loss of beach areas — both of which have been mitigated through an adaptive design process. These include impacts on: i) social cohesion, that will occur because of reduced access to community land used for domestic purposes, bathing and socialising; and ii) fishing livelihoods, which will occur because of a reduction of available locations for the launching fishing boats and the processing of fish. The current design for the revetment includes a green promenade with associated amenities that will be available to be used as community space and for the processing of fish. Additionally, the design will incorporate at least two protected beach areas to act as boat launching and landing sites and associated boat storage space, which will allow fishermen and community members to continue access the bay.

192. No further potential negative social impacts have been identified as a result of the other activities under the project, which are expected to yield a range of social benefits, including strengthened livelihoods, food security and mangrove areas as well as the protection of homes and infrastructure. It should be further noted here that resettlement, land acquisition and any potential impacts on Indigenous People have been investigated and will not occur under the project.

Grievance Redress Mechanism

193. A Grievance Redress Mechanism (GRM) and Complaints Register (CR) have been developed for this project and are described in detail in the ESMP. These two mechanisms are compatible with Liberian legislation and compliant with UNDP and GCF standards and will allow any project stakeholders who have a complaint or feel aggrieved by the project to communicate their concerns and/or grievances through an appropriate process. The GRM and CR will be used at all stages of the project and will provide an accessible, rapid, fair and effective response to concerned stakeholders, especially any vulnerable groups who may lack access to formal legal systems or support.

[145] The Ministry of Mines and Energy will be responsible for obtaining the permit, as the Responsible Party for Activities 1.1 and 1.2.

[146] A number of pollution risks are associated with the use of earth moving vehicles and construction of large-scale infrastructure, including synthetic pollutants, organic waste and construction debris. Further details on potential pollutants are provided in the ESAR (Annex 6).

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

194. A gender assessment and action plan — in line with the objectives of the GCF's Gender Policy and gender guidelines — has been developed by UNDP to inform the design of the proposed project (Annex 8). The document was based on 24 field consultations, as well as available data from studies conducted by the GoL, UN agencies, civil society organisations (CSOs), donor agencies and multilateral development banks.

195. The gender assessment shows that gender inequality persists in Liberia, largely due to patriarchal norms that maintain women's low status and the legacy of violence against women during the civil wars[147]. The country has a low Gender Development Index (GDI), ranking 177 out of 188 countries in 2018. Gender gaps in human development between men and women are evident in inter alia: i) mean years of education — 3.5 years for women and 6.1 years for men; ii) mean income — USD 577 for women and USD 755 for men; and iii) levels of adult HIV — 2.4% for women and 1.8% for men[148]. The burden of family care imposed upon women through the socio-cultural allocation of gender roles restricts their livelihood options, which suggests why approximately 90% of women are employed in the informal sector[149]. This high participation of women in the informal sector limits their social safety nets to adapt to the impacts of climate change.

196. As a result of these gender roles and inequalities, women in Liberia are more vulnerable than men to coastal erosion which causes the destruction of infrastructure, loss of property, displacement, decreases in income, as well as rises in insecurity and gender-based violence. Additionally, coastal erosion contributes to increased gender inequalities by exacerbating the challenges women already face such as family care and food insecurity. For example, stakeholder meetings held in Monrovia from March to April 2019 revealed that women play a central role in the informal fishing sector. Representing 60% of the fishing community, women are mostly involved in drying and selling fish while men are predominantly involved in fishing. Since women sell the fish, they carry the weight of economic fluctuations related to fish stock and quality — both impacted by climate change.

197. From the findings of the gender assessment, a gender action plan (GAP) for the project was developed to ensure gender is mainstreamed in project design and implementation. In particular, the GAP recommends that a women's empowerment approach is employed in the design and implementation of the project, increasing women's skills and economic opportunities. By increasing the climate-resilience of Monrovia's fishing community, the project intentionally targets and benefits women — and by extension, their families and communities. All activities, training and stakeholder consultations under the project will be conducted in a gender-sensitive manner and encourage women's active participation. Activities will contribute to women's empowerment through upskilling of women's groups to pursue climate-resilient and sustainable livelihoods in the Mesurado mangroves, as well as through women's active participation in the mangroves' management. This empowering approach, combined with gender-sensitive activities and stakeholder consultations, will contribute to reducing the gender gap of social, economic and environmental vulnerabilities to climate change in Monrovia.

198. In project-level implementation arrangements, a Gender Specialist will be posted to the Project Management Unit to provide advice and quality assurance oversight, conduct capacity building and to monitor the implementation of gender-related activities and gender-sensitisation. The collaborative relationship between this specialist, UNDP, the Environmental Protection Agency, the Center for all Disasters Management, the Ministry of Gender, Children and Social Protection, and local NGOs will create the cross-sectoral technical perspectives required to ensure the success of this gender-sensitive approach in the project.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

199. The financial management and procurement of this project will be guided by UNDP financial rules and regulations available here.

200. UNDP has comprehensive procurement policies in place as outlined in the 'Contracts and Procurement' section of UNDP's Programme and Operations Policies and Procedures (POPP). The policies outline formal procurement standards and guidelines across each phase of the procurement process, and they apply to all procurements in

147 Republique of Liberia. 2018. Programme Against Sexual and Gender Based Violence and Harmful Traditional Practices.
148 UNDP. 2018. Human Development Report.
149 According to the 2008 report issued to the Committee on the Convention on the Elimination of All forms of Discrimination against Women (CEDAW).

UNDP. The project will be implemented following the National Implementation Modality (NIM) following NIM guidelines available here.

201. The UNDP Enterprise Risk Management (ERM) Policy identifies 'Capacities of the Partners' as a key Strategic Risk to be managed for the success of UNDP's work. The Partner Capacity Assessment Tool (PCAT) was designed to assess the level of risk that is present when UNDP works with Partners to implement programmes and projects. The level of risk is identified by analysing partner capacity and matching project management and oversight with the level of risk assessed. By identifying areas for capacity improvement, the PCAT should also help to reduce future Partner risk levels if the capacity building actions are implemented and sustained. The Partner Capacity Assessment is conducted for all projects and covers executing entities, responsible partners, grant recipients and other partners to ensure that the proposed project partners are not a prohibited organization and do not engage in practices that are inconsistent with UNDP's social & environmental standards and code of ethics.

202. UNDP will ascertain the national capacities of the executing entities by undertaking an evaluation of capacity following the Framework for Cash Transfers to Executing Entities (part of the Harmonized Approach to Cash Transfers - HACT). All projects will be audited following the UNDP financial rules and regulations noted above and applicable audit guidelines and policies.

G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.
☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- ☐ full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- ☐ redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

☒	Annex 1	NDA No-objection letter(s)
☒	Annex 2	Feasibility study - and a market study, if applicable
☒	Annex 3	Economic and/or financial analyses in spreadsheet format
☒	Annex 4	Detailed budget plan
☒	Annex 5	Implementation timetable including key project/programme milestones
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): ☐ Environmental and Social Impact Assessment (ESIA) or ☐ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan
☐	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan
☒	Annex 11	Monitoring and evaluation plans
☒	Annex 12	AE fee request
☒	Annex 13	Co-financing commitment letter, if applicable
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☒	Annex 15	Evidence of internal approval
☒	Annex 16	Map(s) indicating the location of proposed interventions
☐	Annex 17	Multi-country project/programme information
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☒	Annex 20	First level AML/CFT (KYC) assessment
☒	Annex 21	Operations manual (Operations and maintenance)
☐	Annex x	Other references

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter issued by the national designated authority(ies) or focal point(s)

REPUBLIC OF LIBERIA
ENVIRONMENTAL PROTECTION AGENCY
P.O. Box 4024
4th Street Sinkor, Tubman Boulevard,
1000 Monrovia, 10 Liberia

Office of the Executive Director

EPA

ED/EPA-01/000906/19/RL

September 5, 2019

THE GREEN CLIMATE FUND ("GCF")

Re: Funding proposal for the GCF by United Nations Development Programme (UNDP) regarding the Monrovia Metropolitan Coastal Resilience Project (MMCRP)

Dear Sir/ Madam:

I present my compliments and wish to refer to the project "Monrovia Metropolitan Coastal Resilience Project (MMCRP)" in Liberia as included in the funding proposal submitted by the United Nations Development Programme (UNDP) to the Environmental Protection Agency (EPA) on August 31, 2019.

I the undersigned, Hon. Nathaniel T. Blama, Sr., am the duly authorized representative of the EPA, the National Designated Authority of Liberia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Liberia has no-objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with Liberia's national priorities, strategies and plans;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed and acknowledge that this letter will be made publicly available on the GCF website.

Please accept the assurance of my highest esteem and consideration as we strive for environmental sustainability for now and successive generations.

Kind regards,

Nathaniel T. Blama, Sr., *L.H.D., MPA*
EXECUTIVE DIRECTOR/CEO

Website: http://www.epa.gov.lr *Mobile:* +231880707024/+231778059829 *Email:* hwealjr@epa.gov.lr

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Monrovia Metropolitan Climate Resilience Project
Existence of subproject(s) to be identified after GCF Board approval	No
Sector (public or private)	Public
Accredited entity	United Nations Development Programme (UNDP)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Liberia, with a focus on Monrovia Metropolitan Area
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Tuesday, February 2, 2021
Language(s) of disclosure	English
Explanation on language	English is the official language of Liberia, and stakeholders will be able to understand and provide any feedback in English.
Link to disclosure	https://www.lr.undp.org/content/liberia/en/home/library/environment_energy/environmental-and-social-assessment-report.html
Other link(s)	N/A
Remarks	Provision for an ESIA consistent with the requirements for a Category B project is contained in the "Environmental and Social Assessment Report (ESAR)".
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Tuesday, February 2, 2021
Language(s) of disclosure	English
Explanation on language	English is the official language of Liberia, and stakeholders will be able to understand and provide any feedback in English.
Link to disclosure	https://www.lr.undp.org/content/liberia/en/home/library/environment_energy/environmental-and-social-assessment-report.html
Other link(s)	N/A
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the "Environmental and Social Assessment Report (ESAR)".
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A

Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	Stakeholder Engagement Plan/ Tuesday, February 2, 2021
Language(s) of disclosure	English
Explanation on language	English is the official language of Liberia, and stakeholders will be able to understand and provide any feedback in English.
Link to disclosure	https://www.lr.undp.org/content/liberia/en/home/library/environment_energy/mmcrp-stakeholder-engagement-plan.html
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Tuesday, February 2, 2021
Place	Physical copies were made available to the West Point Communities through the local leadership, and were displayed for public scrutiny at the office of the Commissioner within West Point. Address: C/O Hon. C. Wea, Commissioner Waterside Market Township of Westpoint Monrovia Web links (URL) to, and email attachments of, the ESAR and Stakeholder Engagement Plan were shared with the National Designated Authority, Environmental Protection Agency, Ministry of Mines and Energy, Ministry of Public Works, the Ministry of Finance and Development Planning, the African Development Bank, the World Bank, and civil society organizations.
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Tuesday, March 16, 2021

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP160

Proposal name:	Monrovia Metropolitan Climate Resilience Project
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Liberia
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The project addresses impending shoreline regression at West Point, where dwellings and public infrastructure are under threat, including boat launching sites that are critical to fishery-based livelihoods.	Risk of maladaptation needs to be carefully assessed and mitigated when refining and finalizing the engineering design of the revetment construction at West Point.
The Government of Liberia contributes to a considerable amount of co-financing despite its budgetary constraints.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the respective term sheet and addendum XVII, titled "List of proposed conditions and recommendations".

II. Summary of the Secretariat's assessment

2.1 Project background

3. Monrovia, the capital city of Liberia, is vulnerable to the climate change impacts of sea-level rise (SLR) and the increasing frequency of high-intensity storms, both of which contribute to coastal erosion and shoreline retreat. SLR also threatens the sustainability of ecosystem services provided by mangroves in the Mesurado estuary at the centre of the Monrovia Metropolitan Area (MMA), exacerbating the already vulnerable mangroves that are under threat of over-exploitation and urban encroachment. These changes have resulted in change of habitat for economically important fish species and loss of these nursery areas which consequently impact the fishery-based livelihoods of Monrovians.

4. The most vulnerable part of the MMA coast is West Point, one of the most densely populated and poorest areas on the Monrovian coast and the section most vulnerable to climate change-induced acceleration of coastal erosion. Near West Point, the shoreline has regressed by 30 metres in the last decade, leading to loss of 670 dwellings, and threatening public spaces and boat launching sites that are critical to fishery-based livelihoods. It is forecast that coast erosion resulting from climate change will cause further shoreline regression of 190 metres by 2100.

5. The GCF project aims to address this urgent need for adaptation in Monrovia through three interrelated focus areas: (i) coastal protection; (ii) coastal management; and (iii)

diversified climate-resilient livelihoods. In this way, the proposed project will build the long-term climate resilience of coastal communities in Liberia by both addressing immediate adaptation priorities and creating an enabling environment for upscaling coastal adaptation initiatives to other parts of Monrovia and Liberia.

6. The total project cost is USD 25.6 million, all in the form of grants. GCF financing amounts to USD 17.3 million, most of which will be contributed to the revetment construction along the coast near West Point. The remaining cost will be borne by the Government of Liberia (USD 6.8 million in form of grant and in-kind contributions) and UNDP (USD 1.6 million grant).

7. The environmental and social safeguards classification of the project is category B, as informed by the screening performed by UNDP.

2.2 Component-by-component analysis

Output 1: Protection of coastal communities and infrastructure at West Point against erosion caused by sea-level rise and increasingly frequent high-intensity storms (total cost: USD 18.1 million; GCF cost: USD 13.5 million, or 75 per cent)

8. Under this output, GCF grant funding will be used to protect the vulnerable community at West Point through construction of a revetment structure. This revetment structure was selected as the preferred solution because while a “soft solution” in the form of beach nourishment with an associated groyne was considered technically feasible, its sustainability is questionable as the required regular maintenance is not considered feasible in the local context. The revetment structure is expected to prevent sediments and sand being washed away. As per common good practice, appropriate drainage will be included in the detailed design of the structure to allow run-off and prevent localised flooding behind the structure.

9. GCF proceeds will cover 75 per cent of the total cost of this construction and the remaining costs are borne by co-financiers, which will cover the cost of finalisation and validation of the detailed design and construction plans, and rock material for construction of the revetment at West Point.

10. Structural measures for coast protection may cause changes of wave patterns which lead to maladaptation in other areas along the coast. Recognizing this risk, the accredited entity (AE) has confirmed that through detailed modelling it is evident that no effect on the water movement will take place around West Point due to stabilising the coastline with marine boulders.

Output 2: Institutional capacity-building and policy support for the implementation of Integrated Coastal Zone Management (ICZM) across Liberia (total cost: USD 4.1 million; GCF cost: USD 1.6 million, or 40 per cent)

11. This project will develop a national-scale high-resolution multi-criteria vulnerability map and will design a national risk-informed Integrated Coastal Zone Management Plan in consultation with relevant stakeholders, including the private sector. This output will also support cross-sectoral coordination among the 10 government institutions responsible for climate-resilient coast management. Implementing ICZMP in the MMA, and across Liberia, will transform the existing coastal management paradigm and reduce the vulnerability of coastal communities to the impacts of climate change-induced SLR, accelerated coastal erosion and increased incidence of extreme weather events.

Output 3: Protecting mangroves and strengthening gender- and climate-sensitive livelihoods to build local climate resilience in Monrovia (total cost: USD 2.2 million; GCF cost: USD 5.7 million, or 59 per cent)

12. Under this output, the proposed project will put measures in place to reduce the anthropogenic pressure on mangrove ecosystems in the Mesurado Wetland. By reducing this pressure, the health of the ecosystems will be improved and their resilience to sea-level rise and the impacts of climate change will increase, thus increasing the resilience of the communities in Monrovia that rely on these ecosystems for their livelihoods. In addition, reducing the degradation of mangroves will increase the likelihood of their long-term survival and adaptation to climate change through migration. These mangroves are likely to be negatively impacted by climate change-induced sea-level rise which will decrease the area of the Mesurado Wetland in which they can grow.

13. In addition, urban encroachment into the mangrove forests and the harvesting of mangroves for fuelwood, charcoal production and fish smoking have resulted in degradation of the mangrove ecosystems in Monrovia. In order to develop long-term climate resilience within vulnerable, mangrove-dependent communities there is a need to: (i) support community conservation of mangroves in the MMA to secure the provision of critical ecosystem services in the face of observed and expected climate change impacts; and (ii) develop local climate resilient livelihood activities and strengthen market access for small and informal enterprises engaged in these activities. These community level approaches to adaptation will benefit from improved coastal protection measures being implemented under Output 1 and will integrate with the long-term adaptation planning being developed under Output 2, to support transformative change at both institutional and community levels.

Output 4: Project management (total cost: USD 1.2 million; GCF cost: USD 0.8 million, or 75 per cent)

14. Total project management costs (PMCs) amount to USD 1.2 million, to be financed by GCF and UNDP. The total PMCs and the split between GCF and co-financiers comply with the GCF Policy on Fees for Accredited Entities.

III. Assessment of performance against investment criteria

3.1 Impact potential *Scale: Medium to high*

15. Following a feasibility study funded through the GCF Readiness facility, the proposal has established that climate projections based on the Representative Concentration Pathway (RCP) for both RCP4.5 and RCP8.5 show significant impacts on Liberia's coastlines over a range of decision timelines (up to 2100). These impacts include risks to Monrovia's local coastal communities, their livelihoods and the broader coastal infrastructure in the form of (i) degradation of the mangrove ecosystems on which their livelihoods and food security depend and (ii) inundation of vital infrastructure such as boat-launch sites, dwellings and socioeconomic spaces and amenities such as fish markets.

16. In particular, the proposal has articulated well how the city of Monrovia has become highly vulnerable to climate-induced coastline erosion and shoreline retreat as a result of sea level rise and increasing frequency of high-intensity storms and off-shore waves (coastline retreat is estimated to be 110 per cent compared to the baseline). It has also detailed how these impacts, in addition to over-exploitation of its resources for community livelihoods and asset generation, are threatening the Mesurado Wetland, resulting in loss of economically important fishery-based livelihoods of 55,000 Monrovians, of which 46 per cent are women.

17. The project expects the outcomes will contribute to the achievement of the GCF objective of increased climate resilient sustainable development in three ways: (i) actively protecting infrastructure against climate change impacts; (ii) improving climate-responsive coastal planning; and (iii) strengthening local livelihood practices and promoting the

sustainable use of mangrove ecosystems to reduce the climate vulnerability of coastal communities in Monrovia. Outcome 1 is realistically achievable and the plans in the proposal are well constructed. There is concrete action proposed together with evidence that the Government of Liberia is engaged and willing to participate.

18. Outcomes 2 and 3 rely much more on engaging the community and changing habits and the way of life for some Liberians. Outcome 3 in particular relies heavily on community engagement and education to achieve the proposed changes and it is not clear how much community engagement has occurred to date to judge how well the affected communities might view these proposals.

19. Clearly these interventions alone will not yield the optimal results if operational climate information and early warning services over a range of decision timelines (subseasonal to seasonal) are not integrated into the design and implementation of the project (operations and maintenance). In particular, impact-based forecasts related to severe weather (marine forecasts, storm surge, sea level rise, waves/tides, lightning, thunderstorms, flooding and heatwave) will enable early action for resilience of the communities and their livelihoods to climate variability and change.

20. The proposal will therefore benefit from a strategy for enhancing coherence and complementarity between this project and the already approved GCF project "Enhancing Climate Information Systems for Resilient Development in Liberia (Liberia CIS)". It should detail how operational climate information and early warning services will be applied to implementation of the project under consideration. This should include marine forecasts such as storm surge, sea level rise, waves/tides and related severe weather such as flooding, heatwave, storms, lightning, that could impact on the lives and livelihoods of the communities.

3.2 Paradigm shift potential *Scale: Medium*

21. Liberia is endowed with significant natural resources but without external support its ability to effectively manage them will remain a challenge. In addition, climate variability and change is posing significant risks to its coastal resources and the situation will be further exacerbated as climate change gathers pace. A more integrated and comprehensive approach to managing these coastal resources will require the harmonization of Integrated Coastal Zone Management (ICZM) and disaster management strategies and their effective operationalization to manage the scale and intensity of the climate risks described in relation to the impact potential in section 3.1. This will ensure that flood risk issues and associated disasters provoked by maladaptation are better managed through anticipatory actions.

22. The proposal has identified specific interventions through a vulnerability assessment underpinned by a robust climate rationale and extensive stakeholder consultation. It seeks to: (i) under output 1, protect the coastal communities and infrastructure at West Point through the construction of a rock revetment and green promenade; (ii) strengthen institutional frameworks and coordination for implementation of the national adaptation plan (NAP) process through participatory capacity strengthening of Liberia's adaptation-related institutions (namely the Ministry of Mines and Energy, Ministry of Finance and Development Planning and the Ministry of Public Works) and the development of a national ICZM plan; (iii) expand the existing knowledge base for scaling up adaptation through cross-sectoral capacity development under output 2, which includes an action plan for sector-specific implementation of the ICZM plan and the integration of ICZM into relevant by-laws and regulations as well as mainstreaming climate change adaptation into planning, and budgeting processes and systems; and (iv) promoting sustainable co-management of mangrove areas and innovative finance approaches under output 3 to support the local fishery-based livelihoods. This includes training and facilities for small-scale manufacturing and selling of cookstoves to reduce pressure on the mangrove ecosystems.

23. Although these approaches are well-established globally, it is the first time that such an integrated and comprehensive approach will be applied in Liberia. It is clear from the proposal that if it is successfully implemented, new skills will be introduced to the Monrovian community and new business activities will be generated. As a result, it is likely that the proposal will create new jobs and create low-emission and climate-resilient livelihoods for the broader coastal communities.

3.3 Sustainable development potential *Scale: High*

24. The proposal seeks to strengthen adaptative measures to help minimise impacts of climate variability and change through awareness and building institutional and community capacities for implementation of the protective measures, such as the construction of revetments and beaches, to protect against rising sea level, storm surge and beach erosion.

25. A cost-benefit analysis for the economic feasibility of both protective and adaptive measures has reviewed the vulnerability analysis and damage estimation in the do-nothing scenario for five sections of the Monrovia coastline with a return period of 100 years under RCP8.5 and RCP4.5 in 2100. Under the RCP8.5 scenario, the intervention at West point is projected to prevent 150 m of shoreline retreat by 2050 and 360 m by 2100. .

26. For West Point, the cost of inaction will lead to damage of USD 40–48 million by 2100. The interventions would protect the fishery industry that currently supports about 3,000 fisherfolk and 9,000 fishmongers in Monrovia. The support provided for manufacturing eco-friendly cookstoves (Activity 3.4) will support women to engage in activities that will reduce the burden on the mangroves.

27. The proposal is expected to have a positive impact on the environment by reducing coastal erosion and sustaining, and ultimately improving, the nearby wetland areas.

28. The proposal presents evidence that the interventions are likely to improve the health and food security of the community around the wetlands in the long term. Evidence is also provided that the intervention may save some residential areas in the area from inundation and erosion.

29. The proposal also indicates that it will provide the affected communities with additional education and training opportunities, however, it does not indicate that these will necessarily create net income for residents within the proposal's scope or target area. While this is possible it is not specifically indicated.

30. There is expected to be an increase in physical infrastructure as a result of the proposal. However, there needs to be some measure of engagement with, and reassurance provided to, the fisher folk who use the beach at West Point that during and following the installation of the revetment there will always be sufficient room for them to launch and retrieve their boats and clean and smoke their catches.

31. There are clear efforts to ensure equitable gender participation in the contribution to the expected outcomes. It should be noted that the target is for 40 per cent female participation in the construction phase of the revetment but this is likely to be much higher than the current level of female participation in the earthmoving and (heavy) construction industries in Liberia.

3.4 Needs of the recipient *Scale: High*

32. Liberia is classified as a least developed country (LDC) which is vulnerable to climate change ranking fourth in the 2017 Climate Change Vulnerability Index. It is particularly vulnerable in its coastal margins. The future impacts of a number of climate change scenarios have been well presented in the proposal.

33. Limited technical and financial capacity within most government departments is a significant barrier to the implementation of climate change adaptation, and ICZM in particular. The structural economic problems faced by the country as well as financial resources being directed towards the provision of social services, such as education and health, result in limited national sources of financing being available for adaptation interventions.4.

3.5 Country ownership *Scale: High*

34. The proposal is well aligned with the nationally-determined contribution, national adaptation plans, Sustainable Development Goals, Sendai Framework for Action, National Disaster Management Strategy and related national development plans.

35. From the evidence provided by the proposal, it would appear that the Government of Liberia considers the management of coastal erosion and the impacts of extreme climate events to be a high priority. They have developed several national policies and plans in this regard and demonstrated their commitment to the project by providing rock material as in-kind co-finance for Output 1 of the proposed project.

36. The National Policy and Response Strategy on Climate Change (2018) is the central policy underpinning Liberia's response to climate change and serves as the basis for all future adaptation and mitigation planning. It identifies the sectors most vulnerable to climate change impacts and names five enabling pillars to support the safeguarding of these sectors. The proposal will contribute to four of these pillars, specifically cross-sectoral coordination, capacity-building, integrated planning and the development of infrastructure.

37. UNDP is the accredited entity of the project and will provide a qualified team of national and international experts equipped to assist the Government to respond to complex issues such as climate change and green growth. In the proposal, UNDP state that they will ensure that the project adopts a comprehensive, multi-sectoral approach that is centred around results-based activities. The proposal indicates that strong relationships exist between UNDP and local decision-makers, and that UNDP has a proven track record as an impartial provider of technical advice and support.

3.6 Efficiency and effectiveness *Scale: Medium to high*

38. The project seeks to address an urgent adaptation need in Monrovia by the construction of a rock revetment sea defence to protect West Point against climate impacts. The revetment is a preferred solution compared to beach nourishment with groyne due to lower maintenance costs and for sustainability reasons. The infrastructure will protect approximately 1,050 metres of vulnerable coastline and enhance resilience of the livelihoods of about 25 per cent of Monrovia's population at a cost of USD 104 per beneficiary.

39. An expected net present value (ENPV) for the project is estimated to be USD 17.73 million based on a 10 per cent discount rate, and an economic internal rate of return of 21 per cent. The co-financing ratio of about 34 per cent shows a strong commitment of the government despite the major budgetary and economic constraints.

40. The project incorporates best practices and lessons learned from relevant ongoing interventions and initiatives that focus on the protection of vulnerable communities and ecosystems in the coastal zone – particularly two Global Environment Facility (GEF)-funded projects in Monrovia. The Government of Liberia has identified the material for revetment construction and is committed to contributing this to the project and for future coastal protection. The Government's commitment improved the co-financing ratio from initial design and strengthened the case for sustainability and scale-up potential of the intervention. These efforts by the Government will enable the remaining four sites identified and extensively

analysed by the feasibility study to be constructed utilising the lessons and best practices that will be acquired through the implementation of this project.

41. The proposal will benefit from detailing a coherence and complementarity strategy with other coastal infrastructure projects, such as that implemented by the Japan International Cooperation Agency and the World Bank.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

42. **Environmental and social risk category and safeguards instruments**. The project aims to provide adaptive interventions such as institutional development, capacity building, awareness campaign, conduct of assessments and feasibility studies, management and development planning, support for regulations enforcement; and protective interventions such as the construction of about 1,050-meters of revetment wall to protect the most vulnerable section of the coastline from climate change induced erosion in the coastal area of Monrovia. The project interventions will generally target the Mesurado River and St. Paul River estuaries and mangrove areas, as well as in other smaller inlets in the area, and the West Point coastline. The AE has prepared an Environmental and Social Assessment Report (ESAR) and a Stakeholder Engagement Plan (SEP). The ESAR contains detailed description of the environmental and social profile of the project area, the country's regulations relevant to assessing and managing environmental and social impacts of the project, an initial assessment of the environmental and social risks and impacts of the project activities, and a risk mitigation/management plan.

43. The AE has assigned an environmental and social risk (E&S) of Category B for the project. The Secretariat agrees with this categorization and confirms that the project’s E&S risk category is within the level by which the AE is accredited to.

44. **Compliance with GCF’s environmental and social safeguards (ESS) standards.** The project is designed to primarily promote sustainable use of coastal resources, ecological balance while protecting communities. The E&S risks and impacts and how they will be managed are further described below.

45. **ESS 1: Assessment and management of environmental and social risks and impacts.** The project underwent E&S screening using the AE’s Social and Environmental Screening Procedure (SESP). An ESAR was prepared based on the screening and included a sub-assessment of the area such as the mangrove area. The ESAR contains fairly detailed social and environmental characterization of the project site, an analysis of the environmental and social impacts and risks, and an environmental and social management plan (ESMP) which contains an Environmental and Social Mitigation and Monitoring Matrix (ESMMM). The ESAR also covered and addressed all the risks relating to each of the ESS Standards. However, the project will still have to undergo a formal environmental and social impact assessment (ESIA) process in accordance with Liberia's Environmental Protection and Management Law (EPML). The ESAR recommends a number of additional surveys and detailed plans as part of the ESIA, including a Resettlement Action Plan (RAP). The ESAR will thus be upgraded into a full ESIA with additional information, including an updated ESMPs and sub-plans based on these additional data and information.

46. Based on the ESAR review, the key environmental and social risks of the project include: (i) potential for land acquisition and involuntary relocation of residential structures along the alignments of the access road and revetment wall structure, (ii) potential impact in terms of restriction of access to their livelihoods among the coastal communities, particularly the fishing communities along the West Point coast and the small scale/subsistence sand extractors, (iii)

construction-related occupational and community health and safety risks, including potential labor management issues; and (iv) possible unintended long-term adverse induced impacts to adjacent unprotected coastlines and the estuarine due to likely changes in coastline dynamics, including freshwater-marine water interactions, beach scouring fronting the wall, and altered sediment distribution. The detailed ESIA process that will be conducted will revisit these risks and prepare site-specific management measures to avoid and/or mitigate these risks.

47. **ESS 2: Labour and working conditions.** The construction of the revetment wall will require hiring of staff and engagement of contractors. The ESAR has projected about seven expatriates and around 25 local staff to be engaged in the project, plus an undetermined number of local workers. The project and contractor's staff and workers will potentially be exposed to occupational health and safety (OSH) hazards at construction sites, quarry sites and construction routes. The ESAR has assessed that there is a small possibility of residual unexploded ordnance (UXO) in the area. There is also potential risk of worker discrimination including gender-based denial or violation of basic workers' rights and child exploitation.

48. The ESMMM in the ESAR has already mandated strict compliance with OSH standards at construction sites and checks for the presence of UXO prior to any civil works. Project contractors and service providers will also be required to provide a written commitment to equal opportunity hiring, social inclusion and adherence to the prohibition of child labour in accordance with the International Labor Organization (ILO) Convention 138. These broad measures will be fleshed out in more details in the ESIA. The ESIA will further assess OSH compliance issues prevalent in Liberia as well as any labor management issues, including special labour contract arrangements that might violate labour standards and update the ESMMM accordingly.

49. **ESS 3: Resource efficiency and pollution preventions.** The risk of release of air and water pollutants, and hazardous wastes to the environment is assessed to be negligible and limited to accidental spillage of fuel and lubricants, and effluents from batching plants. Dusts and noise may also be generated at the quarry, batching plant, hauling routes and construction sites. To address these issues, the ESMMM includes a detailed waste management plan, quarry management and hauling arrangements. The project may also cause short-term erosion and sedimentation of water bodies, including the beach during construction. In the long term, the project may cause shore and nearshore sediment erosion and deposition change. The ESAR has recognized this risk and has indicated that additional investigation and research will be conducted to evaluate the real extent of the changes and how they might occur and addressed.

50. **ESS 4: Community health, safety and security.** The project will expose the local communities to health and safety hazards during the construction period. There is also the risk of exposure of workers to communicable diseases brought by the influx of labour, and vice versa. The risk of spreading communicable diseases may be exacerbated by the crowded conditions of the settlers in the West Point area. During construction, the local communities may be exposed to safety hazards at the construction sites and construction routes, including heavy equipment traffic, obstructions, sharp objects, boulders and deep excavations. In the long run, unanticipated changes in the ecology resulting from the revetment wall may also trigger emergence of latent diseases, particularly water/vector borne diseases.

51. The ESMP of the ESAR advocated for "strong adherence to standard construction site processes of work segregation from public access, traffic safety measures, and standard health and safety practices" and provision of the "awareness-raising measures on health risks as a matter of standard good practice. Due to the situation in the area, the project should ensure that contractors formulate and adopt an occupational and community health and safety (OCHS) plan which would include community safety management measures, installation of fences, proper lighting and warning signs, construction traffic management, among others, as well as require compulsory medical screening of staff and workers. The project management should also adopt a simple disease surveillance protocol for the community and workers.

52. A further safety risk relates to the possibility of. This risk is amplified by the fact that a large proportion of the population reportedly still suffers from mental illnesses or were mentally or physically traumatized, due to 14 years of intense civil conflict. The project should be cognizant of this risk during the planning and consultation phase, be transparent and strive for equal treatment of groups, and should respond immediately for any perceived unfairness in the allocation of benefits and the social and economic costs burden between groups.

53. **ESS 5: Land acquisition and resettlement.** The ESAR has rated the risks of physical displacement (i.e., loss of dwellings) and economic displacement (i.e. loss of productive assets and entire or portions of livelihood) as "insignificant". While the construction of the revetment wall and promenade, including its access road and quarry site could potentially affect the assets and livelihood of the current occupants of the area, the AE indicated "that the design of the revetment and all proposed construction works is seaward of the houses at West Point, so no physical relocation of communities (housing) is required".

54. The revetment wall may also impact the livelihood of the residents in the area since the revetment and promenade may displace small informal businesses. The main livelihood of the residents in the West Point is also fishing which requires immediate access to gently sloping beach to provide fish landing and space storage of nets and canoes and for other activities and the revetment wall with promenade will constitute a barrier between the settlements and the beach. Moreover, the loss/reduction of the beach area may deprive the community of a vital recreational amenity. To address this, the design provides for two beach fish landing sites, one in the middle of the stretch and another at the southernmost end of the wall. The AE further explained that "early construction of new canoe storage will be required to enable fishers to relocate canoes to allow unencumbered construction traffic movement up and down the shoreline. Close liaison with community and traffic management during construction will be required. To maintain the social functions that the beach currently provides, a total width of 6 m for a promenade along the top of the revetment is accounted for, which is sufficient for recreation and access for pedestrians to enjoy the ocean view."

55. As part of the ESIA, a more detailed assessment of the involuntary resettlement impacts of the revetment wall, access road and quarry site, based on their actual alignments should be made and once detailed engineering designs become available. A resettlement action plan (RAP will also be prepared to address all these impacts including partial and temporary losses to businesses and livelihood should this be identified in the assessment. The potential impacts of project support for regulation of fishing and sand extraction should also be addressed through intensive stakeholder consultations for possible mitigation measures, such as finding of alternative sites and gradual phasing in of the implementation of the policy or regulation, financing support, among others. The ESAR indicates that the RAP will be prepared and implemented should the detailed ESIA indicates this will be required.

56. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** The project clearly aims to conserve the marine and estuarine resources as a means of sustaining the low carbon fisheries. The adaptive interventions will have beneficial impacts to marine ecosystem and estuary most especially the Mesurado Wetland. Among the project interventions are awareness campaigns, capacity building and support for policies and regulations on improved urban waste management, regulations to stop encroachment into the coastal mangroves, fuel harvesting, fishing methods and fish catch regulations for endangered marine fish such as the Cassava Croakers, some shark and ray species, as well as sea turtles and marine mammals such as the sei whale, the blue whale and the Atlantic humpback dolphin. However, some project interventions may have unintended adverse consequences to the marine and estuarine ecosystems should these not be managed well. The revetment wall for instance, will inevitably alter the seawater flow and sediment deposition patterns, freshwater-marine water mixing. These may have long term unintended consequences for the morphology and ecosystem of wetland area. The access road and quarry site may also cause direct damage to

natural habitat if not properly located. Any unintended adverse impacts should be minimized or managed.

57. Direct impacts of the proposed revetment wall to the Mesurado Wetland ecosystem are not expected to be significant, if any. However, impacts on specific areas of the coast, such as the inlet of the Mesurado estuary, could be locally significant if interventions are not specifically designed according to the prevailing dynamic processes. To further gain knowledge of the ecosystem and potential indirect impacts, the ESAR recommends further research under the project as a basis for interventions. Biological surveys must be conducted in the potential disturbance areas prior to the commencement of any construction activities, including in the quarry sites, and areas of high biological density or diversity must be avoided. The design of the project's interventions takes into account the likely coastal process reactions that will result from the construction of the revetment wall. Guidelines for the release of endangered bycatch species must also be agreed with the fishermen. The ESMP, which will be finalised after the ESIA has been conducted in the first year of the project, will include the required habitat monitoring utilising well established scientific methodologies.

58. **GCF Indigenous Peoples Policy and ESS 7:** Indigenous Peoples. Liberia has a multi-ethnic population and in Monrovia where the project is located, the AE has screened that there are no group or groups that are identified as "indigenous" based on the definition of the accredited entity as well as the criteria of GCF. The three (3) fishing communities in the project area trace their origins elsewhere: the Krus are originally from Maryland and Sinoe County, the Fanti from Ghana while the Popo are from Togo. The Togolese Popo fishing group could have been regarded as a minority, but they are integrating rapidly through proximity and intermarriage such that designation as a minority is not useful. Additionally, no group living in the affected communities is socially excluded by factors such as tribal origin, language spoken, historical or cultural influences. Nevertheless, the project should ensure an approach inclusive of all groups. The ESMMM will contain provisions to trigger free, prior informed consent (FPIC) procedures in the event of indigenous peoples being encountered in a pocket of the project-affected areas. Comprehensive culturally appropriate consultations have been carried out during the conduct of the environmental and social assessment (ESA). The consultations shall continue during the conduct of the ESIA and during project implementation in accordance with the SEP. The ESIA should further assess the impacts and benefits of the project to each of these groups. The SEP should clearly identify these various groups and made sure that they will be represented during consultations.

59. **ESS 8: Cultural heritage.** There are a number of historically important sites and landmarks within the general project area such as the landing spot of freed American slaves who resettled to Liberia in the early 1820s; the oldest cotton tree; the abandoned Hotel Africa which hosted the Organization of African Unity conference in 1979; and the Providence Island itself which a major Portuguese trading post. None of these sites however will be affected by the physical activities of the project. The ESAR does not indicate any cultural heritage site present in West Point coastal section where the revetment wall will be constructed. Nevertheless, the ESMMM of the ESAR contains a number of provisions to quickly identify and protect cultural heritage sites. The ESMMM also refers to the Chance Find Procedure for any chance discovery of archaeological artifacts and sites.

60. **Implementation arrangements.** The Environmental Protection Agency will be the Executing Entity (EE) for the project and will establish a Project Management Unit (PMU). The PMU will include specialists in environment, social development, and gender and social inclusion, as well as technical specialists. The PMU will also be responsible for the supervision of the ESMP and will ensure that it is implemented throughout the project cycle. While the service providers and contractors that will be engaged by the project will be responsible for the day to day compliance of the ESMP, a site supervisor will be responsible for the daily environmental

inspections at the construction site. Monitoring of the ESMP implementation will also be conducted.

61. **Stakeholder engagement and grievance redress mechanism.** The ESAR includes a stakeholder engagement plan stating its objectives, contents and key features. It also describes the consultations that have been conducted in developing the project proposal. A more detailed stakeholder engagement plan will be prepared by the Project Management Unit (PMU) after the ESIA has been conducted during the first year of the project. Aside from ensuring meaningful and inclusive consultation, the stakeholder engagement should also ensure that conflict between groups that may arise (e.g. presence of two distinct fishing communities in the area, the Kru and the Panti groups) due to any perceive disproportionate distribution of benefits and costs that may result from the project will be properly addressed.

62. The ESAR also includes a project-level mechanism to register complaints and provide grievance redress at the ground level. The AE's institutional Stakeholder Response Mechanism (SRM) will also be applicable. In line with the GCF indigenous peoples policy, the GCF independent Redress Mechanism and GCF indigenous peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

63. The AE has provided a gender assessment and gender action plan and therefore complies with the requirements of the gender policy of the fund.

64. The gender assessment describes the enabling environment for the pursuit of gender equality in Liberia. Liberia has a relatively strong legal and regulatory framework to support the pursuit of gender equality. It is party to the Convention on the Elimination of all Forms of Discrimination Against Women (CEDAW), the Constitution guarantees all persons, regardless of sex, the enjoyment of fundamental rights and freedom in addition to: the Domestic Relations Law; the Inheritance Act of 1998 (which specifies equal rights in marriage and inheritance under customary and statutory laws); the Rape Law of 2005; and the Anti-Human Trafficking Act of 2005. These laws and treaties are supported by the Liberia Revised National Gender Policy (2018–2022), which promotes gender equality and is supported by the Ministry of Gender, Children and Social Protection for implementation; and the Civil Service Reform Strategy (2008–2011), which is focused on mainstreaming gender and establishing mechanisms for implementing an affirmative action programme, such as the establishment of Gender Focal Points (GFPs) in each Ministry, Agency and Commission Corporation. Liberian employment law prohibits discrimination based on gender and discrimination against women in working conditions and wages.

65. Despite the enabling environment, the assessment indicates that though the Constitution grants equal ownership rights to land and non-land assets to both men and women, Liberia has a dual land tenure system where customary tenure systems generally prevail. Under customary law, women can only access land through their husbands and are unable to inherit land. However, while equal opportunity for all is granted to citizens to engage in governance of the state, several efforts to pass a bill mandating legislative quotas have not yet borne fruit. Women and men have equal rights to access financial services, including credit and bank loans. However, in practice, women often have difficulty accessing credit due to low literacy rates, and/or because they cannot meet the requirements needed to take out a loan. Gender inequality in Liberia varies according to age, status, means of income, county, rural/urban area and traditions, but in general, access to education, healthcare, land, property and justice is more limited for women. These inequalities continue to pose an excessive burden of care on women, maintain unequal power relations in the household and economic disempowerment for women. In addition, women experience lack of access to basic services, rights, opportunities, legal and judicial services, and are subject to low participation in decision-

making in public spheres and institutions. The situation for adolescent girls is critical, with high school dropout due to poverty, teenage pregnancies and sociocultural norms. Rates of sexual and gender-based violence, harmful practices, female genital mutilation, child marriage and teenage pregnancy are all high, while access to sexual and reproductive health rights is low. Members of the LGBTIQ community have limited access to basic rights, services, social freedom, social justice, social economic empowerment and access to health services, and suffer from denial of their fundamental rights to expression, equality and association.

66. The gender action plan provided by the AE fulfils the requirement of the gender policy. It is in line with the findings of the gender assessment and includes baseline, indicators, targets, timelines and budgets. The gender action plan proposed by the AE considers some of the key barriers such as limited access to credit, raw materials, training, marketing, etc.; absence of an enabling environment to contribute effectively to the economy; limited access to productive resources; limited access to decision-making positions in public service; limited participation in more profitable sectors; limited opportunities for economic empowerment; and discriminatory customary practices, stereotypes, norms and biases that continue to prevent equal access to and control over land and land use.

67. The AE will hire a full-time gender expert to mainstream gender throughout the project, and gender expert consultants for specific activities where necessary. At least 20 per cent of project management unit staff will be women involved in effective management and implementation. The Project Steering Committee and other management bodies will be comprised of at least 30 per cent women. The project will put in place mechanisms to ensure women have easy access to report grievances and incidents of gender-based violence. Women are targeted as specific beneficiaries of project interventions and activities which challenge unequal gender relations, gender disparities, inequalities and promote women's empowerment. Activities are designed to contribute to changing norms and behaviour through a number of areas including: gender-responsive training; ensuring equitable access to resources, services and technologies for climate-smart livelihoods; and improving coordination and institutional capacity-building related to gender-transformative climate change adaptation. More specifically, the project will engage women in the design of the coastal defence structures and provide training and employment opportunities related to the construction process. Measures to support equal access to employment, such as provision of childcare, will be implemented. The project will ensure equal participation of women in the development of both an integrated Coastal Management Plan and the Training of Trainer's curriculum which will include social and gender dimensions targeting relevant government ministries involved in Integrated Coastal Zone Management. Activities are designed to ensure women are trained in monitoring of spatial and biophysical coastal information, and that gendered messaging is included in awareness-raising regarding climate change and coastal zone management. The project also takes measures to ensure that women have equal access to a community education and innovation centre that functions as a community knowledge generation and learning hub. It is intended to develop training and to provide capital for women to establish small-scale manufacturing of cookstoves and other livelihood opportunities in the target communities. Lastly, the project will purchase and install cold storage facilities near fish processing sites to support fishmongers, of which 60 per cent will be women.

68. Overall the activities are expected to address challenges of access to services, provide income opportunities, skills and training opportunities that will build the resilience of women to cope with climate change challenges. It will also support the nurturing of an enabling environment that will address sociocultural and gender-based violence issues.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

69. GCF is requested to provide a grant of USD 17 million for climate change adaption in the coastal zone of Monrovia. The project will address its adaptation needs by constructing a rock revetment and diversifying livelihoods. The Government of Liberia is providing USD 2.5 million and USD4.2 million in a form of grant and in-kind, respectively. The AE is providing a grant of USD 1.5 million.

70. Liberia is an LDC. The project expects to have revenue from the manufacturing of cookstoves and collecting fees from the usage of cold storage facilities. However, the AE supports a full grant request as all income will be used to support the sustainability of the project, which also covers the incremental cost of the adaptation measure. The country has economic constraints and limited capacity to borrow from the capital market.

4.3.2. Accredited entity/executing entity (AE/EE) capability to execute the current programme (low risk)

71. UNDP is the AE for the project and has extensive track record of implementing climate change projects. Over the last five years UNDP has implemented a coastal adaptation project, a climate change adaptation project in the agricultural sector, an early warning system project and an extractive industry sustainable development project in Liberia.

72. The Environmental Protection Agency of Liberia (EPA) is the EE for the project. The Acting Executive Director of the EE is appointed as a national designate authority (NDA) to GCF in Liberia. The EE has managed projects with budgets ranging from USD 3 to 7 million financed by external donors. The AE has conducted the capacity assessment of the EE, which resulted in moderate risk.

73. To minimize the risk of corruption, the AE will use the direct cash transfer method where necessary. A majority of GCF resources (87 per cent, USD 13million) will be used for the construction of a rock revetment under Activity 1.2. The AE states that the funds for selected activities, as requested by the Government of Liberia, including Activities 1.1, 1.2, 2.3 and 3.5 due to higher value and high-risk nature, will be channelled directly by UNDP to service providers. Any other funding will be channelled through the EE.

4.3.3. Project-specific risks (medium risk)

74. Project design: To achieve climate impact the project needs to accomplish satisfactory construction and operations of the activities. The quality of construction and operations is influenced by project design, and adherence to international sound engineering practices and to the underlying substantial project feasibility studies. Based on the documents submitted by the AE, the Secretariat has identified certain gaps and deviations in the project design. To mitigate these concerns, the term sheet provides that AE will submit by no later than four months before the anticipated start of any structural work and before seeking any disbursement for such structural work, the following studies to be conducted by engineer(s) with valid certificate(s)/licence(s):

(a) a project-level site-specific hydro-engineering study of the West Point of Monrovia ensuring that no localized or broader flood will be induced by the structural works under this Project, followed by a detailed design of the coastal defence structure incorporating flood risk by designing sufficient drainage capacity to facilitate removal of localized and broader flooding in the project area.

75. **Oversight responsibility of the community stewardship committee (CSC)**: The CSC will manage income from the sale of cookstoves and fees from the usage of cold storage facilities. The CSC members will receive monthly stipends. The AE estimates that the CSC will manage ~USD 20,000 per year which will be used to purchase of materials for the cookstoves and operation and maintenance (O&M) for the cold storage. The financial statements will be

reported to the project management unit (PMU) and the EE is responsible for the oversight of CSC operations. However, in case the expected income is lower and insufficient to pay CSC members' stipends, the project may need some additional financial resources. This amount is expected to be small at about USD 6,000 p.a. and it is advised that the AE keeps some funding aside for such a contingency.

76. **Use of diesel back-up generators**: The cold storage facilities under subactivity 3.5.1 will be solar-powered and may have back-up diesel generators where necessary or battery systems. This activity is entirely financed by the AE's co-financing. The grid connection, battery solutions, O&M plan for solar power, and tariff are not clearly demonstrated in the funding proposal. It is recommended that the AE ensures that the project design pertaining to the composition of solar photovoltaic, battery storage and diesel back-up generators is optimum to minimise the negative climate impact of operating the diesel back-up generators.

77. **O&M plan**: The project estimates the O&M cost of ~USD 50,000 per year following project implementation for a period of six years. The ICZMP process is intended to address institutional responsibilities and budgets for O&M and will be endorsed by the Government as part of its policy processes. Co-financing from the Government does not include the O&M costs. Given that the responsibility for and financing of O&M is not yet decided, the Government's commitment to endorse the O&M plan and allocate required resources is critical.

78. **Stakeholder consent for coastal and land use**: The AE identified the governance risk related to coastal and land use as high risk. The project requires a commitment to coastal and land-use planning polices and actions that are aligned with other actions by the ministry and safeguard communities. The proposal states that the Government's commitment to address land tenure agreements during project implementation has been agreed as a condition of the project. The project would benefit from the confirmation of stakeholder consent for the use of land for the project before GCF disbursement in the term sheet.

79. **Economic analysis**: An economic analysis has been carried out and resulted in an economic internal rate of return (EIRR) of 22 per cent over 50 years. The benefits considered the avoided damages and the increase in income as a result of the project investment. The sensitivity analysis tested three scenarios (i) total cost increase by 15 per cent; (ii) total benefits decreased by 15 per cent; and (iii) total cost increased by 15 per cent and simultaneously total benefits decreased by 15 per cent. The test still shows the EIRRs ranging between 16 and 19 per cent which is higher than the 10 per cent discount rate used.

4.3.4. GCF portfolio concentration risk (low risk)

80. In the case of approval, the impact of this proposal on GCF portfolio concentration, in terms of result area and single proposal, is not material.

4.3.5. Compliance risk rating (medium)

81. As the AE, UNDP confirmed that none of the project activities will be undertaken in any jurisdiction which is subject to or affected by United Nations Security Council (UNSC) resolutions. Furthermore, no individual or entity that is listed on any UNSC sanctions list, including the UN Consolidated Sanctions list, will be involved in any manner with the project or its activities, either as a counterparty, implementer, or beneficiary.

82. UNDP systematically screens all entities being considered as UNDP contractors or partners (including donors) against the Consolidated UN Security Council Sanctions List. This is built into the capacity screening tools for partners, as well as the Standard Operating Procedures (SOPs) for Vendor Management.

83. UNDP also secures contractual representations from its national implementing partners that address risks of funds received for implementation of projects being used to support

individuals or entities associated with terrorism, and inclusion on the Consolidated UN Sanctions Lists of recipients of any amounts provided by UNDP.

84. UNDP confirmed that it has risk-based due diligence requirements for partners and private entities that it intends to engage with. When developing a new project, UNDP ensures that risks are fully identified and considered in the project design and processes, and that adequate and effective measures to prevent, monitor and address such risks are put in place.

85. UNDP advised that a Partner Capacity Assessment (PCA) was undertaken using UNDP's PCA Tool (PCAT) to assess the risk levels associated with the EE overall, and for construction activities, which is the largest cost item for this project. The overall risk level assessed for this project is "low".

86. The result of the assessment recommended UNDP to continue working with the Environmental Protection Agency (EPA) as an implementing partner (UNDP terminology equivalent to GCF executing entity) for the project. The PCAT also has a construction assessment component for projects involving construction. The result of the construction assessment in the PCAT rated EPA "low" risk for construction. This is complemented by the PCAT mechanisms put in place in project design to keep the risks low. These provisions are that EPA will receive technical support from qualified personnel and consultants to be hired by the project, and from qualified government ministries (Ministry of Mines and Energy and the Ministry of Public Works) to ensure construction follows best standards.

87. The PCAT is used in conjunction with the Harmonized Approached to Cash Transfer (HACT) Framework. UNDP's HACT provides a common operational (harmonized) framework for transferring cash to government and non-governmental partners to support a closer alignment of development aid with national priorities and to strengthen national capacities for management and accountability. The HACT Framework comprises the micro assessment, national implementation (NIM) audit, spot checks, and programmatic visits. The micro assessment and spot checks are undertaken by an independent third party.

88. While the micro assessment is for every UNDP Programme Cycle (every four years), project-specific assessment and checks are more frequent (regular spot checks are carried out at least once a year, while specific project audits will be done annually in the case of this GCF project). UNDP has undertaken a HACT micro assessment of the EPA to assess the risks associated with the entity as an implementing partner/executing entity in Liberia.

89. The micro assessment provides an overall assessment of the implementing partner's programme, financial and operations management policies, procedures, systems, and internal controls. It includes:

(a) Review of the implementing partner's legal status, governance structures, and financial viability;

(b) Programme management, organizational structure and staffing, accounting policies and procedures, fixed assets and inventory, financial reporting and monitoring, and procurement; and

(c) Compliance with policies, procedures, regulations and institutional arrangements that are issued both by the Government and the implementing partner.

90. The risk areas covered by the assessment include programme management, organizational structure and staffing, accounting policies and procedures, fixed assets and inventory, financial reporting and monitoring, information systems, and procurement. The overall risk rating is "moderate" across all the risk factors.

91. In addition to the HACT micro assessment, and as part of this project, the UNDP Country Office will undertake the following actions to detect and address any issue related to risks identified in the project:

(a) Spot checks will be carried out once every year by an external independent party and by UNDP;

(b) NIM audits – project audits will be undertaken annually as budgeted for under this project; and

(c) Review of annual workplans to ensure that the planned expenditures are in line with the budgeted project activities.

92. UNDP advised that before transfers are made to the EE, UNDP verifies previous expenditure and activity reports to ensure all previous payments have been used for their intended purposes. Should any incidents be detected during any one of the assessments and audits mentioned above, UNDP will adjust the cash transfer modality to safeguard project funds and address any vulnerabilities related to money laundering, terrorist financing, or prohibited practices among the activities, counterparties, executing entities, or beneficiaries. This will include changing modalities from cash transfers to the EE to direct payments of service providers.

93. UNDP confirmed that the project will not directly or indirectly distribute cash or commodities to beneficiaries, but will rather provide them with services according to the details provided in the description of project activities.

94. The Ministry of Mines and Energy will be the responsible party for the construction and ICZM components of the project, but will not itself receive cash transfers from UNDP for project purposes. The project is also designed such that UNDP makes direct payments, and supervises procurements of large construction and construction design activities.

95. UNDP informed that annual audits, spot-checks and project governance structures will be used to prevent, mitigate, identify, report and remedy any risks, indications and allegations of money laundering, terrorist financing, sanctions violations or prohibited practices. UNDP's procurement and project oversight functions at country, regional and HQ levels will also be used for this purpose.

96. UNDP advised that the project will be subject to UNDP's policies and operational procedures that are contained in its Internal Control Framework and other documents, and will apply them to the project and counterparties. The framework covers assets as well as financial resources to ensure that they are used for their intended purposes. It also defines roles and responsibilities of different individuals, ensuring segregation of duties to avoid conflict of interests, as well as accountability.

97. UNDP advised that the project, through the social and environmental safeguards framework, has provided for a grievance redress mechanism that can be used by all stakeholders.

4.3.6. Recommendation

98. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment		Rationale
Overall programme	Medium	A majority of GCF resources (87%) will be invested in the construction of rock revetment. The AE/EE's coordination to procure service providers to design, implement, operate and maintain the revetment as per sound engineering practices is crucial for the project.
Accredited entity/executing entity capability to implement this programme	Low	
Project-specific execution	Medium	
GCF portfolio concentration	Low	
Compliance	Medium	

4.4 Fiduciary

99. The United Nations Development Programme (UNDP) will serve as the accredited entity (AE) of the project while the national executing entity (EE) for this project is the Environmental Protection Agency of Liberia (EPA).

100. The EE is accountable to UNDP for managing the project, including the monitoring and evaluation of project interventions, achieving project outcomes, and for the effective use of UNDP resources. The Ministry of Mines and Energy (MME), Ministry of Finance and Development Planning (MFDP) and the Ministry of Public Works (MPW) have entered into agreements with the EE to assist in successfully delivering project outcomes and are directly accountable to the EPA as outlined in the terms of their agreement. The EE will delegate to the MFDP and MPW as Responsible Parties – government entities with the technical competence in their respective areas – to ensure that the activity is implemented according to national requirements. A memorandum of understanding (MOU) will be established between the EE and the MFDP and MPW outlining their respective responsibilities under Activity 1.2. No funds will be channelled through the MME and MPW.

101. The funds for selected activities, as requested by the Government of Liberia, including Activities 1.1 and 1.2, 2.3 and 3.5 (due to their higher value and high-risk nature) will be channelled directly to service providers by UNDP as part of UNDP support services to the Government. Any other funding requests will be channelled through the EPA.

102. The Project Management Unit (PMU) will implement the project with support from the EPA and UNDP and will report to the Project Board. The Project Manager will be appointed to lead the PMU which will run the project on a day-to-day basis on behalf of the EPA within the constraints laid down by the Project Board. The Project Manager function will end when the final project terminal evaluation report and other project closure activities and documentation required by GCF and UNDP have been completed and submitted to UNDP.

103. Donor funding, including GCF funding, will be received by the administrative agent (UNDP) on behalf of the EE (Environmental Protection Agency) based on a standard letter of agreement signed between the donors and UNDP and an MOU between UNDP and the EE separately. As per UNDP's National Implementation policy (NIM), the Project Management Unit's capacity assessment determined that funding may be advanced to entities for lower risk activities. UNDP will support the handling and channelling of funds for higher risk activities as a support service provisioned for in the letter of agreement. For the avoidance of doubt, GCF proceeds and the AE's co-financing will not flow through the EE. Instead, UNDP (as administrative agent of the EE) will channel such funding via a direct payments option whereby transfer will be made directly to goods and services providers hired by the AE and upon request of the EE.

104. The financial management and procurement of this project will be guided by UNDP financial rules and regulations. Monitoring, reporting and evaluation arrangements will comply with the relevant GCF policies and accreditation master agreement signed between GCF and UNDP. UNDP will ascertain the national capacities of the EEs by undertaking an evaluation of capacity following the Framework for Cash Transfers to Executing Entities (part of the Harmonized Approach to Cash Transfers/HACT). All projects will be audited following the UNDP financial rules and regulations noted above and applicable audit guidelines and policies.

4.5 Results monitoring and reporting

105. The funding proposal aims to achieve climate change adaptation in the coastal zone of Monrovia, Liberia through coastal protection, effective management and resilient livelihoods.

The project is expected to benefit 250,000 direct beneficiaries – 122,500 female and 127,500 male, and one million indirect beneficiaries – 490,000 female and 510,000 male – in Liberia.

106. Overall, the funding proposal and log frame are assessed to have adequately applied the elements of the GCF results management framework and performance measurement framework. The revised log-frame meets GCF requirements on monitoring and reporting the anticipated results of the intervention. Pertinent information on MoVs, baseline and targets have been provided. Similarly, the log-frame contains clear presentation of project-level results, their indicators and other required information in concise manner. Their diligent application during implementation would facilitate results monitoring, progress reporting, performance assessment and finally their delivery on the ground.

107. However, the recent feedback provided needs to be addressed to improve the overall quality of the final log-frame.

108. Implementation timetable: the implementation timetable has been provided in the required structure and format and would enable progress assessment during the implementation period. The milestones and deliverables are well articulated and reporting timelines are provided as required.

4.6 Legal assessment

109. [The Term Sheet for the proposed project has not yet been reviewed nor agreed. In accordance with Clause 6.01 of the accreditation master agreement, all funding proposals submitted to the Board for consideration shall be accompanied by a term sheet agreed by the accredited entity and GCF, setting out, in summary form, the key terms and conditions relating to the proposed funded activity. This is a preliminary legal assessment subject to revision based on the review and negotiation of the term sheet with the accredited entity.]

110. The Accreditation Master Agreement (the “AMA”), was signed with the accredited entity on 5 August 2016 and it became effective on 23 November 2016.

111. The accredited entity has provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

112. The proposed project will be implemented in the Republic of Liberia, a country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The Secretariat and the Republic of Liberia have not yet engaged in negotiation of an agreement on privileges and immunities.

113. The Heads of the Independent Redress Mechanism (IRM) and Independent Integrity Unit (IIU) have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF are made only after GCF has obtained satisfactory protection against litigation and expropriation in the country or has been provided with appropriate privileges and immunities.

114. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval;

(b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP160

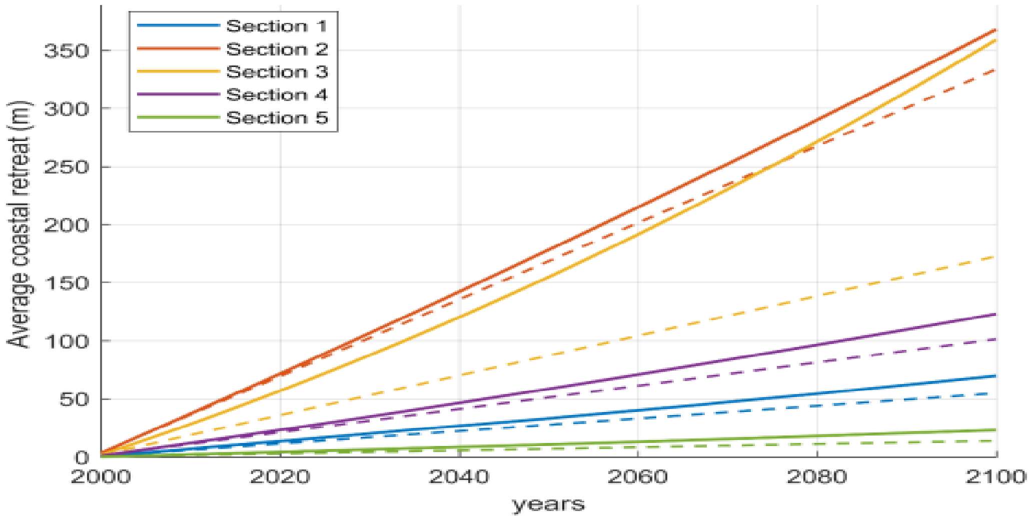
Proposal name:	Monrovia Metropolitan Climate Resilience Project
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Liberia
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential *Scale: Medium to high*

1. Liberia's capital, Monrovia, is extremely vulnerable to the climate change impact of sea-level rise (SLR), which is combining with changes in nearshore wave conditions and sediment flux in the Mesurado Wetland Basin to accelerate a process of gradual coastal erosion, with additional rapid coastal erosion during extreme storm events, which have become more frequent with climate change. In the period from 2000 to 2020, significant shoreline retreat has been recorded in five vulnerable hotspots, with the worst damage on the West Point peninsula, whose artisanal fishing communities lost 670 dwellings and significant infrastructure, as the coast retreated by 30 metres. Using a combination of climate and wave models to understand likely future trends, the project proponents indicate anticipated shoreline retreat in these five hotspots, with and without the impacts of key climate change parameters, and show that West Point is likely to experience the most rapid acceleration as a result of climate change.

Figure 4 from Funding Proposal, showing the actual (2000 to 2020) and anticipated (2020 to 2100) averaged trend in annual coastal retreat for five coastal sections in Monrovia *with climate change* (continuous lines) and *without climate change* (dashed lines) under the scenario RCP 8.5. West Point is Section 3 (shown in yellow).



2. To adapt to the severe impacts of climate change anticipated on Monrovia's coast, the project seeks to change the current approach from a focus on short-term solutions to long-term integrated planning that is climate risk-informed and involves the public sector, private sector

and communities, at all levels of governance. The proposed project is requesting GCF support to address barriers to effective climate change adaptation in the coastal zone of Monrovia, and Liberia more generally, through interventions in three interrelated focus areas: *Component 1* will focus on establishing effective and sustainable coastal protection measures for Monrovia's most vulnerable communities at West Point; *Component 2* will strengthen institutional capacity to manage coastal areas of Liberia in a climate-resilient manner and raise community awareness on climate change impacts and adaptation practices through an Integrated Coastal Zone Management (ICZM) approach; and *Component 3* will contribute to protecting mangroves and strengthening livelihood practices in vulnerable communities across the Monrovia Metropolitan Area – to enhance the adaptive capacity of these communities and these natural ecosystems which, kept intact, can contribute to stabilizing the shoreline and act as important fish breeding grounds.

1.1.1. Adaptation impact

3. The impact potential of the project is considered medium to high. The project will address directly one of the most urgent adaptation needs in Monrovia by constructing a rock revetment to stop the West Point shoreline from regressing further. (The revetment was selected as the preferred solution, because while a 'soft solution' in the form of beach nourishment with an associated groyne was considered technically feasible, the sustainability of this option would be limited, with the high level of regular maintenance required not seen as feasible in the local context.) This will significantly reduce vulnerability of beneficiaries: protecting homes, artisanal fishing livelihoods, and infrastructure of approximately 10,800 people in West Point, avoiding damages of up to USD 47 million to individual and communal property, and securing climate-proof launch sites for fishing boats. The project is expected to directly benefit a total of 250,000 people in the target communities of West Point (through coastal defence and enhanced livelihoods); as well as Topoe Village; Plonkor and Fiamah; and Nipay Town and Jacob's Town (through enhanced livelihoods and improved protection of mangrove ecosystems).

4. The construction of the coastal protection infrastructure will form part of a strategic, cohesive coastal adaptation strategy using an Integrated Coastal Zone Management (ICZM) approach, creating pathways to scale for adapting to climate change along the country's whole coastline. The work with communities in the neighbouring Mesurado Basin to protect mangroves and promote fuel-efficient cooking and fish-smoking will help to support the ICZM efforts. The joint GCF-Government of Liberia investment at West Point will serve as a demonstration site of the value of robust and sustainable coastal protection measures, which will be used to leverage future investments in coastal protection infrastructure for the most vulnerable Liberian communities in all the coastal counties over time. In addition, the comprehensive technical assessments used to inform the design of the West Point revetment will provide a foundation for the design of further coastal protection interventions, and help develop local capacity for construction, inspection and maintenance. The independent Technical Advisory Panel (TAP) notes that keeping mangroves healthy and intact can be seen as contributing to an integrated "grey-green" coastal protection system for the city, and that this concept could be utilized within the ICZM approach and explored further during project implementation.

5. A possible limitation on the project's impact stems from the difficulty of achieving effective land use planning and management in Liberia as the basis for climate risk-informed Integrated Coastal Zone Management – both in the urban coastal context of Monrovia with ongoing in-migration and spread of unplanned informal settlements, and in rural coastal contexts with a dearth of state enforcement, for example to prevent illegal sand mining activity (a directly anthropogenic contributor to coastal erosion outside the metropolitan area). In response to a question on this, the accredited entity (AE) highlighted the existing land-use

policies in Liberia and Monrovia, including zoning ordinances, the National Land-Use and Management Framework and the Liberia Land Rights Policy. It is noted that grounding the ICZM Plan in existing land-use planning frameworks and processes will contribute to the existing enabling environment that these policies are creating for land-use planning, but the scale of the challenges remains daunting.

6. The impact potential of the project would be more clearly demonstrated if the climate rationale were more fully articulated in the funding proposal, using more of the material contained in the Feasibility Study and Vulnerability Assessment to convey the complexity of the climate change parameters that have already begun to impact Liberia's coastline, the evidence for these observed trends, and the anticipated future trends based on climate and wave modelling. The simplification of several complex, interrelated factors down to just two – sea-level rise and increased frequency of intense storm events – does not do justice to the complexity of the processes involved, nor to the work done by the project proponents in understanding these. It is also not clear to the independent TAP whether the AE has exhausted all possibilities for seeking historical data on observed trends in order to validate the use of models from which the anticipated rate of coastal retreat with climate change is derived. It is for these reasons that a condition is presented at the end of the document, involving the development of an additional report, strengthening the climate rationale by providing a clear summary of observed and predicted changes along Monrovia's coastline in relation to the following parameters: sea level rise, sediment transport and exchange, wave conditions, and high-intensity storms; also showing how these parameters are being affected by climate and non-climate drivers, and how they interact to produce shoreline regression – both historical and anticipated. The report will also facilitate the potential replication of this important project in other coastal countries in West Africa experiencing similar climate change vulnerabilities.

7. To provide more detail on the issue in paragraph 6 of this assessment above about conveying the complexity of the climate change parameters: The funding proposal identifies the primary climate change parameters as SLR and increased frequency of intense storms, which are linked to the secondary impact and key hazard of coastal retreat. The Feasibility Study and Vulnerability Assessment, however, indicate a more complex situation – with coastal retreat (both under pre-climate change conditions, and with climate change) being a result of both ongoing gradual structural retreat (related to SLR, sediment deficit and "regular" storm erosion), and also periodic rapid retreat as a result of occasional extreme storm events (whose return interval has become shorter, and is predicted to continue to decrease with ongoing climate change). This is explained well in Section 3.3.1 of the Feasibility Study. The three interlinked factors causing the underlying gradual retreat could have been explained more fully in the funding proposal:

(a) A shift in equilibrium profile due to sea-level rise (Bruun Effect);

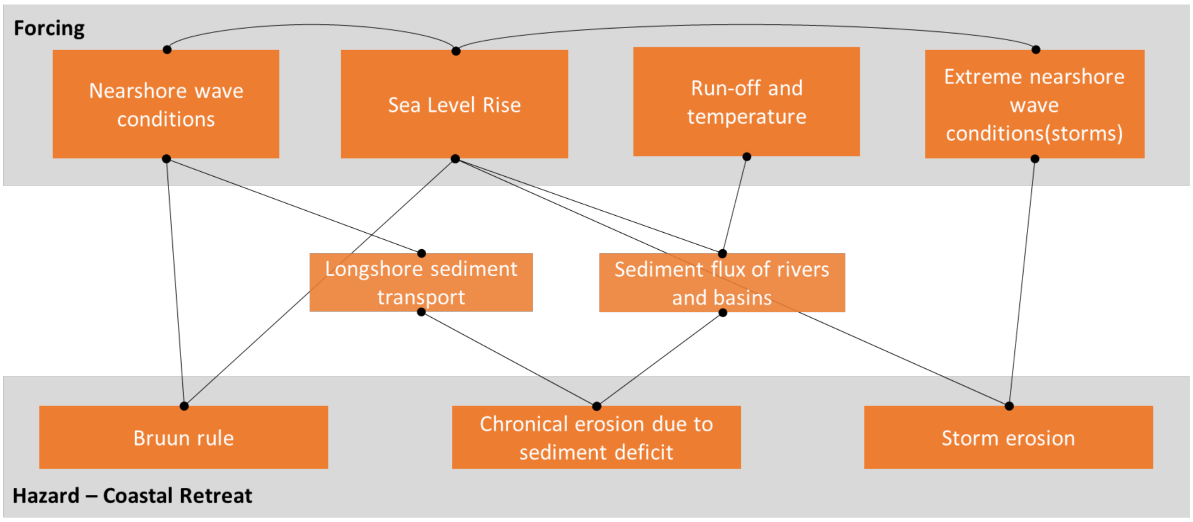
(b) Chronic erosion due to sediment deficit caused by longshore sediment transport and sediment exchange with rivers and estuaries; and

(c) Storm erosion.

8. Paragraph 12 of the funding proposal does indicate that the Feasibility Study undertook modelling of the impact of SLR, waves and sediment transport under future climate conditions. The Vulnerability Assessment uses Global Circulation Models to consider the impact of climate change on wave conditions, run-off and temperature, and portrays a fairly complex picture, as conveyed in the graphic below. The AE has chosen to simplify this complex process in the funding proposal, for example stating in the Executive Summary that "Liberia's capital city, Monrovia, is extremely vulnerable to the climate change impacts of sea-level rise (SLR) and the increasing frequency of high-intensity storms, both of which contribute to coastal erosion and shoreline retreat", and in the section on climate change impacts: "Accelerated coastal erosion, resulting from a combination of sea-level rise and an increase in the frequency of high-intensity

storms" (para 10). It might be more accurate to say that Monrovia is "extremely vulnerable to the climate change impact of sea-level rise, which is combining with changes in nearshore wave conditions, run-off and temperature (all influenced by climate change) to cause shifts in longshore sediment transport patterns along the coast, as well as sediment flux through the two river estuaries and tidal inlet of the Mesurado Wetland Basin. These factors have accelerated the process of gradual coastal erosion, with additional rapid coastal erosion also occurring during extreme storm events, which have become more frequent with climate change."

Figure 10 from the Funding Proposal: The interrelation between forcing mechanisms and coastal retreat



9. In finalizing the independent TAP assessment it has also been observed that the word "forcing" seems to be used in a different sense from the usual sense of referring to processes that alter the global energy balance, for example, in the graphic above, and in the Vulnerability Assessment and the Feasibility Study, e.g. in Section 3.3.2 on "Climate change impacts on forcing mechanisms of coastal erosion" of the Feasibility Study.

10. To provide more detail on the issue in paragraph 6 of this assessment above about using all available historical data, three specific areas where this applies are discussed briefly here: (i) For shoreline regression, it would have been useful to see satellite images, if available, showing the shoreline of West Point in 1988 and 1998 (as well as 2008 and 2018). Data on this must be available at least for a 20year period, since the graph in Figure 4 of the FP (see paragraph 1 above), shows shoreline retreat from 2000. (ii) For sea-level rise, the assessment uses only global models validated with global data, arguing that regional variations are insignificant. Some discussion could have been included on what limited data is available for sea-level rise in Africa, including West Africa – see, for example, Evadzi, P. *et al.* (2019) "West African sea level variability under a changing climate – what can we learn from the observational period?", *Journal of Coastal Conservation*, 23(4). (iii) For validating the wave modelling used, the Vulnerability Assessment states that no historical wave measurements were available. However, a National Oceanic and Atmospheric Administration (NOAA) production hindcast from 2005-2018 was used, in addition to re-analysis data. It could have been made clearer what, if any, actual historical data was used by NOAA in constructing the open source global hindcast data sets, which the AE used to generate data for a point offshore from Monrovia and shown in the Feasibility Study as "Figure 1. Observed impact on wave height last two decades…" particularly since the Feasibility Study states that "Wave height distribution in the area offshore of Monrovia *has increased* [our emphasis] over the past two decades (Figure 1).

11. Figure 4 in the funding proposal is important in conveying the project's impact potential, as discussed in the first paragraph of this assessment. In finalizing the independent TAP assessment, however, it has been observed that Figure 4 has a slightly misleading title and an

error. It refers to “Average coastal retreat between 2000 and 2100” when it could more accurately be captioned: “Actual (2000 to 2020) and anticipated (2020 to 2100) averaged trend in annual coastal retreat”. The West Point lines (Section 3), described as “orange”, should be “yellow”. The narrative accompanying the graph does not explain how the values were arrived at for anticipated retreat in the absence of climate change. It is also noted that the most extreme representative concentration pathway (RCP) 8.5 scenario is used in constructing the graphs associated with the alternative scenario without the project, even though the more moderate RCP 4.5 scenario seems to show similar predictions for accelerated coastal erosion.

12. The funding proposal section on climate change impacts includes a sub-section on precipitation which states that “increased frequency of high-intensity storms will increase flood risk in parts of Monrovia and reduce the ability of fishers to safely operate by making landing sites unsafe and decreasing the number of days where fishers are able to safely operate. The design of the coastal protection measures under Output 1 (Figure 7) include drainage systems to reduce flood risk and protection of boat launching sites for fishers.” We note that, while the rock revetment will be constructed from porous materials so that any water that flows over it can flow back and avoid flooding, as is currently the case with the sandy beach, the presence of the revetment does not seem likely to prevent coastal flooding where topographical features already allow flooding to happen. It is therefore well designed to prevent maladaptation (with extreme storms likely to become more frequent), but the project revetment works are not likely to have a major impact in terms of reducing the hazard of coastal flooding.

1.2 Paradigm shift potential *Scale: Medium*

1.2.1. Innovation

13. The paradigm shift involved in the proposed project is rated as medium overall. The project utilizes an approach to coastal zone planning and management which is innovative in the Liberian context, where land-use planning is not yet well advanced, where cross-sectoral public sector collaboration has not been employed in the coastal zone, where communities’ capacities to participate in coastal governance have not been systematically developed, and where long-term planning for climate risk management has not been widely undertaken. The project enables a crucial shift from short-term ad hoc coastal protection to climate risk-informed planning, involving state and community actors, and embarking on the process of private sector engagement. The project is centred around the use of a known best practice technology to construct coastal protection works to reduce vulnerability of a priority coastal stretch in the capital city. In response to questions from the independent TAP, the project proponents expanded on the issues that should be taken into account in the detailed design phase for the “green promenade” to be constructed along the kilometre-long rock revetment, which has the potential to be highly innovative in the manner in which it caters for the artisanal fishing communities’ socioeconomic needs in the context of evolving climate risks.

1.2.2. Potential for knowledge and learning

14. The project has a strong orientation to knowledge management and learning. This includes activities to: (i) develop ICZM capacity and expertise of technical officials from 10 government institutions, to be trained as trainers for synthesis and analysis of beach profile data, coastal protection feasibility assessments, and maintenance of coastal protection infrastructure; (ii) establish a system for data collection, analysis and management relating to coastal dynamics and ecosystems, helping to increase the evidence base to support ICZM decision-making; (iii) upscale the existing Environmental Knowledge Management System to incorporate information on climate change adaptation and ICZM, with targeted knowledge products developed and disseminated; and (iv) establish an education and innovation centre to

function as a community knowledge generation and learning hub, especially for the manufacturing of energy-efficient technologies. The project will also support sharing of lessons learned and best practice to inform learning and best practice across community, subnational and national levels as well as regional initiatives such as the West Africa Biodiversity and Climate Change WA BiCC programme.

1.2.3. Contribution to the creation of an enabling environment

15. The proposed project will undertake significant work to create an enabling environment in relation to the public sector, though it could do more in relation to the private sector. Component 2 is focused on creating enabling conditions for an integrated approach to coastal management and governance that are necessary to undertake long-term, climate resilient planning in vulnerable coastal areas such as Monrovia. This includes jointly developing a national Integrated Coastal Zone Management Plan and capacitating an ICZM Committee and Cross Sectoral Working Group to implement and update the plan. This will be supported by improving coordination and policy harmonisation among government institutions, and building sectoral and community capacity for the implementation of ICZM at national, county and local levels. The mangrove co-management activities under Component 3 will also be embedded in the strengthened institutional and regulatory framework to create an enabling environment for systematically improving mangrove conservation efforts.

16. The funding proposal states that there will also be a focus on strengthening public-private partnerships for ICZM that will contribute to enabling further private sector involvement and support for coastal adaptation in Liberia in future. The livelihoods support work in Component 3, co-financed by UNDP, takes a small-scale direct approach involving demonstration, rather than a wider market-orientated approach that would utilize project resources to catalyse further private sector investments. The project co-finance from UNDP and Government establishes an education and innovation centre, run by a project-trained Community Stewardship Committee (CSC) which will construct a small-scale manufacturing workshop and build 640 fuel-efficient cookstoves, providing two training workshops per year – one to community members on manufacturing and selling cookstoves, and one to women's groups on business strategies and connecting to partners. Following a query from the independent TAP, the design was adjusted to also include a demonstration of a larger scale fuel-efficient fish-smoking oven (used by women's cooperatives in other West African contexts, with potential to be used by emerging women's Collaborative Associations in Liberia https://ejfoundation.org/news-media/women-are-the-strongest-pillar). Similarly, project co-finance will be used to construct two solar-powered cold storage units to extend the shelf life of fish for sale, with an exit strategy that sees the CSC continuing to run the facilities post-project by charging a small user fee. The addition by the project proponents of some detail on what will be achieved through the annual workshops seems rather ambitious, as one-off events can surely have only a limited effect on business skills development, access to finance, and access to higher value markets for fishmongers.

17. The emphasis in Component 3 on direct use of project funds to set up demonstration operations is a valid approach, but does not appear to have been based on full market and value chain analyses that get to the bottom of the barriers to dissemination of these technologies. More thought could have been given to how the project could catalyse impact beyond a one-off investment – contributing to market transformation at scale in the adaptation context. In response to comments from the independent TAP, the project proponents have decided to maximize the potential of Activity 3.3.3. "*Undertake periodic assessments of climate resilient livelihood practices and new opportunities to inform the design of new options and determine their uptake as a result of the project*" to at least investigate other technologies relevant in the context of adaptation to climate change. These include technologies related to non-climate drivers of coastal ecosystem damage, such as stimulating the Compressed Stabilised Earth Blocks (CSEB)

industry as an alternative to beach and river sand mining; and the agricultural residue fuel-pellet industry, promoting a circular economy while also reducing pressure on mangroves and forests.

1.2.4. Contribution to the regulatory framework and policies

18. The project will enable the development of ICZM policy in Liberia and assist the country to meet its legislative requirement of developing an ICZM Plan and updating it every three years, through the development of two iterations of the Plan during the project lifespan, as well as efforts to achieve the harmonisation of policies and regulations related to ICZM across government institutions. In response to a query from the independent TAP about the difficulty of achieving effective land use planning and management in the Liberian context, where much development is unplanned in practice, the AE responded by highlighting that the ICZM work will be based on and support the implementation of the Land-Use and Management Framework developed by the Liberian Land Authority, as well as existing zoning ordinances in Monrovia and elsewhere. In relation to Component 3, they explained that "*embedding participatory ecosystem management within the broader institutional framework will ensure ongoing support for mangrove conservation and management beyond the project period through the established institutional and regulatory systems. Conversely, incorporating lessons from community-based implementation of ICZM into the planning framework will increase the effectiveness and applicability of the ICZMP going forward.*"

1.2.5. Scalability and replicability

19. The project has potential to leverage additional investment in coastal protection works for adaptation priorities during the project lifetime, enabling replication at other vulnerability hotspots along the coastline of Monrovia and Liberia as a whole. Information generated through development of the high-resolution vulnerability map, as well as lessons learned through implementation of the ICZM Plan, will be shared widely and used to scale up project interventions across all the coastal counties. In Monrovia, in terms of the other two vulnerability hotspots where engineering studies were conducted for the project, New Kru Town is already seeing additional investment, but the Atlantic Shore–US Embassy coastal stretch, where there is high exposure of valuable infrastructure, would be a top priority for replication of approaches and lessons learnt through the proposed GCF project. The Atlantic Shore–JFK Hospital stretch and the Hotel Africa stretch are also important for the city's adaptation strategy.

20. In response to a question on this from the independent TAP, the project proponents decided to include in Component 2 an additional deliverable of a "*financing plan to leverage investment in coastal protection*", along with measures to monitor the effectiveness of the plan. The independent TAP sees this as a good start, but notes that additional efforts will be needed by government and its technical and financial partners to proactively seek new partnerships and unlock investments in specific coastal protection projects. It is appreciated that a reference has been added to the funding proposal to say that collaboration between the Government and development partners initiated during project development "*will be consolidated, including between UNDP, CI* [Conservation International], *the World Bank and JICA* [Japan International Cooperation Agency]*. These partnerships and experience will provide a robust platform for the implementation of other projects in the coastal zone."* The funding proposal notes that, "*Extensive engagement with the private sector during the development and implementation of the ICZMP will strengthen public-private partnerships for coastal management, increasing support for future coastal protection and adaptation initiatives*", though potential private sector partners have not yet been identified.

1.3 Sustainable development potential *Scale: Medium to high*

1.3.1. Environmental co-benefits

21. The project's sustainable development potential overall is rated as medium to high. In terms of the environment, the project is expected to have some benefits in contributing to the protection of ecosystems and biodiversity through Component 3. This involves a small reduction of pressure on Monrovia's mangrove ecosystems, as a fuelwood source, by promoting the manufacture and sale of energy-efficient livelihood products; and making a modest contribution to conserving the mangroves by engaging communities in an effective co-management system, thus supporting mangrove functioning in: (i) stabilising sediments; (ii) providing breeding habitats for a range of marine and terrestrial species; (iii) improving and maintaining water quality; and (iv) acting as carbon sinks. The independent TAP notes that the mangrove co-management agreements will not include explicit enforcement of legislation to prevent damage to mangroves, nor will it include negotiated benefit packages to incentivise protection of mangroves, as with the successful agreements concluded through Conservation International in other parts of Liberia. In response to a question from the independent TAP on this, it is noted that the AE describes the goals of this work as deliberately limited in ambition: "*to do the groundwork of engagement with communities, to support the development of more complex conservation frameworks in future*".

1.3.2. Social co-benefits

22. The project interventions to enhance coastal defence infrastructure (Component 1) will protect coastal communities, reduce mortality and reduce damage to the dense concentrations of economic assets and critical social infrastructure in West Point – including homes, schools, clinics and roads, as well as livelihood assets such as boats and fish processing equipment. The coastal defence infrastructure will also protect hundreds of households from being displaced, thereby avoiding significant social disruption that would occur without the project. The adaptation investment will also allow the Kru and Fanti communities to maintain their way of life centred around the artisanal fishing trade. At the independent TAP's suggestion, the project proponents have made sure that the detailed design of the promenade along the revetment will include provisions for amenities such as benches, fireplaces, sheds and huts, to accommodate a number of social functions previously utilizing the beach.

1.3.3. Economic co-benefits

23. Through the coastal protection defence interventions to be implemented at West Point, the project will reduce replacement costs of economic and social assets that would otherwise be damaged by erosion and storm surges. In addition to this direct benefit, several economic co-benefits are expected through the implementation of ICZM (Component 2) and the promotion of climate-resilient livelihoods (Component 3). Improving management of mangroves in the Mesurado Basin will benefit the fishing sector economically, since these ecosystems act as critical breeding habitats for economically important fish species. Economic benefits will also be generated in the form of short- and long-term work opportunities for the construction, operation and maintenance of the rock revetment structure established by the project, as well as through the Community Stewardship Committee (CSC). Installing cold storage facilities in fishing communities will reduce spoilage of fish, thereby increasing the earning potential of fishery-based livelihoods without increasing demand on fish stocks. Furthermore, establishing facilities for the manufacture and sale of energy-efficient livelihood products will provide income-generation opportunities for community members.

1.3.4. Gender-sensitive development impact

24. Clear efforts have been made to promote gender-responsive livelihood development in the project design, both by providing direct opportunities for women in waged labour – as part

of the construction of the West Point rock revetment – and by supporting livelihood enhancements for female fishmongers and other women. In response to a suggestion from the independent TAP, to enhance not only women's access to fuel-efficient stoves (for household stove-top cooking with pots), but also to new fuel-efficient enclosed ovens for smoking fish (to replace open fires, reduce the fuelwood collection burden, and take pressure off mangroves and forests) the project proponents added to Activity 3.3 an assessment of the possibility of introducing larger scale fuel-efficient kilns (such as the Food and Agriculture Organization's Thiaroye Processing Technique smoking kilns) as part of the alternative livelihood assessment. Such technology could also be introduced as a pilot technology for a pilot women's cooperative structure supported through the education and innovation centre, if the AE and Government wished to further enhance the potential in this area.

25. In response to questions from the independent TAP and the Secretariat about whether it is realistic to have a target of 40 per cent female participation in the construction phase of the revetment, the AE responded to the Panel's satisfaction, noting that "*The quota of 40% women in the wage labour contracted under Activity 1.2 was recommended as part of the Gender Action Plan based on the Gender Assessment (Annex VIII), and is not expected to present a challenge. While there are few women in Liberia in some roles in the construction industry (for example, civil engineers), contracting women as waged labourers does not present the same challenges. Similar projects, like the construction of the revetment at New Kru Town, have successfully employed a quota of women as waged labourers. It is important to the gender-responsive approach of the project to provide equal opportunities to women under this activity as it is a direct income-earning and skills development opportunity which will enhance their ability to work on construction in future. Examples of non-engineering waged labour activities that women can participate in — as stated in the Gender Action Plan — include supervision, security, unskilled labor and catering. In addition, this activity presents an opportuninty to address gender norms that prevent women from engaging in manual labour, and similar occupations.*"

1.4 Needs of the recipient *Scale: High*

1.4.1. Vulnerability of the country, vulnerable groups and gender aspects

26. The needs of the recipient for the proposed project are rated as high. Liberia's population is extremely vulnerable to the impacts of climate change, ranking 4th in the 2017 Climate Change Vulnerability Index. This vulnerability is largely linked to the country's exposure to sea-level rise and extreme weather events along its 560 km coastline. The predicted impact of climate change on shoreline retreat is exacerbated by the high population density and associated concentrations of economic and social assets in the country's coastal zone, as well as the dependence of the population on climate-sensitive livelihoods. Liberia ranks 8th in the world for the percentage of the population living in low elevation coastal zones, with the majority concentrated in the capital city of Monrovia. The target site for costal protection works, West Point, is located on a long, narrow sand spit on the western fringe of the Mesurado Wetland, making it one of the most vulnerable communities to shoreline erosion, which advanced by 30 metres in the past decade, and is projected to accelerate, placing approximately 10,800 people's homes, livelihoods and infrastructure directly at risk over time. This includes 49 per cent of the population which is female, including many women fishmongers whose livelihoods depend on smoking and selling the fish caught along West Point.

1.4.2. Economic and social development

27. The growth of Monrovia's population at 3 per cent per year, coupled with the ongoing shoreline loss, is leading to an increase in the density of the population of West Point and the city more widely, putting considerable pressure on the environment and ecosystems on which

many livelihoods depend. As shown in the funding proposal, the fisheries sector contributes approximately 10 per cent of the national GDP and employs at least 37,000 people in Monrovia alone. The vulnerability of these livelihoods is being exacerbated by the impacts of SLR and the increasing frequency of intense storms. The low-income areas along the coast of Monrovia, including West Point, are home to the traditional artisanal fisherfolk of the Kru and Fanti communities, who derive livelihoods from coastal waters and fish stocks, and 60 per cent of persons active in the fishing sector are women. The mangroves in the Mesurado Wetland serve as critical habitat and nursery grounds for these fish stocks, and nutrient-rich fish serve as the major protein source for much of the Monrovia Metropolitan Area population. The proposed project addresses these vulnerabilities by: building a coastal revetment on 1,050 metres of coastline at West Point – ensuring landings for fishing boats, improving the management of Liberia's coastal zone by developing capacity for ICZM, and supporting the development of gender-sensitive and climate-resilient livelihoods in Monrovia.

1.4.3. Absence of alternative sources of financing

28. Since 1990, Liberia has been classified as a least developed country (LDC) with a 2018 GDP per capita of USD 674 – significantly lower than the LDC average GDP per capita of USD 1,042, according to the funding proposal. In addition, Liberia's Human Development Index (HDI) of 0.465 ranks the country 176th out of 189 countries assessed by the UNDP Human Development Report 2018. Economic and social gains made by Liberia in the post-civil war period from 2000–2010 were severely curtailed by the outbreak of the Ebola virus and subsequent epidemic between 2014 and 2015. According to the United Nations Secretariat (UN/DESA), in 2018 Liberia was more economically vulnerable than the average of all LDCs in terms of the Economic Vulnerability Index; had a Gross National Income index equivalent to only 30 per cent of the LDC average; and had a lower Human Assets Index than the average figure for LDCs. The country is disproportionately reliant on the agriculture, forestry and fishing sectors, and reliance on the export of primary commodities, with little to no value-addition, results in the economy being extremely vulnerable to global commodity price fluctuations and market shocks. This economic vulnerability in turn compromises the ability of the Government of Liberia to deliver basic services such as water, housing, transport and energy to its citizens in a sustainable manner. The necessary emphasis by the Government on basic development needs – including poverty reduction and health – means that it has been difficult to prioritize the funding of climate change adaptation interventions to protect vulnerable communities, despite the urgent need for such interventions. Consequently, the Government does not have the financial capacity to adequately fund adaptation initiatives at the scale required to significantly reduce climate change impacts on vulnerable communities.

1.4.4. Need for strengthening institutions and implementation capacity

29. The Government of Liberia has established several laws, policies, strategies and institutions to address the increasing impact of climate change on the lives and livelihoods of people in Liberia, but limited technical and financial capacity within most government departments is a significant barrier to the implementation of climate change adaptation, and climate risk-informed ICZM in particular. Where adaptation measures have been instituted, financial limitations often constrain their implementation due to: insufficient work facilities; lack of access to technical equipment; lack of training to use and maintain technical equipment; and no data collection, analysis and management facilities. Furthermore, there is limited technical capacity within government institutions and NGOs at both the national and local levels to effectively use and enhance knowledge management systems as well as to disseminate information products to coastal communities for reducing their vulnerability to climate change. The Environmental Protection Agency (EPA) does, however, have a track record of implementing projects with a focus on climate change in Liberia, advised by sector-specific Technical Committees, including those on climate change as well as marine and coastal

ecosystems, and is in a good position to utilize the GCF investment to strengthen institutional capacity.

1.5 Country ownership *Scale: High*

30. Country ownership of the proposed project is rated as high overall. The project has strong alignment with Liberia's climate change strategies and adaptation priorities, as well as broader national development plans. These include: the National Policy and Response Strategy on Climate Change (2018), the Pro-poor Agenda for Prosperity and Development (PAPD, 2018–2023), and the Republic of Liberia Agenda for Transformation: Steps Towards Liberia Rising 2030. Through promoting climate resilient planning and protecting climate-sensitive livelihoods, the project will contribute to the achievement of the objectives of the Liberia National Adaptation Programme of Action (NAPA, 2006), the National Strategic Action Plan for Climate Change and Disaster Risk Management (NSAP), and the Climate Change Gender Action Plan (ccGAP; 2012). In relation to the Paris Agreement, the project will align with one of the three adaptation priorities identified under Liberia's Intended Nationally Determined Contribution (2015), namely the construction of coastal defences to reduce the vulnerability of urban coastal areas.

31. The project was developed following an extensive stakeholder engagement process, beginning with initial institutional and community engagements during the first phase of project design in 2016, prior to the initiation of the Project Preparation Facility-financed feasibility study, identifying broad needs and vulnerabilities within the Monrovia Metropolitan Area, and engaging with relevant government departments to validate the design of the project and ensure alignment with national priorities. In 2019, after the rock revetment was identified as the preferred coastal protection solution, 40 community consultations were conducted with the West Point community, in which the trade-off between coastal protection and changes in access to the beach and sea was discussed in depth. This included consultations to identify a range of desired gender-related development impacts of the project, and to make sure that such impacts were incorporated into the project design. Full support for the selected option was expressed by all participants, and by the Kru and Fanti traditional leadership structures. The detailed design phase will provide a further opportunity for community input on and validation of the proposed alterations to beach fisheries access sites, and plans for the development of public space on the promenade.

32. The national executing entity (EE) for the project is the Government of Liberia, represented by the Environmental Protection Agency (EPA). As the GCF national designated authority for Liberia, the EPA has been involved in the submission of several Concept Notes and Readiness Proposals for the GCF, with a climate information systems simplified approval process project through the African Development Bank approved at B.27, and has a strong track-record of implementing climate adaptation and other projects for the Global Environment Facility. The Ministry of Mines and Energy (MME), as a Responsible Party, will enter into a legally binding memorandum of understanding (MoU) with the EE to assist in successfully implementing project activities. The funds for selected activities, as requested by the Government of Liberia, including the major cost of the infrastructure works at West Point, due to their higher value and high risk nature, will be channelled directly to service providers procured by UNDP, as part of UNDP support services to the Government. The Ministry of Public Works will assist the EE and the MME to implement the infrastructure works. As the AE, UNDP will deliver GCF-specific oversight and quality assurance services for the project. In Liberia, UNDP has led environmental protection and climate change responses through the facilitation of policy formulation, strategic planning, law making, coordination and information sharing. Through this role, UNDP has built strong relationships with decision-makers and proven its strengths as an impartial provider of technical advice and support. In addition to the co-finance letters from UNDP and Government of Liberia, commitment of the Government to the project's

sustainability has recently been confirmed with a letter of financial commitment for the post-project operations and maintenance (O&M). An initial concern, expressed by the independent TAP, that the O&M plan was vague on responsibilities, actions and costs was rectified through the AE's work with Government to produce a new and much more detailed O&M plan, which clarified these matters, as well as plans for capacity transfer through the major contract for the revetment works.

1.6 Efficiency and effectiveness *Scale: Medium to high*

33. The project's overall efficiency and effectiveness is rated at medium to high. The effectiveness of the project's central investment by GCF and the Government of Liberia in the rock revetment at West Point is seen as effective in applying best available practice in coastal engineering, and latest innovations in techniques. The selection of the approach was based on detailed hydraulic studies for the Monrovia Metropolitan Area, using climate models, wave models and morphological models, with a vulnerability assessment covering the anticipated impacts of climate change on coastal structural retreat and storm erosion. A full on-site engineering study considered the alternatives and motivated for the revetment approach, on the basis of its efficacy, cost-effectiveness and light burden in terms of post-project operations and maintenance. The project designs for the recommended rock revetment and promenade, as well as the O&M plan, include clear risk mitigation measures that must be taken to avoid flooding and unrepaired storm damage causing erosion of sediment into scour-holes.

34. The project's cost-effectiveness could be maximized by follow-through on the implementation of a financing plan to be developed through the project, to facilitate new partnerships for financing additional coastal protection measures in vulnerability hotspots, especially the Atlantic Shore–US Embassy coastal stretch in Monrovia. There is also potential during project implementation to leverage further investment in the community mangrove conservation agreements, and in small business development for technologies that take pressure off natural resources and help preserve ecological protection infrastructure. In response to questions from the independent TAP about engagement of the domestic private sector, the AE clarified that "*Relevant private sector partners will be identified by the service provider in consultation with the 10 government institutions. These private sector partners will be targeted because of their location in the coastal zone and the relevance of coastal climate change adaptation to their businesses.*"

35. The total project cost is USD 25.6 million, including GCF grant financing of USD 17.2 million, the bulk of which (70 per cent) will contribute to the revetment construction at West Point. The remaining USD 8.4 million (34 per cent of the total project cost) will be borne by the Government of Liberia (USD 6.8 million in form of grant and in-kind contributions) and UNDP (USD 1.6 million in grants). The government co-financing is considered to be significant, given that Liberia is an LDC with a limited revenue base. As part of this, the Government is making an important contribution of USD 4,5 million worth of rock rubble to be used as a major input to the construction of the West Point rock revetment. GCF has also received a letter of commitment from the Ministry of Mines and Energy to ensure that the costs of approximately USD 62,000 per year for the ongoing O&M of the West Point works, the community education and innovation centre, and the cold storage facilities, will be covered. It is noted that the grant request to GCF is made in the context of coastal adaptation generating largely public goods, and not generating revenue streams from which loans might be repaid.

36. The economic analysis conducted for the project indicates a conservative estimate of the expected net present value (ENPV) of the project at USD 17.73 million, using a 10 per cent discount rate, with an economic internal rate of return of 21 per cent. In addition to estimates for damages avoided, benefits also include an increase in income growth and the social benefit of the project through improved infrastructure. This project proposes the construction of

coastal protection measures along approximately 1,050 metres of exposed and vulnerable coastline as well as measures to protect and enhance the climate resilience of the livelihoods of ~25 per cent of Monrovia's population. The cost of these measures is USD 104 per beneficiary. The independent TAP agrees with the AE that this is a reasonable cost, especially given the fact that the indirect beneficiaries are not included in this calculation.

II. Overall remarks from the independent Technical Advisory Panel

37. The independent TAP recommends this funding proposal for approval with the following condition:

(a) Prior to the first disbursement of the project, the accredited entity shall submit to the Secretariat an additional Report, in a form and substance satisfactory to the Secretariat, which:

  (i) Further strengthens the project's climate rationale by summarizing clearly the observed and predicted changes along Monrovia's coastline in relation to the following parameters: sea level rise, sediment transport and exchange, wave conditions, and high-intensity storms; also showing how these parameters are being affected by climate and non-climate drivers, and how they interact to produce shoreline regression – both historical and anticipated.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP 160)

Proposal name:	Monrovia Metropolitan Climate Resilience Project
Accredited entity:	United Nations Development Programme (UNDP)
Country/(ies):	Liberia
Project/programme size:	Small

Impact potential
We note the iTAP's assessment and rating of the project with respect to its impact potential. The scale of the challenges pertaining to land -use planning is well noted, and the need for grounding the ICZM Plan in existing land-use planning frameworks and processes to contribute to the existing enabling environment that these policies are creating for land-use planning.
Paradigm shift potential
We note and agree with the iTAP's assessment and rating of the project with respect to its paradigm shift potential. Implementation of the project will have a focus on the innovations and partnerships that will strengthen this investment criterion.
Sustainable development potential
The overall rating and assessment is well noted. During implementation, the project will pay attention to increasing the project's co-benefits as indicated in the assessment by iTAP.
Needs of the recipient
We note and agree with the iTAP's assessment of the project on addressing the needs of the recipient both at the national and community levels.
Country ownership
The iTAP's assessment of the project against the "country ownership" criterion is consistent with the process followed in its origination, development and implementation arrangements. This will be maintained during implementation to ensure it continues being owned by the country at all levels.
Efficiency and effectiveness

The overall assessment against this criterion and the medium to high rating is noted, as well as the recommendation to maximize this through the implementation of a financing plan to be developed by the project to facilitate new partnerships for financing. Attention will be paid to the recommendation during implementation.
Overall remarks from the independent Technical Advisory Panel: We thank the iTAP for recommending the project for funding. The comprehensive assessment of the project is well appreciated, as well as the ratings on each of the criteria. During implementation, we will continue working with the Executing Entity and other stakeholders to strengthen the areas that have been highlighted in this assessment. The request by iTAP for an additional annex has been noted, and the summary requested will be prepared based on the analyses undertaken during project design, and included in the feasibility and vulnerability studies and their detailed annexes.

ANNEX 8: GENDER ASSESSMENT AND ACTION PLAN

FOR THE

MONROVIA METROPOLITAN CLIMATE RESILIENCE PROJECT (MMCRP)

September 2019

I - Introduction

The Green Climate Fund (GCF) recognizes the central importance of gender considerations in terms of both impact and access to climate funding and requires a Gender Assessment and Gender Action Plan to be submitted as part of the project-funding proposals that it assesses. The main objective of the Gender Assessment is to screen the gender aspects of the GCF project, and to subsequently strengthen the gender responsive actions within the project "**Monrovia Metropolitan Climate Resilience Project" (MMCRP)**. Therefore, this gender assessment aims to provide an overview of the gender situation in Liberia, with a goal to strengthen the resilience of affected women and men living in "hotspots" of Monrovia Metropolitan Area (MMA) where the country's coastal areas are facing increasing risks of coastal erosion and extreme climate events.

Specifically, the assessment will identify gender issues in Liberia that are relevant to the project and examine potential gender mainstreaming opportunities. Given the fact that gendered climate vulnerability is at the center of the project, the information and design considerations in this Annex should not be considered additional, but rather part of the basis of the proposal, including the Feasibility Study, Stakeholder Engagement and Environmental and Social Management Framework (ESMF) Annexes.

The assessment was based upon conducting field consultations, gathering information from available data from studies conducted by the Government of Liberia, UN and civil society organizations, donor agencies and multilateral development banks. Lessons learned from past studies and assessments on gender in Liberia were also taken into stock. Stakeholders consultations engaging, men, women and youth also raised gender concerns that will be integrated as the project indicators, targets and activities, identifying women as beneficiaries including as leaders and decision-makers.

Gender mainstreaming in this project aims to be both gender responsive, by integrating women's perspectives, and guidance to ensure that existing inequalities are not exacerbated, but also gender transformative, in the sense that it deals with root causes of vulnerability and structural barriers to climate resilience, and challenges the norms around the gendered distribution of labor and constraints in regards to access to resources, capacity building and ability to engage in adaptative measures.

The proposed GCF funded project not only intentionally targets and benefits women, but also considers the intersectional vulnerability: there are some categories that face additional forms of marginalization and stigma due to poverty, health and disabilities, including women, (and women-headed households), youth and children, the elderly, people living with disabilities and marginalized groups such as those with HIV/aids and LGBTIs living in the target areas.

The project design recognizes that SLR and coastal erosion has immediate and long-term impacts on women as described in the consultation reports and in the current the gender analysis. In addition, the project recognizes the effects of changing environmental conditions by supporting climate resilient livelihoods and better integration of gender considerations into the Integrated Coastal Zone Management (ICZM) and into local value chains, in which women are already playing a growing role.

The Gender Assessment intends to provide additional information on the national and local gender context, particularly regarding women's access to resources, their role in decision-making and the gendered aspects of local livelihoods. It provides the basis for, and lessons on which, the Gender Action Plan (which is reflective of the overall project design) has been built.

Overall, the project objective is to enhance the climate resilience of the most SLR-impacted MMA communities that are constantly exposed to storm surges, flooding and salinity ingress through the installation of coastal defence options. These actions will be conducted with a special attention to women - and by extension, their families - and communities who are rendered the most vulnerable.

Specifically, GCF resources will be used to enable all relevant institutions including the Minister of Gender to implement activities in the context of an ICZM that will empower more targeted vulnerable women to become leaders in the adoption of climate change resilient livelihoods and the management of the ICZM options, ensuring that women are involved in the ownership, operationalization and management of the selected options with the technical and financial backstopping support from concerned institutions. Resources will also be used to promote climate resilient livelihood options and participation in their associated value chains for women whose existing livelihoods are particularly vulnerable to salinity ingress. Finally, resources will also be used to strengthen the capacities of national institutions to enable gender-responsive, climate-resilient solutions for both coastal security and livelihoods.

The primary measurable benefits that will be realized as a result of the GCF investment includes

- Provision of assistance to communities in the MMA, with a special focus on women, youth and other vulnerable groups, to pursue climate resilient livelihoods and facilitate their access to associated market linkages, with investment in assets, tools, and training,
- Change of perceptions and norms of gender equality and women's empowerment to induce a transformative change within the communities to improve gender relationship at the family and community levels,
- Inclusion of women in all the national, local and community activities of the ICZM, at management and operational levels,
- Strengthened capacity, coordination and knowledge sharing of national institutions and local communities intervening in the project.

II – Existing Gender inequality and social exclusion in Liberia

Women play a crucial role in Liberia. Not only do they comprise almost half of its population, but they also play important roles at household level, in the rural and urban economies at all levels. The Liberia Revised National Gender Policy (2018-2022) states that "Gender inequality and the marginalization of women in Liberia are perpetuated by socio-cultural perceptions, practices, and stereotypes that support male dominance and the subordination of women"1i . This happens partly through socialization of girls and boys, social norms and customs, which carries into schools, the community and the workplaces.

Gender inequality is not uneven in Liberia. It varies according to age, status, means of income, county rural/urban areas and traditions, but in general, access to education, healthcare, land, property, and justice is limited to women as compared to men. Gender imbalance causes such as excessive burden of

[1] Liberia Revised National Gender Policy (2018-2022), page 5

care on women, illiteracy unequal power relations in the household, lack of access to basic services, economic disempowerment, low participation in decision-making in public spheres and institutions and in the household, lack of access to legal and judicial services, vulnerability to Human Immunodeficiency Virus (HIV) and Acquired Immune Deficiency Syndrome (AIDS), and unequal access to rights and opportunities; the situation is worse for adolescents girls that are often school drop-outs because of poverty and face the issue of teen-age pregnancies.

Gender Inequality Indexes - (GII) and other relevant gender indices

Through the years, several indices have developed to quantify the concept of gender inequality. The United Nations Development Programme uses the Gender Inequality Index (GII) and Gender Development Index (GDI).2 GII is a composite measure that shows inequality in achievement between women and men in reproductive health, empowerment and the labor market while with a measure's achievement in human development in three areas: health, education, and command over economic resources. The GDI considers the gender gaps on human development between men and women. Despite the ratification of the Maputo protocol, its domestication has remained inadequate. With strong patriarchal social norms and entrenched, socially accepted violence against women and girls, the Gender Inequality Index of 0.610 ranks Liberia at 177 out of 188 countries assessed, a decline from earlier years[3]. Gender gaps are evident in mean years of education (3.5 vs. 6.1) and income (US$577 vs. US$755). While patriarchal norms that maintain the low status of women and girls persist, the legacy of violence against women during the civil wars has been normalized.[4] Almost 14 percent [5] of the population living with disabilities confronts stigma and lack of income-earning opportunities. High level of adult HIV prevalence at 2.1 percent[6] (2.4 percent among women, 1.8 percent among men), indicates underlying policy and health system deficiencies and unequal gender relations.[7]

The Global Gender Gap Index (GGGI) of the World Economic Forum examines the gap between men and women in four categories: economic participation and opportunity, educational attainment, health and survival; and political empowerment.[8] Out of 149 countries, Liberia's rank based on GGGI in 2018 is given below[9]:

Description	Score	Rank
Gender Gap Index 2018	0.681	96
Economic participation and opportunity	0.729	41
Educational attainment	0.792	141
Health and survival	0.968	118
Political empowerment	0.236	47

[2] United Nations Development Programme. Human Development Report. http://hdr.undp.org/en/content/table-4-gender-inequality-index.
[3] Human Development Report, UNDP 2016 and 2018; Inequality-adjusted HDI for 2017 is 32 percent lower
[4] Programme Against Sexual and Gender Based Violence and Harmful Traditional Practices, Republic of Liberia, 2018
[5] National Census, Republic of Liberia. 2008
[6] Catch up plan to end AIDS 2017-2020
[7] Gender analysis in UNDP Liberia CPD (2020-24), § 3
[8] World Economic Forum. The Global Gender Gap Report 2018 Country Profiles. http://reports.weforum.org/global-gender-gap-report-2014/economies/#economy=LIB
[9] http://reports.weforum.org/global-gender-gap-report-2014/economies/#economy=LIB

* Inequality = 0.00; Equality = 1.00. Source: The Global Gender Gap Report 2018

The Organization for Economic Cooperation and Development (OECD) developed the Social Institutions and Gender Index (SIGI), a composite index that scores countries: percentage varies to 0 to 100 on 14 indicators grouped into four sub-indices. Liberia scores as follows: 1 - discrimination in the family (60%), 2 -restricted physical integrity (34%) 3 - restricted access to productive and financial resources 4 - (41%) Restricted civil liberties (53%) to measure the discrimination against women in social institutions across 160 countries. The 2019 SIGI value for Liberia equals 47 % suggesting that gender inequality and discrimination against women and girls is high.[10]

Poverty

After emerging from years of conflict, the Ebola crisis and the resultant economic downturn[11], Liberia faces many economic challenges. According to the 2018 UNDAF Liberia Common Country Assessment[12] these economic challenges are: high inflation, declining global commodity prices, growing foreign debt, and import dependency[13], which together slowed down[14] economic growth[15], reducing fiscal space[16] for delivery of essential services. Acute poverty and vulnerability in Liberia are a result of entrenched social, and economic inequalities and exclusion – which resulted in years of conflict[17] and continue to persist due to high levels of centralization in the Capital and a concessions-based growth model.

Liberia's Human Development Index (HDI) has been declining, with the HDI value for 2017 at 0.435, placing Liberia at 181 out of 189 countries [18]. More than half of the country's population [19] lives below the poverty line[20] and marginalization remain deeply entrenched[21]. In 2018[22], 71.2 percent of Liberians experienced multi- dimensional poverty - 33.2 percent severely poor while 20.4 percent "vulnerable" to slipping back into poverty[23].

Weak administrative procedures and limited human and institutional capacities[24] have led to delays in citizens' access to justice, especially in cases of human rights violations and sexual and gender-based violence.

Women economic empowerment and access to resources

Despite the central economic roles of women in households and in the economy in general, women still face some constraints in order to participate effectively in the economic life of the country because of discriminatory norms and practices and inequitable policies, such as:

10 Source: OECD (2019), Gender, Institutions and Development Database, https://oe.cd/ds/GIDDB2019.
11 World Bank, 2018
12 UNDAF Liberia Common Country Assessment, 2018
13 World Bank, 2018
14 8.7 percent (2013) down to 0 percent (2015) and -1.6 percent in 2016
15 IMF Staff 2019 Article IV Mission to Liberia (link)
16 Africa Economic Outlook (2017)
17Truth and Reconciliation Commission (2009), Agenda for Transformation (2012), PAPD 2018
18 Human Development Report, UNDP 2016 and 2018; Inequality-adjusted HDI for 2017 is 32 percent lower
19 4.7million (UN Population Division, 2017)
20 Household Income and Expenditure Survey, 2016
21 Poverty: 71.6 percent in rural areas and 31.5 percent in urban areas
22 Multidimensional Poverty data 2018
23 Human Development Report, UNDP 2016, 2018
24 Liberia Peacebuilding Plan, 2017.

- Limited access to credits, raw materials, training, marketing, etc.
- Absence of enabling environment to contribute effectively to the economy and limited access to productive resources
- Limited access in decision making positions in public service
- Limited participation in more profitable sectors and limited opportunities for economic empowerment
- Discriminatory customary practices, stereotypes, norms and biases that continue to prevent equal access to and control over land and land use

It is important to note that when facing climate hazards, women are more disproportionately affected than men. For example, women may be forced to spend savings during climate shock events, and this loss of savings can subsequently lead a great number of small businesses (women owned and also where women are customers) to collapse. Small businesses predominantly owned by women were largely affected during the economic downturn in Liberia because they were selling perishable food items and many lost their ovens. [25]

Social protection

The main goal of the PAPD is to improve protection of the poorest and most vulnerable population sand groups from poverty, deprivation, and hunger, and enhanced resilience to risks and shocks through a multi-sector approach to support and finance social protection interventions and improve livelihoods and increase employment readiness of extreme poor, youth, and women through increased opportunities for income generation.

Therefore, the situation in SLR affected communities cannot be ignored. Stakeholder meetings held in MMA from March to April 2019 revealed that with affected women involved in drying and selling fish along the sea coast as well as men and youth involved in fishery and petty (small-scale) trading, men and women are frustrated at being neglected, particularly given that the erosion caused by climate change rapidly left most of them homeless. While they do not feel protected at home, there are social measures to protect them if in danger at sea.

However, social protection needs for women and men may vary. Women and men face different risks and vulnerabilities, some specific to their gender and others exacerbated by gender inequalities and discrimination. The design and implementation of social protection programs in Liberia address gender-related constraints, including barriers to women's economic advancement. [26]

In the absence of specific data in relation to gender and social protection, it is clearly understood that women's financial contribution is more important within household survival strategies. However, because women are concentrated in informal and labor-intensive work where they face risks and vulnerabilities (e.g. health risks, interrupted and insecure employment), they have either less or little-to-no resources to save or contribute to pensions or be eligible to some form of a social safety net to tide over times of crisis. Therefore, social security measures are restricted to the small, male-dominated section of the workforce employed in the formal state and private sectors. Social protection is limited for those in the very informal employment sector where women dominate.

[25] Report on progress on the implementation of the NGP, 1Korkoyah and Wreh 2015.
[26] Liberia Demographic Health Survey (LDHS, 2013)

Gender and Employment

Liberia Demographic Health Survey (2013) provides the following figures 53% of women in contrast to 74% of men were employed at the time of the survey. Geographically, by county, women in Gbarpolu (78%) and Bong (72%), were more likely to be employed than women in other counties (40-60%) (LDHS, 2013). While for men, in Gbarpolu (94%), River Cess (94%) and Grand Cape Mount (93%) were more likely to be employed than their female counterparts in other counties (58-87%). Women (59%) and men (82%) in the rural areas are more likely to be employed than those in the urban areas. Employed women (80%) and men (74%) in the lowest quintile are engaged in agricultural activities (LDHS, 2013). The urban and rural differentials show that urban women were concentrated in sales and marketing (66%), compared to rural women who were mostly employed in the agricultural sector (69%)24. Men (22%) in urban areas were employed in sales and marketing. In the rural areas, 69 percent of women and 62 percent of men were engaged in agricultural activities.

By category of employment, women are mostly found in sales and services (49%), followed by agriculture (42%), in contrast to men who were more likely to be employed in agriculture (40%), unskilled manual labor (15%), sales and services (14%) and skilled manual labor (14%). In the professional categories, 4 per cent of women compared to 10 per cent of men had professional, technical or managerial occupations. According to the LDHS (2013), women and men with no education were more likely to be employed (65% and 91%) than their counterparts who had attended school (68-73%), since they could easily access informal employment.

Education is also a hindrance to women's employment in the formal sector, so men are disproportionately clustered in the least productive sectors with 90% employed in the informal sector or in agriculture, compared to 75% of working men. Men are more than three times likely to be employed by the civil service, NGOs, international organizations or public corporations, in mining, in panning, in the manufacturing sector. It is only in agriculture and fisheries that men and women are employed on an equal basis – 1 to 1 ratio.

High participation of women in the informal sector is a consequence of the care economy as the burden of home and family care imposed upon them through the socio-cultural allocation of gender roles. They are more absent in small and medium enterprises due to limited education, lack of capital, credit and land.

While all men and women in the informal sector face similar problems such as absence of social insurance schemes, exploitation, harassment from people in authority (including at border crossings), unsafe work environment, limited finance and business development services, women are more negatively affected due to their over-representation in this sector. For cross-border traders, there is information gap on migration and customs regulations, making women in cross border trade more vulnerable to exploitation, due to limited knowledge about their rights and the laws that protect them.

The Civil Service Reform Strategy (2008-2011) focused on mainstreaming gender and establishing mechanisms for implementing an affirmative action program, drafting and implementing a civil service-wide sexual harassment policy, and the establishment of Gender Focal Points (GFPs) in each Ministry, Agency, and Commission Corporation (MAC). The Decent Work Law passed in 2016 assured the right to minimum remuneration of women domestic workers. It guarantees entitlement of women and men without distinction, exclusion or preference to receive equal remuneration for work of equal or comparable value [27].

It is hoped that the introduction of a new National Employment Policy; the review and revision of the Liberian Labor Code, the transformation of the Liberian Emergency Employment Program (LEEP) into the Liberian Employment Action Plan; the validation of the National Workplace Policy on HIV and AIDS the launch a Women's Entrepreneurship Program involving MSMEs for business skills development, and provision of access to microfinance and functional literacy will make a big difference in Gender and Employment in Liberia.

Health and Reproductive Rights

Liberia faces a few health challenges but more specifically gender norms, roles and relations can influence health outcomes and affect the attainment of mental, physical and social health, and well-being. Fortunately, gender equity is a guiding principle of the NHSWP (2011-21) and improving health outcomes among women and girls is critical to the overall goal. The three objectives of the National Health and Social Welfare Policy and Plan capture the MOHSW's strategy to equitably improve gender equality and the health status of women and girls. To increase access and utilization of services, especially for women and girls at the community level, by expanding access to quality services delivered closer to where people live. Coupled with investments in strengthening human resources, the target is to increase facility-based deliveries by a skilled birth attendant from 22 to 80 percent.

To make services for women and girls more responsive, the cornerstone of 2011-21 NHSWP is the Essential Package of Health Services (EPHS), a gender-sensitive service package with strong linkages to the community and household, as well as gender-specific health messages. The target is for 90 percent of facilities to be fully accredited and providing the EPHS. To provide services that are affordable against a backdrop of ensuring social protection through a combination of prepayment schemes and free services for priority interventions that directly affect the health of women and girls.[28]

Legislation does not restrict the right of women to access contraception and other reproductive health services.[29] Government and INGO health facilities provide free access to sexual and reproductive health services[30] Over 25 percent of adolescents age 15-19 have already had a child.[31] Vaccination rates for infants are higher for girls than boys (42.4% against 36.1%), and infant and child mortality rates in Liberia are slightly higher for boys than girls. The same is true of malnutrition rates.

High maternal mortality and systemic inequities — especially related to health service access and quality – have a major impact on women and girls. The high maternal mortality rate is linked to a high fertility rate, insufficient numbers of skilled birth attendants and low utilization of health facilities for delivery, compounding the risks associated with complications.

[27] Liberia CEDAW Report (2013).
[28] USAID. Country Development Cooperation Strategy (Liberia) – 2013-17, p. 61
[29] US Department of State (2012) in OECD. Social Institutions and Gender Index (SIGI) 2014. Country Profiles http://www.genderindex.org/country/liberia ; OECD. Social Institutions and Gender Index (SIGI)
[30] CEDAW (2009a) p.13 in OECD. Social Institutions and Gender Index (SIGI) 2014. Country Profiles http://www.genderindex.org/country/liberia ; OECD. Social Institutions and Gender Index (SIGI)
[31] DHS (2007) p.53 in OECD. Social Institutions and Gender Index (SIGI) 2014. Country Profiles http://www.genderindex.org/country/liberia ; OECD. Social Institutions and Gender Index (SIGI)

Other factors which also contribute to maternal and newborn mortality are delays in diagnosing problems, deciding to seek medical care, accessibility and affordability, inadequate supply of drugs and equipment, poor nutritional status of pregnant women, and high fertility rates. Some women who survive maternal health complications develop life-long disabilities such as fistulae and secondary infertility. Teenage mortality is high among girls who develop complications during pregnancy and go through caesarean procedures. The situation is even more precarious in communities where there are no medical facilities nor skilled birth attendants to address complications that may arise. Additionally, there are no facilities available to handle mental conditions, such as depression, related to the postpartum. The national contraceptive prevalence rate for family planning is 20% and less than 14% for adolescent girls aged 15–19 years. The unmet need for family planning has virtually remained the same with no significant decrease: from 36% in 2007 to 34% in 2013 (DHS 2013).

In terms of family planning, there is limited participation by men. Access to and knowledge about Sexual and Reproductive Health (SRH) varies between men and women. According to the LDHS (2013), 98 per cent women and 95 per cent of men have knowledge of at least one method of contraception. Teenagers of both sexes either are not aware of or do not practice birth control methods, even though family planning is free. Some of the reasons given for this include cultural misconceptions about family planning, myths that future conception may not be possible, perceptions amongst youths on promiscuity by using family planning, religious background that prohibit family planning, affordability and lack of responsibility.
If this situation is not improved, women will continue to bear the brunt of continue to be vulnerable to emerging health issues/disasters with the lack of or limited access to social protection programs described above.

The delivery of water and sanitation services in Liberia is a key concern in the affected communities visited during consultations. This process affects women and girls' dignity, safety and physical and time burden. The insufficient number of pumps means more time is spent gathering water, sometimes from far away, which affects the well-being of women and girls and the way of life in communities. The unavailability of safe drinking water and poor sanitation services, such as insufficient public toilets and nonfunctioning public common toilet facilities are of great concern. Communities practice open defecation and pollute the beaches. Water and sanitation-related illnesses challenge their health and add a burden to women's caretaking role. In addition, this influences the economic sectors due to the fact that women and children pay high prices from vendors in MMA.[32]

Liberia's coastal communities also face several psychosocial challenges: in addition to economic vulnerability and stress resulting from the reality of living in communities affected by climate-induced erosion, the uncertainty of measures that are going to be put in place by local authorities is a burden. This is even worse for single female-headed households. More than men, the affected women also revealed that the destruction of their productive assets and their houses by the sea has led them to lodge with friends and relatives in the other communities in proximal areas. The use of improper and unsafe places to sleep is causing serious health problems for their families and themselves – including flu, constant cold and malaria. Furthermore, this displacement is connected to teenage pregnancy as well as drug and substance abuse by young men and women which were also rampant phenomenon discussed.

Education

[32] Government of Liberia (2009). Water supply and sanitation policy, p. 6

After years of civil war and the breakout of EVD, Liberia is still in the process of rebuilding its education infrastructure and workforce. The issue of youth education is serious. Based on the Liberia Household Income and Expenditure Survey data collected in 2014, the national literacy rate is estimated to be 66.7 percent, that means, just over two thirds of Liberians can read and write. Residents of urban areas are more likely to be able to read and write than those of rural areas; 76 percent versus 50.1 percent. The gap between men and women is even larger while 80.4 percent of males are reported as literate, only 54.8 percent of females know how to read and write. Therefore, women lack the necessary skills and resources that can provide them with the types of opportunities of their male counterparts.

Liberia Demographic Health Survey (LDHS, 2013) reveals that access to education has increased for both girls and boys (15-19 year) for the population with an increase from 31 percent in 2007 to 41 percent for girls in 2013, and from 36 percent in 2007 to 46 percent for boys in 2013 (LDHS, 2013). However, a gender gap in education persists and shows geographical variations by county, variations in literacy rates by ages and by completion rates, and variations in rural and urban areas. For example, Grand Bassa (66%) and Bong (50%) were cited to have the highest proportion of women/girls who have never attended school. These variations persist even though gender disparities in primary school enrolment have been significantly reduced. The issue of girls dropping out of schools is of concern. Reasons cited for these dropouts include sexual exploitation, rape, teenage pregnancies, forced or early marriages, poverty, boys to girls' preference to education, as well as gender roles stereotyping in educational curriculum and discrimination against the girl child. Discrimination and stigma are noted in the expulsion of pregnant adolescents' girls from school, while the same is not done to the peer boys who have impregnated them. At secondary school level, 10 percent of girls completed secondary school compared to 23 percent of boys. Poverty is also a major factor for girls to drop out of schools.

The Government recognizes the need to close this gap and to promote a drastic and rapid increase in school attendance in rural areas, particularly for girls. This situation was alleviated by providing universal primary education and implementing the National Action Plan 2004-2015: Education for All. The Government has also prioritized girls' education through the National Girls Education Policy to ensure, encourage and support the enrolment and retention of girls in school. The Free and Compulsory Education Law is another important step towards women and girls' equality in education. The Education Reform Act of 2011 aims to enhance access to education from a gender perspective by increasing the number of female teachers and advocating for counseling services for girls, and punitive measures against teachers who sexually abuse and assault students. There are also calls for legislation that will ensure that pregnant girls are not expelled from school.

The situation becomes critical in the coastal communities like new Hotel community, where most adult women and girls are illiterate and lack many of the basic skills to meet up with the challenges of life and empower themselves better. Stakeholder meetings and interviews with communities indicated mothers' worry as to how they can continue to send their children to school since the sea is taking progressively away the public schools' infrastructures and sea level rise and subsequent flooding is a threat to young kids during the raining seasons.

Political participation and governance

Despite Liberia producing the first female president in Africa, women continue to face several challenges running as candidates including fear of violence and intimidation, hostility to their speaking in public, and a lack of funds to run for office. Participation of women in politics and governance remain low - women occupy only 2 out of 30 seats in the Senate and 10 of the 73 seats in the House of Representatives.[33]

In local government, from general town chiefs to county superintendents, women account for fewer than 6% of leadership positions. In the national legislature, women make up around 11.5%. In the Cabinet, or executive branch, women number only 21%. In the Supreme Court, 40% are women with only 5 people on the court. In the decision-making structure in the judiciary, such as circuit court judges, women total just 12.5%.[34]

The Local Government Act enacted in 2018 "affirms the commitment of the Government of Liberia to further the unity of the Republic by providing equal opportunity for all of its citizens to engage in the governance of the state through the devolution of certain administrative, fiscal and political powers and institutions from the national government to local governments". Several efforts to pass a bill mandating legislative quotas have been undertaken but are yet to be successful. The Equal Participation in Politics Act, 2016 has still not been passed into law. If all these laws are passed combined with the will of political parties to institute affirmative action measures, this can significantly increase the number of women who hold public office However, the neutrality of language and lack of incentives and enforcement measures fall short of achieving the desired target. The Laws and related dispensations need to be made explicit and enforceable.

In addition, the Local Government Act includes a gender equity measure which reserves 2 seats in every county council for women and women can also contest for the other seats. In addressing the structural inequity and marginalization of women in the public sector, the Government should develop and adopt policies in all public institutions and government agencies to address gender inequities, including codes of conduct and non-discrimination policies. Although, increasing the number of women's representation is critical, there is also needed to develop a robust gender capacity building program for the different actors at national and local levels.

Decision-making

Women's equitable participation and representation in decision-making strengthens commitments to women's empowerment and gender equality. It also makes for better outcomes related to sustainable development, resilience and adaptation capacity, and overall wellbeing.

The 2013 Liberia Demographic and Health Survey, covering all of Liberia, found that, "most currently married women (58-61 percent) reported that each of three household decisions is made jointly with their husbands. Sixteen percent of women reported that they alone make decisions about their own health care; 19 percent make their own decisions to visit their families and relatives, and 24 percent make their own decisions about major household purchases. More specifically, 30 percent of currently married women who receive cash earnings reported that they alone mainly decide how their earnings are used, while 54 percent say they decided jointly with their husband. Only 15 percent of women reported that their husband mainly decides how their earnings will be used. Younger women are generally more likely than older women to make independent decisions on their earnings. Further, 32

[33] National Election Commission , 2018
[34] 63UN Women (2017). Liberia Fact Sheet: Women's Representation in Various Branches and Levels of Government

percent of urban and 26 percent of rural currently married women reported that they mainly decide how to spend their earnings.[35]

Women in coastal communities are empowered by hardship; as they are not waiting for their partners to attend to domestic care and other family related needs, they are part of the decision in the households and have a say in the family decisions, even though most of the interviewed men responded that they are the heads of the families.

Labor force/ Employment and livelihoods

Liberian employment law prohibits discrimination based on gender[36]. In addition, the "Decent Work Bill" passed into law on June 25, 2016. The bill explicitly excludes discrimination against women in working conditions and wages.[37] Further, pregnant women have the right to three months paid maternity leave,[38] and employers are responsible for paying 100 percent of maternity benefits.[39] But these regulations only apply to women working in the formal sector, and according to the 2008 report issued to the Committee on the Convention on the Elimination of All forms of Discrimination against Women (CEDAW), 90 percent of women are employed in the informal sector.[40] The World Bank considers 67 percent of women in Liberia to be employed.[41]

As of 2014, 47 percent of the total labor force was female[42] although not all those who work are included in the labor force including, for example, unpaid workers, family workers, and students. The labor force size also tends to change seasonally.

Access to resources, land tenure and property rights

According to data from the MGCSP, there are about 367 Women Entrepreneurs who are employers in the trade, commerce and industry sector who are struggling to have access to the capital or assets necessary to begin to expand their businesses. This is in addition to 5,650 women who are listed as employers in small businesses and 1,500 women involved in agricultural entrepreneurship, according to the 2009 Liberia CEDAW report.

Women are the key actors of Informal Cross Border Trade (ICBT), making up approximately 70 percent of the informal cross border traders. Women operating in cross border trade are operating under unbearable conditions because exposed to fraudulent custom charges, extortion, discrimination, and at times sexual harassment and other forms of gender-based violence when crossing the borders. To

[35] Liberia Institute of Statistics and Geo-Information Services (LISGIS), Ministry of Health and Social Welfare [Liberia], National AIDS Control Program [Liberia], and ICF International. 2014. Liberia Demographic and Health Survey 2013. Monrovia, Liberia: Liberia Institute of Statistics and Geo-Information Services (LISGIS) and ICF International.
[36] Committee on the Convention on the Elimination of All forms of Discrimination Against Women (CEDAW) (2008), Consideration of reports submitted by States parties under article 18 of the Convention on the Elimination of All Forms of Discrimination against Women Combined initial, second, third, fourth, fifth and sixth periodic reports of States parties Liberia, CEDAW/C/LBR/6, New York, CEDAW (http://www2.ohchr.org/english/bodies/cedaw/cedaws44.htm - accessed 17 November 2013). p.56 in OECD. Social Institutions and Gender Index (SIGI) (2014 data), p.7 http://www.genderindex.org/sites/default/files/datasheets/LR.pdf
[37] CEDAW (2009c) Summary record of the 902nd meeting. CEDAW/C/SR.902. New York, CEDAW (http://www2.ohchr.org/english/bodies/cedaw/cedaws44.htm - accessed 17 November 2013). p.4 in OECD. Social Institutions and Gender Index (SIGI) (2014 data), p. 7 http://www.genderindex.org/sites/default/files/datasheets/LR.pdf
[38] CEDAW (2009c) p. 4 in OECD, 2014, p. 7.
[39] World Bank (2013) in OECD, 2014, p. 7.
[40] World Bank (2013) in OECD, 2014, p. 7.
[41] World Bank (n.d.) 'Data: labor participation rate in OECD, 2014, p. 7.
[42] World Bank Data accessed 26 September 2016. Source UNESCO http://data.worldbank.org/indicator/SL.TLF.TOTL.FE.ZS?locations=LR&view=chart

improve the conditions of women in informal cross border trade (WICBT), the MGCSP has, through a joint effort with UN Women Liberia and Liberian women traders, formed the Association of Women in Cross Border Trade.

In MMA women when not in fisheries are mostly involved in petty trade which doesn't offer access to quick change in live conditions if not exposed to other opportunities for improvement in their livelihoods. One of the ways to support them is to offer alternatives with vocational skills for them and to the girls to alleviate poverty and support their resilience.

Title 29 of Liberia's Property Law and Article 23 of the Constitution grant equal ownership rights to land and non-land assets to both men and women, discriminatory practices persist.[43] The Land Rights Policy approved by the Land Commission on May 21, 2013 also states that, "the Constitution gives all Liberians 'the right to own property alone as well as in association with others,' which means land ownership is permitted for all Liberians regardless of their identity, whether based on custom, ethnicity, tribe, language, gender, or otherwise. This is a fundamental constitutional principle that has informed every part of the Land Commission's policy recommendation."[44] Further, the government established a Land Commission in 2009 to address conflict over land sales, secure men and women's land tenure, and modernize the country's land law.[45]

While Liberia has a dual land tenure system, based on written law derived from statutes and case law, and on customary law, customary tenure systems generally prevail.[46] Under customary law, women can only access land through their husbands; they cannot themselves inherit land so any disputes considered under customary law may be biased in favor of men.

It is hoped that women's formal access to and control over farmland will increase via the passage in 2018 of the Land Rights Act which ensures women's access to land in order to increase farming and other livelihoods. It is a critical step forward towards improving land security and ensuring the productive use of land assets to reduce poverty and vulnerability, especially owing to its recognition of customary communities as owners of their land which presents a unique opportunity for disadvantaged Liberians in rural areas to be decision makers in the management of their land resources and to benefit from its proceeds. Accelerating the process of accession to the provisions of the Act and ensuring the full functionality of the Land Authority will therefore be a major focus under economic governance.[47]

The Government gives equal opportunity of owning property to everyone regardless of sex and ownership of a house is considered a measure of women's empowerment according to the 2013 Demographic and Health Survey. The DHS indicates that the percentage of women who do not own a

[43] CEDAW (2009a), 'Responses to the list of issues and questions about the consideration of the combined initial, second, third, fourth, fifth and sixth periodic reports Liberia', CEDAW/C/LBR/Q/6/Add.1. New York, CEDAW (http://www2.ohchr.org/english/bodies/cedaw/cedaws44.htm - accessed 17 November 2013); in OECD. Social Institutions and Gender Index (SIGI) (2014 data) http://www.genderindex.org/sites/default/files/datasheets/LR.pdf

[44] Government of Liberia. Land Commission. Land Rights Policy. Approved May 21, 2013

[45] OECD. Social Institutions and Gender Index (SIGI) (2014 data) http://www.genderindex.org/sites/default/files/datasheets/LR.pdf

[46] Hughes, Ailey Kaiser (February 2013) Focus on Land in Africa Brief: Liberia. World Resources Institute and Lands Rural Development Institute in OECD. Social Institutions and Gender Index (SIGI) (2014 data) http://www.genderindex.org/sites/default/files/datasheets/LR.pdf

[47] Republic of Liberia, '2018-2023 Pro-Poor Agenda for Prosperity and Development'(PAPD) - A five-year National Development Plan towards accelerated, inclusive, and sustainable development, September 2018, p.104. / See also Article 15: Equality before the Law and Article 16: Marriage and Family Life of the present report for more details on Women's Land Rights

house is highest in Montserrado (81 percent) out of all the country's counties. By county, the percentage of men who do not own a house is also highest in Montserrado (89 percent).[48]

Women and men have equal right to access financial services, including credit and bank loans. However, in practice, women often have difficulty accessing credit due to low literacy rates, and/or because they cannot meet the requirements needed to take out a loan. Microcredit programmes are provided by NGOs and the government and women are the main beneficiaries. However, most women have found a different means of pulling finance together through savings clubs called, "Susu." Susu clubs are a traditional form of banking and resource mobilization system widely used in Liberia and mostly by low-income earners, including many who have MSMEs. In these savings clubs, a group of people agree to pay a certain amount at the end of each period (daily, weekly or monthly) based on needs. The total amount collected is given to an individual to expand his or her business or to take care of other basic needs. In all the visited communities, women mostly involved in drying and selling fish, as well as those involved in petty trade, find daily "Susu" clubs to be an effective way to save towards better opportunities (either expanding businesses or investment).

Gender-Based Violence

Rates of sexual and gender-based violence (SGBV), harmful practices (HPs), female genital mutilation (FGM), child marriage and teenage pregnancy are all high, while access to sexual and reproductive health rights (SRHR) is low.

Women and girls are the main victims of harmful traditional practices affecting their health, often to the point of permanent physical, psychological and emotional damages, and sometimes even death. Some of these include boy to girl preference, nutritional taboos, early marriages, Female Genital Mutilation (FGM) and Gender-Based Violence (GBV). Health facilities remain insufficiently equipped to deal with GBV, and inaccessible to most survivors. Traumatized women and children who are survivors of different forms of GBV are confronted with the challenge of inadequate access to appropriate counselling and psycho-social support. The prevalence of GBV including the harmful traditional practice such as polygamy has serious repercussions on women and girls and is also often a cause and consequence of HIV among women and girls.

The Liberia EU/UN Spotlight Initiative[49] provides a good situational analysis of the SGBV and Harmful traditional practices in Liberia with regards to the Legislative and Policy Framework, the capacity of related Institutions, Prevention and Social Norms, the current response and the inaccuracy and insufficient data about the nature and the extent of GBV. These are summarized in the in the following sections. While Liberia has ratified key international and regional instruments[50] that seek to address VAWG, implementation is weak or non-existent and significant legislative and regulatory gaps remain. For example, the Children's Law of is not enforced nor operational; the amendment of the Domestic Violence Law failed to ban FGM despite intense debate; and even though the then President issued Executive Order No. 92 prohibiting FGM in January 2018, the practice remains widespread. While the Penal Code criminalizes rape, there is no mention of marital rape. In addition, the sexual and

[48] Liberia Institute of Statistics and Geo-Information Services (LISGIS) - Ministry of Health and Social Welfare (MOHSW). Liberia Demographic and Health Survey 2013. http://microdata.worldbank.org/index.php/catalog/2165
[49] Liberia EU/UN Spotlight Initiative, Programme against SGBV and Harmful practices, 2018
[50] International legal instruments including CEDAW (1979), 1325 NAP (2009), Convention of the Rights of a child (1990); International Convention on the rights of Persons with Disabilities (2006): The Protocol on the Statute of the African Court of Justice and Human Rights, the Kampala Convention.

reproductive rights of minority groups such as sex workers and those who identify as LGBTIQ are not legally recognized.

There is also an urgent need to address the structural gaps in laws and policies through legislative reforms, amendments and repeal of obsolete laws. There is also a need for expeditious enactment of pending bills, such as the Domestic Violence Bill, to ensure effective protection of women's human rights.[51] Some of the major ongoing legal and policy interventions include the anticipated amendment and enactment of the Domestic Violence Bill; implementation of the Law Reform Policy; revision of the Public Health Law and National Family Planning Strategy and the Reproductive, Maternal, Newborn, Child and Adolescent Health[52] Policy; and planned revision of the 2011 National Youth Policy. In addition to the gaps in legislation and regulatory frameworks there is little or no political will and commitment, the lack of public confidence in statutory institutions such as the judiciary and the police leads to underreporting of cases. For example, of the 1,511 cases of rape reported (to all sources, not only police) in 2014/15, only 2% resulted in conviction of the perpetrators.[53] The lack of trust in government structures creates strong social pressure to resolve cases outside formal justice systems, especially when the perpetrator is a family member of the victim or is a wealthy, powerful and influential member of the community.

Current institutional capacities and mechanisms largely remain weak and incapable of efficiently implementing multisectoral programming approaches to plan, implement and monitor interventions to prevent and respond to SGBV, HPs and SRHR-related issues at both national and subnational levels. The most significant obstacle is the absence of adequate institutional budgetary allocation to meet the gender-specific needs of women and men, boys and girls. As a result, there is considerable public mistrust in the delivery of appropriate services to effectively address the structural causes of the high incidence of SGBV and related problems, fostering a culture of impunity.

The Ministry of Gender, Children and Social Protection (MoGCSP) has not received enough funding to prevent and respond to SGBV and HPs and promote realization of SRHR. Other relevant line ministries have also not prioritized any strategic interventions in their sector plans nor allocated budget to complement efforts by MGCSP.

There are several sociocultural factors that pose a challenge to the eradication of GBV in Liberia. In addition to patriarchal norms that maintain the low social status of women and girls, the legacy of the Liberian civil wars – which were characterized by extremely high levels of sexual and other forms of violence against women, with rape being used as a weapon of war – remains. As a result, the nature and level of violence has, to an extent, been 'normalized' in the now post-conflict circumstances,[54] with a cultural and societal acceptance of violence against women and children, especially girls.[55] Data from the most recent Demographic and Health Survey (DHS) in 201356 show, for instance, that 43% of women and 24% of men agree that husbands are justified in beating their wives if they burn the food, neglect the children, go out without telling them or refuse sex.

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[51] Domestic Violence Bill including the FGM Clause; Amendment of the Alien & Nationality Law;

[52] 8https://www.dropbox.com/s/i9zpriiujpj5w2u/Liberia%20RMNCAH%20INVESTMENT%20CASE%202016%20-%202020.pdf?dl=0
[53] MoGCSP Gender-Based Violence Annual Statistical Report of 2015.
[54] Bernath, 2014, Evaluation of the Norwegian Refugee Council's GBV Programme 2009-2014
[55] Evaluation of the Norwegian Refugee Council's GBV Programme 2009-2014.
[56] Demographic and Health Survey (2013). Available at: https://dhsprogram.com/pubs/pdf/SR214/SR214.pdf.

Child marriage, early initiation of sexual activity and teenage pregnancy are common in Liberia: according to the DHS 2013, 24% of women age 25–49 had sexual intercourse by the age of 15 and 78% by age 18, while 37% of women have had their first child by age 18. Statistics reveal that most girls were coerced into their first sexual experience. Rates of teenage pregnancy are also much higher in rural communities than in urban areas. Because there is a strong correlation between child marriage and SGBV occurrence, these young women are at greater risk of impacts including reduced access to SRHR, maternal mortality and morbidity, and health issues such as fistula and uterine prolapse due to early pregnancy.

The most recent UNICEF data from 2008–2011 suggest that 44% of women and girls aged 15–49 are members of a society called Sande.[57] The Sande society is led by traditional leaders called Zoes. The Sande society is the trusted custodian of 'culture' in much of Liberia and has been present in the region for centuries. This society is traditionally believed to inculcate values and teach skills conducive to communal harmony and to prepare children for the rigors of adulthood. It also has a spiritual dimension, though it is not considered to be a religious institution as such, and most Sande and Poro (the male equivalent of Sande) members are also adherents of Christianity or Islam.[58] In order to join the Sande society, which represents a rite of passage towards adulthood, women and girls need to undergo several initiation rituals, the most significant of which is FGM. Most girls who join the Sande society are taken out of normal school and sent to 'bush schools', compromising fulfilment of their right to education.

Regional and national consultations revealed that FGM is not the only HP in Liberia infringing on the rights of women and girls. Other HPs mentioned include forced confession by women to admit crimes they did not commit or else face severe consequences; accusing women of being witches and punishing women for any perceived wrong by undressing them in public.

Moreover, groups that are traditionally left behind[59] are discriminated against, oppressed, stigmatized and abused; and live in constant fear of being attacked and beaten because of their condition, disability and/or sexual orientation. In addition, people with disability, including women and girls, are more at risk of violence in general, including SGBV. The attempts of CSOs and women's groups to advocate for capacity development and institutional change have yet to bear fruit and result in action, especially prevention. Members of the LGBTIQ community have limited access to basic rights, services, social freedom, social justice, social economic empowerment and access to health services; and suffer from denial of their fundamental rights to expression, equality and association. have undergone FGM in Liberia.[60]

Most cases of VAWG are reported by health facilities, NGOs and Liberia National Police (LNP)/Woman and Child Protection Section (WACPS). Of all cases reported by the GBV Unit of MGCSP in 2017, medical treatment was provided to 56.9% of survivors. The most frequent services provided to victims include treatment for sexually transmitted infections, post-exposure prophylaxis, tetanus toxoid vaccination, emergency contraceptives and psychosocial counselling. There is no specialized assistance nor services that are provided to women and girls with disability, refugees or displaced.

[57] Mgbako (2011), Penetrating the Silence in Sierra Leone: A Blueprint for the Eradication of Female Genital Mutilation.
[58] December 2015 report on traditional practices in Liberia, published by United Nations Mission in Liberia (UNMIL).
[59] Lesbian, Gay, Bisexual, Trans and Intersex Persons, women and girls and men and boys living with disabilities,
[60] Recalculation of DHS figure of 66% by UNICEF Division of Data, Research and Policy.

Gender and climate change vulnerability

Availability of data concerning the differential impact of climate change on women and men in Liberia is limited. A Gender and Environment Index (GEI) case study of Liberia[61] observes in 2015 that Liberia has made great progress in recognizing women's role in environmental decision making, as evidenced by numbers of women in national level leadership positions and in civil society. However, discriminatory social norms and practices continue to constrain women's participation and representation. There is need for greater gender responsiveness in policy, and in the translation of policy into action, e.g. improved access to training, education and livelihood options for women as well as engaging men in the pursuit of gender equality. These sorts of actions support women's greater decision-making and strengthen their resilience on climate change issues.

A recent study on "Gender and Social impact of Climate change"[62] in the sectors of Forestry, Agriculture and fisheries has demonstrated that climate change affects women and men differently with women being affected mostly. The study has shown that even though both men and women rely on natural resources for their survival in those sectors, there is an existing wide disparity of the impact of climate change and variation in the adaptative capacities of men and women. Women rely heavily on natural resources for their livelihood, the subsistence of their families, communities, education of their children, health even for their self-empowerment.

Coastal erosion is causing the destruction of infrastructure, loss of property, displacement of people, decrease of income, rise of insecurity and of gender-based violence – all of which increase the vulnerability of women in Liberia and exacerbate the challenges they are already facing.

Nevertheless, as small-scale entrepreneurs, informal petty traders, caretakers in families, breadwinners in households, educators in communities, women play important roles by constantly building resilient households and communities.

In the same manner as the former Agenda for Transformation[63] of 2017, the new five-year national development plan called the Pro Poor Agenda for Prosperity and Development (PAPD) charts a path towards accelerated, inclusive, and sustainable development for the Republic of Liberia. Under its Pillar I, "Power to the people" it also intends to improve the socio-economic and political status and capacity of women in Liberia, with the strategic objectives of improving women's capacity to respond to gender-based violence and harmful traditional practices; increase women's participation in the community decision-making process; and strengthen women's participation in income generation and employment opportunities in agriculture, fisheries, Micro, Small and Medium Enterprises (MSMEs), and the formal sector. The PAPD pillar "Economy and Jobs" (whose goal is to advance the macroeconomic environment enabling private sector-led economic growth, greater competitiveness, and diversification of the economy) pays a special attention to environmental issues and the management of natural resources. However, to make these goals a reality, the Government needs to highlight the different gender issues faced by women and men in vulnerable communities to enable transformative strategic interventions.

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[61] IUCN (2015). Women in environmental decision-making in Liberia: An environment and gender index (EGI) country case study http://www.conservation.org/publications/Documents/CI_Women-in-Environmental-Decision-Making-in-Liberia.pdf
[62] Gender and Social impact of Climate change , Liberia NAP, UNDP/EPA 2018
[63] Republic of Liberia. Agenda for Transformation: Steps Towards Liberia RISING 2030, 2017 http://allafrica.com/download/resource/main/main/idatcs/00080846:1b0f1d46c3e2c30c51658f7f64dcb7b9.pdf

In the meantime, women and men have taken some empirical and pre-emptive actions to reduce the vulnerability and build the resilience of their households and communities against sea level rise. Most of the visited coastal communities that have been in an area for a long time have typically established coping strategies for variable climate and extreme weather. This proposal, in identifying and incorporating gender responsive interventions, seeks to also contribute to social transformation and social justice by reducing economic inequities and existing discriminatory attitudes and practices.

Other kinds of vulnerabilities

During the field visits and consultations for this proposal, children were identified as a vulnerable group. There is a rampant evidence of child labor, children street sellers, beggars and many school dropouts. Girls face more challenges especially within the family: they are at a higher risk of transactional sex, sexual and gender-based violence and forced early marriages. This explains the phenomenon of teen-age pregnancies as assessed above.

Persons with Disabilities (PWDs) face discrimination in education and lack of user-friendly facilities and services; as well as in the job market as a result of social biases and stereotypes associated with disability. Women and children with disabilities suffer double discrimination – first as women and children, second as persons with disabilities and therefore they need special attention. The National Action Plan for Inclusion of Persons with Disabilities in Liberia (2018 – 2022) stresses the need to promote friendly environment and inclusive education for PWDs including promoting their full participation in national development of the country is critical for ensuring growth with equity. It is an expression or elaboration of the UN Convention on the Rights of Persons with Disabilities in practical terms as reflected in the call for PWDs to have i) public accessibility, ii) inclusive education; c) access to employment and livelihood, d) health care, e) independent living and self-determination, and finally f) access to justice and social protection. The National Adaptation Plan (NAP) in Liberia recognizes the special needs of women with disabilities who are further discriminated against because they are women calls for special attention to be given to them.

People Living with HIV and AIDS are also another vulnerable group. Stigma of HIV in communities exists as there is still insufficient understanding of issues related to HIV and AIDS and family members often abandon when those living with HIV/AIDS need their support most. Women particularly bear the brunt of the burden of the disease and require a multi-level support although presently the country has limited “safety nets” to assist and support them. There is a very small number of adolescent-friendly reproductive health services which are sparsely located in just few areas in Monrovia. This acute shortage imposes a high risk of Sexually Transmitted Infections (STIs) and HIV on girls. They are easily also exposed to the risk of unwanted pregnancies which compound the challenges. Fear of negative effects, stigma and discrimination prevent many women from sharing their personal experiences, which has serious implications in accessing treatment and controlling HIV infections.

III - Legal and Administrative Framework to address Gender inequality in Liberia

The Constitution guarantees all persons, regardless of sex, the enjoyment of fundamental rights and freedom. Although there is no official definition of discrimination and discriminatory practices, Liberia is committed to the promotion of the rights of women through various national laws and policies.

Significant progress has been made in passing major legislations in ensuring gender equality and women's empowerment. These include the Domestic Relations Law; the Inheritance Act of 1998 (which specifies Equal Rights in marriage and inheritance under Customary and Statutory Laws); the Rape Law of 2005 (which outlaws gang rape and stipulates life-term sentence for aggressive forms of rape); and the Anti-Human Trafficking Act of 2005, prohibiting trafficking in persons, although it is not explicit on trafficking of women and children for sexual purposes. All these policies and legislations affirm the Government's commitment to address gender inequality and to ensure that women are fully engaged in activities that are of benefit to them and to the nation.

Gender and women empowerment policies

The National Gender Policy (NGP) of 2017 is drafted in line with the Constitution of Liberia which is the supreme law of the land and takes precedence in establishing a framework for the Government to promote unity, liberty, stability, equality, justice and human rights with opportunities for social, economic and political advancement of the whole society, irrespective of gender.

The policy underpins some challenges for the implementation of these laws: one of its strategic actions for 2018-2022 is to "Develop new or review existing laws to eliminate gender insensitive provisions and to harmonize customary and civil laws in conformity with international human rights standards."[64] In line with the NGP, the current 5 years general development plan – the 2018-2023 Pro-poor Agenda for Prosperity and Development - provides to harmonize the 2 systems of law in its short term and mid-term interventions.[65] Customary Law applies mainly to issues of marriage and inheritance and is generally blamed for certain harmful practices, including early marriage and Female Genital Mutilation (FGM). Although the statutory laws prohibit discriminatory practices, they make no specific provisions for protection against discrimination in the private or domestic spheres.

Even where some laws may exist (such as the Penal Code for Rape that has a provision for extensive prison term) women are still challenged with limited access to justice, particularly in the counties and rural areas. Inadequate poor court infrastructure in counties and rural areas, which is further hindered by the destruction of courts during the conflict, inadequate staffing both in law enforcement and adjudication, inadequate capacity of the existing justice system to process caseloads, as well as limited knowledge of rights and negative attitudes of law enforcers. The demand side of justice also needs to be capacitated to understand the law, know their rights, and be positioned to claim and exercise those rights. There is a pressing need for a comprehensive law review and a plan for reform and drafting of new laws. Such efforts could include, among others, the enactment of equality laws to address some existing gaps, review of existing laws and the removal of any discriminatory and contradictory laws in the protection of women's fundamental freedom and rights as enshrined in the Constitution.

[64] Revised National Gender Policy (2018-2022), Ministry of Gender, Children and Social Protection, 2017, Liberia.
[65] Republic of Liberia, '2018-2023 Pro-Poor Agenda for Prosperity and Development'(PAPD) - A five-year National Development Plan towards accelerated, inclusive, and sustainable development, September 2018, pp.108-110.

Liberia is also a party to various international instruments on the promotion of gender equality and women's empowerment. At the global level, treaties, declarations commitments applicable to Liberia include:

- Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW) 1979;
- Optional Protocol on CEDAW;
- Convention on the Rights of Child (CRC) (1990);
- Optional Protocol to the Convention of the Rights of the Child (CRC);
- Convention on the Rights of Persons with Disabilities (2006);
- International Covenant on Civil and Political Rights (1966);
- International Covenant on Economic Social and Cultural Rights (1966);
- UN Security Council Resolution 1325;
- UN Security Council Resolution 1820;
- UN Security Council Resolution 1612 (Children and Armed Conflict);
- Beijing Declaration and Platform for Action (1995);
- International Conference on Population and Development (1994);
- United Nations Declaration on Violence Against Women (1993);
- Millennium Declaration and MDGs (2000);
- Universal Declaration on Human Rights (1948);
- Vienna Declaration and the Plan of Action (1993);
- Sustainable Development Goals 2030.

At the regional level, Liberia is a party to the following instruments (See Annex 2 for detailed description of the commitments and provisions):

- The African Charter on Human and People's Rights on the Rights of Women in Africa
- AU Women's Decade
- African Charter on the Rights and Welfare of the Child
- New Partnership for African Development (NEPAD)
- Solemn Declaration on Gender Equality in Africa (2004)
- The Economic Community of West African States (ECOWAS) Gender Policy
- The Mano River Declaration,
- Maputo Declaration and the Beijing +10 Commitment.

The MoGCSP is the national machinery for promoting gender equality, women's advancement and children's welfare in Liberia. The Ministry also supports the policy and legal landscape to:

- Advise the Government on all matters affecting the development and welfare of women and children;
- Coordinate the Government's gender mainstreaming efforts to ensure that both women's and men's perspectives are central to policy formulation, legislation, resource allocation, planning and outcomes of policies and programs, focusing on gender equality, empowerment of women and development of children; and

- Monitor and report back the impact of national policies and programs on women and children as well as recommend appropriate measures to be taken in mobilizing and integrating women as equal partners with men in the economic, social, political, and cultural development of the country.

The NGP will pay attention to domestication of national, regional, and global development frameworks. The NGP will also ensure that the gender considerations promoted in the international gender equality and women's empowerment instruments are mainstreamed into all sectoral programs. For example, the overarching principle of leave no one behind of the SDGs will be followed through in the same way as the requirements of CEDAW.

On its page 35, the revised policy highlights climate change and degradation of the environment and the grave social and gender implications of ecosystem breakdown should there be a lack of rapidly and adequate action. The gender policy views women as major players.

Liberia Climate Change and gender Action Plan (ccGAP)

In 2012, Liberia put in place a climate change Gender Action Plan (GAP) that is due for revision. Its aim is to ensure that gender equality, is mainstreamed into climate change policies, programmes, and interventions agriculture, coasts, forestry, health, water and sanitation and energy sectors that they are gaps to be filled. It stresses the importance of the fact that it is of upmost importance to deeply understand and analyze men and women's needs, priorities and challenges regarding climate change, and subsequently, draw interventions that are really tailored to both groups.

Consultations with men and women have shown that around 70% of the population living in the identified six hotpots along the coastline of MMA, without any further actions, sea level rise will affect a significant number them and will have a significant impact on their lives and livelihoods especially those identified as the poorest and the most vulnerable groups.

The National Environmental Policy and Response Strategy on Climate change acknowledges the vital role that women play in conservation and the sustainable management of the environment. and aims to ensure their participation in decision making on the implementation of the environmental policy and its activities. It further states that gender mainstreaming shall form an integral part of the basic training, social development, environmental and natural resource management.[66]

The National Policy for Disaster Management

The Government of Liberia has recognized the importance of having measures for the prevention of disasters, or the mitigation of their effects, for preparedness and capacity building to effectively respond to any threatening disaster, and roles and responsibilities of ministries and agencies. To that

[66] The National Environmental Policy and Response Strategy on Climate change, EPA, 2018

effect, an Act establishing the National Disaster Management Agency and a The National Policy for Disaster Management was developed.[67] The policy establishes an institutional and operational framework that will drive successful implementation. Because DRR is multidisciplinary and multi-dimensional, all our development and humanitarian partners are called upon to work with Government in building national and community resilience to disasters within the context of sustainable development. The National Disaster Management Agency (NDMA) is currently developing a National Disaster Management Strategy for Disaster Risk Management in Liberia as well as a Guide for Emergency Response and Preparedness. Gender equality is a strong planning principle.

This policy recognizes the important role played by women in development and the burden they carry and suffer during disasters. Consequently, all activities implemented by the government and all its partners before, during and after disasters, will proactively and consciously include participation of women and other vulnerable groups namely women, children, People Living With HIV/AIDS, the elderly, disabled and other vulnerable people at national, county, district and community/chiefdom levels. . The policy further stresses that all DRR plans including contingency/preparedness plans shall have a gender strategy for emergency response informed by differentiated needs.

The proposed actions are:

- Strengthen Women's Security in Crisis:
- Expand Women's Participation, representation and Leadership and Build their skills and confidence individually and within networks.
- Promote Gender Equality in DRR by Incorporate gender analysis in the assessment of disaster risks, impacts and needs.
- Ensure Gender-Responsive Recovery by providing equal economic opportunities for women including access to assets, such as land and credit, social protection and sustainable livelihoods, transportation, shelter and health care during and after disasters.
- Develop Capacities and skills of men and women for Social Change before, during and after disasters reduce vulnerability to natural hazards, achieve equitable post-disaster reconstruction, build social cohesion and ban norms that are against.

IV – Lessons learned from climate resilience and gender programmes in Liberia

Liberia is still in the process of investing significant human and financial resources to both mitigation and adaptation priorities and continues to improve in programmatic design and practice for climate change. In parallel, the New Revised Gender and PWD policies to tackle gender inequality and social exclusion are led by the Ministry of Gender, Children and Social Protection (MoGCSP), which are relevant to addressing the needs and priorities of marginalized communities in the country. These policies and its related programmes still need to integrate climate change adaptation programmes for the country's most vulnerable.

At the same time, financing is still to be mobilized to address the climate change adaptation needs of vulnerable households, and the existing one are more oriented in financing infrastructures development and research. There is also an urgent need for better coordination of stakeholders and

[67] The National Disaster Management Agency, 2012

integration of efforts addressing multiple facets of resilience and many social factors. The MMCRP seeks to alleviate gender inequality, as well as focus on building climate resilience, women's economic empowerment, improvement of gender norms and relationship at the family and community levels. In order to achieve a transformative impact, the interventions of this project must carefully incorporate these elements and lessons learned that consultations with officials of key entities, and stakeholders' consultations have brought forward:

- Capacity building of community members (men, women, youth) living in in the MMA hotspots is critical to minimize risks.
- Motivation and awareness raising activities should continue for project ownership and success. Communities should understand that capacity building and abstaining from developing aggravating factors, such as sand mining should be the prime objective of adaptation and disaster management strategies, rather than relief and response.
- Capacity building of the institutions working on climate change adaptation should continue
- The generation of knowledge, knowhow, resources (material and money through savings, emergency funds and emergency plans and emergency response units) at the community level, and maintenance of community assets are essential to address immediate emergency needs.
- Indeed preparedness, the capacity to respond in emergencies by communities themselves should be promoted because as early response from external actors during disasters is most of the time, inadequate or slow, and their strategies are not sustainable.
- Partnerships and coordination between Government Ministries, as well as between Government agencies, Non-Governmental Organizations, local authorities and community leaders should be reinforced.
- Ensuring the participation of women, girls, youth, religious leaders, opinion leaders, minorities, and persons with disabilities in planning and implementation of climate resilience is critical.
- Adaptive economic opportunities for year-round employment and earning opportunities for community men and women, especially for the youth, are critical for their awareness and livelihood adaptability.
- Livelihood support to vulnerable women, especially widows, single mothers, women heads of households and their family's needs is necessary to be followed up to ensure retention of assets and their productive use
- The social issues discussed during stakeholders' consultations also form development challenges, this project it is critical to support initiatives that will improve the social well-being.
- From a gender perspective, the following recommendations could be made:
- Ensure women's voices as well as men's voices are heard at all stages of adaptation and disaster management (from preparedness to recovery) planning.
- Increase access to training for economic activities, which are flood/drought/salinity resilient (as appropriate in the area).
- Maximize the diversification of livelihoods options to increase economic security.
- Promote equal access for women and men to the benefits and opportunities of the project.
- Increase training in financial skills to plan for crisis and maximize marketing of goods before crisis.

- Ensure women's safety in shelters, latrines and public spaces.
- Ensure access to women's specific and gender sensitive accessible medical care.
- Champion men in the communities and households to reduce women's burden of unpaid work and assist in some domestic chores
- Ensure new livelihoods for women do not involve a huge increase in time unless other household members take on other time burdens for responsibilities women currently hold.

It has been proved that providing livelihood resilience support to poor women

- brings positive impact on their status within the household and in the community at large,
- helped to develop their confidence to move around the community and network with other women,
- yields greater levels of respect for women in the community, and
- selection brings pride for women within their family and improves intra-household relationships as well as community harmony.

Therefore, in such a complex setting, the main challenge in this project design is to holistically address the different needs challenges and aspirations of women and men and capture them in the action plan.

V - Recommendations

5.1 – Gender mainstreaming

In preparation of this project, UNDP conducted stakeholder consultations with a wide range of men and women in April 2019. These consultations were held in 22 coastal communities in MMA and the focus group discussions are included as part of the feasibility study. One of the key aspects of the feasibility study was to better understand the current gender relationships at the household and community levels. the gendered vulnerabilities to coastal hazards. This has clearly demonstrated that addressing gender inequality at the household and community levels is essential for building resilience. A compilation of the results of the 6 focus group discussion (FGD) findings is form part of an additional annex to this proposal.

The different roles, priorities, aspirations and needs of men and women, and the ways in which both perspectives are valuable must be recognize and effectively utilized these gender-in designing the project interventions. It is also important to recognize the invaluable role that women can pay in resource management and tap on their strengths and capabilities, to shift the perception from "victims" to agents of change. By so doing, gender dimensions will be integrated for gender responsiveness in implementation and to obtain gender transformative results.

The gender analysis, through document review, stakeholder engagement, and consultation showed the need for:

- Engagement for the different stakeholders to consider gender Equality and women empowerment as a factor for success,
- Assessment of gender relations, roles and responsibilities, resource use and management, decision making and power relations in targeted communities in order to have relevant gender-

related project activities, capable to respond to the growing threats of coastal erosion and extreme climate events,

- Gender mainstreaming the Monrovia Metropolitan Climate Resilient Project (MMCRP) and constantly adopting an inclusive, participatory and right-based approach in its implementation;
- Integration of sex and age disaggregated data and indicators to establish a baseline by which areas of focus are made relevant and progress measured;
- Consideration of both gender and women specific interventions to incorporate into the coasted Gender Action Plan.

5.2 - Project design

The project intends to propose some components and actions to create a paradigm shift that transforms vulnerable communities in the MMA suffering from SLR, to agents of change in climate change adaptation, with greater access to improved services, resources and productive assets.

Men and women are targeted but including women as specific beneficiaries of the interventions, will address challenges of unequal gender relations, gender disparities, inequalities and women empowerment. It will contribute to change of norms and behavior change through gender responsive training. The project also presents many advantages: ensuring equitable access to resources, services and technologies for climate-smart livelihoods, improve coordination and institutional capacity building at the intersection of gender-transformative climate change adaptation. Overall, the strategy and action areas are based on an empowerment approach for women and consistent with the GoL's Gender Strategy and Action plan, aiming to transform gender relations in the target communities.

Furthermore, livelihood interventions supported by the project are not only climate resilient, chosen based on their resistance to increasingly SLR and vulnerability to extreme weather, but also able to provide significant economic gains, as well as the opportunity to better integrate women into existing value chains in the light of changing environmental conditions.

From the gender analysis these are the key approaches that should be accounted for:

1 - Creating an enabling environment for women's advancement: The consultations have identified norms and social constraints which may contribute to more gendered vulnerability in a changing climate. Bringing men and women together in this project at all stages may contribute to shift community attitudes, perceptions and behavior towards women's status and roles, eventually creates an enabling environment for women to thrive better. This will be done through sensitization, partnering with men's collective work, and letting women and girls raise their voice against discrimination and violence.

2 - Increasing women's voice and agency: The project will attempt to improve women's participation and provide more opportunities to their social empowerment. This will be achieved by helping women participate in the identification of needs and gaps in coping with climatic conditions as well as identification of coping mechanisms and solutions for better management of climate impact. By strengthening women's voices and increasing their agency the project can bring extensive resilience dividends - not only for the women themselves, but also for their families and contributing towards the communities' overall wellbeing. It is expected that gender-responsive training and interventions and women and adolescent girls' active engagement will ensure the effectiveness of the programme.

3 - Proactively addressing challenges: If the project intends to positively shift power relations, social norms, and stereotypes it is clear that it will raise resistance from the community. There may be social conflict and perhaps SGBV, especially when women are the primary beneficiaries of the project. In the design of interventions, scrutiny must be observed to select opportunities that would be considered socially suitable for, appropriate, and preferred by women. Strong transformative change could then be achieved if it is accompanied by community sensitization activities in addressing norms and accompanied by advocacy around acceptance of what could be considered as “appropriate” alternative livelihoods and skills for women.

4 - Increasing women’s skills and capacities: This project is designed to improve women’s climate resilience and intends to use a comprehensive adaptation strategy that engages women directly in decision-making, in the management of resources, and in capacity building for the successful pursuit of climate-resilient livelihoods. The project includes components on sustainable alternative livelihoods support and market linkages as well as management of costal defense options. This will be accomplished by developing women’s awareness, technical skills, and resource management capacities regarding various facets of adaptation. This will in turn contribute to the improved health and wellbeing of women and their families, resulting in lower occurrence of diseases and improved food security. Additionally, women will be trained in the management of coastal defence solutions, choice of economic options, contingency planning for livelihoods and disaster management as far as climate resilience is concerned.

5 - Increasing women’s economic opportunities: Promoting women’s capacity and resilience in the context of a changing climate includes diversifying their economic opportunities. This can be accomplished in the vulnerable communities through provision of necessary assets and infrastructure, capital, skills and linkage to markets. By so doing, women will be able to manage critical resources, gain access to assets and increase their savings in the short and long terms.

6 - Beneficiary Selection: In the hotspots of MMA, the project targets the most vulnerable beneficiaries, particularly women and, where appropriate, adolescent girls. These are the groups who are particularly affected by the loss of productivity or livelihoods due to sea level rise and other observed and projected climate impacts. Among these hotspots, the beneficiary selection was refined through consultations with targeted communities and local government institutions, and households that face intersectional marginalization were identified. These include the poorest households with the least capital and assets to pursue adaptive livelihoods, female-headed households, households headed by adolescent girls married early and are solely responsible for household income, households with disabilities or chronic illness, and households with other marginalized groups. Orientation meetings in the project areas on its objectives, importance and role of women and marginalized groups in community-based climate change adaptation will ensure clarity and transparency. In this process a tiered grievance mechanism will be established and available to target communities, to signal issues with beneficiary selection if they arise.

6 - Establishment of gender-responsive grievance mechanism: The purpose of a grievance mechanism is to minimize possible conflict and to address it, if it occurs. For the implementation of this project, a robust, gender-sensitive grievance mechanism will be put in place which will allow women beneficiaries to report any incidences of social conflict, discrimination or possible increase of GBV arising from their involvement in project activities. The grievance mechanisms facilitate demand-driven accountability of social protection programmes and call for the civil society organizations to play a clear and distinctive role for the communities' benefit. The GRM is described in detail in Annex 6 of this project - Environmental and Social Management Framework.

Additionally, there are pre-existing mechanisms regarding GBV in the MMA, including a Gender Unit in local police stations and a referral path mechanism to UN Women, UNFPA, and UNICEF. Additionally, the Spotlight initiative, sponsored by UN and EU, is focused on addressing GBV. There focus includes institutional work and providing support to survivors.

5.3 - Project implementation

For the project implementation, the collaborative relationship between UNDP, EPA, NDMA, Ministry of Gender and well-screened local NGOs selected through a rigorous process is necessary. This will create, under the supervision of gender professionals, valuable experience, country ownership at national and community levels and the cross-sectoral technical perspectives required to ensure the success of this approach.

5.4 - Effective project management

For the operationalization of these recommendations, there is need to include a Gender Advisor in the Project Support Unit to provide advice, quality assurance oversight, capacity building and monitor the right implementation of gender related activities. Other elements to take into consideration are:

- At least 20% of project management unit staff will be women involved in effective management and implementation.
- The Project Steering Committee and other management bodies will comprise at least 30% women. Gender Officers will be international P3 or P4.
- Initial gender-specific workshops will be conducted to introduce Gender Action Plan to stakeholders at national and community / local level.
- The project management unit will monitor the impact of the project on women and adolescents' girls, reporting results through collection and analysis of sex and age disaggregated data as baseline and this on a regular basis.
- Gender sensitive trainings will be conducted for all management and implementation staff.

5.5 - Monitoring and evaluation

Through onset analysis, data have been collated to establish a baseline. These data shall be monitored against throughout implementation and evaluation. The analysis identified the differences between men and women within at-risk populations. Also, as during the feasibility study, during project implementation, qualitative assessments will be conducted on the gender-specific benefits that can be directly associated to the project. This will be incorporated in the annual Project Implementation Report, Mid-Term Report, and Terminal Evaluation. In order to monitor and evaluate progress of the project, the quantitative and qualitative indicators could be used and measured as such:

Quantitative outcomes:

- Number of men and women employed from the jobs created by the project.
- Number of men and women benefiting from training opportunities and awareness raising sessions.

- Number of men and women benefiting from knowledge management and information dissemination.
- Number of men female-headed households and other vulnerable groups as beneficiaries.
- Number of men and women benefiting from improved or alternative livelihoods.
- Number of business development services component targeting coastal women entrepreneur groups.
- Number of opportunities to generate additional income to all affected communities.
- Number of incentives given to women to respond to address their family's basic needs, such as better health and nutrition, linking to response to climate induced disasters (sea erosion) and sea level rise.
- Number of training and educational activities related to climate change, adaptative strategies, fish conservation, entrepreneurship skills.

N.B.: These figures will be disaggregated by sex and age.

Qualitative outcomes:

- Satisfaction for inclusive management of the coastal defence options.
- Existence of capacities at national, local and community level for ICZM.
- Existence of efficient coordination mechanisms for ICZM both at national, local and community levels.
- Perception on increase of income for men and women in the communities.
- Perception of improved wellbeing and health in the families and communities.
- Level of appreciation of improved self-esteem, empowerment and security of women in the communities as a result of training opportunities for women in decision-making and leadership.
- Increase of women involvement and participation of women in public and project decision-making as a result of initiation of women economic empowerment.
- Level of awareness-raising and improved perception of gender equality.
- Degree of change of social norms in favor of women and girls' empowerment.

VI – Proposed Gender Action Plan

The purpose of a Gender Action Plan, tailored to the Liberia Monrovia Metropolitan Area (MMA) context, is to operationalize the constraints and opportunities for women and men that were identified during the gender analysis at national and at MMA community level, towards fully integrating them into the project design, providing the framework for a gender-responsive and socially inclusive project.

This section is an overview of some proposed interventions that seek to address the gender-differentiated vulnerabilities and resilience in the coastal communities in the Monrovia Metropolitan Area (MMA) to ensure a gender-responsive project.

For each of the activities of the project an amount to support gender specific interventions has been allocated, as a subset of the overall project budget. **In all UNDP and GCF projects, at least 15 % of the total amount of the budget should be dedicated to gender equality and women empowerment.**

These are the approaches that are recommended to make the project a gender-responsive:

Overall

- Awareness raising of all stakeholders on sea level rise issues for the success of the MMCRP and institutional support,
- Encouraging gender-responsive opportunities for improved livelihoods,
- Increase of women's participation in decision-making in all the ICZM institutions
- Consideration of differentiated needs of women and other marginalized groups in building climate resilience,
- Filling the gap of the absence of sex and age disaggregated data,
- Organization of dialogue with all stakeholders on aspects of climate resilience adaptation, social inclusion and gender equality,
- Devoting gender technical expertise for guidance in implementation and monitoring of gender results to hold individuals and institutions accountable for gender equality related outcomes: this means inclusion or 01 or 02 Gender positions in the PMU team.

1 - For institutional strengthening and awareness raising of all stakeholders on ICZM:

- Improve institutional capacity, coordination, accountability and oversight (capacity of project executing and implementing agencies in addressing gender issues in climate change programming).
- Developing the capacity of all stakeholders (institutions and communities) on inclusive, participatory and sustainable development and integrated coastal zone management.
- Raise awareness of women and youth in the existence and use of early warning weather systems.
- Finding ways of empowering communities for the sustainable management of integrated coastal zone.
- Build on the projects, structures and initiatives being rolled out by the GoL and other development partners, in order to maximize the use of resources, and for greatest efficiency and effectiveness.
- Assess how gender is currently being mainstreamed in differing Ministries and sectors, to most effectively develop needs assessments, enable planning, and be effective in monitoring and evaluation.

2 - Awareness raising on gender equality and women empowerment issues

The following form the gender mainstreaming components of the project under this awareness-raising approach:

- Organise gender trainings for all relevant stakeholders.
- Improve active participation, equitable representation, and voice (being heard) of women, youth, people living with disabilities, as well as men in all areas of decision-making related to building more resilient coastal communities (including technical, economic, social, accessibility etc.).
- Engage men and boys in community discussions to develop “gender champions” who can support efforts to address gender-based violence in the household and communities (including workplaces, schools, etc.) and who can act as peer mentors for others in the community.
- Support women’s access to literacy, services, information, finances, and enjoyment of benefits from the eco-stove production included in the project equally as men (with men, boys understanding value of sharing tasks such as childcare, etc.).

3 – Working towards a change of norms and of negative perceptions on gender

Discussing climate change related issues with communities is a good opportunity to:

- Hold dialogues with community heads, religious leaders and female activists to identify the norms and harmful traditional attitudes/practices and barriers that restrict women’s participation with the purpose to establish an environment conducive to participation (committees involving men and women, girls and boys).
- Raise awareness on responsible parenthood and shared responsibilities in households’ activities, which encourage family members to help reduce the mothers and girls’ unpaid burden of work, and childcare.
- Promote positive social norms (community awareness and orientation on role of women and girls in general and in climate adaptation in particular) through training sessions and separate community theatre-based activities.
- Engage the communities in activities promoting safety, security for women and fight against SGBV.

4 - For Targeting and selection of beneficiaries

- Women head of households are the primary beneficiaries.
- Targeting other groups suffering other intersectional marginalization will be a priority.
- Ensuring women have proposed livelihoods options that relieve the burdens and reduce the time of unpaid care work is a priority.

5 - Targeted initiatives and trainings for women’s economic empowerment.

This can be achieved via the following:

- Analysis of the gendered division of labor (gender-differentiated responsibilities and needs).
- Ensuring that working conditions are gender responsive at the homestead and community levels.
- Ensuring equal pay for equal work, and access to the administrative, technical and managerial positions in the project.
- Promoting advocacy and awareness regarding gender-based violence. This may be done via partnering with the local Gender Unit(s) at the police station(s) and through raising awareness

about the referral path to UN Women, UNFPA, and UNICEF. Additionally, the project can partner with the Spotlight Initiative Project, sponsored by UN and EU, which is focused on addressing GBV through institutional work and providing support to survivors.
- Providing gender ombudsmen for complaints related to sexual harassment and other forms of GBV.
- Improving and providing access to livelihoods and improvement of basic social facilities.
- Developing/adapting financial and other community/business organizations (cooperatives, fisher associations, conservation and rehabilitation committees, etc.) that provide women as well as men with access to productive resources, services, and information (including credit) to maintain that they need to contribute to cleaning, rehabilitating, and conserving the wetlands and mangroves.
- Developing livelihood options that build on the different interests and needs as well as address the different challenges facing women, men and youth in the coastal communities (either same livelihoods they are doing now including fisheries or skilling up for new economic opportunities).
- Ensuring any location for resettlement also includes a hospital or health facility as well as community school and playground for children. These are important to women, men, boys, and girls and to the community.
- Providing women trainers and women-exclusive training sessions, including flexible times, and provision of household training for women-headed households as required.
- Integrating of men and community elders in community trainings that addresses women's participation, rights and norms around equal access to arising opportunities.

6 - For continuous learning:
- Take a continuous learning approach that incorporated the perspectives and experiences of women and refined interventions with the collection of sex-disaggregated data.
- Research on the underlying gender-based social dynamics and preferences related to technology adoption (for both water and livelihoods) to avoid project failure and to ensure that women can benefit from resources/assets made available to them.
- Monitoring of changing power dynamics and attitudes at the intra-household and community levels and monitoring possible increase in conflict as a result of project interventions.

7 - Encouraging gender-responsive opportunities for improved livelihoods

- Improving and providing access to livelihoods and improvement of basic social facilities
- Developing/adapting financial and other community/business organizations (cooperatives, fisher associations, conservation and rehabilitation committees, etc.) that provide women as well as men with access to productive resources, services, and information (including credit) that they need to contribute to cleaning, rehabilitating, and conserving the wetlands and mangroves.
- Developing livelihood options that build on the different interests and needs as well as address the different challenges facing women, men and youth in the coastal communities (either same livelihoods they are doing now including fisheries or skilling up for new economic opportunities).

- Ensuring any location for resettlement also include a hospital or health facility as well as community school and playground for children. These are important to women, men, boys, and girls and to the community.

8 - For ownership of assets:

- Encourage women to have better control over their productive assets (finance, productive skills, and technology) through women led groups.
- Facilitate training in rights, fair working conditions and negotiation.
- Conduct monitoring of productive assets and revenues, to ensure that revenues are kept in the hands of women and targeted beneficiaries.

9 - For market integration:

As women are key players in fisheries, agriculture, and informal sector activities (petty trade), this proposal seeks to address the impacts of climate change on women's livelihoods as well as those of men as well as identify enhanced livelihood opportunities

- Link with financial institutions to inform women on access to finance, financial management training, and market prices for women.
- Provide conservation trainings on alternatives to mangrove fuel wood, given mangrove is fuel wood for drying fish
- Give preference for employment in fishery activities.
- Partner with relevant institutions/ NGOs to propose TOT training opportunities for women in skills they might not otherwise consider.

10 - For gender-responsive grievance mechanism:

- Support the development of gender sensitive (and marginalized group sensitive) grievance mechanism, which allows women easy and unrestricted access, including provision of female GRM focal points in communities.
- Ensure that the Grievance mechanism is accessible to everyone and ensure that the mechanism is full responsive to gender in Liberia. Those operating mechanisms must be gender aware and gender sensitive.
- Ensure that there are defined SOPs for addressing concerns of women and the relevant persons will be trained on these SOPs.The grievance mechanism must be project specific and therefore focused on employment in construction, for example. It must be operational from the beginning of the project.
- Integrate the sex-disaggregated data and gender-sensitive information collected during this proposed project into regular updates of this gender sensitive (and marginalized group sensitive) grievance mechanism to ensure it is as gender-responsive and as accessible as possible.
- Results from the survey included in 1.2.1 of the GAP (to assess all stakeholder's satisfaction with the project) may highlight unsatisfied stakeholders. These unsatisfied persons may submit concerns via the grievance mechanisms, and this should be made explicitly clear such that this feedback is used to appropriately adjust project design and implementation.

The Gender Action Plan (GAP) below is aligned to the activities of the LogFrame of the project application. It suggests actions to arrive at a gender-responsive, inclusive (Leaving No One Behind) and sustainable implementation of the MMCRP projecfic indicators are proposed to measure and track progress on gender responsiveness at relevant activity level. These should be incorporated into the detailed M&E plan which will be developed at the start of implementation. These indicators will contribute to the regular collection of age and sex-disaggregated data and will help to measure gender progress throughout the project's implementation. A table showing the timeline for the activities and milestones in this GAP is provided below. The total budget for GAP activities is US$1,763,210, which is ~7% of the total project budget.

Objective	Sub activities	Indicator	Responsible Institutions	Proposed Budget
Output 1. Protection of coastal communities and structure at West Point against erosion caused by sea-level rise and increasingly frequent high-intensity storms				
Activity 1.1: Prepare construction plan and finalise technical design specifications for coastal defence structure at West Point.	1.1.2 Conduct 1 stakeholder consultations to present the final designs to all relevant stakeholders, including communities, NGOs, development organizations and government.	Baseline: 0 Target: 1 Indicator: Number of stakeholder consultations conducted that include women leaders participation in validating the final technical designs and construction plan for the installation and management of coastal defence solution or engineering options for West Point considering youth, people living with disabilities, etc. Baseline: 0 Target: at least 50% Indicator: Percentage of women (including by age group), youth, elderly and	UNDP, EPA and the MME, MPW, NMDA, and the MoGCSP Monrovia City Council.	Cost included under Activity 1.1 $ **5,000**

		people living with disability participating mobilized in / being represented in community consultations/studies on coastal defences in terms of opportunity to speak, voice heard on coastal defence for West point Baseline: 0 Target: 1 Indicator: Number of meetings with women and men in the communities to understand the potentially different gender-based needs and challenges around coastal defense Baseline: 0 Target: 1 data collection survey Indicator: Data collected on sex and age-disaggregated information about women-run businesses and women's groups in MMA		
Activity 1.2: Construct coastal defence structure to protect West Point against climate change-induced coastal erosion.	1.2.1. Include in the TORs of the construction company the requirements to be in conformity with a socially-sensitive design incorporating the needs of all inhabitants an also conduct a satisfaction survey after Y2.	Baseline: No Target: Yes Indicator: A gender and social responsive final design of the coastal defence structures considering gender	UNDP, EPA and the MME, NMDA, MSD and the MoGCSP Monrovia City Council The contracted construction company	Cost included under Activity 1.2 **$ 7,000**

		needs and constraints for social inclusion Baseline: 0 Target: 1 survey Indicator: A perception survey report on satisfaction of the installation of coastal defence solutions and enjoyment of benefits, disaggregated by sex and age group.		No cost for the survey (done by the company)
	Ensure equitable opportunities for men, women, youth – selected by the communities - to participate in, and benefit from, income-earning opportunities during the construction of the coastal defence infrastructure.	Baseline: 0% Target: at least 40% Indicator(s): Women to constitute at least 40% of waged labor on project-related activities. [68][69] Baseline: 0% Target: at least 40% Indicator(s): Youth[70] constitute at least 40% of waged labour on project-related activities.	UNDP, EPA and the MME, NMDA, MSD and the MoGCSP. Leaders of the 4 areas of the project Work force in targeted communities	No cost

[68] Project-related activities that women may engage in apart engineering, are supervision, security, unskilled labor, catering to feed the workers, etc.
[69] Equal pay for equal work will be required at all times.
[70] Aged 18-24.

		Baseline: 0% Target: at least 30% Indicator(s) At least 30% of waged labour on project-related activities are from female-headed households.		
	Provide training to casual workers employed under Activity 1.2. on safety and security during the construction and operational phases in West Point[71]. (Yr. 3, Yr. 4)	Baseline: 0% Target: at least 40% Indicator: Percentage of women nominated by the communities and trained for casual work Baseline: 0 % Target: 100% Indicator: Percentage of women trainees involved in safety, security, surveillance, maintenance and protection activities	MME, MPW, EPA The contracted construction company. Monrovia City Council Technical team of relevant institutions	Cost included under Activity 1.2 **$ 14,600**
	Provide support for community gender-responsive initiatives that support childcare and domestic duties for women wage labourers to	Baseline: 0 % Target: 100% Indicator: Percentage of women engaged in wage labour supported		Cost included under Activity 1.2 **$ 25,000**

[71] Safety and security training will be included in the employment contracts. Safety monitoring will be part of contractor's work SOPs and safety equipment should be women-specific and women-friendly.

	participate in construction of revetment for duration of the construction.	(in their additional domestic duties, for example) during their project-related labour.		
Output 2: Institutional capacity building and policy support for the implementation of Integrated Coastal Zone Management (ICZM) across Liberia.				
Activity 2.1. Develop an Integrated Coastal Zone Management Plan for Liberia	Activity 2.1.1 Develop ICZM Plan for Monrovia Metropolitan Area and establish an ICZM committee and CSWG to mainstream ICZM across relevant government ministries	Baseline: 3 [72] Target: 5 Indicator: Number of additional policy/programs in relevant sectors with evidence of integrating gender in climate vulnerability assessment and promoting gender equality in climate finance	UNDP, EPA and the MME, NMDA, Monrovia City Council MoGCSP. Technical sub-units in concerned Ministries	Cost already included in Activity 2.1 **$ 379,100**
	2.1.2. Contract a national Gender and Social impact service provider to support the trainer-of-trainers, collect sex and age disaggregated data, coordinate with relevant ministries gender issues in developing the ICZMP	Baseline: 0 Target: 1 Indictor: Evidence brought by the gender and social provider, that the future coastal zone management plan and investment process is gender responsive and socially inclusive	UNDP, EPA and the MME, NMDA, MSD and the MoGCSP.	Cost already included under Activity 2.1 (Cost for one expert: $150,000 /2) **$ 75,000**

[72] Reference is made of the Liberia Gender and Climate Change Strategy, the MoGCSP National Gender Strategy, the EPA National Policy and Response Strategy Gender etc.

	2.1.3 The national Gender and Social impact provider will develop ToT training material, train and mentor staff in MACs and disseminate policies and actions on 'Gender Sensitive Climate Change Action'	Baseline: 0 Target: 1 ToT training material compiled Indicator(s): Evidence of inclusion in the curricula of modules on "Gender and household methodologies" for sustainable Integrated Coastal Zone Management and climate resilience Baseline: 0 Target: at least 30% of MAC staff[73] Indicator: percentage of women staff by sex and age in institutions, including the MoGCSP, that were capacitated and had received information to effectively monitor, implement, coordinate and enforce climate resilient ICZM within their mandate (evidence brought by the national Gender and Social impact service provider)	UNDP, EPA and the MME, NMDA, MSD and the MoGCSP.	Cost already included in Activity 2.1 **$ 4,500**
Activity 2.2. Capacitate the Cross-Sectoral Working group to mainstream	2.2.3 Facilitate the collaboration between the gender and social impact provider and the MoGCSP to develop a TOT training curriculum to include	Baseline: 0 Target: 1		Cost already included in Activity 2.2

[73] This target will be increased if there are more women in the MAC.

ICZM into relevant government sectors through a Training-of-Trainers approach.	social and gender dimensions in addition to technical, administrative and operational elements for training of trainers. .	Indicator: Availability of a multi-disciplinary training curricula for members of CSWG to use across sectors key institutions Baseline: 0 Target: 1 Indicator: Availability of sex and age disaggregated data on trainees in institutions that were capacitated and had received information on how to mainstream ICZM in their daily work	UNDP, EPA, MPW and the MME, NMDA, MSD and the MoGCSP Technical units of relevant entities	**$ 116,360**

2.3. Strengthen the asset base and technical capacity of the ICZMU for the collection of spatial and biophysical coastal information to support the implementation of the ICZMP	2.3.2 Request the contracted specialist firm in the Tors, to organise 2 information sessions on the process of buoy-based data collection and knowledge generation for EWS in the 4 areas	Baseline: 0 Target: 2 information sessions Indicator: Attendance lists with sex and age disaggregated data of technicians that were capacitated and had received information on ICZM and <u>using</u> existing early warning system and weather Baseline: 0 Target: at least 30% Indicator: Percentage of women technicians recorded via attendance lists above.	EPA, NDMA, LMS and ICZMU Technical staff EPA UNDP NGOs	Cost already included
Activity 2.4. Strengthen the existing Environmental Knowledge Management System (EKMS) to act as a platform for awareness-raising and sharing of climate-risk informed ICZM approach	2.4.3. Include in the TORs of the contracted national service provider to develop, print and disseminate gender sensitive and socially inclusive knowledge products on ICZM and the EKMS targeting the 10 implementing ministries for ICZM as well as municipalities	Baseline: 0 % Target: 100 % Indicator: The design and pictures in all posters, flyers and brochures do not foster exclusion, discrimination, gender-bias and stereotypes	UNDP, EPA and the MME, NMDA, MSD and the MoGCSP.	Cost already included in Activity 2.4 **$ 35,000**

Activity 2.5. Conduct an awareness-raising campaign for coastal communities in focus areas on climate change impacts and adaptation practices.	2.5.1 Contract a national media firm with expertise in communications, climate change and gender to design awareness-raising campaigns on climate change impacts and adaptation practices (to be included in the TORs)	Baseline: 40%[74] Target: 80% of targeted populations Indicator: Percentage of population disaggregated by sex and age in coastal communities with awareness in climate change impacts and adaptation practices Baseline: 40% Target: at least 80% of female-headed households in the areas Indicator: Percentage of female-headed households where at least one member of the household has awareness on climate change impacts and adaptation practices	UNDP, EPA and the MME, NMDA, MSD and the MoGCSP.	Cost already included in Activity 2.5 **$ 125,000**
	2.5.2 Host quarterly radio programmes on climate change impacts, vulnerability and gender-responsive adaptation practices in coastal areas.	Baseline: 10 % Target: 04 per year Indicator: Total number of broadcasts that include women's experiences, their LEK and adaptation.[75]	UNDP, EPA Contracted consultant NGOs Local radios	Cost already included in Activity 2.5 **$ 9,600**

[74] Some awareness raising activities are already on-going
[75] Other climate change induced issues such as water salinity, sand mining, sanitation, SGBV, safety and security, etc. could be discussed in the radio talk shows

	2.5.3 Establish, train and employ community-based knowledge sharing groups in each of the 4 communities targeted under the project for 5 years (beginning in year 2)	Baseline: 0 Target 1: at least 50% of trainers are women Target 2: at least 50 % of employees are women Indicator 1: Percentage of trainers in project-supported activities are women. Indicator 2: 8 women employed for the 4 communities.	UNDP, EPA and the MME, NMDA, MSD and the MoGCSP Contracted consultant NGOs.	Cost already included in Activity 2.5 **$ 100,800**
	2.5.5 Host 3 community meetings per year in each of the 4 areas to raise awareness and share information on social and gender dimensions of climate change as well as sustainable and innovative adaption strategies to support ongoing awareness-raising activities, one of which will be on social protection, the rights of women and girls as well as SGBV in addition to climate change vulnerability and adaptive strategies.	Baseline: 0 Target: at least 90 % of populations in the areas Indicator: Percentage of women and men with figures disaggregated by age and sex having participated in sharing their experiences in sustainable and innovative adaption strategies and are aware of the impact of negative gender norms and SGBV	UNDP, EPA and the MME, NMDA and the MoGCSP Contracted consultant NGOs. CSC members.	Cost already included in Activity 2.5 **$ 210,000**
Output 3. Protecting mangroves and strengthening gender- and climate-sensitive livelihoods to build local resilience in Monrovia				

Activity 3.1: Establish a community education and innovation centre to function as a community knowledge generation and learning hub, repository on climate change adaptation practices and to host community activities under Output 3.	3.1.1. Locate, renovate and equip existing structure situated on community or municipal-owned land in West Point to act as education and innovation centres to upskill communities on climate-resilient livelihoods and good adaptation practices.	Baseline: 0 Target: 1 Indicator: 1 operational Education and Innovation Centre that are gender-friendly and accessible by PWD[76]	UNDP, EPA and the MME, NMDA, and the MoGCSP CSC members	Cost already included in Activity 3.1 **$ 200,000**
	3.1.1 Equip the buildings with robust furniture and facilities, considering access for people with disabilities (PWD) and the needs of vulnerable people and women.	Baseline: 0 Target: 1 Indicator: 01 infrastructures Gender and PWD friendly equipment procured and are available	Contracted companies and suppliers, UNDP, and MoGCSP	Cost already included in Activity 3.1 **$ 100,000**
	3.1.2. Establish and commission the ten-person Community Stewardship Committee (CSC) with representatives from each of the four areas.	Baseline: 0 Target: 5 Indicator: Number of nominated women (out of 10 people on the CSC) and effectively seating in the 4 elected bodies and receiving stipends	UNDP, EPA and the MME, NMDA, MSD and the MoGCSP Community leaders	Cost already included in Activity 3.1 **$ 30,000**
Activity 3.2. Establish community-led co-management agreements to ease	3.2.3 Organise awareness-raising campaign designed and rolled out on sustainable Community-based Natural	Baseline: 0 Target: 1 eco-stove startup Indicator:	The contracted company	Cost already included in Activity 3.2 **$ 82,000**

[76] The Centre will include gender-sensitive facilities such as separate toilets for men and women, spaces for children to sleep to facilitate attendance of women. It will include youth-sensitive facilities such as a small kitchen for child's food preparation. It will also include facilities – such as ramps and guides – for wheelchair-bound and partially-sighted people.

anthropogenic pressure on mangroves in the MMA.	Resource Management (CBNRM) and the co-management agreements.	The knowledge products are gender sensitive in their design and depict the use of mangroves by different social groups and sectors	UNDP, EPA the MoGCSP NGOs and CBOs.	
Activity 3.3. Conduct annual assessments to evaluate the project-induced changes in mangrove degradation, community perceptions and awareness of climate change impacts, adaptation options and mangrove ecosystems throughout the project implementation period.	3.3.2. Contract a national service provider with expertise in the development of alternative, climate-resilient livelihoods **to assess and establish** a baseline of existing livelihood strategies as well as investigating and developing livelihoods for the benefit of women and other vulnerable people in the four target communities.	Baseline: 0 Target: 1 Indicator: A comprehensive survey report that describes existing livelihood adaptative strategies used by men and women and other vulnerable in the 4 areas in the use of mangroves for livelihood. This survey will include considerations beyond eco-stoves and cold storage. Any useful findings from the survey will be integrated into training. This will fall under the role of the Community Stewardship Committee and feed into the Education and Innovation Centre after the project.	UNDP, EPA and the MME, NMDA, MSD and the MoGCSP, Consultant.	Cost already included in Activity 3.3 **$ 80,000**
Activity 3.4. Establish small-scale manufacturing	3.4.1. Contract a national firm or service provider to provide equipment (such as moulds) and initial material	Baseline: 100[77] Target: at least 740[78] eco-stoves Indicator:	UNDP, EPA and the and the MoGCSP Firm / Service provider	Cost already included in Activity 3.4

[77] EPA has been supporting local initiatives on this.
[78] The project will provide resources for 640 eco-friendly cookstoves to be manufactured, split evenly over Year 2 and Year 4.

facilities and develop training material to capacitate community members to manufacture and sell eco-stoves to support alternative climate-resilient livelihoods.	inputs as well as training on the construction of energy efficient stoves through a training-of-trainers approach.	Number of eco-stoves manufactured by community members trained at the education and innovation centre. Baseline: 2% Target: 60% - [79] Indicator: Percentage increase in the number of households using eco-stoves in urban MMA		US$ 70,000
	3.4.2 Request the national firm to organised 01 training per year at the Education and Innovation Centre for women to engage in entrepreneurial development in financing, marketing and selling eco-friendly products to support the uptake of energy efficient products and climate-resilient diversified livelihoods.	Baseline: 0 Target: 1 training per year Indicator(s): Number of trainings per year at the Education and Innovation Centre. Baseline: 0 Target: 125 (50% of total) Indicator(s): Increase in the number of women trained with improved income, access to markets after producing eco-stoves and to finance /credit facility that is socially accessible. Baseline: 0 Target: 40% Indicator:	UNDP, EPA and the MoGCSP CSC members NGOs Micro-finance institutions.	Cost already included in Activity 3.4 **$14,750**

[79] Reference to Liberia NDC the number is eco-stoves to be distributed in urban areas is 280,583 by 2030.

		Proportion of people involved in training at the Education and Innovation Centre who are youth (ages 18-24). Baseline: 0 Target: at least 30% Indicator: Proportion of people involved in training at the Education and Innovation Centre who are from female-headed households. Baseline: 0 Target: 1 starter kit and set of equipment provided to each woman trained Indicator(s): Number of starter kits and equipment provided by the selected national firm/service provider.		
Activity 3.5. Purchase and install low-maintenance eco-friendly cold storage facilities near fish processing sites to reduce pressure on	3.5.1 Include in the TORs of the national consultant with expertise on gender and livelihoods to conduct an assessment on the differentiated use and impact of the cold stores by fishmongers in West Point.	Baseline: 0 Target: 42% women, 28% men of the total 70% of fishmongers Indicator: Percentage of men and women fishmongers in the communities that were able to engage in **innovative ways** of fish	UNDP, EPA, NAFFA and the MoGCSP Consultant	Cost already included **$ 9,500**

mangroves and increase market efficiency.		processing because of the cold storage facility.		
	3.5.2 Include in the TORs of the firm in charge of developing the cold storage to organise trainings for the members of the Community Stewardship Committee (CSC) in West Point to operate and manage the solar powered cold facilities, cold chain and fish conservation after the construction of eco-friendly cold storage facilities	Baseline: 0 % Target: 70% of the fishmongers (42% women and 28% men) Indicator: Number of women fishmongers in the community that uses the affordable cold store facility, for conservation and that have increased their income.	UNDP, EPA, NAFFA and the MoGCSP The contracted firm	**Cost already included $ 70,000**

Timeline indicating activities, targets and milestones for the proposed activities in the GAP. This timeline is aligned with the implementation timetable of the project.

Sub-activity	Year 1				Year 2				Year 3				Year 4				Year 5				Year 6			
	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4	Q 1	Q 2	Q 3	Q 4
1.1.2 Conduct 1 stakeholder consultations to present the final																								

designs to all relevant stakeholders, including communities, NGOs, development organizations and government.																								
1.2.1. Include in the TORs of the construction company the requirements to be in conformity with a socially-sensitive design incorporating the needs of all inhabitants an also conduct a satisfaction survey after Y2.																								
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1.2.1. Provide training to casual workers employed under Activity 1.2. on safety and security during the construction and operational phases in West Point[80]. (Yr. 3, Yr. 4)																								

[80] Safety and security training will be included in the employment contracts. Safety monitoring will be part of contractor's work SOPs and safety equipment should be women-specific and women-friendly.

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curriculum to include social and gender dimensions in addition to technical, administrative and operational elements for training of trainers.																								
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Agency: United Nations Development Programme - Liberia Country Office
UNDP-Liberia, Pan African Plaza, 1st Street,
Sinkor, Tubman Boulevard,
P. O. Box 10-274
Liberia, West Africa.

Author: Chantal Kingue Ekambi (Mrs.)
Gender Specialist/ Consultant
Date: September 2019

Contact details:
Email: chantal.ekambi@undp.org
Telephone Number: +2310770003965
Skype: priso.ndine